

Compact Heat Exchangers

Third Edition

W. M. Kays

Dean of Engineering, Stanford University

A. L. London

Professor of Mechanical Engineering, Stanford University



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Compact Heat
Exchangers

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• **Kerala:** Ponary House, Paudua Raod,
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Nomenclature

Most of the nomenclature is defined as it is introduced or else is obvious from the context of its use. However, it is summarized here for convenience.

Any consistent dimensioning system may be used. All the heat transfer and flow-friction parameters are presented in nondimensional form so that a shift to a preferred system of dimensions presents no complications.

Roman Letter Symbols

A	Exchanger total heat transfer area on one side
A_c	Exchanger minimum free-flow area, or pA_{fr} for matrix surfaces
A_f	Exchanger total fin area on one side
A_{fr}	Exchanger total frontal area
A_k	Cross-sectional area for longitudinal conduction
a	Plate thickness
a	Short side of a rectangular flow passage
b	Plate spacing
b	Long side of a rectangular flow passage
C	Flow-stream capacity rate (Wc_p)
C_c	Flow-stream capacity rate of cold-side fluid
C_h	Flow-stream capacity rate of hot-side fluid
C_L	Coupling-liquid capacity rate
C_{min}	Minimum of C_c or C_h
C_{max}	Maximum of C_c or C_h
C_r	Rotor capacity rate of a rotating periodic flow exchanger (rotor mass times specific heat times r/h)
C_r^*	Rotor capacity-rate ratio (C_r/C_{min}), dimensionless
$\bar{C}_r^*$	($C_r^* \theta_r / \theta_{d,min}$), dimensionless
$\bar{C}$	Fluid heat capacity within exchanger ($C\theta_d$)
$\bar{C}_{min}$	$C\theta_d$ for minimum-capacity-rate fluid
$\bar{C}_w$	Wall total heat capacity (exchanger core mass times specific heat of core material)
$\bar{C}_w^*$	$\bar{C}_w / \bar{C}_{min}$, dimensionless
c	Specific heat
c_p	Specific heat at constant pressure
c_v	Specific heat at constant volume

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D	Inside diameter of a circular tube
D_h	Hydraulic diameter of any internal passage ($D_h = 4r_h = 4A_cL/A$)
d	Outside diameter of a tube in a tube bundle, crossed-rod matrix, or a pin in pin-fin surface
E	Friction power expended per unit of surface heat transfer area [see Eq. (1-2)]
F_G	Correction factor to log-mean rate equation, dimensionless
f	Mean friction factor, defined on the basis of mean surface shear stress [Eq. (1-6)]
f	Fuel-air ratio
f_z	Local friction factor, defined on the basis of local surface shear stress
f_{app}	Apparent mean friction factor [Eq. (6-6)]
G	Exchanger flow-stream mass velocity (W/A_c)
g_c	Proportionality factor in Newton's second law
H/C	Hydrogen-carbon ratio for hydrocarbon fuels
h	Unit conductance for thermal-convection heat transfer
K_c	Contraction loss coefficient for flow at heat exchanger entrance [Eq. (5-1)], dimensionless
K_d	Momentum flux correction factor, [Eq. (6-7)], dimensionless
K_e	Expansion loss coefficient for flow at heat exchanger exit [Eq. (5-2)], dimensionless
k	Unit thermal conductivity
L	Total heat-exchanger flow length; also flow length of uninterrupted fin
l	Fin length from root to center
M	Molecular weight
m	A fin effectiveness parameter $\sqrt{2h/k\delta}$, $\sqrt{4h/kd}$
m	Exponent in Eq. (4-2)
m_o	Slope of operating line (C_c/C_h), dimensionless
n	Number of passes in a multipass heat exchanger
n	Exponent in Eq. (4-1)
P	Pressure
p	Porosity of a matrix surface, dimensionless
q	Heat transfer rate
q''	Heat flux, heat transfer rate per unit of surface area
R	Universal gas constant
R	Heat transfer resistance
R_c	Resistance on the cold-fluid side of a heat exchanger
R_h	Resistance on the hot-fluid side of a heat exchanger
R^*	Heat transfer resistance ratio, (R on C_{min} side)/(R on C_{max} side)
r	A radial coordinate
r_h	Hydraulic radius (A_cL/A) (or $pA_{fr}L/A$ for matrix surfaces)
r_i	Inner radius of an annulus or inner radius of a circular fin
r_o	Outer radius of an annulus or outer radius of a circular fin
r^*	r_i/r_o
T	Absolute temperature
t	Temperature to any arbitrary scale
t_c	Cold-fluid-side temperature
t_h	Hot-fluid-side temperature
U	Unit overall thermal conductance
V	Velocity
V	Volume

v	Specific volume
W	Mass flow rate
X	Parameter in the log-mean rate equation approach to heat exchanger design
X	Parameter in Fig. 3-6
X_c	Specific-heat correction factor for humidity and products of combustion
X_d	Density correction factor for humidity and products of combustion
x	Axial flow coordinate
x°	Axial flow coordinate (x/L), dimensionless
x_l	Longitudinal-tube pitch ratio in a circular tube bank (Fig. 7-5), dimensionless
x_t	Transverse-tube pitch ratio in a circular tube bank (Fig. 7-5), dimensionless
Y	Parameter in Fig. 3-6
Z	Parameter in the log-mean rate equation approach to heat exchanger design
Z	Influence coefficient for annulus heat transfer [Eqs. (6-1) and (6-2)]

Greek Letter Symbols

α	Ratio of total transfer area on one side of the exchanger to total volume of the exchanger For matrix surfaces $\alpha = A/A_f L$ for either one side or both sides
α°	Aspect ratio of a rectangular flow passage (b/a), dimensionless
β	Ratio of total heat transfer area on one side of a plate-fin heat exchanger to the volume between the plates on that side
Δ	Denotes difference
δ	Fin thickness
δ	Denotes difference
ε	Exchanger effectiveness, dimensionless [Eq. (2-6)]
ε_p	Effectiveness of one pass of a multipass heat exchanger, dimensionless
ε_f°	Outlet-fluid temperature response to a step change in one of the fluid inlet temperatures, dimensionless
ε_w°	Wall temperature response at the fluid outlet section to a step change in fluid inlet temperature, dimensionless
Γ	Parameter defined by either Eq. (2-15) or Eq. (2-19), dimensionless
Γ'	Parameter defined by Eq. (2-16), dimensionless
λ	Longitudinal conduction parameter defined by Eq. (2-25), dimensionless
η_f	Fin temperature effectiveness, dimensionless [Eq. (2-4)]
η_o	Total surface temperature effectiveness, dimensionless [Eq. (2-3)]
Φ	Indicates "function of"
σ	Ratio of free-flow area to frontal area, A_c/A_f , dimensionless
μ	Viscosity coefficient
ρ	Density
ω	Absolute humidity
τ_0	Unit surface shear stress
θ	Time
θ	Angular position coordinate in a circular tube (see Fig. 6-1)
θ_d	Dwell time, exchanger residence time for a fluid (L/V)
$\theta_{d,\min}$	θ_d for the $C_{\min}$ fluid
$\theta_{d,\max}$	θ_d for the $C_{\max}$ fluid
θ_r	Rotor rotation period for a periodic-flow heat exchanger
θ°	Generalized time parameter for a direct-transfer exchanger ($\theta/\theta_{d,\min}$), dimensionless

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θ_d^*	Dwell time ratio ($\theta_{d,\min}/\theta_{d,\max}$), dimensionless
θ_R^*	Generalized time parameter for a periodic-flow exchanger (θ/θ_r)

Dimensionless Groupings

Re	Reynolds number ($4r_h G/\mu$), a flow modulus
Re_d	Reynolds number (dG/μ)
St	Stanton number (h/Gc_p), a heat transfer modulus
Nu	Nusselt number ($h4r_h/k$), a heat transfer modulus
Pr	Prandtl number ($\mu c_p/k$), a fluid properties modulus
N_{tu}	Number of heat transfer units of an exchanger, a heat transfer parameter ($AU/C_{\min}$); more formally defined by Eq. (2-7)
$C_{\min}/C_{\max}$	Flow-stream capacity-rate ratio [$(Wc_p)_{\min}/(Wc_p)_{\max}$]

Subscripts

<i>a</i>	Air side
<i>av</i>	Average
<i>c</i>	Cold-fluid side of heat exchanger
<i>h</i>	Hot-fluid side of heat exchanger
<i>i</i>	Refers to inner surface of an annular passage or inner radius of a circular fin
<i>L</i>	Coupling liquid in a liquid-coupled heat exchanger
<i>m</i>	Mean conditions, defined as used
<i>o</i>	Refers to conditions at surface, or specifically to inner surface of an annular passage or inner radius of a circular fin
<i>p</i>	Refers to one pass of a multipass heat exchanger
<i>r</i>	Matrix rotor
<i>w</i>	Wall; water side
<i>x</i>	Local conditions
∞	Conditions far downstream
<i>ii</i>	Conditions at inner surface of an annular passage when the inner surface alone is heated
<i>oo</i>	Conditions at outer surface of an annular passage when the outer surface alone is heated
1,2	Indicate different sides of the heat exchanger; inlet and outlet conditions
max	Maximum
min	Minimum
<i>lma</i>	Log mean average

The following symbols and systems of units are used:

	English	SI
Mass	lb _m (pound mass)	kg (kilogram)
Force	lb _f (pound force)	N (newton, kg/(m·s ²))
Length	ft, in (foot, inch)	m (meter)
Time	s, h (second, hour)	s (second)
Thermal energy	Btu (British thermal unit)	J (joule, N·m)
Power	Btu/s, Btu/h, hp (horsepower)	W (watt, J/s)
Temperature	°F, °R (degree Fahrenheit, degree Rankine)	K, °C (kelvin, degree Celsius)
Pressure	lb _f /ft ²	Pa (pascal, N/m ²)

Under the **English System**, the proportionality factor in Newton's second law becomes $g_c = 32.2 \text{ lb}_m \cdot \text{ft}/(\text{lb}_f \cdot \text{s}^2)$, and the universal gas constant becomes $R = 1,545 \text{ ft} \cdot \text{lb}_f/(\text{lb}_m \cdot \text{mol} \cdot \text{R})$. In the **SI**, g_c has the value of unity and is dimensionless. The universal gas constant is $R = 8,314 \text{ J}/(\text{kmol} \cdot \text{K})$.

Note the dimensions consistently employed for the following properties, coefficients, and variables:

	English	SI
Density ρ	lb_m/ft^3	kg/m^3
Viscosity μ	$\text{lb}_m/(\text{h} \cdot \text{ft})$	$\text{Pa} \cdot \text{s}, \text{N} \cdot \text{s}/\text{m}^2$
Thermal conductivity k	$\text{Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{ft})$	$\text{W}/(\text{m} \cdot \text{K})$
Specific heat c_p	$\text{Btu}/(\text{lb}_m \cdot ^\circ\text{F})$	$\text{kJ}/(\text{kg} \cdot \text{K})$
Heat transfer coefficient h	$\text{Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})$	$\text{W}/(\text{m}^2 \cdot \text{K})$
Heat flux q''	$\text{Btu}/(\text{h} \cdot \text{ft}^2)$	W/m^2
Mass flow rate $\dot{W}$	lb_m/h	kg/s
Molecular weight M	$\text{lb}_m/(\text{lb}_m \cdot \text{mole})$	kg/kmol

Table A-15 in App. A contains a set of conversion factors for shifting from one system of units to another, including some of the commonly used archaic units.

Preface to the Reprint Edition

Since the 1984 edition went out of print about three years ago the authors have received a continuous stream of letters and phone calls, primarily from people engaged in design of compact heat exchangers, or in design studies, all wanting to know how they could purchase or otherwise get their hands on a copy. It soon became evident that there exists really no adequate substitute for this compilation of design data.

This reprint is essentially the same as the previous 1984 edition (with errors corrected) and thus does not include any data obtained since that date, although the amount of such recent data appears to be small. So it does still include the largest collection of compact heat exchanger data available in the world today and is still unique in that all of these data are presented in a consistent manner so that comparisons can be readily made. A large portion of these data was obtained in a single test system which simply adds to comparability as well as reliability. The availability of a large collection of data for heat transfer surfaces with relatively small geometric differences also makes it feasible to infer with some confidence the behavior of other, non-tested, heat transfer surfaces by interpolation.

Until a more complete compilation of data becomes available it is our hope that this edition will satisfy the needs of heat exchanger designers in all those various fields involving heat transfer to and from gases where light weight and small size are of critical importance.

W. M. Kays

Preface to the Third Edition

It is now 19 years since the second edition of *Compact Heat Exchangers* was published. In the intervening years manufacturing techniques have been developed to fabricate some of the older compact surface configurations using high-temperature materials, new surface configurations with superior flow characteristics have been manufactured, and the area per unit volume has been increased substantially. The research program which generated the original data on which this book is based was slowed down and eventually terminated, but a number of the newer surfaces were tested using essentially the same techniques as were used for the earlier work.

In the meantime new research on some of the theoretical solutions for flow in the simple geometries has rendered obsolete some of the solutions presented in the second edition. The same thing can be said of the solutions for the transient behavior of heat exchangers.

The availability of new data and more modern solutions suggested that the time was appropriate for a new edition. This edition does not differ radically from its predecessor but it does contain the basic test data for 11 new surface configurations, including some of the very compact ceramic matrices. In addition to modernization of the theoretical solutions and correlations for simple geometries, and the transient solutions, a number of other improvements will be found. Finally, the slow conversion (at least in the United States) to the *Système Internationale* (SI) system of units suggested that the time had come to make that conversion in *Compact Heat Exchangers*. Since the English system is apparently destined to disappear only slowly in the United States, it was decided to introduce a dual system of units in the new edition. So all dimensions are given in both systems, and the fluid properties in the Appendix are likewise presented in both systems.

A unique feature of *Compact Heat Exchangers* has always been that virtually all of the basic test data originate from a single research program under the supervision of the authors. There is thus no question about the comparability of the test results of one surface to another. In recent years additional data have been obtained by others, but the authors have chosen to maintain the original tradition so that there is almost complete internal consistency.

W. M. Kays

A. L. London

Preface to the Second Edition

For many years the only generally available basic heat transfer and flow-friction data of sufficient accuracy for heat-exchanger design was for flow through and over banks of circular tubes. The need for small-size and lightweight heat exchangers in all varieties of powered vehicles from automobiles to spacecraft, as well as in a multitude of other applications, has resulted in the development of many heat transfer surfaces that are much more compact than can be practically realized with circular tubes. In addition, many of these surfaces possess other characteristics that are superior to circular tubes. However, lack of basic heat transfer and flow-friction design data, and a lack of understanding of the basic mechanisms involved, for a long period of time restricted their use to heat exchangers that could be developed by cut-and-try methods. It ultimately became apparent that rationally optimized heat-exchanger design, the development of new surfaces of superior characteristics, and the development of methods of fabrication of compact surfaces for high-temperature service could only take place after the basic characteristics of the already existing surfaces were known and understood.

Recognizing the need for such data the U.S. Navy Bureau of Ships initiated in 1945 a test program at the Naval Engineering Experiment Station, Annapolis, Maryland. In 1947, the Office of Naval Research, in cooperation with the Bureaus of Ships and Aeronautics, extended this work by establishing a similar program at Stanford University. Later the Atomic Energy Commission joined in support.

A number of manufacturers provided test cores for these investigations, and the authors acknowledge especially the cooperation of the Harrison Radiator Division of General Motors, Lockport, New York; the Modine Manufacturing Company, Racine, Wisconsin; The Trane Company, LaCrosse, Wisconsin; The AiResearch Manufacturing Company, Los Angeles, California; and The Ferrotherm Company, Cleveland, Ohio.

Most of the test cores were of low-temperature construction employing soldering or brazing techniques. However, the primary objective of this program was to investigate the effects of geometry on convective heat transfer and flow-friction performance, with the hope that the geometrical advantages would provide incentive for the development of high-temperature fabrication techniques and of new superior surfaces. Since the first publications of the results of the program, both kinds of developments have indeed occurred.

The American Society of Mechanical Engineers published the first results of the program in 1951 in a monograph entitled *Gas Turbine Plant Heat Exchangers—Basic Heat Transfer and Flow Friction Design Data*, by W. M. Kays, A. L. London, and D. W. Johnson. In 1955, *Compact Heat Exchangers*, by W. M. Kays and A. L. London, was published; it contained a considerable additional body of basic data from the test program, as well as data from other investigators. Following the publication of *Compact Heat Exchangers*, the test program was continued, and new test cores were obtained, some of which were developed directly as a result of the earlier work. This second edition of *Compact Heat Exchangers* contains all the new basic data that have been obtained, as well as extensive revisions and additions to the chapters on analytic solutions for flow in tubes, an extension of the chapter on heat exchanger design theory, and a new chapter on the transient behavior of heat exchangers; various other sections have been brought up to date in the light of more recent research. The basic data section has been expanded to include the characteristics of 25 new surfaces, and this section, reporting the characteristics of more than 90 surfaces, remains the real core of the book.

Although too numerous to name specifically, the authors take this opportunity to acknowledge the assistance over the past 15 years of the approximately 60 Stanford University mechanical engineering students who participated in various phases of the test program. Without their assistance this book could never have been written.

W. M. Kays

A. L. London

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1

Introduction

The design of a heat exchanger involves a consideration of both the heat transfer rates between the fluids and the mechanical pumping power expended to overcome fluid friction and move the fluids through the heat exchanger. For a heat exchanger operating with high-density fluids, the friction-power expenditure is generally small relative to the heat transfer rate, with the result that the friction-power expenditure is seldom of controlling influence. However, for low-density fluids, such as gases, it is very easy to expend as much mechanical energy in overcoming friction power as is transferred as heat. And it should be remembered that in most thermal power systems mechanical energy is worth 4 to 10 times as much as its equivalent in heat.

It can be readily shown that for most flow passages that might be used for the heat transfer surfaces of an exchanger, the heat transfer rate per unit of surface area can be increased by increasing fluid-flow velocity, and this rate varies as something less than the first power of the velocity. The friction-power expenditure is also increased with flow velocity, but in this case the power varies by as much as the cube of the velocity and never less than the square. It is this behavior that allows the designer to match both heat transfer rate and friction (pressure-drop) specifications, and it is this behavior that dictates many of the characteristics of different classes of heat exchangers.

If the friction-power expenditure in a particular application tends to be high, the designer can reduce flow velocities by increasing the number of flow passages in the heat exchanger. This will also decrease the heat transfer rate per unit of surface area, but according to the above relations the reduction in heat transfer rate will be considerably less than the friction-power reduction. The loss of heat transfer rate is then made up by increasing the surface area (lengthening the tubes), which in turn also increases the friction-power expenditure, but only in the same proportion as the heat transfer surface area is increased.

In gas-flow heat exchangers the friction-power limitations generally force the designer to arrange for moderately low mass velocities. Low mass velocities, together with the low thermal conductivities of gases (low relative to most

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liquids), result in low heat transfer rates per unit of surface area. Thus large amounts of surface area become a typical characteristic of gas-flow heat exchangers. Gas-to-gas heat exchangers may require up to 10 times the surface area of condensers or evaporators or liquid-to-liquid heat exchangers in which the total heat transfer rates and pumping-power requirements are comparable. For example, a regenerator for a gas-turbine plant, if it is to be effective, requires several times as much heat transfer surface as the combined boiler and condenser in a steam power plant of comparable power capacity.

These considerations have led to the development of many ways to construct heat transfer surfaces for gas-flow applications in which the surface area density is large. Such surfaces will be referred to here as *compact heat transfer surfaces*. Several typical compact heat transfer surface arrangements are illustrated in Fig. 1-1.

Perhaps the simplest and most common surface arrangement for a two-fluid heat exchanger is the circular tube bundle shown in Fig. 1-1*a*. This arrangement, of course, has long been used for both high- and low-density fluids, but the only way in which surface area density can be substantially increased is to decrease the diameter of the tubes. Fabrication difficulties and cost place a rather severe limitation on what can be accomplished in this direction, and large heat exchangers with tube diameters of less than $\frac{1}{4}$ in. (0.006 m) are rare.

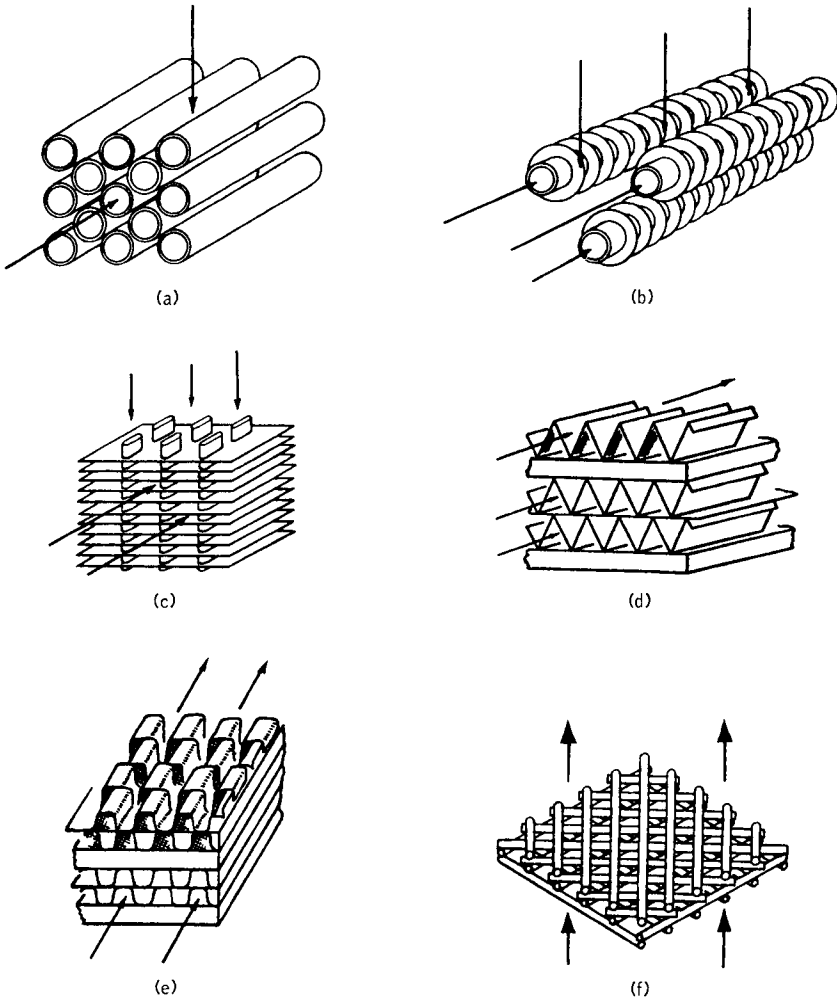
An effective way to increase surface area density is to make use of secondary surfaces, or fins, on one or both fluid sides of the surface. Figure 1-1*b* illustrates a finned circular tube surface in which circular fins have been attached to the outside of circular tubes. Such an arrangement is frequently used in gas-to-liquid heat exchangers where optimum design demands a maximum of surface area on the gas side. Fins *could* be used in a liquid-to-liquid heat exchanger, or on the liquid side of a gas-to-liquid heat exchanger, but here another difficulty arises. The low friction-power requirement characteristic of high-density fluids, together with the relatively high thermal conductivity of liquids, results in high convection heat transfer rates in any optimum design (high heat transfer coefficients). If fins are employed, the high heat transfer rates must be conducted along the fins, and the conduction resistance may destroy all or most of the advantage of the extra surface area gained (see the discussion of fin effectiveness in Chap. 2).

Another popular variation of the finned-tube arrangement is shown in Fig. 1-1*c*. Here the tubes are illustrated as flat, but they can also be circular.

In compact gas-to-gas heat exchangers, large area density is desirable on *both* fluid sides, and a method for accomplishing this objective with fins is illustrated by the plate-fin arrangement of Figs. 1-1*d* and *e*. The heat exchanger is built up as a sandwich of flat plates bonded to interconnecting fins. The two fluids are carried between alternate pairs of plates and can be arranged in either counter-flow or crossflow, which provides an added degree of flexibility in this arrangement.

Figure 1-1*e* also illustrates another variation; the fins can be interrupted rather than continuous, an arrangement which alters the basic convection heat transfer and flow-friction characteristics in a manner that will be discussed presently.

Fig. 1-1 Some typical examples of compact heat exchanger surfaces.



In the periodic-flow-type heat exchanger, energy is transferred by convection and stored in a matrix, from which it is later given up to the other fluid. Figure 1-1f illustrates one such compact matrix, which could be built up of stacks of solid rods or stacks of wire screens. Matrices can also be constructed using stacks of plates and fins or simply packed bundles of tubes. Some of the most common matrices are made using glass ceramic materials.

An interesting and important feature of the compact heat transfer surfaces illustrated in Fig. 1-1 can be demonstrated if the heat transfer rate per unit of surface area is plotted as a function of the mechanical power expended to

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overcome fluid friction per unit of surface area. Such a plot for three different surfaces is shown in Fig. 1-2. The heat transfer rate for a unit of area and for one degree of temperature difference is merely the heat transfer coefficient h evaluated for some particular set of fluid properties from

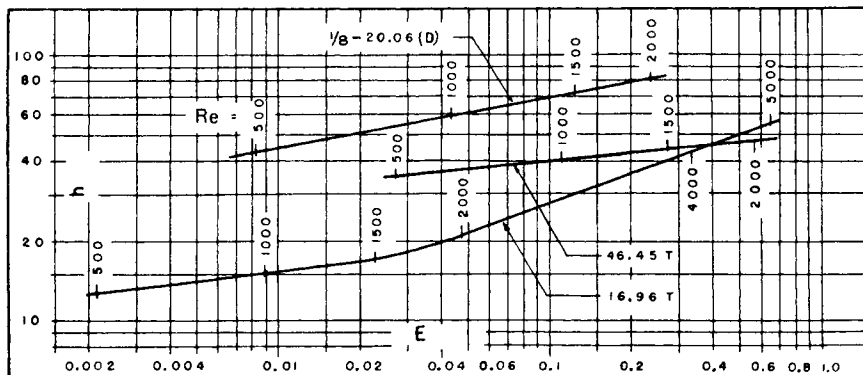
$$h = \frac{c_p \mu}{Pr^{2/3}} \frac{1}{4r_h} (St Pr^{2/3}) Re \quad (1-1)$$

The friction power expended per unit of surface area can be readily evaluated as a function of the Reynolds number, the friction factor, and the specified fluid properties from

$$E = \frac{1}{2g_c} \frac{\mu^3}{\rho^2} \left(\frac{1}{4r_h} \right)^3 f Re^3 \quad (1-2)$$

A plot of h versus E can be prepared once the basic convection heat transfer and friction characteristics are known as functions of the Reynolds number. Any particular surface arrangement is then represented by a single curve on a plot such as that in Fig. 1-2 [for fluid properties of air at 1 atm and 500°F (260°C)].

Fig. 1-2 A comparison of heat transfer and friction-power characteristics of three compact surfaces on a unit of surface area basis. Dimensions are in Btu/(h · ft² · °F) for h and in hp/ft² for E . Geometrical descriptions of the surfaces are provided in Chap. 9.



The interesting feature of this plot is the very wide difference in friction-power expenditure for a given heat flux for different surfaces, or conversely, the smaller difference in heat flux for a given friction-power expenditure. At the beginning of this chapter the important influence of friction-power expenditure on heat exchanger design was discussed, and in gas-flow heat exchangers it is the necessity to minimize friction power that forces the use of large amounts of surface area. This in turn has resulted in the development of more compact heat

transfer surfaces, but it is apparent from Fig. 1-2 that another way to minimize friction power is to select surfaces that plot “high” on a heat transfer–friction power plot, such as Fig. 1-2.

It should be emphasized that selection of a surface configuration for a particular application is by no means as simple as this, for there are many additional considerations. However, other things being equal, three heat exchangers designed for identical thermal and pressure-drop performance and using the three surfaces represented on Fig. 1-2 will have quite different volumes and weights, and the smallest and lightest will be based on the surface configuration designated $\frac{1}{8}$ -20.06(D), the upper curve. However, this design will probably have a significantly larger frontal area, even though volume is smallest, and this may or may not be desirable.

A surface which has a basic characteristic of high heat flux relative to friction-power expenditure will be termed a *high-performance surface*. There remains the question of why some heat transfer surfaces have this characteristic.

It should first be noted that compactness itself leads to high performance. A compact surface has small flow passages, and the heat transfer coefficient h always varies as a negative power of the hydraulic diameter of the passage. Thus compact surfaces tend, by their very nature, to have high heat transfer coefficients; this leads to high-performance curves on the heat transfer–friction power plot, despite the influence of small hydraulic diameter on the friction power, as might be noted in Eq. (1-2).

In addition to the influence of small hydraulic diameter, however, high-performance characteristics can be obtained by any modification of the surface geometry that results in a higher heat transfer coefficient at a given flow velocity. One of the most widely used ways of increasing conductance is to interrupt the wall surfaces so that the boundary layers can never become thick. Figure 1-1 illustrates such an interrupted surface. Finned surfaces lend themselves especially conveniently to such treatment. Surface interruption also increases friction factors, but a small increase in heat transfer coefficient can more than offset a large friction factor increase, because flow velocity can then be decreased and friction power varies with as much as the cube of the velocity.

Other methods of obtaining high performance by change of flow-tube geometry include the use of curved or wavy passages in which boundary-layer separation is induced. A tube bank in which a fluid flows normal to the tubes is a high-performance surface, since a new boundary layer builds on each tube, and the result is much higher heat transfer coefficients than can be obtained by flow of the same velocity through the inside of the tubes. Various types of inserts (*turbulence promoters*) are frequently used inside a tube to increase the heat transfer coefficient, but this scheme is not nearly as effective as directly interrupting the boundary-layer sublayers on the heat transfer surface.

It is clear that compact high-performance heat transfer surfaces can be manufactured in a very large variety of geometrical configurations and that, for many applications, the most desirable surfaces are those of very complex geometry. Only for the geometrically simple surface does a completely analytic treatment to establish the basic characteristics appear feasible; for all others, the

6 Compact Heat Exchangers

basic characteristics can be reliably established only by experiments and by making use of model laws to extend the range of applicability of the results.

The primary objectives of this book are to present the results of a very large number of such experiments, employing a single consistent method of presentation. Chapter 10 contains these experimental data in tables and in 132 graphs, covering a sufficiently wide range of geometrical configurations so that it is possible to deduce at least approximately the characteristics of many others merely by interpolation. The complete set of geometries covered is assembled for quick reference and comparison in Figs. 9-1 to 9-17.

An interesting feature of interrupted-fin surfaces is that their range of usefulness tends to be limited to a range of Reynolds numbers which for a smooth tube would extend from the high laminar region to the low Reynolds number turbulent region. Typically this implies Reynolds numbers (based on hydraulic diameter) from about 500 to 15,000, which of course includes the transition region. Outside of this range the interrupted-fin surfaces tend to lose their advantage and smooth passages serve just as well. It develops that a great many gas-flow applications fall in this range of Reynolds numbers for reasons having to do with the amount of friction power it is economical to expend relative to the heat transferred (this is associated with minimizing thermodynamic irreversibility), and with the practicability of manufacturing small, compact flow passages. With the advent of the manufacturability of very compact matrices for periodic-flow heat exchangers, made often from glass ceramics, the Reynolds number range of interest tends to extend to quite low laminar-flow Reynolds numbers. In such cases the interrupted-fin "high-performance" surfaces no longer show an advantage, and simple smooth passages are generally preferred. Since the flow passages tend to have very large length-to-diameter ratios, the flow is usually "fully developed," and the simple analytic solutions for fully developed flow become applicable.

SOURCES OF DATA

The great bulk of the experimental data presented here was obtained directly from two research programs. The first took place at the U.S. Navy Engineering Experiment Station and was supported by the U.S. Navy Bureau of Ships. A description of the experimental apparatus and the method of data reduction is given in London and Ferguson [1]. The second program was at Stanford University under the joint sponsorship of the Office of Naval Research, the Bureau of Ships, the Bureau of Aeronautics, and the Atomic Energy Commission. The experimental apparatus is described by Kays and London [2]. Actually, the experimental systems were quite similar, and cross-checks using the same test cores indicated that they yielded very consistent results.

The matrix surfaces were tested by a totally different technique. The test system and method of reduction of data are described by Wheeler [3].

The complete test data are available in a series of project reports. Only the summarized data are presented here.

A certain amount of data from other investigators is also included and is referenced as it appears. In general, the authors have preferred to include only data obtained by using approximately the same procedures as the projects mentioned above, so that a truly consistent treatment is possible. In some cases, the authors have reevaluated the data of others, again for consistency of treatment.

A considerable number of analytic convection solutions pertinent to compact heat exchangers have been included, and the sources of these are referenced as they appear.

All the experimental data presented have been obtained from experiments using air ($Pr = 0.7$) as a working fluid, since the major interest in compact heat exchangers is for gas-flow applications. The applicability of the data to fluids outside the gas Prandtl number range may be open to some question and depends upon the particular geometry and what is known of the influence of the Prandtl number. The analytic solutions are more general in this regard and, for some cases, cover the complete Prandtl number spectrum ranging from the liquid metals through gases, water, and viscous liquids.

METHOD OF PRESENTATION OF BASIC TEST DATA

One of the present objectives is to use a common treatment of basic heat transfer and flow-friction design data for all the surfaces considered, so as to avoid the confusion often encountered with a large number of arbitrarily defined parameters. It is quite feasible to use the same definition of parameters like friction factor or hydraulic diameter for such widely differing surface geometries as flow through tubes and flow normal to banks of tubes; the resulting simplification for the heat exchanger designer is quite apparent. For all the data presented the following format is employed.

The basic heat transfer and flow-friction performance data for each surface are presented in both tabular and graphical form as

$$StPr^{2/3} = \phi_1(Re) \quad (1-3)$$

$$f = \phi_2(Re) \quad (1-4)$$

The Prandtl number was not a test variable, but the two-thirds power of the Prandtl number is included as an approximation over a moderate range of Prandtl numbers and should be at least adequate for all gases. A large number of the surfaces considered are of the interrupted-fin variety, with a laminar boundary layer on a major part of the surface. The analytic solutions for laminar-boundary-layer heat transfer indicate that the Prandtl number enters as about the two-thirds power for the Prandtl number range 0.5 to 15. For turbulent flow in tubes, the available analytic solutions suggest that in the gas range the Prandtl number enters as nearer the one-half power, but the two-thirds power has been retained here for consistency and will lead to little error in the gas range of Prandtl numbers (0.5 to 1.0). For laminar flow in long tubes, the Prandtl number effect is closer to a 1.0 power, but for finite tubes of the length typically

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employed in heat exchangers, the two-thirds power is again near correct. The two-thirds power is thus a reasonable compromise and does allow presentation of the complete characteristics of a surface on a single graph where the Reynolds number range covers both laminar and turbulent flow.

There are a few exceptions to the above method of presentation. The Nusselt number (Nu) is employed as a heat transfer parameter in place of the Stanton number (St) in the analytic solutions given in Chap. 6. In the summaries of analysis and experiment in Chap. 7, the length-to-diameter ratio of a tube, L/D or $L/4r_h$, is employed as an additional parameter in certain cases. Also, a temperature ratio, T_w/T_m , is employed in some cases in the summaries of Chap. 7. The reasons for using a temperature ratio are fully discussed in Chap. 4, where the effects of temperature-dependent fluid properties are considered.

The mass velocity G in the Stanton and Reynolds numbers is, in all cases except matrix surfaces, evaluated on the basis of the *minimum* free-flow area A_c , regardless of where this minimum occurs in the passage. Thus $G = W/A_c$. In the case of matrices, $G = W/pA_F$, where the porosity p and the frontal area A_F combine to yield an "effective" free-flow area pA_F corresponding to A_c .

The Reynolds number is based on a hydraulic diameter defined as

$$\frac{D_h}{L} = \frac{4r_h}{L} = 4 \frac{A_c}{A} \quad (1-5)$$

where L is the flow length of the heat exchanger, A_c is the flow cross-sectional area, and A is the total heat transfer area. For flow normal to tube banks, L is an equivalent flow length measured from the leading edge of the first tube row to the leading edge of a tube row that would follow the last tube row, were another tube row present.

For any cylindrical tube, Eq. (1-5) reduces to the conventional definition of hydraulic diameter, i.e., four times *cross-sectional area divided by wetted perimeter*. For a tube of circular cross section, $4r_h$ is simply tube diameter. The use of this parameter as the geometry dimension in the Reynolds number in no way implies that the basic heat transfer and flow-friction performance of different geometric configurations can be correlated thereby. In fact, the heat transfer performance for flow normal to banks of circular tubes with differing patterns can be better correlated by using the outside tube diameter in the Reynolds number if correlation is what is desired. However, the present objective is a simple and consistent treatment, and the above definition of hydraulic diameter can be applied to any kind of interior heat transfer surface without ambiguity, providing that one does not attempt to apply the test data from one surface to a geometrically dissimilar surface. Furthermore, in heat exchanger design work, $A/A_c = L/r_h$ is a directly useful parameter in its own right.

Friction factor is defined on the basis of an equivalent shear force in the flow direction per unit of heat transfer (or friction) area. Whether this equivalent shear force is a true viscous shear or is primarily a pressure force, as in the case of tube banks, is of no consequence. For most of the surfaces considered it is a combination of viscous shear (skin friction) and pressure force (form drag), but

there is no reason in design work to attempt to separate these effects. With such a definition of the friction factor, a common treatment of all types of surfaces is possible. Thus

$$\tau_0 = \frac{f}{\rho} \frac{G^2}{2g_c} \quad \text{or} \quad f = \frac{\rho \tau_0}{G^2/2g_c} \quad (1-6)^\dagger$$

For flow through cylindrical tubes, f in Eq. (1-6) is the same as the conventional Fanning friction factor and is also identical to the conventional drag coefficient for flow along flat surfaces. To determine pressure drop in a heat exchanger core, there are other effects besides flow friction, and a complete force-momentum equation, including the friction factor term, is given in Chap. 2. It is again emphasized that this definition of the friction factor, and the integrated form of the force-momentum equation as given in Chap. 2, applies equally well to flow through tubes and flow normal to banks of tubes of any type.

The fluid properties in all the experimentally determined basic heat transfer and flow-friction design data were evaluated at a bulk average fluid temperature. The effects of temperature-dependent fluid properties are discussed in Chap. 4.

SUMMARY OF CONTENTS

Chapter 2 contains a summary of the design theory necessary to predict the heat transfer and pressure-drop behavior of a heat exchanger. Included are the solutions, in both tabular and graphical form, for a large number of heat exchanger flow arrangements, presented as effectiveness, ε , versus number of heat transfer units, N_{tu} . Equations for the evaluation of heat-exchanger-core pressure drop are also presented.

The transient behavior of heat exchangers is considered in Chap. 3. Included are graphs from which thermal lags can be evaluated for several common types of heat exchangers and also for ducting or piping.

The effects of temperature-dependent fluid properties on heat transfer and flow friction in gas-flow heat exchangers are discussed in Chap. 4. Included are recommendations for corrections to be made for these effects during design.

Chapter 5 contains graphs of heat-exchanger-core abrupt contraction and expansion pressure-loss coefficients for various types of surface geometries. These are used to evaluate pressure losses at the core entrance and exit arising from boundary-layer separation.

A number of analytic solutions for heat transfer and flow friction in smooth tubes are summarized in Chap. 6. Extensive data on the concentric-circular-tube annulus are presented, including procedures for handling asymmetric heating from the two surfaces of an annulus. Rectangular and triangular tubes are also treated, although somewhat less completely. The analytic solutions presented in

$^\dagger g_c$ in Eq. (1-6) is the proportionality factor in Newton's second law. In the SI system of dimensions g_c has the value of 1.0 and is dimensionless. In English units $g_c = 32.2 \text{ (lb}_m \cdot \text{ft)/(lb}_f \cdot \text{s}^2\text{)}$.

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Chap. 6 are not restricted to the Prandtl number range of gases, as are the experimental data which form the main body of this book, and thus may be used for liquids, including liquid metals.

For certain heat transfer surfaces of simple flow geometry, sufficient data have been obtained, both experimental and analytic, so that a generalized treatment can be presented. Summaries of data for gas flow through circular tubes and rectangular tubes of various aspect ratio, flow normal to banks of staggered circular tubes, and flow through stacked-screen matrices are presented in Chap. 7. Where applicable, these data are recommended for design purposes rather than the experimental data of Chap. 10, except for those cases where the design surface is identical to a particular surface considered in Chap. 10.

In Chap. 8 a brief description is given of the test apparatus and methods used to obtain the data in Chap. 10.

Chapter 9 contains a summary of all the pertinent geometrical data for the test surfaces of Chap. 10. These data are presented in tables; diagrams of all the test surfaces are also shown.

Chapter 10 comprises the main body of this book. Basic heat transfer and flow-friction test data for a wide variety of compact high-performance surfaces are presented in 132 graphs.

Appendix A contains a compilation of fluid properties useful in application of compact heat transfer surfaces.

Appendix B contains several illustrative heat exchanger analyses to demonstrate a procedure for using the design data.

Appendix C presents in detail the derivation of some of the effectiveness- N_{tu} relationships presented in Chap. 2.

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2. Kays, W. M., and A. L. London: Heat Transfer and Flow Friction Characteristics of Some Compact Heat Exchanger Surfaces — Part I: Test System and Procedure, *Trans. ASME*, vol. 72, pp. 1075–1085, 1950.
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2

Heat Exchanger Thermal and Pressure-Drop Design

The purpose of this chapter is to summarize exchanger heat transfer and flow-friction performance theory. Designers use this theory in conjunction with the basic design data reported in Chaps. 6, 7, and 10 to size the heat exchanger core for a specified heat transfer duty and pressure drop. This theory also provides designers with the equations to use in optimizing their designs on the basis of whatever criteria they may select for this purpose.

Details of the derivation of the final equations are omitted here but are included for some cases in Appendix C to illustrate the general method of approach. Where available, literature references are provided for the cases not covered in Appendix C. Major emphasis is placed on the basic principles and the conceptual ideas employed, together with the engineering significance of the final relationships. These are presented in the form of graphs and tables useful for design and also in algebraic form when feasible.

Appendix B contains applications of the information presented in this chapter to illustrate the prediction of performance of gas-turbine-plant heat exchangers. Throughout this section frequent references are made to the gas-turbine-plant regenerator and intercooler applications, as these are excellent examples of heat exchangers requiring a refined design. It needs to be emphasized, however, that the techniques and procedures presented here are broadly applicable to the design of all classes of unfired heat exchangers.

To summarize briefly, this chapter contains a consideration of the heat transfer theory of three types of exchanger systems, as follows:

1. The conventional *direct-transfer type*, in which the two fluids exchanging thermal energy are separated by the heat transfer surface.
2. The *liquid-coupled indirect-transfer type*, which consists essentially of two direct-transfer units coupled with a pumped heat transfer medium. The transfer fluid circulates between the hot-fluid exchanger, where the thermal energy is picked up, and the cold-fluid exchanger, where the thermal energy is used to heat the cold fluid. The conventional automobile engine block and air-cooled radiator system is a common example of a liquid-cou-

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pled indirect-transfer exchanger; in this case, heat is transferred indirectly from the combustion products to cooling air.

3. The *periodic-flow type*, like the common Ljungstrom air preheater, consists of a matrix heat transfer surface which is rotated so that an element is periodically passed from the hot to the cold flow streams and back again. As the hot fluid passes through the matrix, the fluid is cooled and the matrix is heated. In the cold-side part of the cycle the cold fluid is heated and the matrix cooled.

After the development of the heat transfer theory, the prediction of flow pressure drop is considered. Then a short treatment follows on how one can go about the task of surface selection for a given application based on a coupling of the heat transfer and pressure-drop behavior.

Some considerations of the design of flow headers are provided. Also, for convenience, useful geometrical relations for flow-area, heat transfer area and detailed core geometry are provided:

The chapter is concluded with a methodology for sizing a core for given heat transfer and pressure-drop performance.

A COMPREHENSIVE DESIGN PROCEDURE

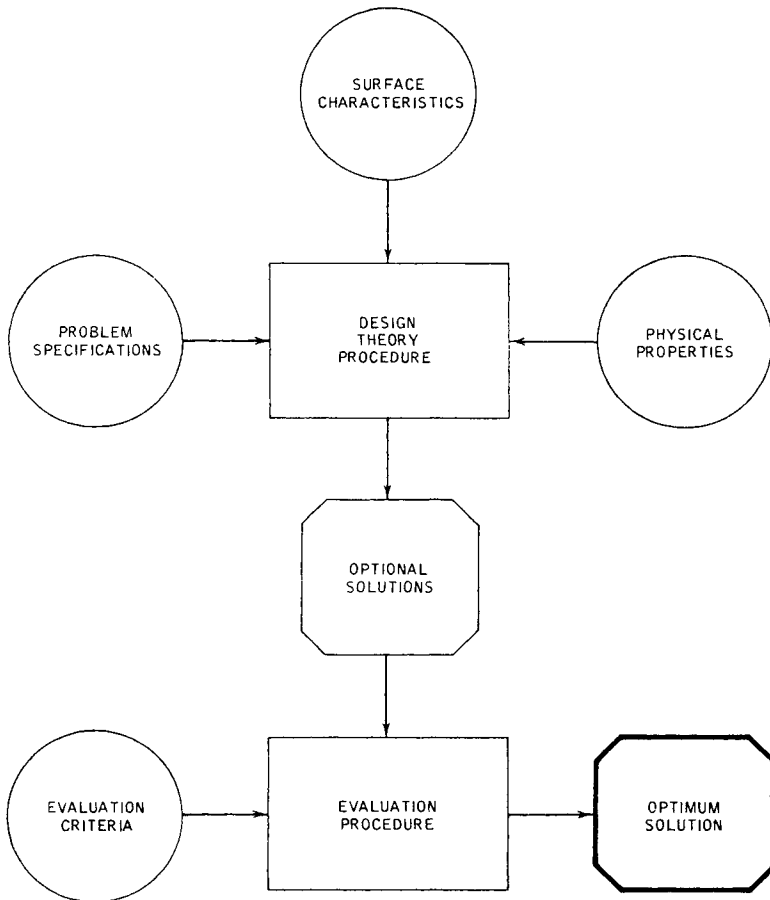
The methodology of arriving at an optimum heat exchanger design is a complex one, not only because of the arithmetic involved, but more particularly because of the many qualitative judgments that must be introduced. To illustrate the design procedure in a schematic fashion, Fig. 2-1, modified from a presentation by Mason [1], is given.

The *inputs to the design-theory procedure* include, along with the problem statement or specifications, the surface heat transfer and flow-friction design characteristics and information on physical properties. These last two inputs are considered in Chaps. 4 to 10 and Appendix A. The problem statement may specify a consideration of different exchangers, as, for instance, periodic-flow and direct-transfer types. Several or many surface geometries may be used. Some options may be allowed in the physical properties; for example, the matrix material to be used in a periodic-flow-type exchanger.

The design-theory procedure is the main subject of this chapter. It can be set up in a computer program; if it is, the *output* may be a large number of *optional solutions*. Some of these optional solutions may represent an estimate of what a competitor may offer and others may represent customers' suggestions. These optional solutions, plus evaluation criteria, serve as the inputs to the *evaluation procedure*.

The evaluation procedure is in a large measure qualitative; brazing furnace size, shipping limitations, delivery dates, company policy, and an estimate of the strength of the competition are all examples of qualitative evaluation criteria. In contrast, trade-off factors may be developed to quantitatively weigh the relative costs of pressure drop, weight, heat transfer performance, and leakage (in the case of the periodic-flow exchangers). A rather startling example of such trade-off factors is the estimate that a saving of 1 kg is worth \$5,000 in a commercial aircraft application.

Fig. 2-1 Methodology of heat exchanger design.



The final output is an *optimum design* or, possibly, several such designs to submit to the customer. Alternatively, the final output may be used to formulate a new problem input statement for a parametric study leading to an optimum overall system rather than just an optimum heat exchanger based on somewhat arbitrary initial specifications of requirements.

Clearly, a comprehensive design procedure of any generality cannot be presented in a monograph of this type. What is attempted here is to provide the quantitative inputs of surface characteristics and a sampling of typical physical properties information, together with a systematic development of the design theory. The methodology presented in Fig. 2-1 will be considered again prior to the illustrative procedure for sizing a heat exchanger that concludes this chapter.

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HEAT TRANSFER IN DIRECT-TYPE EXCHANGER

Exchanger Variables—Dimensional and Nondimensional

For the conventional two-fluid heat exchanger, the parameters relating to heat transfer performance are as follows:

U = overall conductance for heat transfer, $\text{Btu}/(\text{h} \cdot ^\circ\text{F} \cdot \text{ft}^2)$ of A , or $\text{W}/(\text{m}^2 \cdot \text{K})$

A = surface area on which U is based, ft^2 or m^2

$\left. \begin{matrix} t_{h,\text{in}} \\ t_{h,\text{out}} \end{matrix} \right\}$ = hot-fluid terminal temperatures, $^\circ\text{F}$, K , or $^\circ\text{C}$

$\left. \begin{matrix} t_{c,\text{in}} \\ t_{c,\text{out}} \end{matrix} \right\}$ = cold-fluid terminal temperatures, $^\circ\text{F}$, K , or $^\circ\text{C}$

$C_h = (Wc_p)_h$ = hot-fluid capacity rate, $\text{Btu}/(\text{h} \cdot ^\circ\text{F})\dagger$, W/K , or $\text{W}/^\circ\text{C}$

$C_c = (Wc_p)_c$ = cold-fluid capacity rate, $\text{Btu}/(\text{h} \cdot ^\circ\text{F})\dagger$, W/K , or $\text{W}/^\circ\text{C}$

Flow arrangement = counterflow, parallel flow, crossflow, parallel counterflow, or combinations of these basic arrangements

The interrelation of these parameters provides the basis for the heat transfer aspects of exchanger design.

The significance of all the foregoing variables is self-evident, with the exception of the overall conductance U . This term comes from an overall *heat transfer rate equation* which combines the convective and conductive mechanisms responsible for the heat transfer from the hot to the cold fluid into a single equation similar to Ohm's law for the steady-state flow of electrical current.

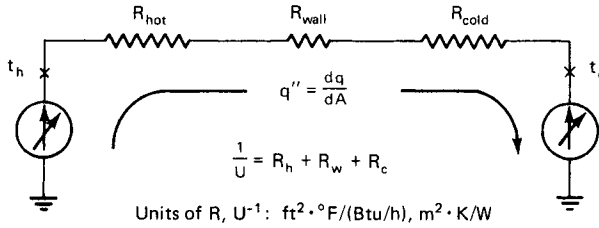
$$\frac{dq}{dA} = U(t_h - t_c) \quad (2-1)$$

here dq/dA is the heat flux per unit transfer area [$\text{Btu}/(\text{h} \cdot \text{ft}^2)$ or W/m^2] at a section in the exchanger where the temperature difference is $(t_h - t_c)$. From this relation, it is evident that U is an overall thermal conductance based on a temperature potential $(t_h - t_c)$ and a unit transfer area. The reciprocal of U is an overall thermal resistance which can be considered as having the following series components:

1. A hot-side convection component, including the temperature ineffectiveness of the extended surface or fin area on this side
2. A wall-conduction component
3. A cold-side convection component, including the temperature ineffectiveness of the extended area on this side
4. Fouling factors to allow for scaling in service on both the hot and cold sides

$\dagger$ For gas flow, c_p denotes the constant-pressure specific heat in distinction to c_v , the constant-volume specific heat. For liquids (and solids), no such distinction is needed.

Fig. 2-2 A thermal circuit representation for the heat transfer rate equation, Eq. (2-1).



A thermal circuit expressing this idea, but omitting the fouling factors for simplicity, is presented in Fig. 2-2. Working equations for the overall thermal resistance follow:

$$\frac{1}{U_h} = \frac{1}{\eta_{o,h} h_h} + \frac{a}{(A_w/A_h)k} + \frac{1}{(A_c/A_h)\eta_{o,c} h_c} \quad (2-2)$$

$$\frac{1}{U_c} = \frac{1}{\eta_{o,c} h_c} + \frac{a}{(A_w/A_c)k} + \frac{1}{(A_h/A_c)\eta_{o,h} h_h}$$

where U_h is based on a unit of hot-side total area (including fin or extended area) and U_c is based on a unit of cold-side total area. A_w denotes the average wall area; $\eta_{c,h}$ and $\eta_{o,c}$ denote the temperature effectiveness of the total area A_h and A_c , respectively. It is evident from Eqs. (2-2) that $U_h A_h = U_c A_c$.

The convective heat transfer coefficients h_c and h_h are complex functions of the surface geometry, fluid properties, and flow conditions. Except for some of the geometrically simple cases, the engineer generally relies on model experiments to establish these coefficients. In Chaps. 6, 7, and 10, these coefficients are presented in graphical form employing nondimensional parameters. In Chap. 4, consideration is given to the effects of temperature-dependent fluid properties on the heat transfer coefficients.

If no extended surface is employed on either side, both $\eta_{o,h}$ and $\eta_{o,c}$ are unity, $A_w \approx (A_h + A_c)/2$, and Eq. (2-2) can be simplified accordingly. However, where extended surface is employed, temperature gradients along the fins extending into the fluid reduce the temperature effectiveness of the surface, and η_o is less than unity as a consequence; η_o is a weighted average of the 100 percent effectiveness of the prime surface and the less than 100 percent fin surface η_f . Thus

$$\eta_o = 1 - \frac{A_f}{A} (1 - \eta_f) \quad (2-3)$$

Gardner [2] gives relations for η_f for a number of fin geometries. For many of the heat transfer surfaces considered here, the relation for the straight fin with constant-conduction cross section may be used to a good approximation.

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$$\eta_f = \frac{\tanh ml}{ml} \quad (2-4)$$

where

$$m = \sqrt{\frac{2h}{k\delta}} \quad \text{for thin sheet fins} \quad (2-5)$$

$$m = \sqrt{\frac{4h}{kd}} \quad \text{for circular pin fins}$$

When the fin extends from wall to wall, the effective fin length l is half the wall spacing. This relation is shown graphically in Fig. 2-13. Also shown is η_f for an annular circular fin of the type quite commonly employed on the outside of circular tubes. The circular-fin solution is plotted as a family of curves going in the limiting case to the behavior of a straight fin [Eq. (2-4)].

In the usual gas-to-gas or gas-to-liquid heat exchanger, the wall-resistance component in Eq. (2-2) may be neglected relative to the fluid-side resistances. In a gas-to-liquid exchanger, such as a water-cooled intercooler, the air-side resistance component is usually much greater than that on the water side and is said to control the heat transfer. For a gas-to-gas exchanger, the two resistances may be of comparable magnitude.

The heat transfer rate equation (2-1) must be combined with an *energy equation*, equating the loss of enthalpy of the hot fluid to the gain of enthalpy of the cold fluid, in order to relate the heat exchanger variables listed at the beginning of this discussion. These variables are too numerous to permit ready graphical description of their relation. However, they may be judiciously grouped into a smaller number of nondimensional parameters which do allow such a representation. The nondimensional groupings selected as most convenient and possessing the most readily visualized physical significance will now be named and defined.

EXCHANGER HEAT TRANSFER EFFECTIVENESS

$$\varepsilon = \frac{q}{q_{\max}} = \frac{C_h(t_{h,\text{in}} - t_{h,\text{out}})}{C_{\min}(t_{h,\text{in}} - t_{c,\text{in}})} = \frac{C_c(t_{c,\text{out}} - t_{c,\text{in}})}{C_{\min}(t_{h,\text{in}} - t_{c,\text{in}})} \quad (2-6)$$

where $C_{\min}$ is the smaller of the C_h and C_c magnitudes.

NUMBER OF EXCHANGER HEAT TRANSFER UNITS

$$N_{\text{tu}} = \frac{AU_{\text{av}}}{C_{\min}} = \frac{1}{C_{\min}} \int_0^A U \, dA \quad (2-7)$$

where A is the same transfer area as used in the definition of U . In design work U can generally be treated as constant.

CAPACITY-RATE RATIO

$$\frac{C_{\min}}{C_{\max}} \quad (2-8)$$

where $C_{\min}$ and $C_{\max}$ are, respectively, the smaller and the larger of the two magnitudes C_h and C_c .

In general, it is possible to express

$$\varepsilon = \Phi \left(N_{tu}, \frac{C_{\min}}{C_{\max}}, \text{flow arrangement} \right) \quad (2-9)$$

as revealed by the graphs of Figs. 2-14 to 2-30.

These nondimensional parameters are not only useful in allowing a compact graphical presentation of exchanger performance but also possess a readily grasped physical significance. The effectiveness ε compares the actual heat transfer rate, $q = C_h(t_{h,in} - t_{h,out}) = C_c(t_{c,out} - t_{c,in})$, to the thermodynamically limited, maximum possible heat transfer rate as would be realized only in a counterflow heat exchanger of infinite heat transfer area, namely, $q_{\max} = C_c(t_{h,in} - t_{c,in})$ if $C_c < C_h$ or $C_h(t_{h,in} - t_{c,in})$ if $C_h < C_c$. Thus ε possesses the significance of effectiveness of the heat exchanger from a thermodynamic point of view.

Note that, given the operating conditions $t_{h,in}$, $t_{c,in}$, C_h , and C_c , the magnitude of ε completely defines the heat transfer performance. If $C_h = C_{\min}$, then $\varepsilon = (t_{h,in} - t_{h,out}) / (t_{h,in} - t_{c,in})$, which is a "temperature effectiveness" for cooling the hot fluid. But if $C_c = C_{\min}$, then $\varepsilon = (t_{c,out} - t_{c,in}) / (t_{h,in} - t_{c,in})$, which is the temperature effectiveness for heating the cold fluid. However, the general definition of effectiveness, Eq. (2-6), is *not* a temperature effectiveness, but rather a "heat transfer effectiveness," and ambiguity will be avoided if this definition is strictly adhered to.

The number of heat transfer units N_{tu} is a nondimensional expression of the "heat transfer size" of the exchanger. Examination of Fig. 2-14, as an example, demonstrates the asymptotic character of the ε - N_{tu} relation for a given capacity-rate ratio. When the N_{tu} is small the exchanger effectiveness is low, and when the N_{tu} is large, ε approaches asymptotically the limit imposed by flow arrangement and thermodynamic considerations. The manner in which the transfer area and overall conductance enter into the N_{tu} expression, Eq. (2-7), emphasizes the costs of attaining a large N_{tu} (and hence high effectiveness) in terms of capitalization, space, and weight for transfer area A , or in terms of an increased flow-friction power requirement to obtain lower convective resistances for higher U .

From Eq. (2-2) for the case of negligible wall resistance, an interesting and useful relationship between N_{tu} and the Stanton numbers for each of the fluid streams can be derived.

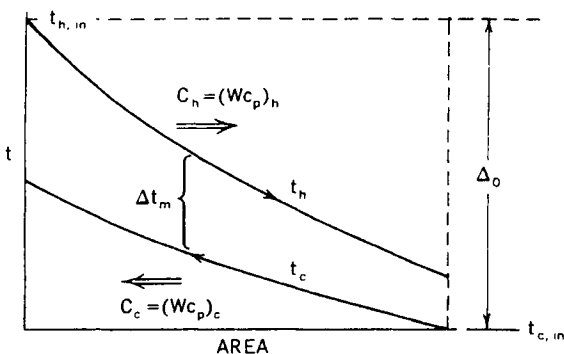
$$\frac{1}{N_{tu}} = \frac{1}{\eta_{o,h}(L/r_h)_h(C_h/C_{\min})St_h} + \frac{1}{\eta_{o,c}(L/r_h)_c(C_c/C_{\min})St_c} \quad (2-10)$$

where $C_{\min}$ is the smaller of the two capacity rates C_h and C_c .

The third nondimensional parameter, the capacity-rate ratio $C_{\min}/C_{\max}$, is simply the ratio of mass flow rate times specific heat capacity for the two streams. These products can be considered as flow stream thermal-capacity rates, i.e., energy-storage rate in the stream per unit of temperature change.

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Fig. 2-3 Fluid temperature conditions in a counterflow heat exchanger.



The relationship implied in Eq. (2-9) is a key one in the presentation of exchanger heat transfer performance. The argument leading to it follows. Consider the presentation in Fig. 2-3 of the fluid temperature condition in a counterflow heat exchanger, with $C_h < C_c$ ($C_h = C_{\min}$, $C_c = C_{\max}$). The heat transfer rate equation (2-1) may be written in integrated form

$$q = UA \Delta t_m \quad (2-1a)$$

where Δt_m is a suitably averaged mean temperature difference ($t_h - t_c$). Two additional expressions may be written for q , based on energy balance considerations

$$q = C_h(t_{h,in} - t_{h,out}) \quad (2-11)$$

$$q = C_c(t_{c,out} - t_{c,in}) \quad (2-12)$$

As previously noted, the maximum possible heat transfer rate is limited by flow rates and inlet fluid temperatures to

$$q_{\max} = C_{\min}(t_{h,in} - t_{c,in}) = C_h \Delta_o$$

Normalizing Eqs. (2-1a), (2-11), and (2-12) with $q_{\max}$ yields

$$\varepsilon = \frac{UA}{C_h} \frac{\Delta t_m}{\Delta_o} \quad (2-1b)$$

$$\varepsilon = \frac{t_{h,in} - t_{h,out}}{\Delta_o} \quad (2-11a)$$

$$\varepsilon = \frac{t_{c,out} - t_{c,in}}{\Delta_o} \frac{C_c}{C_h} \quad (2-12a)$$

The mean temperature difference Δt_m is some sort of a mean value between the terminal magnitudes ($t_{h,in} - t_{c,out}$) and ($t_{h,out} - t_{c,in}$). Moreover, the flow arrangement should be expected to influence the averaging procedure. Now from Eqs.

(2-11a) and (2-12a),

$$\begin{aligned} t_{h,in} - t_{c,out} &= (t_{h,in} - t_{h,out}) - (t_{c,out} - t_{c,in}) + (t_{h,out} - t_{c,in}) \\ &= \varepsilon \Delta_o - \varepsilon \Delta_o \frac{C_h}{C_c} + (1 - \varepsilon) \Delta_o = \left(1 - \varepsilon \frac{C_h}{C_c}\right) \Delta_o \end{aligned}$$

Similarly,

$$t_{h,out} - t_{c,in} = (1 - \varepsilon) \Delta_o$$

Thus it is apparent that

$$\frac{\Delta t_m}{\Delta_o} = \Phi \left(\varepsilon, \frac{C_h}{C_c} \right)$$

and, combining with Eq. (2-1b),

$$\varepsilon = \Phi \left(N_{tu}, \frac{C_h}{C_c} \right)$$

Removing the restrictions specified in Fig. 2-1 of $C_h < C_c$ and a counterflow arrangement suggests the validity of Eq. (2-9).

ε - N_{tu} Relations

ε - N_{tu} relations described by Figs. 2-14 to 2-29 will now be considered.

Counterflow (Fig. 2-14) See Appendix C for the derivation. The algebraic relation is

$$\varepsilon = \frac{1 - e^{-N_{tu}(1 - C_{min}/C_{max})}}{1 - (C_{min}/C_{max})e^{-N_{tu}(1 - C_{min}/C_{max})}} \quad (2-13)$$

where $N_{tu} = AU/C_{min}$ is always based on the *minimum* capacity rate. Calculated results for the preparation of Fig. 2-14 are listed in Table 2-2. Note that for all capacity-rate ratios the effectiveness approaches unity for large magnitudes of N_{tu} . This is a direct consequence, of course, of the definition of ε [see Eq. (2-6) and accompanying text]. Note further that the smaller the capacity-rate ratio, the higher the effectiveness for a given N_{tu} . Two limiting cases of Eq. (2-13) are of particular interest for gas-turbine heat exchanger design, namely, $C_{min}/C_{max} = 0$ and 1. The first case approximates the situation for a water-cooled intercooler, where $C_{water} \gg C_{air}$, and the second case is descriptive of a regenerator, where $C_{gas} \approx C_{air}$. For these conditions, Eq. (2-13) reduces to

$$\varepsilon = 1 - e^{-N_{tu}} \quad (2-13a)$$

for

$$\frac{C_{min}}{C_{max}} = 0$$

and to

$$\varepsilon = \frac{N_{tu}}{1 + N_{tu}} \quad (2-13b)$$

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for
$$\frac{C_{\min}}{C_{\max}} = 1$$

Note that for an evaporator or condenser, $C_{\min}/C_{\max} = 0$, because if one fluid remains at constant temperature throughout the exchanger, its specific heat, and thus its capacity rate, is by definition equal to infinity.

Parallel Flow (Fig. 2.15) The equation for this relation is

$$\varepsilon = \frac{1 - e^{-N_{tu}(1+C_{\min}/C_{\max})}}{1 + C_{\min}/C_{\max}} \quad (2-14)$$

Computed results for the preparation of Fig. 2-15 are summarized in Table 2-3. Note that, unlike the counterflow case, the asymptote for effectiveness is less than unity except for $C_{\min}/C_{\max} = 0$. For $C_{\min}/C_{\max} = 1$ the maximum possible effectiveness in parallel flow is only 50 percent, or one-half that of counterflow. For $C_{\min}/C_{\max} = 0$, Eq. (2-14) reduces to

$$\varepsilon = 1 - e^{-N_{tu}} \quad (2-14a)$$

which is identical to Eq. (2-13a) for counterflow. It is thus evident that, as far as performance is concerned, a counterflow intercooler ($C_{\text{water}} \gg C_{\text{air}}$), evaporator, or condenser possesses no advantage over a parallel-flow unit. However, for $C_{\min}/C_{\max} = 1$, Eq. (2-14) reduces to

$$\varepsilon = \frac{1 - e^{-2N_{tu}}}{2} \quad (2-14b)$$

and it is seen from Fig. 2-28, where both Eqs. (2-13b) and (2-14b) are plotted, that a parallel-flow gas-turbine-plant regenerator ($C_{\text{gas}} \approx C_{\text{air}}$) has substantially lower performance for $N_{tu} > 0.7$ ($\varepsilon > 40$ percent).

Crossflow, Both Fluids Unmixed (Fig. 2-16) In this type of crossflow heat exchanger, each fluid stream is assumed to have been divided into a large number of separate flow tubes for passage through the heat exchanger with no cross mixing. The effectiveness is based on the mixed mean temperature of the outlet fluids. The analytic solution for this arrangement cannot be expressed in closed form.

Table 2-4 and Fig. 2-16 are based on a series solution by Mason [3]. As for all previous flow arrangements, for $C_{\min}/C_{\max} = 0$

$$\varepsilon = 1 - e^{-N_{tu}}$$

Note that all the curves of Fig. 2-16 approach $\varepsilon = 1$ asymptotically, as was the case for counterflow. However, for all $C_{\min}/C_{\max} > 0$, the effectiveness for a given N_{tu} is less than for counterflow, with the greatest difference occurring for $C_{\min}/C_{\max} = 1$ (see Fig. 2-28).

Crossflow, One Fluid Mixed, the Other Unmixed (Fig. 2-17) See Appendix C for the derivation. This flow arrangement is described schematically in Fig. 2-17;

one fluid is considered to flow through separate tubes so that cross mixing is nil, while the other fluid is perfectly cross-mixed. The ε - N_{tu} equations are as follows:

$$\begin{aligned} \text{For} \quad C_{\max} &= C_{\text{unmixed}}, C_{\min} = C_{\text{mixed}}: \\ \varepsilon &= 1 - e^{-\Gamma C_{\max}/C_{\min}} \end{aligned} \quad (2-15)$$

$$\text{where} \quad \Gamma = 1 - e^{-N_{tu}(C_{\min}/C_{\max})}$$

$$\begin{aligned} \text{For} \quad C_{\max} &= C_{\text{mixed}}, C_{\min} = C_{\text{unmixed}}: \\ \varepsilon &= \frac{C_{\max}}{C_{\min}} (1 - e^{-\Gamma' C_{\min}/C_{\max}}) \end{aligned} \quad (2-16)$$

$$\text{where} \quad \Gamma' = 1 - e^{-N_{tu}}$$

Here, as before, $N_{tu} = AU/C_{\min}$.

Computed results from these equations are summarized in Table 2-5. For the limiting case of $C_{\min}/C_{\max} = 0$, both Eqs. (2-15) and (2-16) reduce to

$$\varepsilon = 1 - e^{-N_{tu}}$$

the same as for the counterflow and parallel-flow cases, emphasizing again that flow arrangement is unimportant where one capacity rate is very much greater than the other. Note also from Fig. 2-17 that, if the option is allowed, it is better to mix the fluid with the minimum capacity rate in preference to the fluid with the maximum capacity rate.

Crossflow, Both Fluids Mixed (Fig. 2-18) This is an uncommon flow arrangement in which each fluid temperature is a function of only one spatial coordinate. Such an arrangement can be obtained only by baffling both fluids. One reason this arrangement is of interest is that it is a case where it is possible to obtain a *decrease* in effectiveness with increasing N_{tu} .

The solution can be presented in closed algebraic form

$$\varepsilon = \frac{N_{tu}}{N_{tu}/(1 - e^{-N_{tu}}) + (C_{\min}/C_{\max})N_{tu}/(1 - e^{-N_{tu}(C_{\min}/C_{\max})}) - 1} \quad (2-17)$$

Computed results from this equation are in Table 2-6.

Multipass Overall-Counterflow Arrangements, Fluids Mixed between Passes See Appendix C for the derivation. It is possible to obtain solutions for overall-counterflow multipass configurations in rather simple algebraic form if it is postulated that the fluids are "mixed" between passes and that the total N_{tu} is equally distributed between passes of the same basic arrangement in an overall-counterflow configuration

$$\varepsilon = \frac{[(1 - \varepsilon_p C_{\min}/C_{\max})/(1 - \varepsilon_p)]^n - 1}{[(1 - \varepsilon_p C_{\min}/C_{\max})/(1 - \varepsilon_p)]^n - C_{\min}/C_{\max}} \quad (2-18)$$

where n is the number of identical passes in the overall-counterflow arrangement and ε_p is the effectiveness of each pass, a function of N_{tu}/n and the basic

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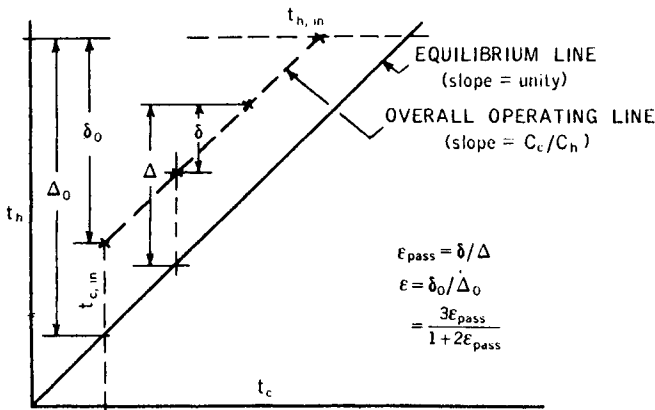
flow configuration of the pass. Note that the individual passes can be *any one* of the basic flow arrangements.

For the special case of $C_{\min}/C_{\max} = 1$,

$$\varepsilon = \frac{n\varepsilon_p}{1 + (n-1)\varepsilon_p} \quad (2-18a)$$

The method used in deriving Eq. (2-18) is illustrated in Appendix C. It may be described as one of synthesis, in that the overall effectiveness of the complex system is related to the component effectiveness of each pass. An *operating-line-equilibrium-line technique*, similar to that frequently used in the analysis of mass transfer systems, can be employed very effectively to show the algebraic relations in terms of simple geometrical relations. This is illustrated in Fig. 2-4 for $C_{\min}/C_{\max} = 1$, $\varepsilon_p = 0.30$, and $n = 3$; $\varepsilon = 0.562$ from Eq. (2-18a).

Fig. 2-4 Operating-line and equilibrium-line representation for $C_c/C_h = 1$, $n = 3$, $\varepsilon_p = 0.30$.



Multipass Overall-Counterflow Heat Exchangers with “Unmixed” Crossflow Passes Figure 2-19 was prepared from Eq. (2-18a) assuming “unmixed” crossflow passes, for which ε_p is given in Fig. 2-16. Since Eq. (2-18a) was employed, it is assumed that there is complete mixing between passes. In contrast, Fig. 2-20, from the results presented by Stevens et al. [4], is based on one fluid mixed between passes and the other fluid unmixed. Figure 2-21, based on the same reference, shows the performance of an overall-counterflow arrangement with crossflow passes where there is no mixing either within passes or between passes. The difference between these three sets of curves is not great, and thus the stipulation of mixing between passes, as in Eq. (2-18), is not highly restricting.

Figures 2-19, 2-20, and 2-21 demonstrate that a close approach to counter-

flow behavior can be realized by multipassing and that a large number of passes are often not warranted.

Counterflow with Crossflow Headers (Fig. 2-22) A difficulty arises in building a counterflow heat exchanger, using very compact plate-fin heat transfer surfaces, in the separation of the two fluids at the ends of the heat exchanger. Some kind of crossflow sandwich header must be used, and the usual procedure is to extend the plates to form one or another of the two patterns shown in Fig. 2-22. Then the fins in the end sections are brazed in such a manner as to turn the flow and separate the fluids as in a crossflow heat exchanger.

The result is a composite heat exchanger with a true counterflow central core, but with crossflow headers. One can anticipate that the performance, based on the *total* heat transfer surface area, would be inferior to that of a simple counterflow arrangement but superior to a simple crossflow arrangement. Frequently in gas-flow applications the flow frontal area must be so large that a substantial portion of the total heat transfer surface area (large portion of the total N_{tu}) is in the crossflow header section, and it is thus desirable to have a reasonably accurate solution for such a composite heat exchanger.

A numerical solution for this flow arrangement is provided by Kays et al. [5]. General behavior is pictured in Fig. 2-22, and Table 2-7 summarizes their results in more detail.

Parallel-Counterflow, Shell Fluid Mixed (Fig. 2-23) This basic flow arrangement is described schematically in Fig. 2-23. It is one of the most common arrangements for shell-and-tube construction. The algebraic relation, from Wright [6] and Bowman et al. [7], is

$$\varepsilon = \frac{2}{[1 + (C_{\min}/C_{\max})] + \sqrt{1 + (C_{\min}/C_{\max})^2(1 + e^{-\Gamma})/(1 - e^{-\Gamma})}} \quad (2-19)$$

where, in this case

$$\Gamma = N_{tu} \sqrt{1 + \left(\frac{C_{\min}}{C_{\max}}\right)^2}$$

As for all the previous flow arrangements, this equation will reduce to Eq. (2-13a) for the case of $C_{\min}/C_{\max} = 0$. Calculated results for the preparation of Fig. 2-23 are listed in Table 2-8. Note from this table and from Fig. 2-23 that for $C_{\min}/C_{\max} > 0$, the effectiveness for an infinite area ($N_{tu} = \infty$) is always less than unity.

Equation (2-19) was derived for one shell pass and two tube passes. However, four, six, eight, etc. tube passes yield results which are numerically so close to the two-tube-pass situation that nothing is to be gained by presenting these more complex expressions.

Multipass Overall-Counterflow Heat Exchangers with Parallel-Counterflow Passes Figures 2-24, 2-25, and 2-26 show the effect of multipassing the basic

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parallel-counterflow arrangement. These are some of the more popular arrangements where shell-and-tube construction is employed. The results are based entirely on Eq. (2-18), using the effectiveness of each pass, ε_p , as evaluated from Eq. (2-19).

Split-Flow Heat Exchangers (Fig. 2-27) This rather uncommon flow arrangement is described schematically in Fig. 2-27. The ε - N_{tu} characteristics were obtained from the work of Iqbal and Stachiewiez [8]. The solution is presented in closed form in this paper. However, it is too complex algebraically to be presented here. The tabulations in Tables 2-9a and b were obtained from the computer program results of Iqbal and Stachiewiez.

Like the crossflow both-fluids-mixed arrangement of Fig. 2-18, it is possible to obtain a decrease in effectiveness with an increasing N_{tu} . There is a small difference in performance between the situations $C_{shell} = C_{min}$ and $C_{tube} = C_{min}$. The difference is negligible for $N_{tu} < 5$ but becomes more significant in the higher N_{tu} range where an increase in N_{tu} results in lower effectiveness.

Effect of Flow Arrangement As has been pointed out, all flow arrangements have the same ε - N_{tu} relationship for $C_{min}/C_{max} = 0$, and for all other magnitudes of C_{min}/C_{max} the effectiveness for counterflow is highest. The maximum difference in performance is for the case of $C_{min}/C_{max} = 1$. To illustrate this difference, Figs. 2-28 and 2-29 were prepared. A study of these graphs will provide an idea of relative area requirements for heat exchangers of different flow arrangements and, in conjunction with Fig. 2-19, leads to the conclusion that, for effectiveness of the order of 80 percent, pure counterflow or multipass counterflow arrangements must be employed for $C_{min}/C_{max} = 1$ direct-transfer-type exchangers.

Effect of $C_{min}/C_{max} < 1$ for a Regenerator (Fig. 2-30) Most gas-turbine regenerators operate in the region $C_{min}/C_{max} = 0.90$ to 0.97 because of the influence of the combustion products on both the specific heat and gas-flow rate. For design purposes, it is convenient to use the simpler ε - N_{tu} relations for $C_{min}/C_{max} = 1$ and make an approximate adjustment for the departure of this magnitude from unity. Figure 2-30 was prepared for this purpose. As cross-counterflow arrangements (with unmixed passes) will fall between the extremes of counterflow on the one hand and single-pass crossflow on the other, interpolation may be used.

Log-Mean Rate Equation Compared to ε - N_{tu} Approach

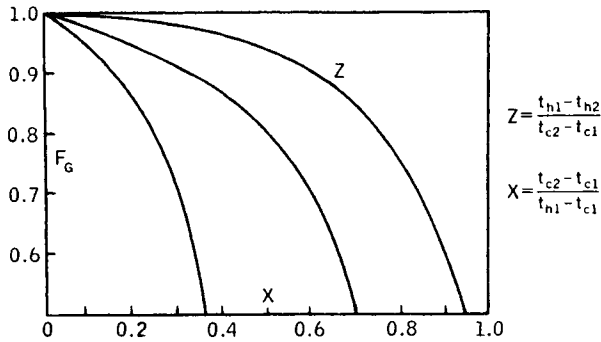
A relation commonly used in heat exchanger design is a log-mean rate equation of the form [9]

$$q = UAF_G \Delta t_{\log\text{-mean}}$$

where q is the overall heat transfer rate (in Btu per hour or watts) for the exchanger, and $\Delta t_{\log\text{-mean}}$ is the log-mean temperature difference of the two fluids calculated as for the case of true counterflow. If the exchanger is actually a

counterflow unit, the nondimensional factor F_G is unity. For all other flow arrangements, F_G is less than unity. This factor has been calculated for a variety of flow configurations and is presented in graphical form by Bowman et al. [7]. Figure 2-5 indicates the parameters employed. The purpose of the following text is first to demonstrate the one-to-one correspondence between the log-mean rate equation and the ε - N_{tu} approach, and second, to present some arguments in favor of the latter method.

Fig. 2-5 Flow configuration correction factors for log-mean rate equation (not to scale). Flow arrangement specified.



Suppose one considers two exchangers, a reference counterflow exchanger and the exchanger in question. Each is to operate with the same U , inlet fluid temperatures, and flow rates, and the area of the reference exchanger is to be proportioned so that it has the same q . It then follows that $\Delta t_{\log\text{-mean}}$ will be the same for the two exchangers and

$$F_G = \frac{A_{\text{counterflow}}}{A} \leq 1$$

or

$$F_G = \frac{N_{tu}(\text{counterflow for the same } \varepsilon)}{N_{tu}(\text{actual})}$$

Examination of X and Z leads to the conclusions that

$$X = \varepsilon \quad \text{for } C_c = C_{\min}$$

or

$$X = \varepsilon \left(\frac{C_h}{C_c} \right) \quad \text{for } C_h = C_{\min}$$

and

$$Z = \frac{C_c}{C_h} = \text{either } \frac{C_{\min}}{C_{\max}} \text{ or } \frac{1}{C_{\min}/C_{\max}}$$

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It is clear, then, that a one-to-one correspondence exists between the two sets of parameters.

The arguments in favor of the ϵ - N_{tu} approach follow:

1. The effectiveness is a simply defined, overall performance parameter having *thermodynamic* significance like an efficiency factor; it should therefore stand alone as a dependent variable and not appear directly in the abscissa and indirectly in the ordinate.
2. The log-mean rate equation misleadingly simplifies the notion of what is involved in heat exchanger design theory, since the implication is that only a rate equation is required. In reality, of course, both the rate equation and energy-balance principles are involved, but the latter is hidden in the F_G factor.
3. The ϵ - N_{tu} approach simplifies the algebra involved in predicting the performance of complex flow arrangements.
4. The more cogent arguments are related to ease of use in design work. To illustrate, consider the following two typical problems:
 - a. Given U , C_c , C_h , and the terminal temperatures, to determine the necessary A .
 - b. Given A , U , C_c , C_h , and $t_{h,in}$ and $t_{c,in}$, to determine outlet temperatures.

A detailed comparison of the steps for both problems is made in Table 2-1. For the first problem, both approaches give a straightforward solution. However, the ϵ - N_{tu} approach avoids the tedium involved in evaluating $\Delta t_{\log\text{-mean}}$. The ϵ - N_{tu} approach offers even greater advantages for the second problem, as can be seen from the table; the ϵ - N_{tu} approach is straightforward, whereas the log-mean Δt approach involves successive approximations.

Table 2-1 Comparison of ϵ - N_{tu} and Log-Mean Δt Approaches

ϵ - N_{tu} Approach	Log-mean Δt Approach
Problem (a)	
1. Calculate ϵ from the given terminal temperatures. Also calculate $C_{\min}/C_{\max}$. 2. Use the ϵ-N_{tu} curve for the particular flow arrangements and $C_{\min}/C_{\max}$ to evaluate N_{tu}. 3. Calculate A from$A = N_{tu} \frac{C_{\min}}{U}$	1. Calculate X and Z from the given terminal temperatures. 2. Use the $F_G(X, Z)$ curve for the particular flow arrangement to get F_G. 3. Calculate $\Delta t_{\log\text{-mean}}$ from the terminal temperatures. 4. Calculate A from$A = \frac{q}{UF_G \Delta t_{\log\text{-mean}}}$where q is evaluated from the terminal temperatures and the appropriate capacity rate.

Table 2-1 Comparison of ε - N_{tu} and Log-Mean Δt Approaches (*continued*)

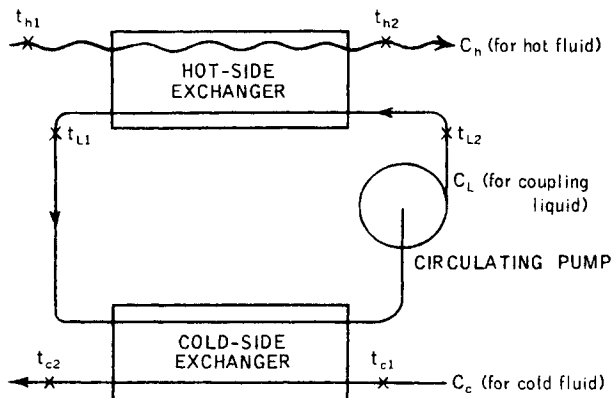
Problem (b)	
1. Calculate N_{tu} from the given information. Also calculate C_{min}/C_{max}. 2. Use the ε-N_{tu} curve for the particular flow arrangement and C_{min}/C_{max} to evaluate ε. 3. Calculate q from $q = C_{min}(t_{h,in} - t_{c,in})\varepsilon$ and terminal temperatures from $(t_{c,out} - t_{c,in})C_c = q$ $(t_{h,in} - t_{h,out})C_h = q$	1. Calculate Z from $Z = C_c/C_h$. 2. Assume terminal temperature so as to evaluate $X_{1st\,approx}$. 3. Get $F_{G,1st\,approx}$ from curve. 4. Evaluate $\Delta t_{log-mean,1st\,approx}$ 5. Determine $q_{1st\,approx}$ from the log-mean rate equation. 6. Calculate terminal temperature to compare with the assumptions of step 2. 7. Repeat until satisfactory agreement is obtained.

HEAT TRANSFER IN LIQUID-COUPLED INDIRECT-TYPE EXCHANGERS

The liquid-coupled indirect-transfer-type exchanger system is pictured in Fig. 2-6. This system consists of two direct-type exchangers coupled together by the circulation of a satisfactory heat transfer medium, such as water or a liquid metal.

The principal advantages are (1) since the hot-fluid flow area is not coupled directly with the cold-fluid flow area, a less awkward heat exchanger shape may result, particularly if there is as much as a 6:1 disparity in flow densities, as in the gas-turbine-regenerator application. (2) If the hot and cold fluids are gases, the liquid coupling will, in general, allow for a better and more compact machinery arrangement because of the simpler gas-ducting situation.

Fig. 2-6 Liquid-coupled indirect-transfer type of heat exchanger.



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The disadvantages are (1) greater total transfer area requirement, of the order of 10 to 20 percent for the gas-turbine regenerator; (2) additional complications of the coupling-liquid circuit; (3) the lack of a universally satisfactory heat transfer medium.

ε - N_{tu} Relations

The theory of the liquid-coupled indirect-transfer-type exchanger is developed by London and Kays [10]. It is based on a synthesized solution employing an operating-line–equilibrium-line representation, using for the cold-side unit (t_L , t_c) coordinates, for the hot-side unit (t_h , t_L) coordinates, and for the overall performance (t_h , t_c) coordinates, all on the same plane. The results follow:

For $C_L > C_c > C_h$,

$$\varepsilon = \frac{1}{1/\varepsilon_h + (C_h/C_c)/\varepsilon_c - C_h/C_L} \quad (2-20a)$$

For $C_L > C_h > C_c$,

$$\varepsilon = \frac{1}{1/\varepsilon_c + (C_c/C_h)/\varepsilon_h - C_c/C_L} \quad (2-20b)$$

For $C_c > C_h > C_L$,

$$\varepsilon = \frac{1}{C_h/C_L(1/\varepsilon_c + 1/\varepsilon_h - 1)} \quad (2-20c)$$

For $C_h > C_c > C_L$,

$$\varepsilon = \frac{1}{C_c/C_L(1/\varepsilon_c + 1/\varepsilon_h - 1)} \quad (2-20d)$$

For $C_c > C_L > C_h$,

$$\varepsilon = \frac{1}{1/\varepsilon_h + C_h/C_L(1/\varepsilon_c - 1)} \quad (2-20e)$$

For $C_h > C_L > C_c$,

$$\varepsilon = \frac{1}{1/\varepsilon_c + C_c/C_L(1/\varepsilon_h - 1)} \quad (2-20f)$$

In these relations ε is the overall effectiveness for transfer from the hot to the cold fluids, and ε_c and ε_h are the component effectiveness of the cold-side and hot-side direct-transfer exchangers.

For the special case of $C_h = C_c = C$, as approximated in the gas-turbine-regenerator application, the applicable foregoing equations reduce to two.

For $C_h = C_c = C > C_L$,

$$\varepsilon = \frac{C_L/C}{1/\varepsilon_c + 1/\varepsilon_h - 1} \quad (2-21a)$$

For $C_h = C_c = C < C_L$,

$$\varepsilon = \frac{1}{1/\varepsilon_c + 1/\varepsilon_h - C/C_L} \quad (2-21b)$$

If the further specification is made that the coupling-liquid capacity rate $C_L = C_c = C_h = C$, the resulting equation is

$$\varepsilon = \frac{1}{1/\varepsilon_c + 1/\varepsilon_h - 1} \quad (2-22)$$

It is important to note that in Eqs. (2-20a) to (2-22), the overall and component effectiveness are defined, as throughout this chapter, by the temperature change for the fluid having the minimum capacity rate. Thus, for example, if $C_L > C_h > C_c$,

$$\begin{aligned} \varepsilon &= \frac{C_c(t_{c,2} - t_{c,1})}{C_c(t_{h,1} - t_{c,1})} \\ \varepsilon_h &= \frac{C_h(t_{h,1} - t_{h,2})}{C_h(t_{h,1} - t_{L,2})} \\ \varepsilon_c &= \frac{C_c(t_{c,2} - t_{c,1})}{C_c(t_{L,1} - t_{c,1})} \end{aligned}$$

Optimum Conditions of Operation

The foregoing relationships can be used to provide design criteria for optimizing the coupling-liquid flow rate and the heat-transfer-area distribution between the hot-side and cold-side units [10]. For the case of $C_h = C_c = C$, the following criteria result:

1. The coupling liquid C_L should be kept within the fairly narrow range $0.95 < C_L/C < 1.20$, with unity being the optimum.
2. The heat-transfer-area distribution should be kept within the rather broad range

$$\frac{0.75}{U_c/U_h} < \frac{A_c}{A_h} < \frac{2.0}{U_c/U_h}$$

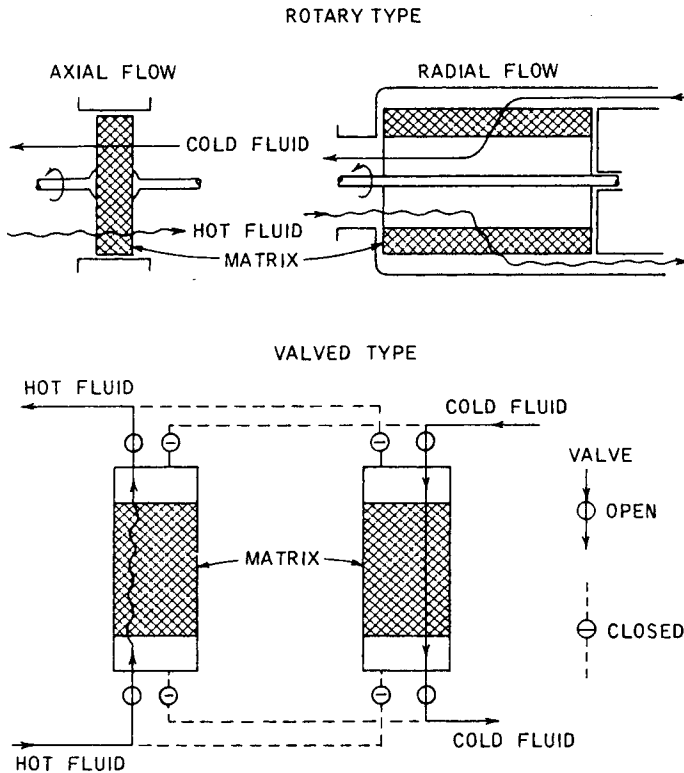
If the cold-side area is less expensive than the hot-side area, it would pay to unbalance the design to the right. If the machinery arrangement is such as to make it desirable to reduce the cold-side volume, the design may be unbalanced toward the left with no significant loss in performance.

HEAT TRANSFER IN PERIODIC-FLOW-TYPE EXCHANGERS

The two types of periodic-flow exchangers are the rotary and valved types, as described in Fig. 2-7. In the rotary type, the rotation of the porous matrix provides a flow of the solid phase from hot-side to cold-side streams in a regular,

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Fig. 2-7 Periodic-flow exchanger types.



periodic manner. Thus the matrix is alternately heated and cooled, and in this manner heat is transferred indirectly from the hot to the cold fluids. The relationship of the periodic-flow type to the previously considered liquid-coupled indirect-transfer type is evident.

The valved type of exchanger has two identical matrices, each one functioning alternately as either a hot-flow or a cold-flow matrix by means of a periodic switching of the quick operating valves. In this exchanger, the mass of the two matrices divided by the period of valve operations corresponds to the mass of the single matrix in the rotary type times the speed of rotation. The heat exchanger design theory will apply equally to both types, once this correspondence on the matrix mass rate is recognized.

There are three major advantages of the periodic-flow heat exchanger relative to the direct-transfer type

1. A much more compact heat transfer surface can be employed. For example, a 24-mesh screen matrix has an area density of the order of $1,000 \text{ ft}^2/\text{ft}^3$ ($3,300 \text{ m}^2/\text{m}^3$).

2. The heat transfer surface in general is substantially less expensive per unit of transfer area.
3. Because of the periodic flow reversals, there are no permanent flow-stagnation regions and, consequently, the surface tends to be self-cleaning. The self-cleaning feature has been well demonstrated by Ljungstrom-type air preheaters used in central station plants employing the very lowest grades of coals.

The major disadvantages are

1. There is some mixing of the hot and cold fluids because of leakage and carryover.
2. If the fluids are at different pressures, as in the gas-turbine regenerator, the sealing problem is a difficult one for the rotary type.

ε - N_{tu} Relations

The theory for this type of exchanger is summarized by Coppage and London [11]. Because of mathematical difficulties, computer solutions are required. Lambertson [12] and Bahnke and Howard [13] provide the most comprehensive results. Figures 2-31 to 2-38 and Tables 2-10 to 2-15 were compiled from these sources.

The nondimensional parameters used in these presentations parallel those used heretofore, with some additions and modifications as follows:

$$\varepsilon = \Phi \left(N_{tu,o}, \frac{C_{min}}{C_{max}}, \frac{C_r}{C_{min}} \right) \quad (2-23)$$

Here the modified number of transfer units is by definition

$$N_{tu,o} = \frac{1}{C_{min}} \left[\frac{1}{(1/hA)_c + (1/hA)_h} \right]$$

Alternatively, to gain the same form as Eq. (2-10) and its relation to the thermal circuit of Fig. 2-2,

$$\frac{1}{N_{tu,o}} = \frac{1}{(C_h/C_{min})N_{tu,h}} + \frac{1}{(C_c/C_{min})N_{tu,c}}$$

The matrix capacity rate is the matrix mass rate times the specific heat of the solid. For the rotary type,

$$C_r [\text{Btu}/(\text{h} \cdot ^\circ\text{F}) \text{ or } \text{W/K}] = (\text{rev/h})(\text{matrix mass})(c_{\text{solid}})$$

For the valved type, the mass of both the identical matrices would be used together with the valve cycles per hour (with the period being the interval "valve on to off to on"). Note that the modified number of transfer units $N_{tu,o}$ parallels that for the direct-transfer-type exchanger where, for the case of negligible wall resistance and all prime surface ($\eta_o = 1$), Eq. (2-2) yields

$$\frac{1}{AU} = \left[\left(\frac{1}{hA} \right)_c + \left(\frac{1}{hA} \right)_h \right]$$

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The range covered in the ε - $N_{tu,o}$ graphs (Figs. 2-31 to 2-38) and tables (Tables 2-9 to 2-14) are as follows:

For $C_{\min}/C_{\max}$	1.0, 0.95, 0.90, 0.80, 0.70, 0.50
For $C_r/C_{\min}$	1 to ∞
For $N_{tu,o}$	0 to 100 for some tables 0 to 10 on most graphs

The higher $N_{tu,o}$ ranges are of interest in cryogenic heat exchanger applications. However, because of the asymptotic approach of effectiveness toward unity on the usual ε - $N_{tu,o}$ graph (Figs. 2-31 to 2-36), it is more revealing to graph $(1 - \varepsilon)$ versus $N_{tu,o}$ on log-log coordinates, as in Fig. 2-37 for $C_{\min}/C_{\max} = 1$ and Fig. 2-38 for $C_{\min}/C_{\max} = 0.95$.

In addition to the nondimensional parameters noted in Eq. (2-23), the ratio of conductances

$$(hA)^{\circ} = \frac{hA \text{ on the } C_{\min} \text{ side}}{hA \text{ on the } C_{\max} \text{ side}}$$

enters into the analysis. The ε - $N_{tu,o}$ curves in Figs. 2-31 to 2-38 are all based on $(hA)^{\circ} = 1$. However, it is revealed by computer calculations [12, 13] that the influence of this parameter is quite negligible for the ranges of $(hA)^{\circ}$ shown on the graphs in Figs. 2-31 to 2-36. The largest error is two points on ε , which occurs on the $C_{\min}/C_{\max} = 0.50$ graph (Fig. 2-36) at $C_r/C_{\min} = 1.0$. This error magnitude is halved for $C_r/C_{\min} = 1.5$. For the $C_{\min}/C_{\max} = 1$ graph (Fig. 2-31), the maximum error, again at $C_r/C_{\min} = 1.0$, is only of the order of 0.2 points on ε .

Influence of Matrix Speed

A simple empirical formulation for the influence of $C_r/C_{\min}$, for $\varepsilon \geq 90$ percent, is

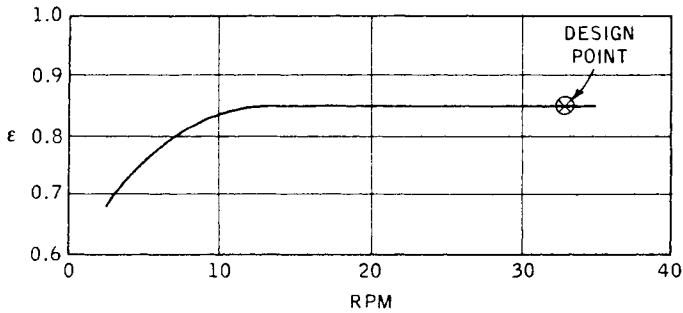
$$\varepsilon \approx \varepsilon_{\text{counterflow direct type}} \left[1 - \frac{1}{9(C_r/C_{\min})^{1.93}} \right] \quad (2-24)$$

The right-hand bracket accounts for the matrix speed. The counterflow direct-transfer exchanger effectiveness is a function only of $N_{tu,o}$ and $C_{\min}/C_{\max}$; it may be determined either from Eq. (2-13) or Fig. 2-14. For higher values of effectiveness ($\varepsilon > 90$ percent), an ineffectiveness graph, $(1 - \varepsilon)$ versus $N_{tu,o}$ (Figs. 2-37 and 2-38), is recommended in place of Eq. (2-24).

Matrix speed is significant not only because of the direct influence of C_r or ε , as indicated by Eq. (2-24), but also because of a *carryover loss* associated with the void volume of the matrix. For minimum carryover, it would be desirable to operate at low rotative speed, but this results in a lower ε . Another advantage of low-speed operation is the reduction of seal wear if rubbing seals are employed.

After a preliminary matrix design is accomplished, Eq. (2-24) can be employed to investigate the influence of operation at off-design speeds. Figure 2-8 indicates such an extrapolation. It is evident that the design rotative speed was

Fig. 2-8 Influence of rotative speed on performance of periodic-flow exchanger.



much higher than necessary and that a reduction of speed from 31 r/min (corresponding to $C_r/C_c = 10.8$) to 15 r/min could be made.

Matrix speed has a substantial influence on the periodic fluctuation of the matrix temperature. This subject is covered by Mondt [14]. It may be shown by elementary considerations that the average matrix double amplitude is given by

$$\delta t_{m,av} = \frac{\varepsilon}{C_r/C_{min}} (t_{h,in} - t_{c,in})$$

Rotating Seals and Switch Valving

A very important aspect of the mechanical design of a periodic-flow heat exchanger is the provision for obtaining flow periodicity with minimum leakage. The subject is beyond the scope of this monograph. Design problems with respect to rotating seals are discussed by Hrynyszak [15], and the fluid mechanics of clearance seals are covered by Harper [16].

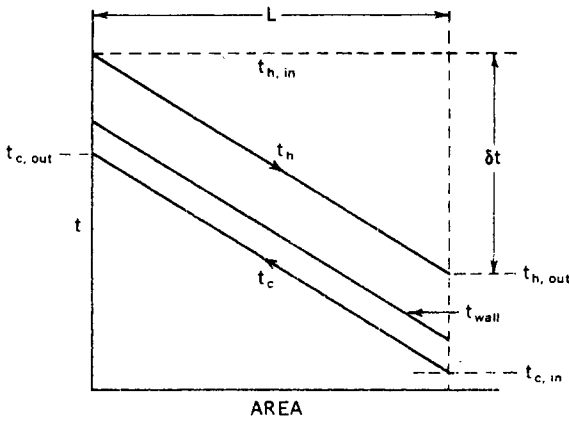
EFFECT OF LONGITUDINAL CONDUCTION IN HEAT EXCHANGERS

The exchanger theory presented to this point is all based on the idealization that there is no longitudinal conduction (in the flow direction), either in the solid wall or in the fluid. Fluids generally have a low thermal conductivity (liquid metals excepted), but the wall conductivity may be quite high. Consequently, only wall-conduction effects will be considered in the following treatment. The influence of longitudinal conduction is to reduce exchanger effectiveness for a given number of transfer units, and this reduction may be quite serious in exchangers with short flow lengths designed for high effectiveness ($\varepsilon > 90$ percent).

The following simplified analysis yields a rough approximation of the influence of longitudinal conduction. Consider the temperature conditions pictured in Fig. 2-9 for a direct-transfer-type exchanger with $C_{min}/C_{max} = 1$. The temper-

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Fig. 2-9 Temperature distribution in a counterflow heat exchanger with $C_c = C_h$.



ature difference δt shown for the hot fluid is of the same magnitude for the cold fluid and also for the wall. Then the wall temperature gradient is $\delta t/L$, with L being the flow length. If the wall cross-sectional area for longitudinal conduction is designated A_k , the longitudinal heat transfer by conduction is of the order

$$q_k \approx k A_k \frac{\delta t}{L}$$

In contrast, the convection heat transfer rate is given by energy-balance considerations as

$$q = C_c \delta t = C_h \delta t$$

and then

$$\frac{q_k}{q} = \frac{(k/L)A_k}{C}$$

Now, if q_k tends to reduce the actual transfer q , q_k/q is of the order $\delta \varepsilon/\varepsilon$, where $\delta \varepsilon$ is the decrement in effectiveness; thus

$$\frac{\delta \varepsilon}{\varepsilon} \approx \frac{(k/L)A_k}{C_{\min}} = \lambda \quad (2-25)$$

where λ , the nondimensional conduction parameter, is defined as shown. This simply determined result provides an approximation of the actual effect of longitudinal conduction. When a more complete analysis is attempted, the solutions are sufficiently complex to require computer calculation. Fortunately, the comprehensive results of Bahnke and Howard [13] are available, and Figs. 2-39 and 2-40 were prepared from this reference. In these graphs, exchanger

ineffectiveness $(1 - \varepsilon)$ is presented as a function of N_{tu} (or $N_{tu,o}$) and λ for the two magnitudes of C_{min}/C_{max} , 1.0 and 0.95, which are of primary interest in the gas-turbine-regenerator application. While the original paper [13] was concerned only with the periodic-flow-type regenerator, the periodic-flow and direct-transfer types have identical behavior for $C_r/C_{min} = \infty$. Moreover, it is shown that it is only necessary to have $C_r/C_{min} > 5$ for the periodic-flow behavior to very closely approach the $C_r/C_{min} = \infty$, or direct-transfer-type, behavior. This result is suggested by the previous curves, Figs. 2-31 to 2-36, which show very little difference between $C_r/C_{min} = 5$ and ∞ . Note that, as in the case of Figs. 2-37 and 2-38, the ineffectiveness $(1 - \varepsilon)$ is a more useful dependent parameter than the effectiveness ε directly, particularly for $\varepsilon > 90$ percent. The logarithmic coordinates are useful in properly emphasizing the importance of the longitudinal conduction effect. For instance, it is readily apparent from Figs. 2-39 and 2-40 that if the desired $\varepsilon = 98.5$ percent

C_{min}/C_{max}	N_{tu}		% increase of N_{tu}
	For $\lambda = 0$	For $\lambda = 0.01$	
1.00	66	190	188
0.95	29.5	40	36

The rather striking percent of increase in N_{tu} can be readily scaled from the graph.

This illustration also serves to emphasize the strong influence of C_{min}/C_{max} for high- ε heat exchangers. Here only a 5 percent increase in the capacity-rate ratio from 0.95 to 1.00 results in approximately a 124 percent and a 375 percent increase in the required number of transfer units for $\lambda = 0$ and 0.01, respectively.

CORE PRESSURE DROP

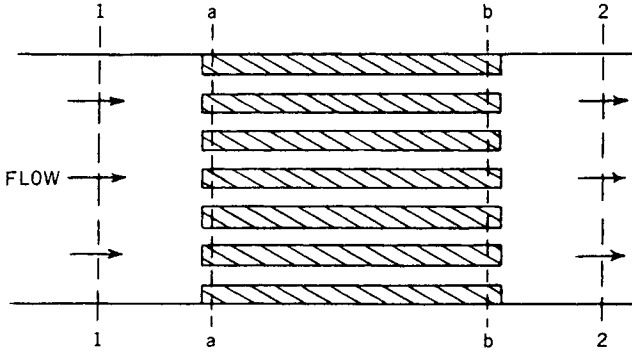
In the design of liquid-to-liquid heat exchangers, accurate knowledge of the friction characteristics of the heat transfer surface is relatively unimportant because of the low power requirement for pumping high-density fluids. For gases, however, because of their lower density, the friction power per unit mass flow rate is greatly multiplied. Thus, to the designer, the friction characteristic of the surface assumes an importance equal to that of the heat transfer characteristic. The friction characteristic needed is the flow-friction factor f , which is reported in Chaps. 6, 7, and 10 as a function of flow geometry and the Reynolds number.

The Pressure-Drop Equation

Figure 2-10 shows the flow system to be considered. For gas-flow heat exchanger applications, the pressure changes from sections 1 to a and from b to 2 are very small relative to the total pressure; thus $v_a \approx v_1$ and $v_b \approx v_2$.

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Fig. 2-10 Heat exchanger core model for pressure-drop analysis. G is based on the minimum free-flow area in the core.



Then, by definition of the entrance and exit loss coefficients K_c and K_e [Eqs. (5-1) and (5-2)], and by an integration of the momentum equation through the core, the relation for the flow-stream pressure-drop calculation for most heat exchanger cores is

$$\frac{\Delta P}{P_1} = \frac{G^2}{2g_c} \frac{v_1}{P_1} \left[\underbrace{(K_c + 1 - \sigma^2)}_{\text{entrance effect}} + \underbrace{2 \left(\frac{v_2}{v_1} - 1 \right)}_{\text{flow acceleration}} + \underbrace{f \frac{A}{A_c} \frac{v_m}{v_1}}_{\text{core friction}} - \underbrace{(1 - \sigma^2 - K_e) \frac{v_2}{v_1}}_{\text{exit effect}} \right] \quad (2-26a)^\dagger$$

However, for flow normal to tube banks or through wire matrix surfaces, as might be employed in periodic-flow-type exchangers, entrance and exit loss effects are accounted for in the friction factor, and the equation becomes (with K_c and $K_e = 0$),

$$\frac{\Delta P}{P_1} = \frac{G^2}{2g_c} \frac{v_1}{P_1} \left[\underbrace{(1 + \sigma^2) \left(\frac{v_2}{v_1} - 1 \right)}_{\text{flow acceleration}} + \underbrace{f \frac{A}{A_c} \frac{v_m}{v_1}}_{\text{core friction}} \right] \quad (2-26b)$$

The porosity p replaces σ for matrix surfaces.

For multipass arrangements, losses in the return headers must be accounted for separately, as must any losses in inlet and exit headers and associated ducting.

It is worth noting that $A/A_c = L/r_h$, by definition of the hydraulic radius, Eq.

$^\dagger g_c$, the proportionality factor in Newton's second law, is equal to 32.2 (lb_m/lb_f)ft/s² in English units and equal to unity (dimensionless) in SI units.

(1-4); further

$$\frac{G^2}{2g_c} \frac{v_1}{P_1} = \frac{(V_1^2/2g_c)}{(P_1/\rho_1)} = \frac{\text{dynamic head}}{\text{static head}}$$

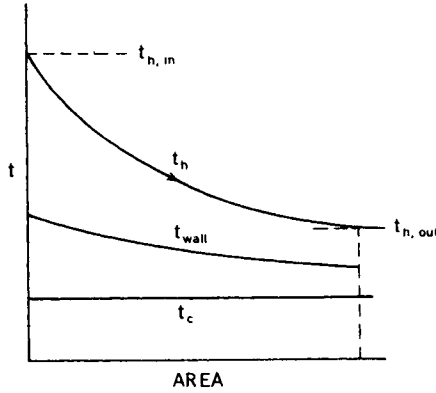
where $V_1 = G/\rho_1$ is the flow velocity entering the core, based on the minimum free-flow area, which defines G .

The correct mean specific volume to be used in Eqs. (2-26a and b) is

$$v_m = \frac{1}{A} \int_0^A v \, dA \quad (2-27)$$

Consider the flow temperature conditions pictured in Figs. 2-9 and 2-11. For a magnitude of unity for $C_{\min}/C_{\max}$ (Fig. 2-9), flow-stream temperatures vary

Fig. 2-11 Temperature distribution in a heat exchanger of any flow arrangement with $C_c \gg C_h$.



linearly with area in a true counterflow arrangement, and also to a good approximation for any flow arrangement other than parallel flow. Consequently,

$$\frac{v_m}{v_1} \approx \frac{P_1}{P_{av}} \frac{T_{av}}{T_1}$$

or

$$v_m \approx \frac{v_1 + v_2}{2} \quad (2-27a)$$

where T_{av} and P_{av} are arithmetic averages of the terminal magnitudes. In contrast, if the wall temperature is essentially uniform (Fig. 2-11), as in the water-cooled intercooler, the condenser, or the evaporator, Eq. (2-27) reduces to

$$\frac{v_m}{v_1} \approx \frac{P_1}{P_{av}} \frac{T_{lma}}{T_1} \quad (2-27b)$$

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where P_{av} is the arithmetic average of the terminal magnitudes and T_{lma} is related to the log-mean temperature difference between the fluid with the changing temperature and the constant fluid temperature Δt_{lma} by

$$T_{lma} = T_{const} \pm \Delta t_{lma} \quad (2-28)$$

Here, for the conditions shown in the figure, the plus sign is used and

$$\Delta t_{lma} = \frac{(t_{h,in} - t_c) - (t_{h,out} - t_c)}{\ln [(t_{h,in} - t_c)/(t_{h,out} - t_c)]} = \frac{t_{h,in} - t_{h,out}}{N_{tu}} \quad (2-29)$$

A similar expression with t_c in place of t_h would be used if t_h were essentially constant and t_c varied ($C_h \gg C_c$) with the minus sign used in Eq. (2-28).

The mass velocity G in Eq. (2-26a) is based on the *minimum free-flow area*, consistent with the definition of friction factor employed here.

Entrance and exit effects in Eq. (2-26a) normally provide only a small contribution to the overall pressure drop in the usual exchanger design because, since A/A_c is quite large, the core-friction term controls the magnitude of ΔP . Consequently, high accuracy in the evaluation of K_e and K_e is not required. Chapter 5 (Figs. 5-2 to 5-5) reports magnitudes of the entrance and exit loss coefficients as functions of flow geometry and Reynolds number.

The friction factor f , as used in Eq. (2-26a), is affected by the variation of fluid properties ρ and μ over the flow cross section, as well as variations in the flow direction. Chapter 4 deals with this problem and provides the correct f to use in the pressure-drop equations.

The Core-Velocity Equation

Equations (2-26a and b) provide means for evaluating pressure drop given the core velocity and geometry; however, there is no direct relationship of the pressure drop to heat transfer performance in these equations. The usual design problem provides a specification of both pressure drop and heat transfer performance, and it is useful to evaluate a first approximation for the core velocity in order to obtain a preliminary size for the core. An approximate equation for this purpose is

$$\frac{V_1^2/2g_c}{P_1/\rho_1} \approx \left(\frac{\Delta P/P}{N_{tu}} \right)_{\text{one side}} \frac{\rho_m}{\rho_1} \frac{St}{f} \eta_o \quad (2-30a)$$

Here, presumably, the designer is in a position to specify the allowable pressure drop ($\Delta P/P$) on each side and the necessary $N_{tu, \text{one side}}$ defined by

$$N_{tu, \text{one side}} = \left(\eta_o \frac{L}{r_h} St \right)_{\text{one side}}$$

which is related to the overall number of heat transfer units by Eq. (2-10). The necessary magnitude can be estimated by working backward from the given ε to the necessary overall N_{tu} , estimating or specifying a desirable distribution of heat transfer resistances, hot and cold sides, and then using Eq. (2-10) to approximate the N_{tu} , on each side, as defined above. The ratio St/f contains the

core-surface characteristics for a given fluid. Figure 2-41 is a presentation of this characteristic as a function of the Reynolds number for several rather different surface geometries.† The important points to note at this time are that (1) there is relatively small variation in St/f for, say, a twofold range of Re ; (2) there is only about a sixfold range St/f for markedly different surfaces; and (3) the scale of the surface is of minor importance and enters primarily via the Reynolds number. Thus one can readily estimate a magnitude of St/f to use in Eq. (2-30) for the purpose of calculating the core flow velocity. Moreover, from the first approximation flow velocity the Re can be evaluated for a better second approximation of St/f for an iterative calculation for V_1 .

Since the velocity varies as the square root of St/f , the sixfold extreme range of this characteristic means that there can be only a $2\frac{1}{2}$ -fold range of velocities possible (based on free-flow area) by surface-type selection alone. Only a very small change in V_1 is possible by using finer or grosser surfaces of a particular type. Further, a graph such as Fig. 2-41 allows ready selection of surface types which will produce small flow frontal areas, if this is a design requirement.

Equation (2-30) is readily derived from Eq. (2-26) by neglecting the entrance and exit effects along with the flow-acceleration pressure-drop term. Thus the V_1 calculated from Eq. (2-30) tends to be slightly on the high side.

SURFACE SELECTION METHODOLOGY

In the schematic representation of the methodology of heat exchanger design, Fig. 2-1, surface characteristics are pictured as a key input to the design theory procedure. Figure 2-12 is an elaboration of Fig. 2-1, with two steps replacing the single surface characteristic, namely, surface selection (two or more cases) and surface characteristics including heat transfer, flow friction, and geometrical characteristics of the surface selected. The following treatment deals briefly with a methodology the designer may use in selecting surfaces for a case-by-case exchanger design study leading to the optional solutions indicated in Fig. 2-12.

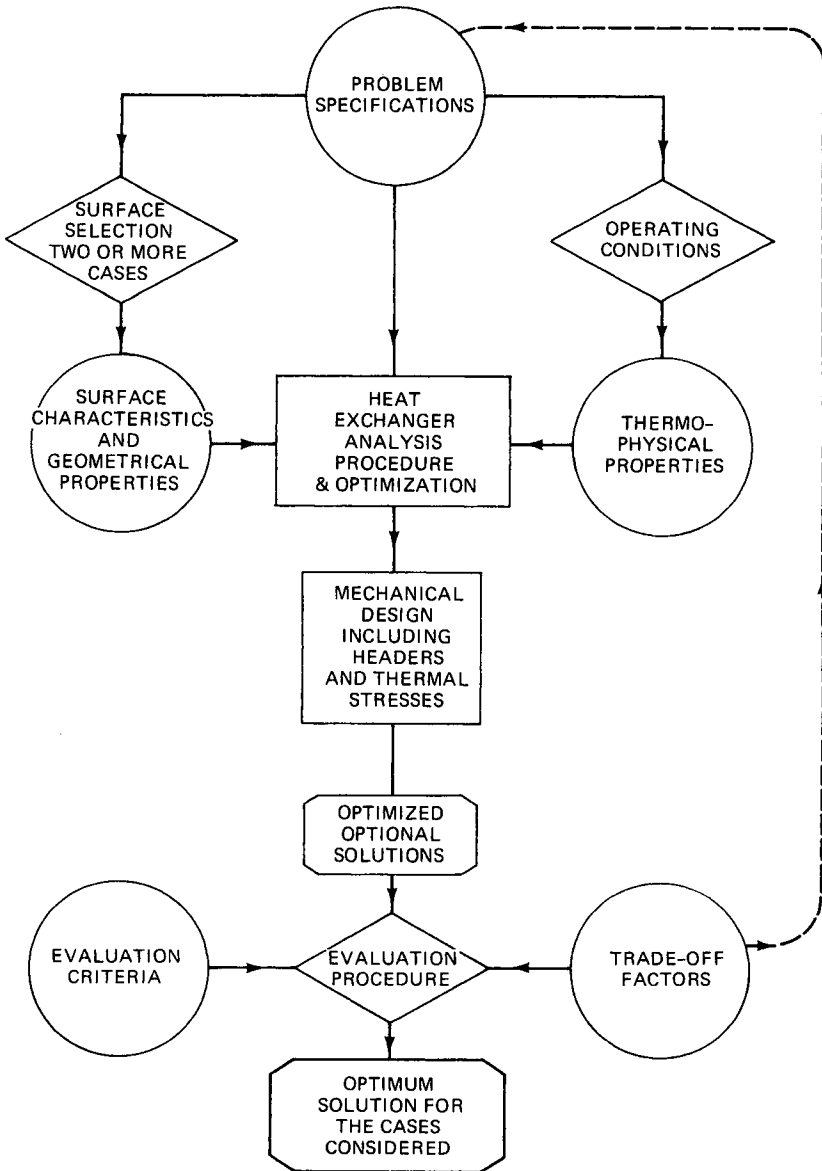
As previously mentioned in the discussion of Eq. (2-30a), graphs of St/f versus Re or $StPr^{2/3}/f$ versus Re , as in Fig. 2-41, provide a means of selecting a surface geometry which results in an exchanger of smaller flow area. To emphasize this point, consider the following reformulation of Eq. (2-30a), using the mass velocity G as a replacement for V_1 .

$$G = \frac{W}{A_c} \approx \left[2g_c \frac{P_1}{v_m} \eta_o \left(\frac{\Delta P/P_1}{N_{tu}} \right)_{\text{side}} \left(\frac{St}{f} \right) \right]^{1/2} \quad (2-30b)$$

It is evident that the surface with the higher St/f will have the smaller flow area A_c . Thus a *surface flow area "goodness factor"* method of selection is provided by a graph such as Fig. 2-41. Recall, however, that the frontal-area to flow-area ratio, $1/\sigma$, is involved in the translation of A_c to the core frontal area.

† Figure 2-41 employs an ordinate $(St/Pr^{2/3})/f$. To convert to St/f , as desired for Eq. (2-30), multiply by 1.27 for gaseous fluids with $Pr \approx 0.70$.

Fig. 2-12 Methodology of heat exchanger design and optimization.



A companion goodness factor will now be considered, namely, a *heat transfer surface area* (or a *core volume* or a *core mass*) *goodness factor*. Equations (1-1) and (1-2) and Fig. 1-2 will suffice for this consideration. A surface geometry selection that falls high on a graph of h versus E (for a given set of fluid properties c_p , μ , Pr , ρ , and a specified hydraulic diameter $4r_h$) will have the smallest heat transfer area requirement. Similarly, a graph of $h\beta$ versus $E\beta$ will provide a selection method for surface geometries which will yield small volume and, usually, small mass cores.

FLOW DISTRIBUTION AND HEADER DESIGN†

One disadvantage in the use of highly compact surface geometries is that the resulting core shapes are characterized by large flow frontal area and short flow lengths for the gas-flow path. The common automobile radiator is a close-at-hand example—a frontal area of approximately 3.3 ft^2 (0.31 m^2) and an air flow length of 1.2 in (0.0305 m) for a 220 hp (164 kW) automobile engine! In this application, the large frontal area can be accommodated by a front mounting on the vehicle. However, heat exchangers in other installations must be located in ducting, and the flow header‡ configurations have a definitive influence on the engine system envelope geometry.

Figure 2-42a describes a folded-core concept used to reduce header volume. However, if the pressure drop across the core is not uniform, as is the case for the Fig. 2-42a configuration, the flow distribution over the heat transfer surfaces will not be uniform, and a penalty in heat exchanger performance is the result. From this viewpoint, the design objective is to provide for acceptably uniform flow through the core with an acceptable header geometry and flow-stream mechanical energy losses. *Uniformity* of flow distribution over the core face is the dominating function of the headers. The importance of header design is emphasized in the schematic representation of the methodology of heat exchanger design (Fig. 2-12), where it is listed in the mechanical design box. A poor header configuration may actually result in a lower overall pressure drop, but the penalty in heat transfer performance can be very large.

London et al. [17] considered three header configurations, as shown in Fig. 2-42b to d. They provided the theory needed to specify the exit and inlet header shapes and sizes which yield the desired uniformity of the pressure drop, ΔP_{matrix} , over the core. Experimental results support the predictions quite well (± 5 percent nonuniformity).

As illustrated in the numerical example of Ref. 17, the parallel-flow configuration has a header pressure loss (total-to-total), exclusive of core pressure drop, equal to 2.47 times the inlet velocity head based on the velocity u_i shown in Fig. 2-42b. In contrast, the free-discharge configuration has an inlet velocity head

† The following considerations relate to the fluid flow aspects of header design. Heat transfer behavior of headers was considered previously for the geometry described in Fig. 2-22 and the associated text.

‡ The header is also referred to as the tank, the manifold, the box, or the distributor.

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loss of unity, and the counterflow configuration has 0.595 times the inlet velocity head loss, only one-quarter that of the parallel-flow configuration! Clearly, if the option is available, a counterflow header arrangement is the preferred geometry. Of the total header loss, about 74 and 50 percent, respectively, for the parallel and counterflow arrangements is chargeable to the inlet header and the balance to the outlet header. The outlet header loss is largely associated with the nonuniform velocity distribution shown as $u_o(y)$ in Fig. 2-42*b* and *c*. Of course, for the free-discharge header, all the loss is chargeable to the inlet header.

While the foregoing results are based on a uniform inlet velocity u_i , as shown in Fig. 2-42*b* to *d*, experimental results [17] demonstrate that header performance is relatively insensitive to inlet velocity maldistribution.

The following expressions provide the geometrical description of the inlet headers in terms of the header section area variation shown on Figs. 2-42*b* to *d*.

PARALLEL-FLOW INLET HEADER

$$\frac{A}{A_i} = \frac{(1 - X)(A_o/A_i)}{\left[\left(\frac{\rho_i}{\rho_o} \right) \frac{\pi^2}{4} X^2 + \left(\frac{A_o}{A_i} \right)^2 \right]^{1/2}} \quad (2-31)$$

COUNTERFLOW INLET HEADER

$$\frac{A}{A_i} = \frac{(A_o/A_i)}{(\rho_i/\rho_o)^{1/2}(\pi/2)} = 1 \quad (2-32a)$$

FREE DISCHARGE INLET HEADER

$$\frac{A}{A_i} = (1 - X) \quad (2-33)$$

The exit headers for the first geometries have a uniform section A_o , a boxlike geometry.

Examination of Eq. (2-31) for the parallel-flow geometry indicates that A_i (at $X = 0$ for the inlet header) can be selected independently of A_o (for the exit header). In contrast, for the counterflow geometry, which is also boxlike, i.e., of uniform section $A = A_i$, Eq. (2-32*a*) indicates that A_i/A_o is fixed.

$$\frac{A_i}{A_o} = \left[\left(\frac{\rho_i}{\rho_o} \right)^{1/2} \left(\frac{\pi}{2} \right) \right]^{-1} \quad (2-32b)$$

As a point of interest, Eq. (2-33) for the free-discharge inlet header is derived from Eq. (2-31) by allowing A_o/A_i to become very large, as suggested by the upper drawing in Fig. 2-42*d*.

Reference 17 provides, in addition to the header geometries, expressions for pressure and velocity variations. Also provided are the derivations for the following head-loss expressions chargeable to the headers.

PARALLEL-FLOW HEADER

$$\frac{\Delta P_t}{h_i} = 1.467 \frac{h_o}{h_i} + 1 \quad (2-34)$$

COUNTERFLOW HEADER

$$\frac{\Delta P_t}{h_i} = 1 - \frac{4}{\pi^2} = 0.595 \quad (2-35)$$

FREE-DISCHARGE HEADER

$$\frac{\Delta P_t}{h_i} = 1 \quad (2-36)$$

where ΔP_t denotes the loss in “total” pressure and h_i and h_o denote the inlet and outlet velocity heads, or dynamic pressures, respectively.

If the ΔP_t is small, say less than 10 percent, compared to the matrix ΔP , the header design will not impact greatly on flow distribution. However, frequently ΔP_t may be higher than 30 percent of the matrix ΔP , and then the header design becomes very important.

USEFUL RELATIONS FOR SURFACE AND CORE GEOMETRY

Certain geometrical relations are necessary in the application of the basic heat transfer and flow-friction data to the design problem. One particular set and form of these relations that has proved to be convenient is given here.

The dimensions for the various surfaces in Tables 9-1 to 9-5 apply to the equations below. These dimensions are, in effect, part of the *basic design data* for the surface selected as indicated in the Fig. 2-12 methodology. They include the following:

b = plate spacing (plate-fin surfaces only), ft or m

r_h = flow-passage hydraulic radius $A_c L/A$, ft or m

β = ratio of total transfer area of one side of exchanger to volume between plates of that side (plate-fin surfaces only), ft^2/ft^3 or m^2/m^3

α = ratio of total transfer area of one side of exchanger to *total* exchanger volume (given only for matrix, tubular, and finned-type surfaces), ft^2/ft^3 or m^2/m^3

p = porosity, void volume/total volume (matrix surfaces only)

In addition, for the plate-fin surfaces, the plate thickness a , ft or m, must be separately specified.

The following geometrical factors are required as a design result for *each* of the *two sides* of the complete heat exchanger core:

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A = total transfer area of one side of exchanger, ft² or m²

A_c = free-flow area of one side, ft² or m²

A_{fr} = frontal area of one side, ft² or m²

L = flow length on one side, ft or m

V = total exchanger volume, ft³ or m³

α = ratio of total transfer area of one side of exchanger to *total* exchanger volume (this is already part of given basic data for matrix tubular and finned-tube surfaces), ft²/ft³ or m²/m³

σ = ratio of free-flow to frontal area of one side of exchanger

The equations below give the relations between surface and core factors for one side of the exchanger. Subscript 1 refers to any one side, and 2 refers to the other side; factors without a subscript are common to both sides. The same relations apply to side 2 with the exchange of subscripts 1 for 2 and 2 for 1.

$$r_h = L \left(\frac{A_c}{A} \right) \quad (2-37a)$$

$$\sigma_1 = \left(\frac{A_c}{A_{fr}} \right)_1 = \left(\frac{Ar_h}{LA_{fr}} \right)_1 = \frac{(Ar_h)_1}{V} = (\alpha r_h)_1 \quad \text{(except for matrix surfaces)} \quad (2-37b)$$

$$\sigma_1 = \frac{b_1 \beta_1 r_{h1}}{b_1 + b_2 + 2a} \quad \text{(plate-fin surfaces only)} \quad (2-37c)$$

$$\alpha_1 = \frac{A_1}{V} = \left(\frac{A}{LA_{fr}} \right)_1 = \left(\frac{\sigma}{r_h} \right)_1 \quad \text{(except for matrix surfaces)} \quad (2-37d)$$

$$\alpha_1 = \frac{b_1 \beta_1}{b_1 + b_2 + 2a} \quad \text{(plate-fin surfaces only)} \quad (2-37e)$$

$$A_{c1} = (\sigma A_{fr})_1 = \left(\frac{Ar_h}{L} \right)_1 = \left(\frac{A\sigma}{L\alpha} \right)_1 \quad \text{(except for matrix surfaces)} \quad (2-37f)$$

$$\left(\frac{r_h}{L} \right)_1 = \left(\frac{A_c}{A} \right)_1 = \left(\frac{\sigma}{L\alpha} \right)_1 \quad \text{(except for matrix surfaces)} \quad (2-37g)$$

$$p = \frac{A_c}{A_{fr}} = \alpha r_h \quad \text{(matrix surfaces only)} \quad (2-37h)$$

$$\alpha = \frac{p}{r_h} \quad \text{(for matrix surfaces only)} \quad (2-37i)$$

$$A_c = p A_{fr} = \alpha r_h A_{fr} = A \frac{r_h}{L} \quad \text{(matrix surfaces only)} \quad (2-37j)$$

$$A = \frac{\alpha}{p} L A_c \quad \text{(matrix surfaces only)} \quad (2-37k)$$

$$A = \alpha L A_{fr} \quad (2-37l)$$

The following relations† apply for square-mesh crossed-rod matrices only:

$$p = 1 - \frac{\pi}{4x_t} \quad (2-37m)$$

$$\sigma = \frac{(x_t - 1)^2}{x_t^2} \quad (2-37n)$$

$$\alpha d = \frac{\pi}{x_t} \quad (2-37o)$$

$$\frac{r_h}{d} = \frac{x_t}{\pi} - \frac{1}{4} = \frac{1}{4} \frac{\text{Re}}{\text{Re}_d} \quad (2-37p)$$

PROCEDURE FOR SIZING A HEAT EXCHANGER

The methodology of heat exchanger design was previously considered in relation to Figs. 2-1 and 2-12. It is the purpose of this section to consider in some detail the workings of the *heat exchanger analysis procedure* box of Fig. 2-12.

Two broad categories of *problem specification* are as follows. Given the core geometry, the flow rates, and the entering fluid temperatures, what is the *rating* of the heat exchanger? That is, what heat transfer rate and exchanger effectiveness is predicted and what are the resulting outlet temperatures? Illustrations of this rating procedure are covered in Appendix B. The second category, which is the major subject of this section, is termed the *sizing problem* in distinction to the *rating problem*. The sizing problem asks the following question: What is the size of the core, given the flow rates and their entering and *leaving* temperatures? These in turn establish the desired heat transfer rate and exchanger effectiveness. Clearly, one problem is the inverse of the other. The sizing problem is the *design* problem, while the rating problem is a *performance* prediction for a specified design.

The complete design of a heat exchanger involves a whole set of considerations, as indicated by Fig. 2-12. In this section the term *design* will be used in a very restricted sense. Suppose it is assumed that not only are the performance characteristics completely established but also that the general type of heat exchanger has been selected, the flow arrangement (counterflow, crossflow, etc.) has been chosen, and the heat transfer surface configurations for the two fluid sides have been selected. The problem then is to determine the dimensions of the overall heat exchanger.

Note that, if a plate-fin type of heat exchanger is chosen, the designer can in principle select the surface configurations for the two fluid sides completely independently. This, in fact, is one of the virtues of the plate-fin construction. The majority of the surface configurations considered in this book are of this type. As an example, one might select surface number 11.1 from Fig. 9-4 for one

† For matrix surfaces, α , r_h , and p apply to the condition of perfect stacking, i.e., no separation between layers.

fluid side, and surface number 11.5-3/8W from Fig. 9-7 for the other side. Why one would select a particular pair of surfaces for a particular application is another question, discussed previously and not of concern here.

As a general rule, tubular surface configurations (for example, surface number CF-8.72(c) from Fig. 9-10, or surface number 9.68-0.87R from Fig. 9-13) are so constructed that both fluid surfaces are specified, or else restricted to a narrow range of possibilities, as soon as one surface is selected. Thus the designer has somewhat less freedom in surface selection.

For rotating periodic-flow heat exchangers, the same surface is exposed to each of the two fluids, so only one surface can be selected.

In any case, for present purposes it is assumed that the surfaces have been established, as well as the flow arrangement and the desired operating parameters. Under these conditions, it is possible to size a simple crossflow heat exchanger, or a multipass crossflow heat exchanger, to yield a prescribed heat exchanger effectiveness and independently prescribed pressure drops on *both* fluid sides. Similarly, it is possible to size a rotating periodic-flow heat exchanger for prescribed effectiveness and two independently prescribed pressure drops. On the other hand, for a simple counterflow heat exchanger, it is possible to size for prescribed effectiveness and for the pressure drop on only *one* of the two fluid sides. The pressure drop on the other side becomes a dependent variable. It can be changed only by selecting a different surface for that side. Note then that the counterflow heat exchanger does not provide quite as much design flexibility as either the simple or multipass crossflow exchanger, although it does in general yield a higher effectiveness for a given amount of surface area (i.e., a given N_{tu}).

The procedure for sizing any one of these heat exchangers is almost inevitably an iterative one and thus lends itself very conveniently to computer implementation. To illustrate such a procedure, a simple crossflow arrangement will be considered. It will then be shown how a multipass crossflow heat exchanger can be easily derived from a single-pass crossflow design. A procedure for designing a counterflow heat exchanger can then be easily inferred, subject to the restriction that only one fluid pressure drop can be independently prescribed. This procedure is also very easily adapted to the design of a rotating periodic-flow heat exchanger.

Figure 2-43 shows the geometry of a single-pass crossflow heat exchanger. It will be assumed that each pass is unmixed. The two fluids will be designated by subscripts h and c . The two fluid flow rates, W_h and W_c , are specified, as well as all four terminal temperatures and the pressure drop for each fluid. The problem is to determine the three dimensions a , b , and c , and thus the volume and (usually) the weight of the heat exchanger. Note that a is the flow length for the hot fluid, b is the flow length for the cold fluid, and c is a "no-flow" dimension that might be scaled up or down if the flow rates are changed in proportion to each other, all other specifications remaining the same.

Note that the frontal area for the hot fluid (i.e., the cross section facing fluid h) is bc . Similarly, the frontal area for the cold fluid is ac .

The first step, after having chosen the two surfaces, is to assemble the geometric characteristics of the surface pair:

$$\begin{array}{ll}
 r_{h,h} & r_{h,c} \\
 \sigma_h & \sigma_c \\
 \alpha_h & \alpha_c \\
 (A_f/A)_h & (A_f/A)_c \\
 (A_w/A)_h & (A_w/A)_c \\
 \delta_h & \delta_c
 \end{array}$$

The equations in the preceding section can be used to determine the α 's and the σ 's.

The next question is what to use for an iteration variable. A very convenient variable is the mass velocity G for each side of the heat exchanger, since it is easy to make a reasonable initial estimate of both G 's, and the design will converge rapidly by a simple adjustment of the G 's. Note that G is the mass flow rate W divided by the flow cross-sectional area A_c . A_c is in turn related to the frontal area A_{fr} through σ . Thus, when G_h and G_c are specified, the frontal areas bc and ac are fixed. The complete dimensions of the heat exchanger are then established when the volume V is determined.

A first estimate of G_h and G_c can be made using the following approximate relation:

$$G \approx \sqrt{\left(\frac{\text{StPr}^{2/3}}{f}\right)\left(\frac{\Delta P}{N_{tu}}\right) \frac{g_c}{v_m \text{Pr}^{2/3}}} \quad (2-38)$$

This relation is derived from the core-velocity equation (2-30b) by treating η_o as unity (neglecting fin ineffectiveness) and taking

$$N_{tu,side} \approx 2N_{tu}$$

In this last approximation it is implied that in Eqs. (2-2) or (2-10) the wall resistance is negligible and the convective heat transfer resistances are approximately equally distributed on the two sides of the heat exchanger.

As previously discussed, the ratio $\text{StPr}^{2/3}/f$ tends to be reasonably constant over a 2:1 Reynolds number range for most surfaces and does not vary greatly from surface to surface, as can be seen in Fig. 2-41. It can be estimated from the basic data curves for the surfaces in question, but a figure of 0.3 provides a good starting point in any case, since this is merely the initial estimate for an iterative procedure. ΔP is the desired pressure drop for the side in question, and N_{tu} is determined for the desired heat exchanger effectiveness and the C_{min}/C_{max} , derived from the given flow rates (see Fig. 2-14 or Table 2-3).

With initial estimates of G_h and G_c , the following variables can be evaluated in succession:

- | | | |
|----------------------------|-------------------------|--|
| 1. Re_h | Re_c | from the definition of Reynolds number |
| 2. $(\text{StPr}^{2/3})_h$ | $(\text{StPr}^{2/3})_c$ | from the basic data curves |
| 3. f_h | f_c | from the basic data curves |
| 4. h_h | h_c | from (2) and the G 's |
| 5. $\eta_{f,h}$ | $\eta_{f,c}$ | from Eq. (2-4) |
| 6. $\eta_{o,h}$ | $\eta_{o,c}$ | from Eq. (2-3) |
| 7. $(UA)_h$ | or $(UA)_c$ | from Eq. (2-2) |
| 8. A_h | or A_c | from (7) and N_{tu} |

9. V		from A and α
10. ac	bc	from the G 's, W 's, and σ
11. a, b, c		from V and ac and bc
12. $(K_c, K_e)_h$	$(K_c, K_e)_c$	from $\sigma_h, \sigma_c, Re_h, Re_c$
13. $(\Delta P/P_1)_h$	$(\Delta P/P_1)_c$	from Eq. (2-26a)

These pressure-drop results are then compared with those specified for the design, and new estimates of G_h and G_c are made according to the following approximation, which comes directly from Eq. (2-26a):

$$G \propto \sqrt{\Delta P} \quad (2-39)$$

The procedure is then repeated until the two pressure drops are within some predetermined fraction of the specified pressure drops. The convergence is generally quite rapid, often no more than three or four iterations. It can be done by hand calculator but obviously lends itself very well to a computer program.

A multipass cross-counterflow heat exchanger can be designed using a simple modification of the same procedure. Examine the two heat exchangers shown in Fig. 2-43. Both have the same volume and therefore the same amount of surface area, if the surface configurations are the same. Both have the same frontal areas and therefore the same free-flow areas. If the flow rates are the same, both have the same G 's. The total flow length for each fluid is the same, and except for any pressure drop associated with the turning of the cold fluid between passes, the pressure drops for each fluid must then be the same. UA must be the same, and thus so must N_{tu} . In fact, the only difference is the effectiveness, which is higher for the two-pass case, since the applicable effectiveness solution is Fig. 2-20 or Eq. (2-18) rather than Fig. 2-16. It is apparent, then, that a multipass exchanger can be designed as a single-pass exchanger, but using the multipass value of N_{tu} , and then the overall dimensions can be transformed as in Fig. 2-43.

Numerical examples for the exercise of the foregoing procedure for the design, or sizing, problem can be readily formulated from any of the four illustrative rating-type problems presented in Appendix B. Recall that the problem specification box of Fig. 2-12 would start for this exercise with one of the performance result sets derived in Appendix B and then, presumably, the sizing procedure, after several iterations, would result in the core dimensions pictured in the appropriate Appendix B sketch, Fig. B-1, B-3, B-4, or B-5.

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Table 2-2 Counterflow Exchanger Performance

Exchanger effectiveness (ϵ) as a function of capacity-ratio ratio ($C_{\min}/C_{\max}$) and number of heat transfer units (N_{tu})

[illegible]

Table 2-3 Parallel-Flow Exchanger Performance

Exchanger effectiveness (ϵ) as a function of capacity-rate ratio (C_{min}/C_{max}) and number of heat transfer units (N_{tu})

N_{tu}	ϵ for indicated capacity-rate ratios, C_{min}/C_{max}				
	0	0.25	0.50	0.75	1.00
0	0	0	0	0	0
0.25	0.221	0.215	0.208	0.202	0.197
0.50	0.393	0.372	0.352	0.333	0.316
0.75	0.528	0.487	0.450	0.418	0.388
1.00	0.632	0.571	0.518	0.472	0.432
1.25	0.713	0.632	0.564	0.507	0.459
1.50	0.777	0.677	0.596	0.530	0.475
1.75	0.826	0.710	0.618	0.544	0.485
2.00	0.865	0.734	0.633	0.554	0.491
2.50	0.918	0.765	0.651	0.564	0.497
3.00	0.950	0.781	0.659	0.568	0.498
3.50	0.970	0.790	0.663	0.570	0.499
4.00	0.982	0.795	0.665	0.571	0.500
4.50	0.989	0.797	0.666	0.571	0.500
5.00	0.993	0.799	0.666	0.571	0.500
∞	1.000	0.800	0.667	0.571	0.500

Table 2-4 Crossflow Exchanger with Both Fluids Un-Mixed

Crossflow (both fluids un-mixed) exchanger effectiveness (ϵ) as a function of capacity-rate ratio (C_{min}/C_{max}) and number of heat transfer units (N_{tu})

N_{tu}	ϵ for indicated capacity-rate ratios, C_{min}/C_{max}				
	0.00	0.25	0.50	0.75	1.00
0.00	0.000	0.000	0.000	0.000	0.000
0.25	0.221	0.215	0.209	0.204	0.199
0.50	0.393	0.375	0.358	0.341	0.326
0.75	0.528	0.495	0.466	0.439	0.413
1.00	0.632	0.588	0.547	0.510	0.476
1.25	0.714	0.660	0.610	0.565	0.523
1.50	0.777	0.716	0.660	0.608	0.560
1.75	0.826	0.761	0.700	0.642	0.590
2.00	0.865	0.797	0.732	0.671	0.614
2.50	0.918	0.851	0.783	0.716	0.652
3.00	0.950	0.888	0.819	0.749	0.681
3.50	0.970	0.915	0.848	0.776	0.704
4.00	0.982	0.934	0.869	0.797	0.722
4.50	0.989	0.948	0.887	0.814	0.737
5.00	0.993	0.959	0.901	0.829	0.751
6.00	0.997	0.974	0.924	0.853	0.772
7.00	0.999	0.983	0.940	0.871	0.789
∞	1.000	1.000	1.000	1.000	1.000

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Table 2-5 Crossflow Exchanger with One Fluid Mixed

Crossflow (one fluid "mixed," the other fluid "unmixed") exchanger effectiveness (ϵ) as a function of capacity-rate ratio ($C_{\text{mixed}}/C_{\text{unmixed}}$) and number of heat transfer units (N_{tu})

N_{tu}	ϵ for indicated capacity-rate ratios, $C_{\text{mixed}}/C_{\text{unmixed}}$							
	0	0.25	4.00	0.50	2.00	0.75	1.333	1.000
0	0	0	0	0	0	0	0	0
0.25	0.221	0.215	0.213	0.209	0.209	0.204	0.204	0.198
0.50	0.393	0.375	0.375	0.358	0.357	0.341	0.341	0.325
0.75	0.528	0.495	0.494	0.465	0.463	0.463	0.435	0.410
1.00	0.632	0.587	0.585	0.545	0.542	0.505	0.503	0.469
1.25	0.713	0.658	0.654	0.605	0.600	0.556	0.552	0.510
1.50	0.777	0.714	0.706	0.652	0.644	0.594	0.589	0.540
1.75	0.826	0.758	0.747	0.689	0.677	0.623	0.616	0.562
2.00	0.865	0.793	0.778	0.715	0.702	0.645	0.636	0.579
2.50	0.918	0.844	0.820	0.760	0.736	0.677	0.663	0.601
3.00	0.950	0.879	0.846	0.789	0.756	0.697	0.679	0.613
3.50	0.970	0.903	0.861	0.808	0.768	0.710	0.689	0.621
4.00	0.982	0.920	0.870	0.823	0.776	0.718	0.695	0.625
4.50	0.989	0.933	0.876	0.834	0.780	0.724	0.698	0.628
5.00	0.993	0.942	0.880	0.841	0.783	0.728	0.700	0.630
∞	1.000	0.982	0.885	0.865	0.787	0.736	0.703	0.632

Table 2-6 Crossflow Exchanger with Both Fluids Mixed

Crossflow (both fluids mixed) exchanger effectiveness (ϵ) as a function of capacity-rate ratio ($C_{\text{min}}/C_{\text{max}}$) and number of heat transfer units (N_{tu})

N_{tu}	ϵ for indicated capacity-rate ratios, $C_{\text{min}}/C_{\text{max}}$					
	0	0.20	0.40	0.60	0.80	1.00
0	0	0	0	0	0	0
0.20	0.181	0.178	0.175	0.172	0.169	0.166
0.60	0.451	0.431	0.412	0.395	0.378	0.362
1.00	0.632	0.593	0.557	0.523	0.491	0.462
1.40	0.753	0.698	0.647	0.599	0.555	0.515
1.80	0.835	0.767	0.703	0.645	0.591	0.543
2.00	0.865	0.792	0.723	0.660	0.603	0.552
2.20	0.889	0.812	0.739	0.672	0.611	0.557
2.60	0.926	0.841	0.761	0.687	0.621	0.563
3.00	0.950	0.860	0.774	0.695	0.625	0.565
3.50	0.970	0.875	0.783	0.700	0.626	0.563
4.00	0.982	0.884	0.787	0.700	0.624	0.559
4.50	0.989	0.888	0.789	0.698	0.621	0.555
5.00	0.993	0.890	0.788	0.695	0.617	0.551

Table 2-7 Counterflow Exchanger with Crossflow Headers

ϵ for $C_{\min}/C_{\max} = 0.20$								
N_{tu} , counter- flow	N_{tu} , crossflow							
	0	1	2	3	4	5	6	7
0	0	0.597	0.811	0.903	0.946	0.969	0.982	0.989
1	0.605	0.825	0.914	0.955	0.975	0.986	0.992	0.995
2	0.834	0.923	0.961	0.979	0.989	0.993	0.996	0.998
3	0.927	0.966	0.983	0.991	0.995	0.997	0.998	0.999
4	0.967	0.985	0.992	0.996	0.998	0.999	0.999	1.000
5	0.985	0.993	0.997	0.998	0.999	0.999	1.000	1.000
6	0.995	0.997	0.998	0.999	1.000	1.000	1.000	1.000
7	0.998	0.999	0.999	1.000	1.000	1.000	1.000	1.000
ϵ for $C_{\min}/C_{\max} = 0.40$								
N_{tu} , counter- flow	N_{tu} , crossflow							
	0	1	2	3	4	5	6	7
0	0	0.564	0.759	0.848	0.898	0.928	0.948	0.961
1	0.577	0.783	0.871	0.916	0.942	0.959	0.970	0.978
2	0.795	0.887	0.930	0.954	0.968	0.977	0.983	0.988
3	0.897	0.940	0.962	0.975	0.982	0.987	0.991	0.993
4	0.945	0.968	0.979	0.986	0.990	0.993	0.995	0.996
5	0.970	0.982	0.989	0.992	0.995	0.996	0.997	0.998
6	0.985	0.990	0.994	0.996	0.997	0.998	0.998	0.999
7	0.991	0.995	0.997	0.998	0.998	0.999	0.999	0.999
ϵ for $C_{\min}/C_{\max} = 0.60$								
N_{tu} , counter- flow	N_{tu} , crossflow							
	0	1	2	3	4	5	6	7
0	0	0.532	0.708	0.792	0.842	0.875	0.898	0.916
1	0.552	0.739	0.821	0.867	0.896	0.917	0.931	0.943
2	0.754	0.843	0.888	0.914	0.932	0.945	0.954	0.962
3	0.854	0.902	0.928	0.944	0.955	0.963	0.969	0.974
4	0.908	0.937	0.953	0.963	0.970	0.976	0.980	0.983
5	0.940	0.959	0.969	0.976	0.980	0.984	0.986	0.988
6	0.963	0.973	0.980	0.984	0.987	0.989	0.991	0.992
7	0.975	0.982	0.987	0.989	0.991	0.993	0.994	0.995

(continued)

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Table 2-7 Counterflow Exchanger with Crossflow Headers (*continued*)

ϵ for $C_{\min}/C_{\max} = 0.8$								
N_{tu} , counter- flow	N_{tu} , crossflow							
	0	1	2	3	4	5	6	7
0	0	0.503	0.660	0.736	0.783	0.815	0.838	0.856
1	0.512	0.694	0.768	0.811	0.840	0.860	0.876	0.889
2	0.712	0.793	0.835	0.861	0.880	0.894	0.905	0.914
3	0.805	0.852	0.878	0.896	0.908	0.918	0.926	0.932
4	0.860	0.891	0.908	0.920	0.929	0.936	0.942	0.947
5	0.895	0.917	0.929	0.938	0.945	0.950	0.954	0.958
6	0.922	0.936	0.945	0.951	0.956	0.960	0.963	0.966
7	0.940	0.950	0.957	0.961	0.965	0.968	0.971	0.973
ϵ for $C_{\min}/C_{\max} = 1.00$								
N_{tu} , counter- flow	N_{tu} , crossflow							
	0	1	2	3	4	5	6	7
0	0	0.476	0.615	0.682	0.723	0.752	0.773	0.790
1	0.500	0.649	0.714	0.751	0.777	0.795	0.810	0.821
2	0.667	0.739	0.775	0.798	0.815	0.828	0.838	0.846
3	0.750	0.792	0.816	0.831	0.843	0.852	0.860	0.866
4	0.800	0.828	0.844	0.855	0.863	0.870	0.876	0.880
5	0.834	0.853	0.865	0.873	0.880	0.885	0.889	0.893
6	0.856	0.872	0.881	0.887	0.892	0.897	0.900	0.903
7	0.875	0.886	0.893	0.899	0.903	0.906	0.909	0.912

Table 2-8 Parallel-Counterflow Exchanger with Shell Fluid Mixed

Parallel-counterflow (one fluid mixed) exchanger effectiveness (ϵ) as a function of capacity-rate ratio ($C_{\min}/C_{\max}$) and number of heat transfer units (N_{tu})

N_{tu}	ϵ for indicated capacity-rate ratios, $C_{\min}/C_{\max}$				
	0.0	0.25	0.50	0.75	1.00
0.00	0.000	0.000	0.000	0.000	0.000
0.25	0.221	0.215	0.209	0.204	0.198
0.50	0.393	0.375	0.357	0.340	0.324
0.75	0.528	0.494	0.463	0.434	0.407
1.00	0.632	0.584	0.540	0.500	0.463
1.25	0.714	0.652	0.597	0.546	0.500
1.50	0.777	0.705	0.639	0.579	0.526
1.75	0.826	0.744	0.670	0.603	0.544
2.00	0.865	0.775	0.693	0.620	0.557
2.50	0.918	0.816	0.724	0.642	0.572
3.00	0.950	0.841	0.741	0.654	0.579
3.50	0.970	0.855	0.751	0.660	0.582
4.00	0.982	0.864	0.757	0.663	0.584
4.50	0.989	0.869	0.760	0.665	0.585
5.00	0.993	0.872	0.762	0.666	0.585
∞	1.000	0.877	0.764	0.667	0.586

Table 2-9a Split-Flow Exchanger, $C_{tube} = C_{\min}$

Split-flow exchanger with shell fluid mixed. Exchanger effectiveness (ϵ) as a function of capacity-rate ratio ($C_{\min}/C_{\max}$) and number of transfer units (N_{tu}) for $C_{tube} = C_{\min}$

N_{tu}	ϵ for indicated capacity-rate ratios, $C_r/C_{\min}$					
	0	0.2	0.4	0.6	0.8	1.0
0	0	0	0	0	0	0
0.5	0.393	0.378	0.364	0.350	0.337	0.324
1.0	0.632	0.583	0.557	0.523	0.491	0.462
1.5	0.777	0.718	0.663	0.613	0.566	0.524
2.0	0.865	0.791	0.722	0.660	0.603	0.552
2.5	0.918	0.833	0.755	0.683	0.619	0.562
3.0	0.950	0.857	0.771	0.693	0.624	0.564
3.5	0.970	0.870	0.778	0.695	0.623	0.561
4.0	0.982	0.876	0.779	0.693	0.619	0.556
4.5	0.989	0.878	0.777	0.689	0.614	0.551
5.0	0.993	0.877	0.773	0.683	0.607	0.545
5.5	0.996	0.875	0.768	0.676	0.601	0.538
6.0	0.998	0.872	0.762	0.670	0.594	0.532
6.5	0.999	0.868	0.756	0.663	0.588	0.526
7.0	0.999	0.864	0.750	0.656	0.581	0.520
7.5	0.999	0.860	0.744	0.650	0.575	0.515
8.0	1.000	0.856	0.738	0.644	0.569	0.510
8.5	1.000	0.852	0.732	0.638	0.564	0.505
9.0	1.000	0.849	0.727	0.632	0.559	0.500
9.5	1.000	0.846	0.722	0.627	0.554	0.496
10.0	1.000	0.843	0.718	0.622	0.549	0.492

Table 2-9b Split-Flow Exchanger, $C_{\text{shell}} = C_{\text{min}}$

Split-flow exchanger (shell fluid mixed) effectiveness (ϵ) as a function of capacity rate ($C_{\text{min}}/C_{\text{max}}$) and number of heat transfer units (N_{tu}) for $C_{\text{shell}} = C_{\text{min}}$

N_{tu}	ϵ for indicated capacity-rate ratios, C_r/C_{min}					
	0	0.2	0.4	0.6	0.8	1.0
0	0	0	0	0	0	0
0.5	0.393	0.378	0.364	0.350	0.337	0.324
1.0	0.632	0.593	0.557	0.523	0.492	0.462
1.5	0.777	0.718	0.664	0.613	0.567	0.524
2.0	0.865	0.792	0.723	0.661	0.603	0.552
2.5	0.918	0.835	0.757	0.685	0.620	0.562
3.0	0.950	0.861	0.775	0.697	0.626	0.564
3.5	0.970	0.876	0.785	0.701	0.627	0.561
4.0	0.982	0.884	0.789	0.702	0.624	0.556
4.5	0.989	0.889	0.791	0.700	0.620	0.551
5.0	0.993	0.892	0.790	0.697	0.615	0.544
5.5	0.996	0.893	0.789	0.693	0.609	0.538
6.0	0.998	0.893	0.787	0.689	0.604	0.532
6.5	0.999	0.893	0.785	0.685	0.599	0.526
7.0	0.999	0.892	0.782	0.680	0.593	0.520
7.5	0.999	0.892	0.780	0.676	0.588	0.515
8.0	1.000	0.891	0.777	0.672	0.583	0.510
8.5	1.000	0.890	0.774	0.668	0.578	0.505
9.0	1.000	0.889	0.771	0.664	0.573	0.500
9.5	1.000	0.888	0.769	0.660	0.569	0.496
10.0	1.000	0.887	0.766	0.656	0.564	0.492

Table 2-10 Periodic-Flow Exchanger, $C_{\min}/C_{\max} = 1.0$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,0}$)

$N_{tu,0}$	ϵ for the indicated matrix capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.5	2.0	3.0	5.0	10.0	∞
0	0	0	0	0	0	0	0	0
0.5	0.322	0.326	0.328	0.330	0.332	0.333	0.333	0.333
1.0	0.467	0.478	0.485	0.491	0.496	0.499	0.500	0.500
1.5	0.548	0.566	0.576	0.586	0.594	0.598	0.599	0.600
2.0	0.601	0.623	0.636	0.649	0.659	0.664	0.666	0.667
2.5	0.639	0.665	0.679	0.694	0.705	0.711	0.713	0.714
3.0	0.667	0.696	0.712	0.728	0.740	0.746	0.749	0.750
3.5	0.690	0.721	0.738	0.755	0.767	0.774	0.777	0.778
4.0	0.709	0.741	0.759	0.776	0.789	0.796	0.799	0.800
4.5	0.724	0.758	0.776	0.794	0.807	0.814	0.817	0.818
5.0	0.738	0.772	0.791	0.809	0.822	0.829	0.832	0.833
5.5	0.749	0.785	0.803	0.821	0.834	0.842	0.845	0.846
6.0	0.759	0.796	0.814	0.832	0.845	0.853	0.856	0.857
6.5	0.768	0.805	0.824	0.842	0.855	0.862	0.865	0.867
7.0	0.776	0.814	0.833	0.850	0.863	0.870	0.874	0.875
7.5	0.784	0.822	0.840	0.858	0.871	0.878	0.881	0.882
8.0	0.790	0.829	0.847	0.865	0.877	0.884	0.888	0.889
8.5	0.796	0.835	0.854	0.871	0.883	0.890	0.894	0.895
9.0	0.802	0.841	0.859	0.876	0.888	0.895	0.899	0.900
9.5	0.807	0.846	0.864	0.881	0.893	0.900	0.904	0.905
10.0	0.811	0.851	0.869	0.886	0.898	0.904	0.908	0.909
20.0	0.865	0.906	0.922	0.935	0.943	0.948	0.951	0.952
50.0	0.914	0.951	0.962	0.970	0.975	0.978	0.980	0.980
90.0	0.935	0.969	0.977	0.982	0.986	0.987	0.988	0.989
100.0	0.939		0.979	0.984		0.989	0.989	0.990
500.0	0.974		0.995	0.996		0.998	0.998	0.998

Table 2-11 Periodic-Flow Exchanger, $C_{\min}/C_{\max} = 0.95$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,o}$)

$N_{tu,o}$	ϵ for the indicated matrix capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.5	2.0	3.0	5.0	10.0	∞
0	0	0	0	0	0	0	0	0
0.5	0.325	0.329	0.331	0.333	0.335	0.336	0.336	0.336
1.0	0.471	0.483	0.490	0.497	0.502	0.505	0.506	0.506
1.5	0.554	0.573	0.584	0.594	0.602	0.607	0.608	0.609
2.0	0.608	0.632	0.645	0.659	0.669	0.675	0.677	0.678
2.5	0.647	0.674	0.690	0.706	0.717	0.723	0.726	0.727
3.0	0.676	0.707	0.723	0.741	0.753	0.760	0.763	0.764
3.5	0.699	0.732	0.750	0.768	0.781	0.789	0.792	0.793
4.0	0.718	0.753	0.772	0.790	0.804	0.811	0.815	0.816
4.5	0.734	0.770	0.790	0.809	0.822	0.830	0.833	0.835
5.0	0.748	0.785	0.805	0.824	0.838	0.846	0.849	0.850
5.5	0.759	0.798	0.818	0.837	0.851	0.859	0.862	0.864
6.0	0.770	0.809	0.829	0.848	0.862	0.870	0.874	0.875
6.5	0.779	0.819	0.839	0.858	0.872	0.880	0.884	0.885
7.0	0.787	0.828	0.848	0.867	0.881	0.889	0.892	0.893
7.5	0.794	0.836	0.856	0.875	0.889	0.896	0.900	0.901
8.0	0.801	0.843	0.863	0.882	0.895	0.903	0.906	0.908
8.5	0.807	0.850	0.870	0.888	0.901	0.909	0.912	0.914
9.0	0.813	0.855	0.876	0.894	0.907	0.914	0.918	0.919
9.5	0.818	0.861	0.881	0.899	0.912	0.919	0.923	0.924
10.0	0.823	0.866	0.886	0.904	0.917	0.924	0.927	0.928
20.0	0.877		0.940	0.954		0.968	0.971	0.972
50.0	0.926	0.969	0.978	0.987	0.992	0.994	0.995	0.996
90.0	0.947	0.985	0.993	0.996	0.998	0.999	0.999	0.999
100.0	0.950		0.994	0.997		0.999	1.000	1.000

Table 2-12 Periodic-Flow Exchanger, $C_{\min}/C_{\max} = 0.90$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,0}$)

$N_{tu,0}$	ϵ for the indicated matrix capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.5	2.0	3.0	5.0	10.0	∞
0	0	0	0	0	0	0	0	0
0.5	0.327	0.331	0.334	0.336	0.338	0.338	0.339	0.339
1.0	0.476	0.489	0.496	0.503	0.508	0.511	0.512	0.513
1.5	0.560	0.580	0.591	0.603	0.611	0.616	0.617	0.618
2.0	0.616	0.641	0.655	0.669	0.680	0.686	0.688	0.689
2.5	0.655	0.684	0.700	0.717	0.729	0.736	0.739	0.740
3.0	0.685	0.717	0.735	0.753	0.766	0.774	0.777	0.778
3.5	0.708	0.743	0.762	0.781	0.795	0.803	0.806	0.807
4.0	0.727	0.764	0.784	0.804	0.819	0.826	0.830	0.831
4.5	0.744	0.782	0.803	0.823	0.838	0.846	0.849	0.850
5.0	0.757	0.797	0.818	0.839	0.853	0.862	0.865	0.866
5.5	0.769	0.810	0.831	0.852	0.867	0.875	0.879	0.880
6.0	0.780	0.822	0.843	0.864	0.878	0.887	0.890	0.892
6.5	0.789	0.832	0.853	0.874	0.888	0.896	0.900	0.902
7.0	0.797	0.841	0.862	0.883	0.897	0.905	0.909	0.910
7.5	0.804	0.849	0.871	0.891	0.905	0.913	0.917	0.918
8.0	0.811	0.856	0.879	0.898	0.912	0.920	0.923	0.925
8.5	0.817	0.863	0.884	0.904	0.918	0.926	0.929	0.931
9.0	0.823	0.869	0.891	0.910	0.923	0.931	0.935	0.936
9.5	0.828	0.874	0.896	0.915	0.928	0.936	0.939	0.941
10.0	0.833	0.880	0.901	0.920	0.933	0.940	0.944	0.945
20.0	0.887		0.954	0.968		0.981	0.984	0.985
40.0	0.925		0.984	0.992		0.997	0.998	0.998
100.0	0.956		0.999	0.999		1.000	1.000	1.000

Table 2-13 Periodic-Flow Exchanger $C_{\min}/C_{\max} = 0.80$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,o}$)

$N_{tu,o}$	ϵ for the indicated matrix capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.5	2.0	3.0	5.0	10.0	∞
0	0	0	0	0	0	0	0	0
0.5	0.332	0.337	0.339	0.341	0.343	0.344	0.345	0.345
1.0	0.486	0.500	0.507	0.515	0.521	0.524	0.525	0.525
1.5	0.573	0.595	0.607	0.619	0.629	0.634	0.636	0.636
2.0	0.630	0.657	0.673	0.689	0.701	0.707	0.710	0.711
2.5	0.670	0.702	0.721	0.739	0.753	0.760	0.763	0.764
3.0	0.701	0.737	0.757	0.777	0.792	0.800	0.803	0.804
3.5	0.725	0.763	0.785	0.806	0.822	0.830	0.834	0.835
4.0	0.744	0.785	0.808	0.830	0.846	0.855	0.858	0.860
4.5	0.760	0.804	0.827	0.849	0.866	0.874	0.878	0.880
5.0	0.774	0.819	0.843	0.866	0.882	0.891	0.894	0.896
5.5	0.786	0.832	0.856	0.879	0.895	0.904	0.908	0.909
6.0	0.797	0.844	0.868	0.891	0.907	0.915	0.919	0.921
6.5	0.806	0.854	0.879	0.901	0.917	0.925	0.929	0.930
7.0	0.814	0.863	0.888	0.910	0.925	0.933	0.937	0.939
7.5	0.821	0.871	0.896	0.918	0.933	0.941	0.944	0.946
8.0	0.828	0.879	0.903	0.925	0.939	0.947	0.951	0.952
8.5	0.834	0.886	0.910	0.931	0.945	0.952	0.956	0.957
9.0	0.839	0.892	0.916	0.936	0.950	0.957	0.961	0.962
9.5	0.845	0.897	0.921	0.941	0.955	0.961	0.965	0.966
10.0	0.849	0.902	0.926	0.946	0.959	0.965	0.968	0.970

Table 2-14 Periodic-Flow Exchanger, $C_{\min}/C_{\max} = 0.70$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,o}$)

$N_{tu,o}$	ϵ for the indicated matrix capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.50	2.00	3.00	5.00	10.00	∞
0	0	0	0	0	0	0	0	0
0.5	0.337	0.342	0.344	0.347	0.349	0.350	0.350	0.350
1.0	0.496	0.510	0.519	0.527	0.533	0.537	0.538	0.538
1.5	0.586	0.609	0.622	0.636	0.646	0.652	0.654	0.655
2.0	0.644	0.674	0.691	0.709	0.722	0.729	0.732	0.733
2.5	0.685	0.720	0.740	0.761	0.776	0.784	0.787	0.788
3.0	0.716	0.755	0.777	0.800	0.816	0.825	0.828	0.829
3.5	0.740	0.782	0.806	0.830	0.847	0.856	0.860	0.861
4.0	0.759	0.804	0.829	0.854	0.871	0.880	0.884	0.886
4.5	0.775	0.823	0.848	0.873	0.890	0.900	0.904	0.905
5.0	0.789	0.838	0.864	0.889	0.906	0.915	0.919	0.921
5.5	0.800	0.851	0.878	0.903	0.919	0.928	0.932	0.933
6.0	0.810	0.863	0.890	0.914	0.930	0.939	0.943	0.944
6.5	0.819	0.873	0.900	0.924	0.939	0.948	0.951	0.953
7.0	0.827	0.882	0.909	0.932	0.947	0.955	0.959	0.960
7.5	0.834	0.890	0.916	0.939	0.954	0.961	0.965	0.966
8.0	0.840	0.897	0.923	0.946	0.960	0.967	0.970	0.971
8.5	0.846	0.903	0.929	0.951	0.965	0.971	0.974	0.975
9.0	0.851	0.909	0.935	0.956	0.969	0.975	0.978	0.979
9.5	0.856	0.914	0.940	0.960	0.972	0.978	0.981	0.982
10.0	0.860	0.919	0.944	0.964	0.976	0.981	0.984	0.985

Table 2-15 Periodic-Flow Exchanger, $C_{\min}/C_{\max} = 0.50$

Exchanger effectiveness (ϵ) as a function of matrix capacity-rate ratio ($C_r/C_{\min}$) and modified number of heat transfer units ($N_{tu,o}$)

$N_{tu,o}$	ϵ for the indicated capacity-rate ratios, $C_r/C_{\min}$							
	1.0	1.25	1.50	2.00	3.00	5.00	10.00	∞
0	0	0	0	0	0	0	0	0
0.5	0.348	0.353	0.356	0.359	0.361	0.362	0.362	0.362
1.0	0.516	0.532	0.542	0.552	0.559	0.563	0.564	0.565
1.5	0.610	0.637	0.652	0.669	0.681	0.687	0.690	0.691
2.0	0.669	0.704	0.725	0.746	0.762	0.770	0.773	0.775
2.5	0.710	0.752	0.776	0.800	0.818	0.828	0.832	0.833
3.0	0.740	0.787	0.813	0.840	0.859	0.869	0.873	0.874
3.5	0.763	0.814	0.842	0.870	0.889	0.899	0.903	0.905
4.0	0.782	0.835	0.865	0.893	0.912	0.922	0.926	0.927
4.5	0.797	0.853	0.883	0.911	0.930	0.939	0.943	0.944
5.0	0.809	0.867	0.898	0.926	0.944	0.952	0.956	0.957
5.5	0.819	0.879	0.910	0.937	0.954	0.963	0.966	0.967
6.0	0.828	0.890	0.920	0.947	0.963	0.970	0.973	0.975
6.5	0.836	0.898	0.929	0.955	0.970	0.977	0.979	0.980
7.0	0.843	0.906	0.937	0.961	0.975	0.981	0.984	0.985
7.5	0.849	0.913	0.943	0.967	0.980	0.985	0.987	0.988
8.0	0.855	0.919	0.949	0.972	0.983	0.988	0.990	0.991
8.5	0.860	0.924	0.954	0.976	0.986	0.991	0.992	0.993
9.0	0.864	0.929	0.958	0.979	0.989	0.992	0.994	0.994
9.5	0.868	0.934	0.962	0.982	0.991	0.994	0.995	0.996
10.0	0.872	0.937	0.965	0.984	0.992	0.995	0.996	0.997

Fig. 2-13 Heat transfer effectiveness of straight and circular fins.

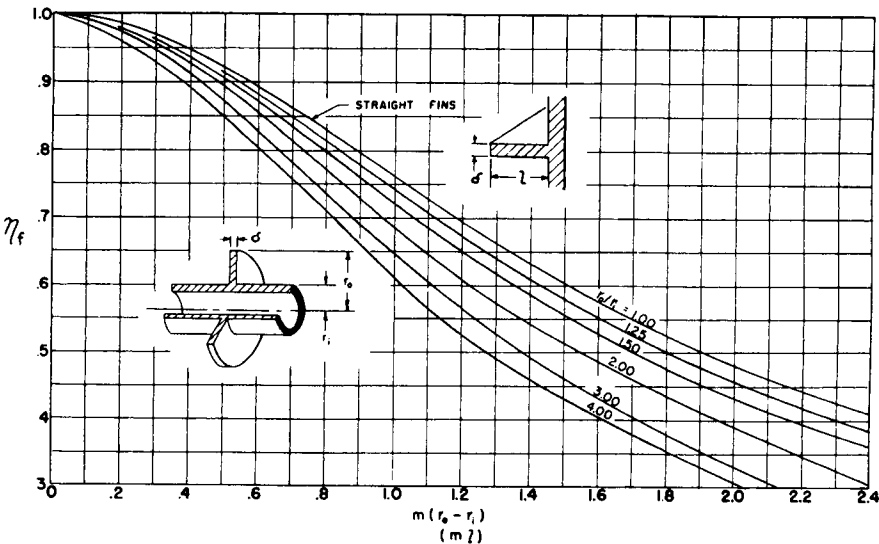


Fig. 2-14 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; counterflow exchanger.

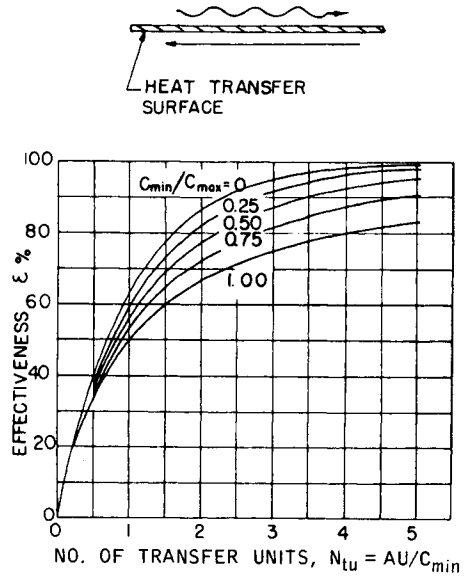


Fig. 2-15 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; parallel-flow exchanger.

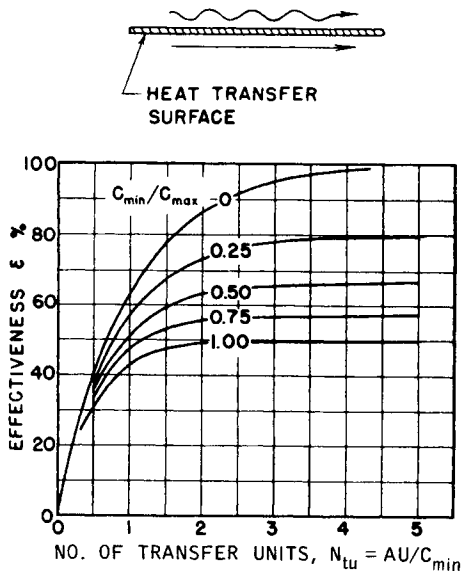


Fig. 2-16 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; crossflow exchanger with fluids unmixed.

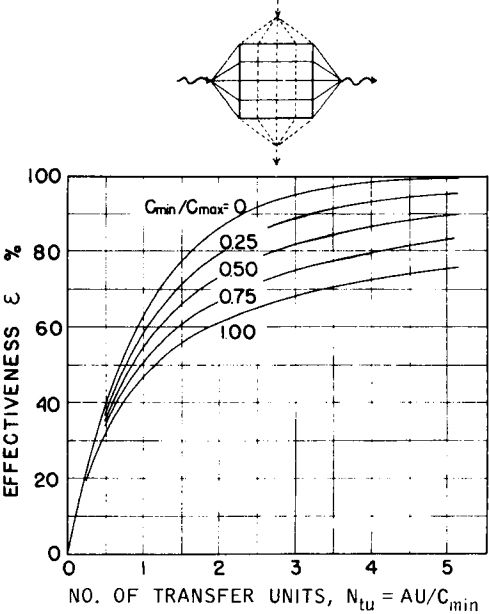


Fig. 2-17 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; crossflow exchanger with one fluid mixed.

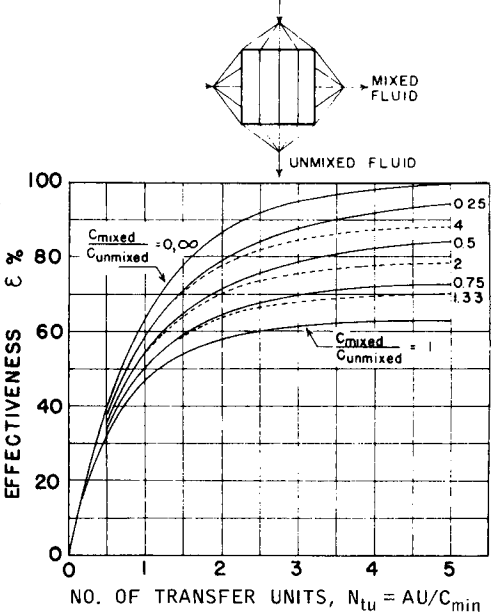


Fig. 2-18 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; crossflow exchanger with both fluids mixed.

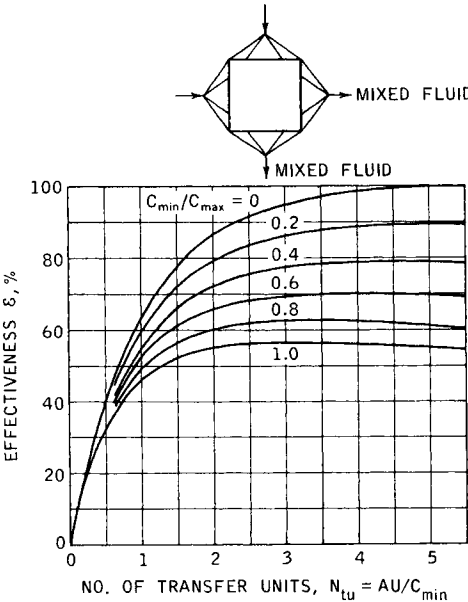


Fig. 2-19 Heat transfer effectiveness as a function of number of transfer units and number of passes; multipass cross-counterflow exchanger, $C_{min}/C_{max} = 1$; unmixed flow within passes, both fluids mixed between passes.

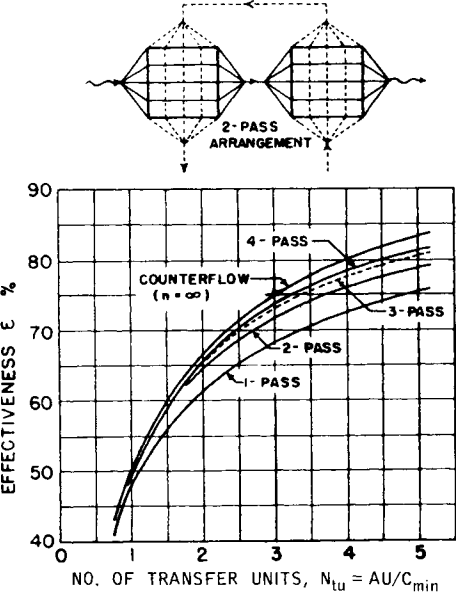


Fig. 2-20 Heat transfer effectiveness as a function of number of transfer units and number of passes; multi-pass cross-counterflow exchanger, $C_{\min}/C_{\max} = 1$; unmixed flow within passes, one fluid unmixed between passes.

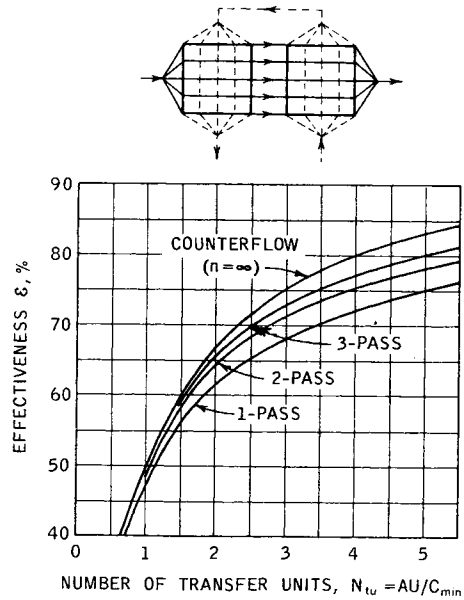


Fig. 2-21 Heat transfer effectiveness as a function of number of transfer units and number of passes; multi-pass cross-counterflow exchanger, $C_{\min}/C_{\max} = 1$; unmixed flow within passes, both fluids unmixed between passes.

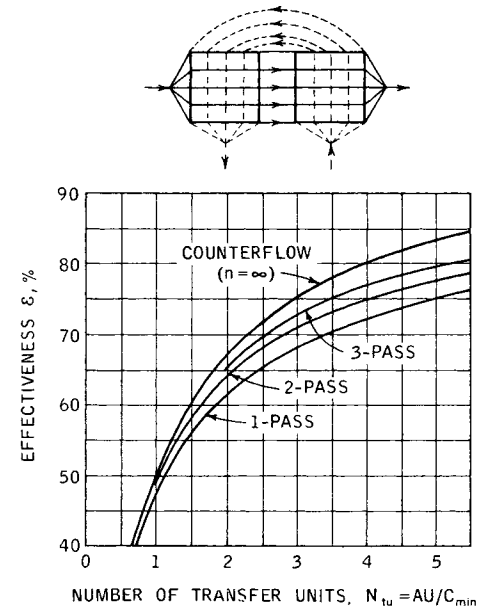


Fig. 2-22 Heat transfer effectiveness as a function of counterflow number of transfer units and crossflow number of transfer units; counterflow exchanger with crossflow headers.

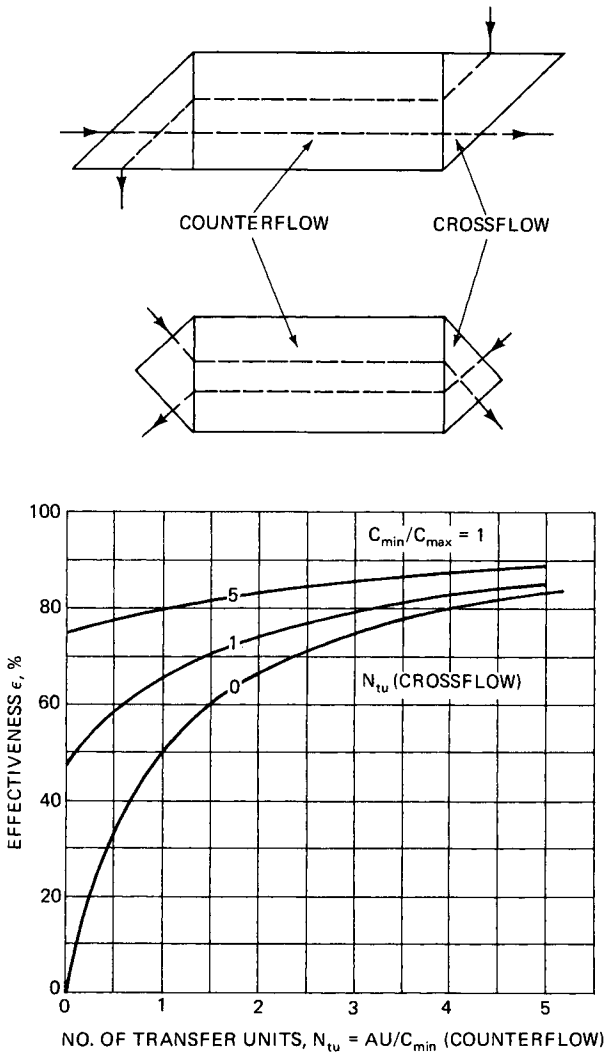


Fig. 2-23 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; 1-2 parallel-counterflow exchanger.

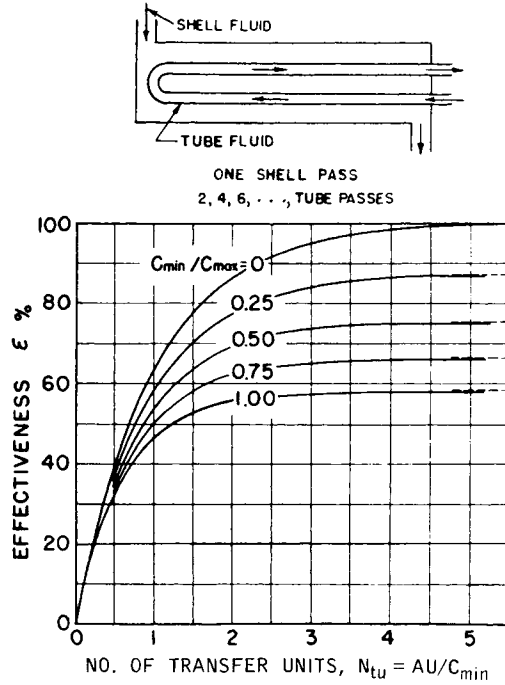


Fig. 2-24 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; 2-4 multipass counter-flow exchanger.

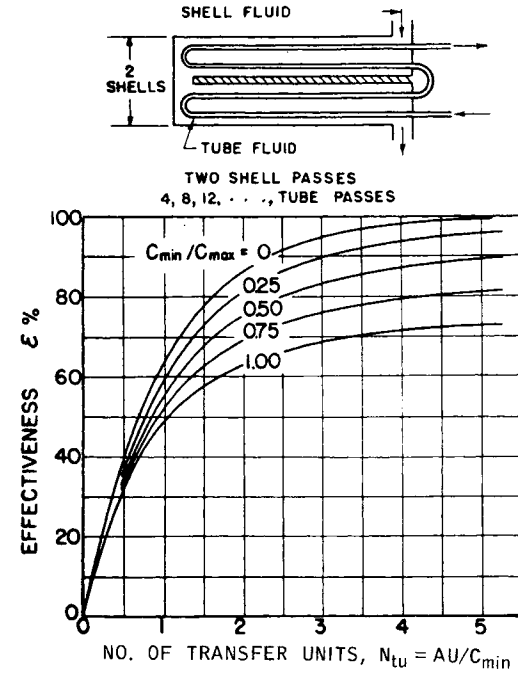


Fig. 2-25 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; 3-6 multipass counterflow exchanger.

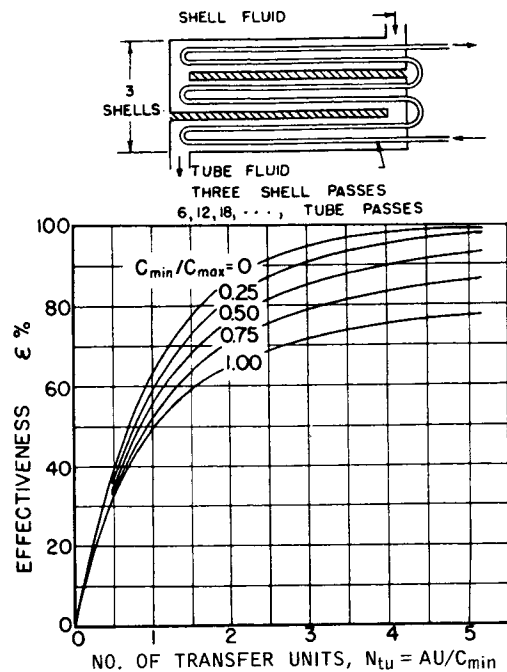
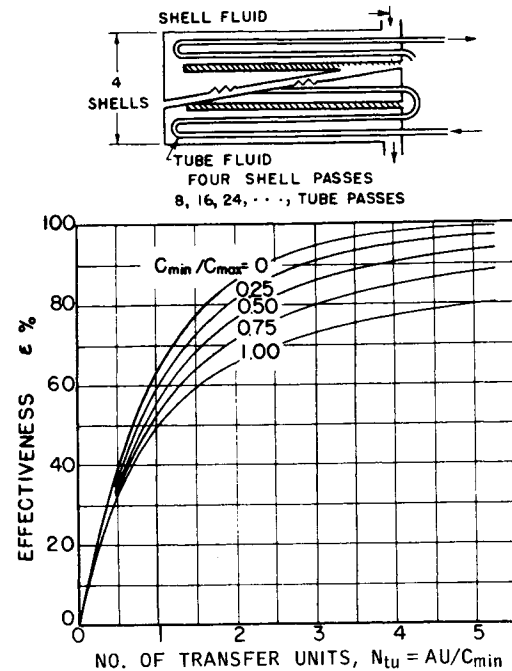


Fig. 2-26 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; 4-8 multipass counterflow exchanger.



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Fig. 2-27 Heat transfer effectiveness as a function of number of transfer units and capacity-rate ratio; split-flow exchanger, shell fluid mixed.

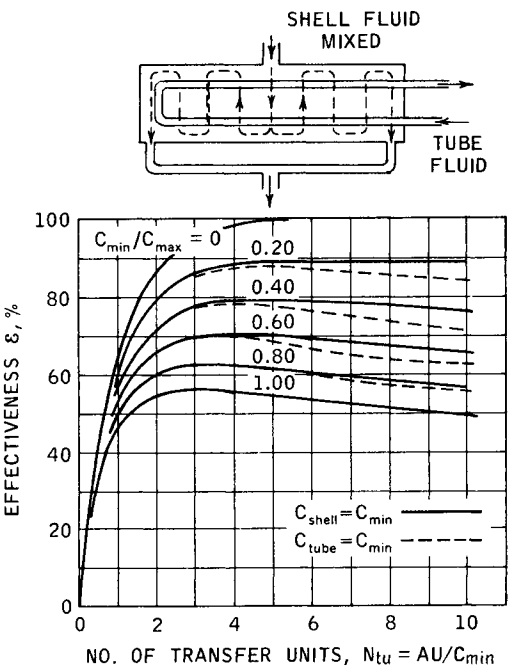


Fig. 2-28 Heat transfer effectiveness as a function of number of transfer units; effect of flow arrangement for $C_{min}/C_{max} = 1$.

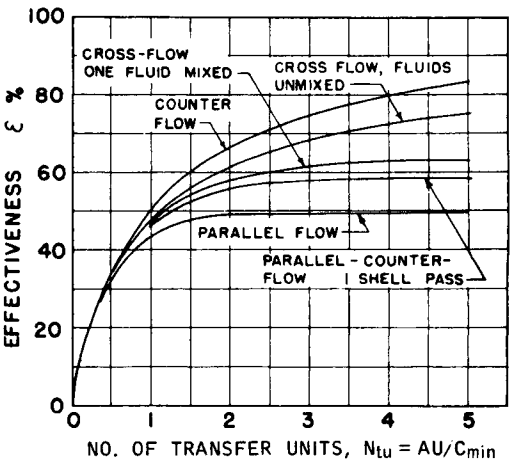


Fig. 2-29 Heat transfer effectiveness as a function of number of transfer units; multipass counterflow exchanger (parallel-counterflow passes); effect of number of shell passes for $C_{\min}/C_{\max} = 1$.

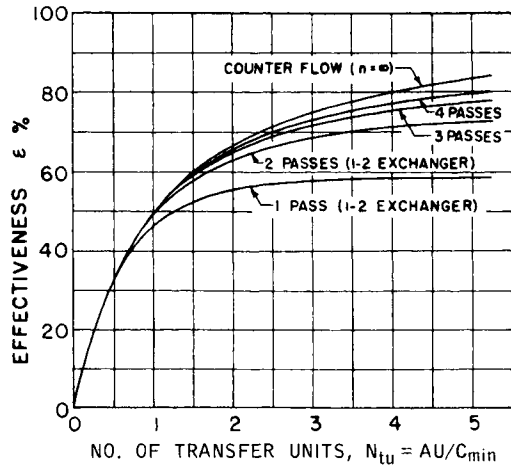


Fig. 2-30 Heat transfer effectiveness as a function of number of transfer units; effect of $C_{\min}/C_{\max}$.

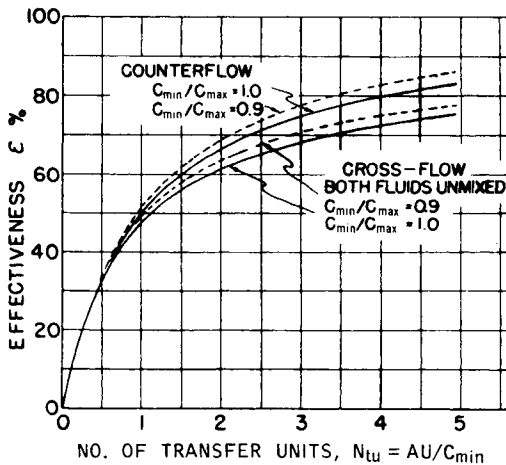


Fig. 2-31 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 1$.

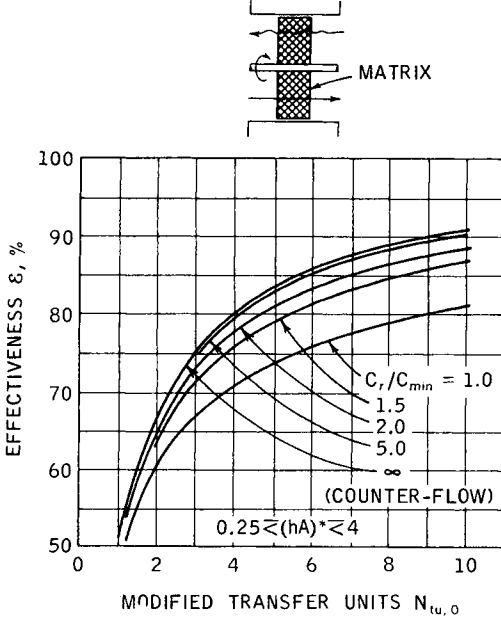


Fig. 2-32 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.95$.

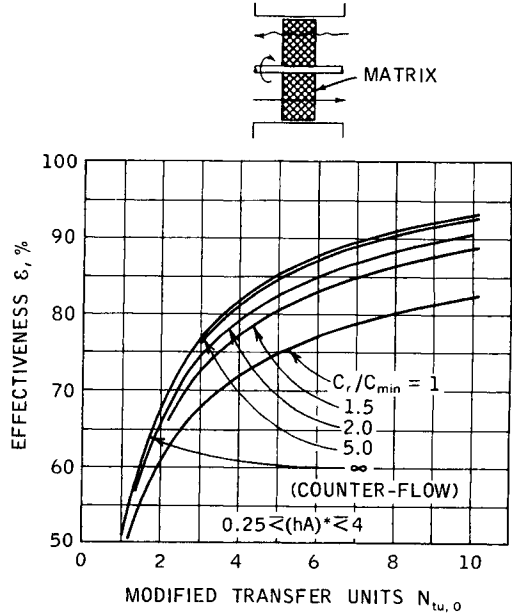


Fig. 2-33 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.9$.

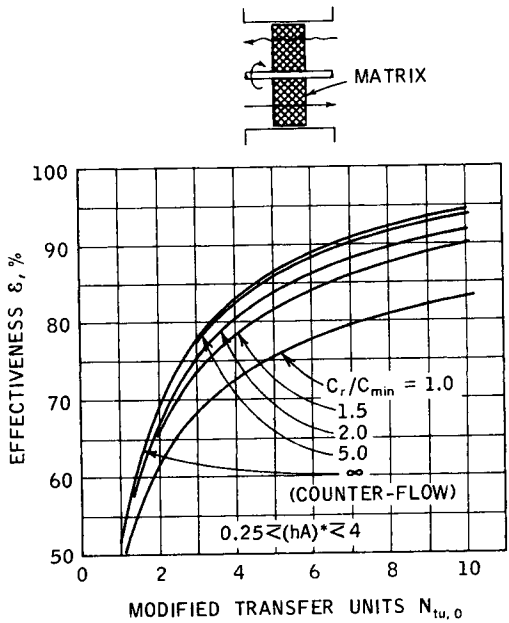


Fig. 2-34 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.8$.

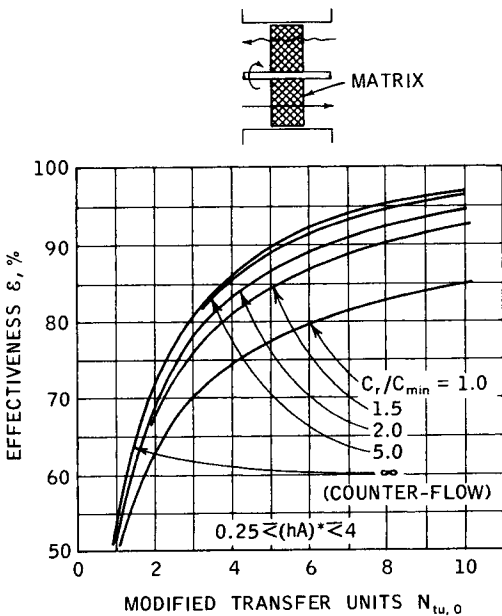


Fig. 2-35 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.7$.

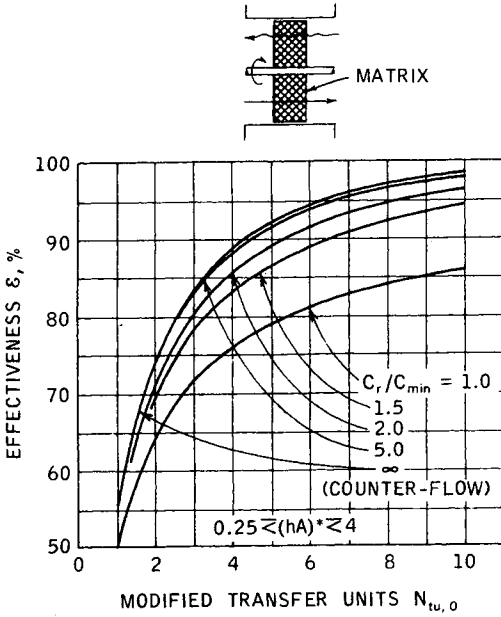


Fig. 2-36 Heat transfer effectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.5$.

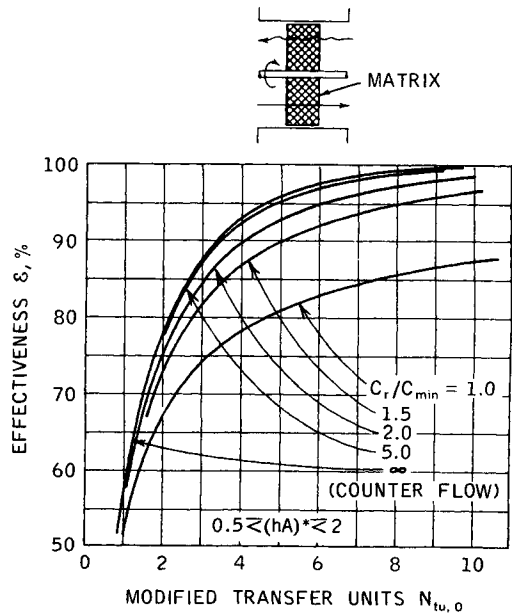


Fig. 2-37 Heat transfer ineffectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 1$.

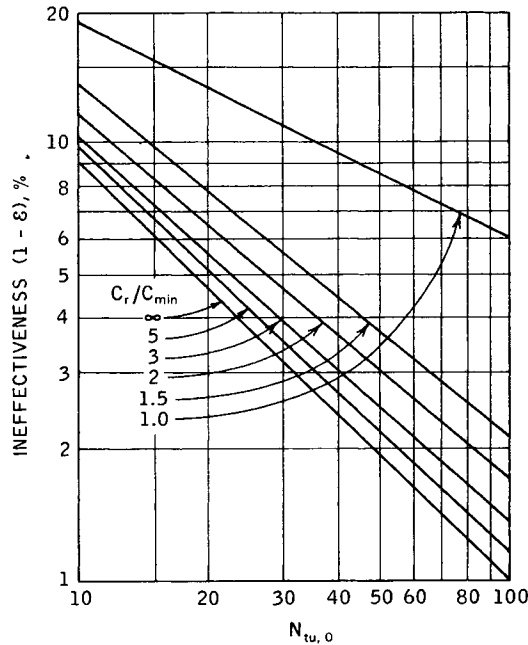


Fig. 2-38 Heat transfer ineffectiveness as a function of number of transfer units and matrix capacity-rate ratio; periodic-flow exchanger performance for $C_{\min}/C_{\max} = 0.95$.

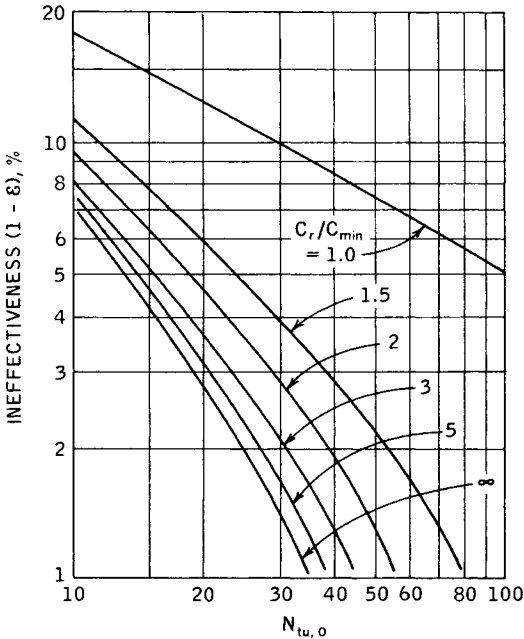


Fig. 2-39 Effect of longitudinal thermal conductivity on the performance of periodic-flow and direct-transfer exchangers, $C_{\min}/C_{\max} = 1$, $C_r/C_{\min} > 5$.

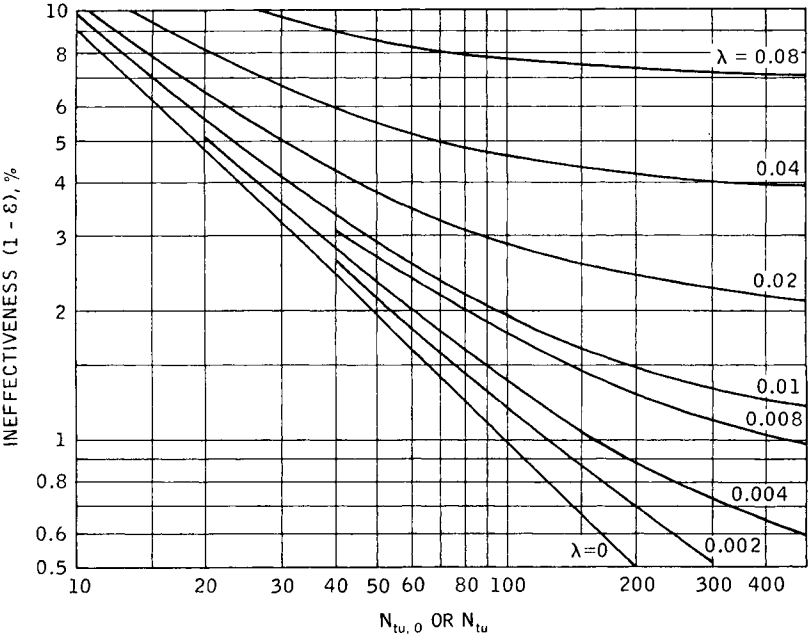


Fig. 2-40 Effect of longitudinal thermal conductivity on the performance of periodic-flow and direct-transfer exchangers; $C_{\min}/C_{\max} = 0.95$, $C_r/C_{\min} > 5$.

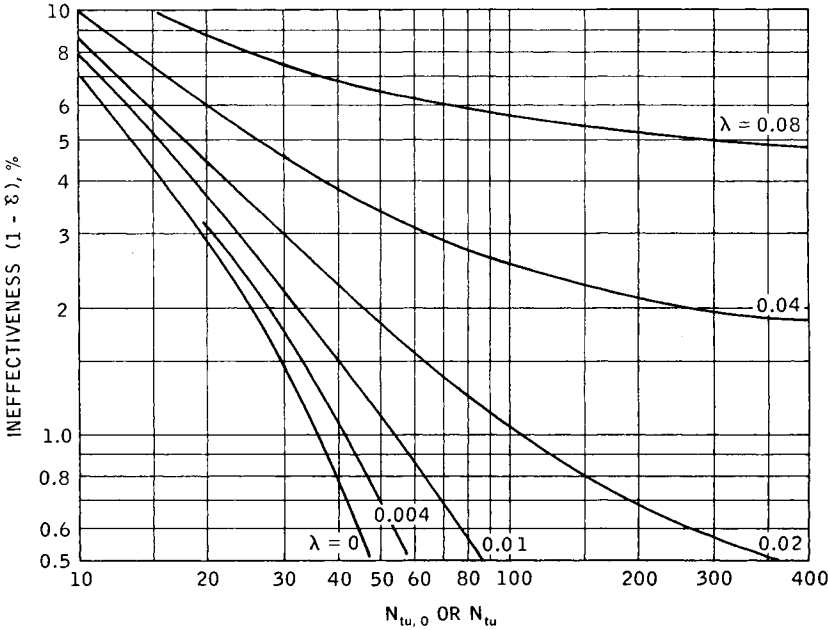


Fig. 2-41 Ratio of Stanton number to friction factor for a number of illustrative cases.

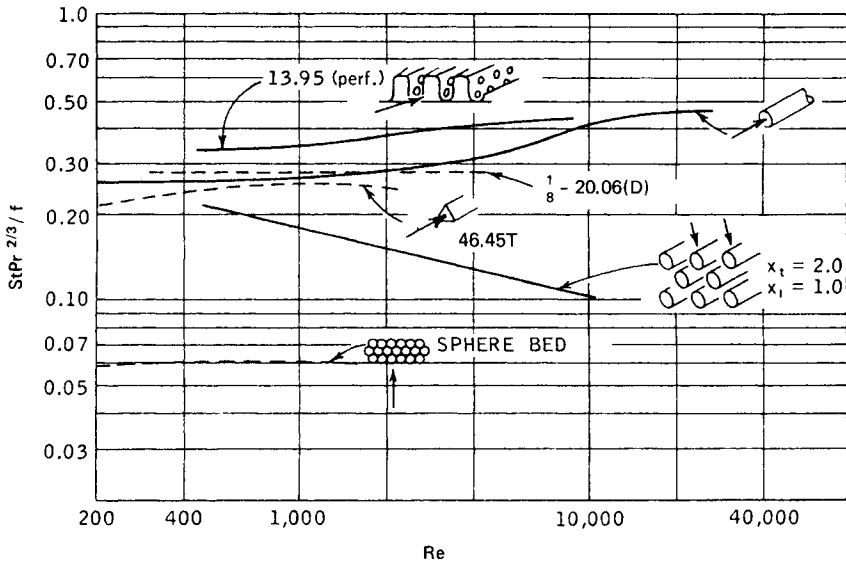
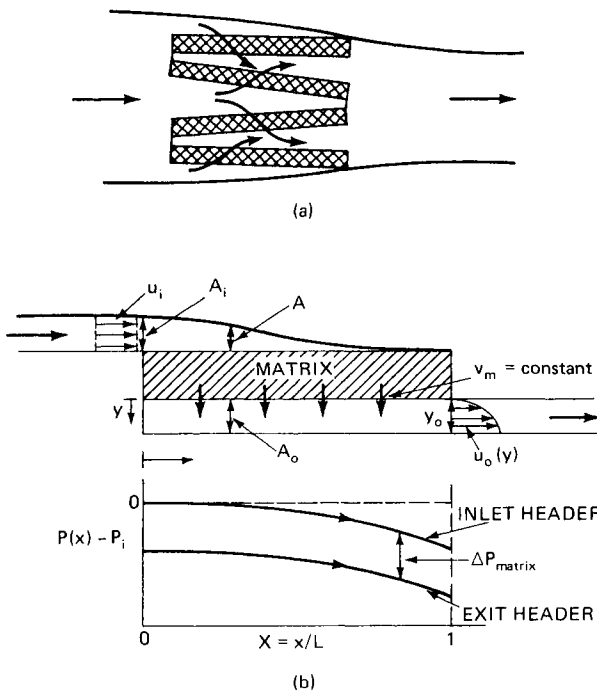


Fig. 2-42 Header configurations. (a) Folded core; (b) parallel-flow; (c) counterflow; (d) free-discharge.



(continued)

Fig. 2-42 (continued)

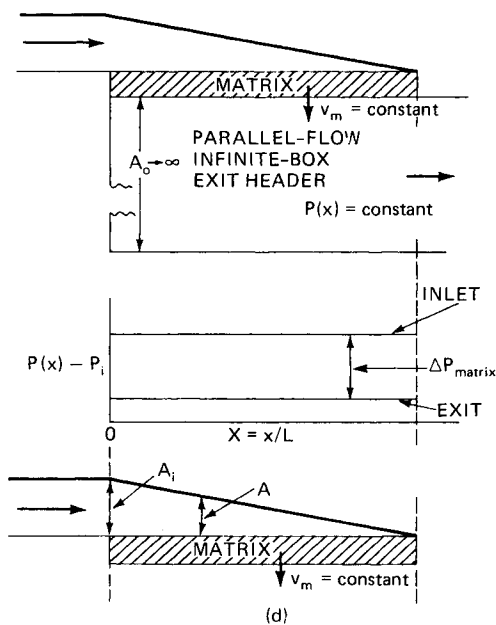
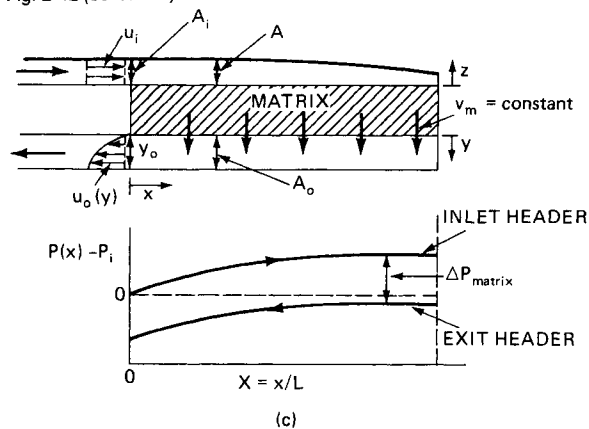
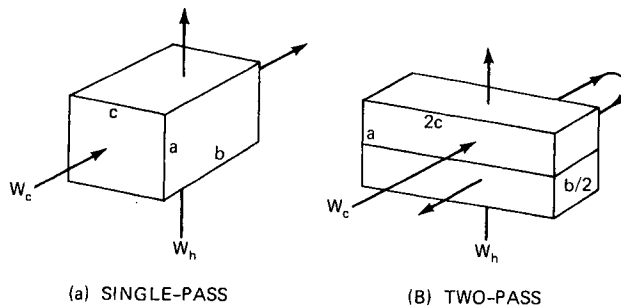


Fig. 2-43 Crossflow and two-pass cross-counterflow related designs. (a) Single-pass; (b) two-pass.



3

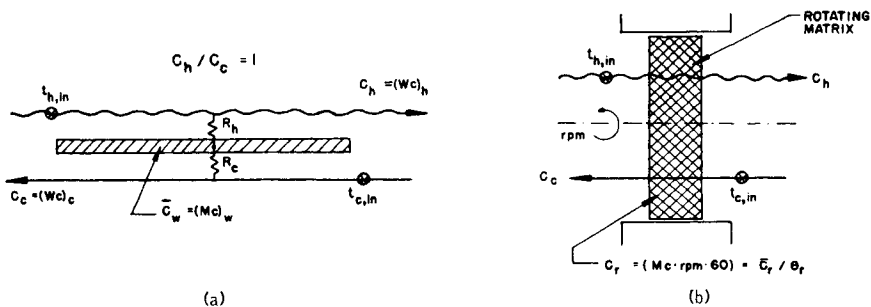
The Transient Response of Heat Exchangers

In a heat exchanger the outlet-fluid temperature response to a change in one of the inlet-fluid temperatures is not instantaneous. The lag is influenced by the thermal capacitances of the fluids and the solid wall, as well as the resistances to heat transfer. The response behavior of gas-turbine-plant heat exchangers is of particular interest to the designer of the fuel-control system, because the shaft power output is influenced by the lag introduced by the exchangers. In process plants and air-conditioning systems, as well as power plants, the transient response of the heat exchangers is needed by the controls engineer.

SUMMARY OF SOLUTIONS

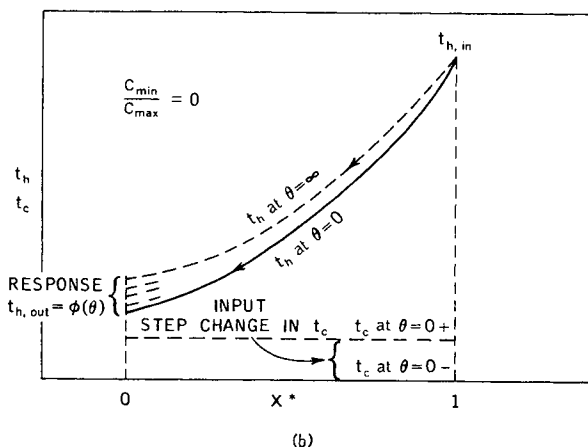
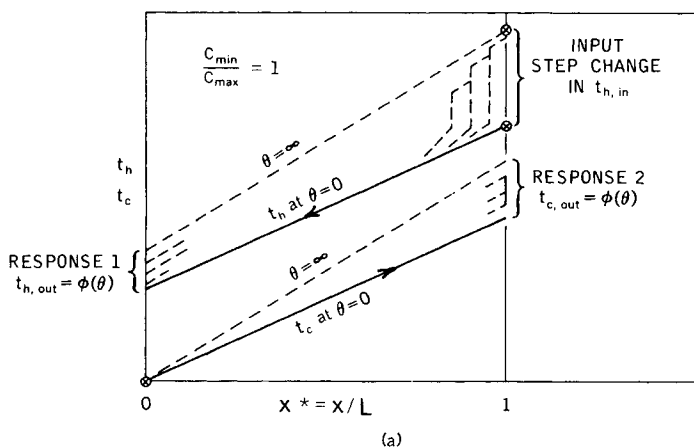
In Fig. 3-1, schematic descriptions are provided for both direct-transfer and periodic-flow types of counterflow exchangers. The inputs to be considered are step changes in either $t_{h,in}$ or $t_{c,in}$, and the transient response outputs are the changes in $t_{c,out}$ and $t_{h,out}$. The flow stream and matrix *capacity rates* C_c , C_h , and C_r are described, along with the *capacitances* $\bar{C}_w$ and $\bar{C}_r$.

Fig. 3-1 Schematic descriptions for the transient response of (a) direct-transfer and (b) periodic-flow types of counterflow exchangers.



Solutions for 16 transient-response problems are provided in papers by Cima and London [1], London, et al. [2], London, et al. [3], Rizika [4], and Myers, et al. [5]. These 16 problems are described schematically in the five illustrations of Fig. 3-2. Some of these sketches describe several problems; for instance, Fig. 3-2a serves for two direct-transfer-type exchanger problems and also for two periodic-flow-type exchanger problems. A detailed description of each problem is given in Table 3-1, Summary of Solutions.

Fig. 3-2 Descriptions of the problems under consideration in terms of input and response. (a) Regenerators, direct-transfers, and periodic-flow types; (b), (c), (d), intercoolers and precoolers; (e) the insulated duct.



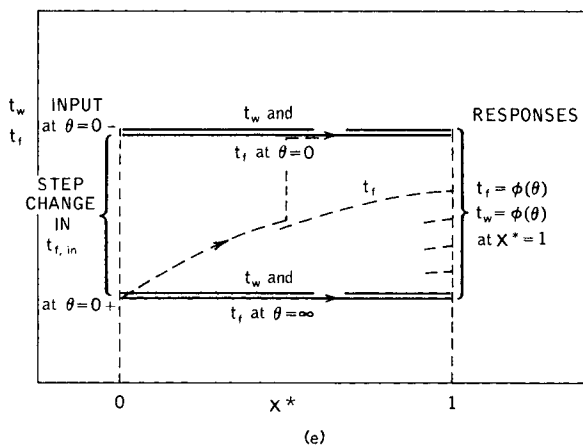
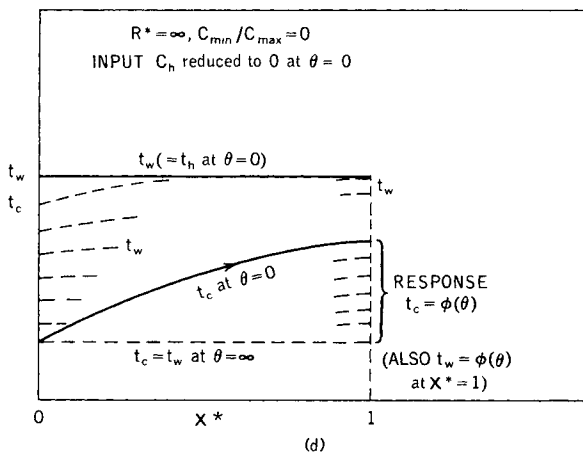
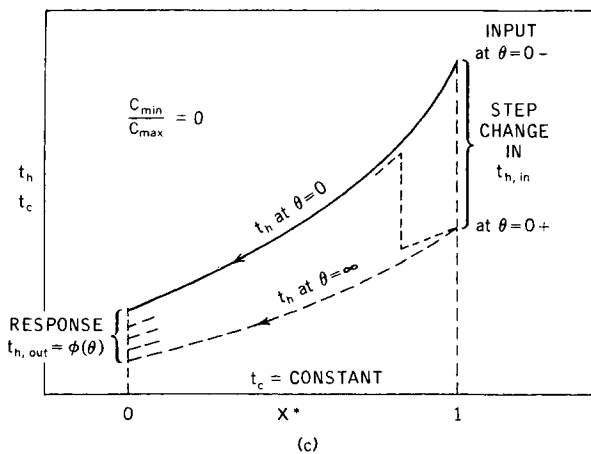


Table 3-1 Summary of Solutions

Solution	Independent parameters			Figure refs. (Prob. type) [Graph. results]	Literature references
	Fixed	Partially restricted	Unrestricted		
1	$\frac{C_{\min}}{C_{\max}} = 1$ $x^* = 1$	$\bar{C}_w^* > 100$	$\theta^*, N_{tu}, \theta_d^*, R^*$	(Direct-transfer $\varepsilon_{f,2}^*$ Fig. 3-2a) [Fig. 3-3]	1, 2, 3, 12†
2	$\frac{C_{\min}}{C_{\max}} = 1$ $x^* = 0$	$\bar{C}_w^* > 100$ $\frac{1}{4} \leq R^* \leq 4$	$\theta^*, N_{tu}, \theta_d^*$	(Direct-transfer $\varepsilon_{f,1}^*$ Fig. 3-2a) [Fig. 3-3]	3, 12†
3	$\frac{C_{\min}}{C_{\max}} = 1$ $x^* = 1$	$\bar{C}_r^* > 100$	$\theta_R^*, N_{tu,0}, \theta_d^*, R^*$	(Periodic-flow $\varepsilon_{f,2}$ Fig. 3-2a) [Figs. 3-4, 3-5]	3
4	$\frac{C_{\min}}{C_{\max}} = 1$ $x^* = 0$	$\bar{C}_r^* > 100$ $\frac{1}{4} \leq R^* \leq 4$ $2 \leq C_r^* \leq 20$	$\theta_R^*, N_{tu,0}, \theta_d^*$	(Periodic-flow $\varepsilon_{f,1}^*$ Fig. 3-2a) [Fig. 3-4]	3
5	$\frac{C_{\min}}{C_{\max}} = 1$ $R^* = 1$ $\theta_d^* = 1$ $x^* = 1$		$\theta^*, N_{tu}, \bar{C}_w^*$	(Direct-transfer $\varepsilon_{f,2}^*$ Fig. 3-2a) [Fig. 3-3]	1
6	$\frac{C_{\min}}{C_{\max}} = 1$ $\bar{C}_w^* = 0$ $\theta_d^* = 1$ $x^* = 1$		θ^*, N_{tu}	(Direct-transfer $\varepsilon_{f,2}^*$ Fig. 3-2a) [Fig. 3-3]	1
7	$\frac{C_{\min}}{C_{\max}} = 0$ $x^* = 0$	$\theta^* \leq 1$	$N_{tu}, \bar{C}_w^*, R^*$	(Direct-transfer Fig. 3-2b) [Fig. 3-6]	2,4
8	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = \infty$ $x^* = 0$		$\theta^*, N_{tu}, \bar{C}_w^*$	(Direct-transfer Fig. 3-2b) [Fig. 3-7]	2
9	$\frac{C_{\min}}{C_{\max}} = 0$ $\bar{C}_w^* = 0$ $x^* = 0$		θ^*, N_{tu}, R^*	(Direct-transfer Fig. 3-2b) [Fig. 3-7]	2
10	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = 0$ $x^* = 0$		$\theta^*, N_{tu}, \bar{C}_w^*$	(Direct-transfer Fig. 3-2b) [Fig. 3-7]	2
11	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = 1$ $x^* = 0$	$\bar{C}_w^* > 100$	θ^*, N_{tu}	(Direct-transfer Fig. 3-2b) [Fig. 3-8]	5

Table 3-1 Summary of Solutions (*continued*)

12	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = 1$ $N_{tu} = 3$ $x^* = 0$		$\theta^*, \bar{C}_w^*$	(Direct-transfer Fig. 3-2b) [Fig. 3-9]	5
13	$\frac{C_{\min}}{C_{\max}} = 0$ $N_{tu} = 3$ $x^* = 0$	$\bar{C}_w^* > 100$	θ^*, R^*	(Direct-transfer Fig. 3-2b) [Fig. 3-10]	5
14	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = 1$ $x^* = 0$		$\theta^*, N_{tu}, \bar{C}_w^*, R^*$	(Direct-transfer Fig. 3-2c) [Fig. 3-11]	5
15	$\frac{C_{\min}}{C_{\max}} = 0$ $R^* = \infty$ $x^* = 1$		$\theta^*, N_{tu}, \bar{C}_w^*$	(Direct-transfer Fig. 3-2d) [Figs. 3-12, 3-13, 3-14]	2
16	$x^* = 1$		$\theta^*, N_{tu}, \bar{C}_w^*$	(Insulated-duct Fig. 3-2e) [Figs. 3-12, 3-13, 3-14, 3-15]	2, 6

† A recent reference by Romie [12] presents finite-difference solutions which eliminate the indicated constraints on $C_{\min}/C_{\max}$ and C_w^* .

Each of the problems is referenced to the appropriate sketch of Fig. 3-2 by the notation "(Prob. type)" in the next to last column and to the graphical results by the notation "[Graph. results]." The second and third columns describe the fixed and partially restricted parameters, which in some degree characterize the problem. The unrestricted parameters are given in the fourth column. These are the independent coordinates of the graphs, presenting the transient response as the dependent variable. Some of these solutions are purely analytical, namely solutions 7 to 10 and 15 and 16 of Table 3-1. The others were derived either from digital computer finite-difference calculations from the differential equations (solutions 3 and 4), or from experiments with an electromechanical analog (solutions 1, 2, 5, and 6). Techniques of dimensional analysis were used to form the nondimensional groupings used to correlate the results in a reasonably compact graphical presentation.

Solutions 11 to 13, Figs. 3-8 to 3-10, were calculated from the set of equations of Table 3-2, recommended by Myers et al. [5]. These equations tie to the exact solution of Rizika [2] at $\theta^* = 1$, matching both the magnitude of ϵ_f^* and its time derivative. They represent an empirical extrapolation, the validity of which has been well confirmed by the finite-difference solutions of the exact differential equations [5].

Solution 14, Fig. 3-11, also comes from the Myers et al. paper. It is a quite complete analytical solution, details of which are provided in the appendix of Ref. 5. Unfortunately, functions of Bessel functions are required, making direct

Table 3-2 Solution for Fig. 3.2b Response for $\theta^* \geq 1$

$$\begin{aligned}\varepsilon_f^* &= 1 - U \exp \left[-\frac{a(\theta^* - 1)}{\bar{C}_w^*} \right] \\ U &= 1 - \frac{\{1 - e^{-Z}[Y \sinh(Z/Y) + \cosh(Z/Y)]\}}{1 - e^{-N_{tu}}} \\ Z &= \frac{N_{tu}(1 + R^*)(1 + R^* + \bar{C}_w^*)}{2R^*\bar{C}_w^*} \\ Y &= \left[1 - \frac{4R^*\bar{C}_w^*}{(1 + R^* + \bar{C}_w^*)^2} \right]^{-1/2} \\ a &= \frac{2N_{tu}(1 + R^*)\bar{C}_w^*Ye^{-Z} \sinh(Z/Y)}{U(1 + R^* + \bar{C}_w^*)(1 - e^{-N_{tu}})}\end{aligned}$$

Note: Solutions 11 to 13 of Table 3-1, Figs. 3-8 to 3-10, were prepared from this equation set.

Source: Ref. 5.

calculations somewhat inconvenient to make, so the equation form of the solution is not reported here.

While 15 solutions apply to the two-fluid heat exchanger, the sixteenth is for what might be described either as an insulated duct through which fluid is flowing or as a porous cylinder or matrix with flow in the axial direction. The transient response of the insulated duct may be of direct interest to the designer. The response of the porous cylinder is of decided interest in heat transfer research, as transient techniques are often used to establish the basic heat transfer characteristics of compact matrix surfaces [6, 7]. The responses are given for both the solid and the fluid temperatures, and also for the maximum slope of the response curve. These results are given graphically in Figs. 3-12 to 3-15 and more accurately in Tables 3-3 and 3-4. Many of the screen- and sphere-bed-matrix heat transfer characteristics reported in Chap. 7 were obtained using the maximum slope solution of Fig. 3-15 and Table 3-4 with the transient technique developed by Locke [6]. The tabular information for solution 16 was obtained from computer programs from several sources (Refs. 6, 8, and 9).

THE NONDIMENSIONAL PARAMETERS

A key requirement in the use of Table 3-1 and the associated graphs is an understanding of the significance of the nondimensional parameters employed. Because the governing differential equation and boundary conditions are available [2], it is possible to formulate an adequate set of nondimensional parameters by purely formal methods. In fact, many such sets may be formulated, each being convertible to the others. The particular parameters selected for this presentation are considered to have the following advantages: (1) They are readily understandable and nameable; (2) many of them have already been used

to express the steady-state heat exchanger performance covered in Chap. 2; (3) the independent parameters are readily controlled; that is, one may be varied while the others are kept constant for analog or computer program experimentation.

Direct-Transfer-Type Exchangers

The set of nondimensional parameters is as follows:

$$\varepsilon_{f,1}^{\circ}, \varepsilon_{f,2}^{\circ}, \varepsilon_f^{\circ} = \Phi\left(N_{tu}, \frac{C_{\min}}{C_{\max}}, x^{\circ}, R^{\circ}, \bar{C}_w^{\circ}, \theta_d^{\circ}, \theta^{\circ}\right) \quad (3-1)$$

where $\varepsilon_{f,1}^{\circ}$, $\varepsilon_{f,2}^{\circ}$, ε_f° are the dependent outlet-fluid *temperature responses* to the step input of one of the fluid inlet temperatures (see Fig. 3-2a to d).

$N_{tu} = AU/C_{\min} = 1/(R_c + R_h)C_{\min}$ is the exchanger *number of heat transfer units*, as used in Chap. 2, a nondimensional expression of exchanger size.

$C_{\min}/C_{\max}$ is the *capacity-rate ratio* of the two fluids, as used in Chap. 2. If $C_c < C_h$, the capacity-rate ratio is C_c/C_h ; otherwise it is C_h/C_c .

x° expresses the *position* in the exchanger. For this consideration $x^{\circ} = 0$ or 1, corresponding to the fluid outlet sections.

$R^{\circ} = (R \text{ on unstepped side})/(R \text{ on stepped side})$ is the heat transfer *resistance ratio* (see Fig. 3-1).

$\bar{C}_w^{\circ} = \bar{C}_w/\bar{C}_{\min}$ is the *wall-capacitance* parameter.†

$\theta_d^{\circ} = (\theta_d \text{ for the } C_{\min} \text{ fluid})/(\theta_d \text{ for the } C_{\max} \text{ fluid})$ is the *dwell-time ratio* or exchanger residence-time ratio for the two fluids.

$\theta^{\circ} = \theta/(\theta_d \text{ for the } C_{\min} \text{ fluid})$ is the *generalized time* parameter.

In forming ε_f° , the fluid-temperature change from the initial temperature is normalized with respect to the ultimate change at time equals infinity. Thus ε_f° is zero at the initial instant and tends toward unity as time increases. $\varepsilon_{f,1}^{\circ}$ denotes the response on leaving the exchanger of that fluid which had the step input imposed on it at entrance to the exchanger; while $\varepsilon_{f,2}^{\circ}$ denotes the response of the other fluid on leaving the exchanger (see Fig. 3-2a). When it is obvious that the response of only one of the fluids is of interest, the subscript 1 or 2 is omitted, as in Fig. 3-2b to d.

It is to be noted that N_{tu} , $C_{\min}/C_{\max}$, x° , and R° should all appear in a complete solution for steady-state behavior. While x° and R° do not appear explicitly in the Chap. 2 solutions, implicitly $x^{\circ} = 0$ or 1, and R° would enter if the heat exchanger surface temperature was required.

For the solutions in hand, $C_{\min}/C_{\max}$ is specialized to be either unity (solutions 1, 2, 5, and 6 of Table 3-1) or zero (solutions 7 to 15 of Table 3-1). Thus, with $x^{\circ} = 0$ or 1, Eq. (3-1) reduces to

$$\varepsilon_{f,1}^{\circ}, \varepsilon_{f,2}^{\circ}, \varepsilon_f^{\circ} = \Phi(N_{tu}, R^{\circ}, \bar{C}_w^{\circ}, \theta_d^{\circ}, \theta^{\circ}) \quad (3-2)$$

Further reduction in the number of independent parameters is achieved either by specification of fixed magnitudes (for example, $\bar{C}_w^{\circ} = 0$, $\theta_d^{\circ} = 1$, as in solution

† $\bar{C}_{\min} = (C\theta_d)$ for the $C_{\min}$ fluid.

6 of Table 3-1) or by partially restricting the range (for example, $\bar{C}_w^* > 100$, $\frac{1}{4} \leq R^* \leq 4$, as in solution 2 of Table 3-1).

Periodic-Flow-Type Exchanger

This set of nondimensional parameters in a considerable degree parallels the set just described for the direct-transfer exchanger:

$$\varepsilon_{f,1}^*, \varepsilon_{f,2}^* = \Phi \left(N_{tu,o}, \frac{C_{\min}}{C_{\max}}, x^*, R^*, C_r^*, \theta_d^*, \frac{\theta_r}{\theta_{d,\min}}, \theta_R^* \right) \quad (3-3)$$

where $\varepsilon_{f,1}^*, \varepsilon_{f,2}^*$ are the dependent fluid *temperature responses* to the step input (see Fig. 3-2a). The instantaneous fluid temperature is the average over the flow cross section at an instant.

$N_{tu,o} = 1/[(R_c + R_h)C_{\min}]$ is the modified *number of transfer units* for the exchanger (as used in Chap. 2), a nondimensional expression of exchanger size.

$C_{\min}/C_{\max}$ is the *capacity-rate ratio* of the two fluids, as used in Chap. 2. If $C_c < C_h$, $C_{\min}/C_{\max} = C_c/C_h$.

x^* expresses the *position* in the exchanger. The magnitudes of $x^* = 0$ and 1 are of primary interest as they correspond to the fluid outlet sections (see Fig. 3-2a).

R^* is R_c/R_h if the step input is on $t_{h,in}$, or it is R_h/R_c if the step input is on $t_{c,in}$. R^* is the *heat transfer resistance ratio*.

$C_r^* = C_r/C_{\min}$ is the *rotor capacity-rate ratio* (as used in Chap. 2) or the *matrix-switching capacity-rate ratio* for a valved periodic-flow exchanger. (See Fig. 2-6 for a description of the rotary and valved types of periodic-flow exchangers.)

$\theta_d^* = (\theta_d \text{ for the } C_{\min} \text{ fluid})/(\theta_d \text{ for the } C_{\max} \text{ fluid})$ is the *dwelt-time ratio*, or exchanger residence-time ratio for the two fluids.

$\theta_r/\theta_{d,\min} = (\text{rotor period})/(\theta_d \text{ for the } C_{\min} \text{ fluid})$ is the nondimensional *rotor period*.

$\theta_R^* = \theta/\theta_r$ is the *generalized time parameter*.

As for the direct-type exchanger, $\varepsilon_{f,1}^*$ and $\varepsilon_{f,2}^*$ are normalized-fluid outlet responses that have magnitudes of zero initially and tend toward unity for large times. $\varepsilon_{f,1}^*$ is the response of that fluid leaving the exchanger which had imposed on it the step input on entering the exchanger, while $\varepsilon_{f,2}^*$ denotes the response of the other fluid. (See Fig. 3-2a for an example in which the step input is imposed on the hot fluid. The definitions, however, are general enough to apply to the other situation, in which the step input is imposed on the cold fluid.)

Note that the dependent response ε_f^* and the independent parameters $N_{tu,o}$, $C_{\min}/C_{\max}$, x^* , R^* , and θ_d^* have their counterparts in Eq. (3-1) for the direct-transfer-type exchanger. Further, the counterpart of $\bar{C}_w^*$ in Eq. (3-1) is matched by the combination, from Eq. (3-3),

$$C_r^* \frac{\theta_r}{\theta_{d,\min}} = \frac{\bar{C}_r}{\bar{C}_{\min}} = \bar{C}_r^*$$

The generalized time parameters θ° of Eq. (3-1) and θ_R° of Eq. (3-3) are related by

$$\theta_R^\circ = \frac{\theta}{\theta_r} = \frac{\theta/\theta_{d,\min}}{\theta_r/\theta_{d,\min}} = \frac{\theta^\circ}{\theta_r/\theta_{d,\min}}$$

All the independent parameters of Eq. (3-3), exclusive of the generalized time θ_R° , would appear in a complete steady-state solution for the periodic-flow exchanger. The steady-state solutions of Chap. 2 do include $x^\circ = 1$ or 0 implicitly and would include R° if the solutions were complete enough to account for matrix temperatures. Matrix-temperature envelopes are considered by Mondt [10].

For periodic-flow-type exchanger solutions 3 and 4 of Table 3-1, $C_{\min}/C_{\max} = 1$ and $x^\circ = 0$ or 1 are specified as fixed. Furthermore, θ_d° and $\theta_r/\theta_{d,\min}$ turn out to be not significant for

$$\bar{C}_r^\circ = \left(\frac{C_r^\circ \theta_r}{\theta_{d,\min}} \right) > 100$$

an imposed partial restriction on the solutions. Consequently, Eq. (3-3) is reduced substantially to

$$\varepsilon_{f,1}^\circ, \varepsilon_{f,2}^\circ = \Phi(N_{tu,o}, R^\circ, C_r^\circ, \theta_R^\circ) \quad (3-4)$$

Insulated Duct

For solutions 15 and 16 of Table 3-1, the set of nondimensional parameters is as follows:

$$\varepsilon_f^\circ, \varepsilon_w^\circ = \Phi(N_{tu}, \bar{C}_w^\circ, \theta^\circ) \quad (3-5)$$

where ε_f° and ε_w° are the *fluid* and *wall temperature responses*, respectively, at the fluid outlet section.

$N_{tu} = hA/C$ is the *number of heat transfer units*.

$\bar{C}_w^\circ = \bar{C}_w/\bar{C} = \bar{C}_w/C\theta_d$ is the *wall-capacitance* parameter.

$\theta^\circ = \theta/\theta_d$ is the *generalized time* parameter.

As for the two-fluid exchangers of the direct-transfer and periodic-flow types, ε_f° and ε_w° are normalized responses that have an initial magnitude of zero and tend toward unity for large times.

These solutions apply to a porous cylindrical matrix heated or cooled by an axial flow stream, as well as to an insulated duct. Usually the N_{tu} can be expected to be less than unity for the insulated duct (Fig. 3-12) but may be quite large for the porous matrix (Figs. 3-13 and 3-14).

Figure 3-15 provides the maximum slope of the ε_f° curves of Fig. 3-14.

RESPONSE TO A FLOW-RATE CHANGE

To this point attention has been given to an outlet-fluid temperature response due to an inlet-fluid temperature step input. Solution 15 (Fig. 3-2d) is an

exception. Here it is supposed that $C_{\max} (\gg C_{\min})$ is suddenly reduced to zero; then the $C_{\min}$ outlet temperature (at $x^* = 1$) and the wall temperature (at $x^* = 1$) will have the same behavior as for an insulated duct or porous matrix (Figs. 3-12 to 3-14). This is an extreme example of a flow-rate change, and the lag is substantial, being of the order

$$\theta_{90} = (2\bar{C}_w^* + 1)\theta_d$$

for a 90 percent response (from Fig. 3-14 and $N_{tu} = 3$). For a $C_{\min}/C_{\max} = 1$ exchanger, if both flow rates are changed simultaneously, it is equivalent to a step change in N_{tu} . Note that N_{tu} increases with a reduction in flow. Under these conditions, the wall temperatures do not change greatly, and the lag of the fluid outlet temperatures is shown to be small [1], of the order

$$\theta_{90} \approx \theta_d$$

Generally, the lag due to a flow-rate change is substantially smaller than the lag due to the accompanying inlet-fluid temperature changes.

RESPONSE TO AN ARBITRARY INPUT OR A PERIODIC INPUT

The governing differential equations for both direct-transfer and periodic-flow types of regenerators are linear, at least within the usual idealizations [2]. Consequently, an extension of the ε_f^* results to other than a step input can be accomplished readily by a standard technique known as Duhamel's method; this procedure is illustrated in Ref. 2. The controls engineer may be interested in the *frequency response*, the amplitude and phase angle of the outlet response due to a fixed-frequency-sine-wave periodic input. This information as a function of frequency can be extracted from the transient responses reported here by several standard techniques [11].

ILLUSTRATIVE EXAMPLE

In Table 3-5, a 75 percent effective direct-transfer regenerator is compared to a 91 percent effective periodic-flow-type unit. These design effectivenesses are considered to be typical in practical gas-turbine-plant applications.

It is worth noting that the $\varepsilon_{f,1}^*$ response is of lesser importance than $\varepsilon_{f,2}^*$ for high-effectiveness exchangers, as the temperature change associated with $\varepsilon_{f,1}^*$ is proportional to the steady-state *ineffectiveness* $(1 - \varepsilon)$ and is much smaller in magnitude than the temperature change associated with $\varepsilon_{f,2}^*$, which is proportional to the steady-state *effectiveness* ε .

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Table 3-3 Insulated Duct or Porous Matrix Solution

Wall temperature response ϵ_w^*											Fluid temperature response ϵ_f^*										
τ^*	N_{tu}^*										τ^*	N_{tu}^*									
	0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0		0.2	0.4	0.6	0.8	1.0	1.2	1.4	1.6	1.8	2.0
0	0	0	0	0	0	0	0	0	0	0	0	0.819	0.670	0.549	0.449	0.368	0.301	0.247	0.202	0.165	0.135
0.5	0.337	0.289	0.247	0.211	0.181	0.154	0.132	0.112	0.096	0.062	0.5	0.885	0.781	0.688	0.604	0.530	0.464	0.406	0.354	0.309	0.369
1.0	0.562	0.499	0.442	0.391	0.346	0.305	0.269	0.237	0.208	0.183	1.0	0.927	0.855	0.785	0.718	0.654	0.594	0.539	0.487	0.439	0.394
1.5	0.712	0.650	0.592	0.538	0.488	0.441	0.399	0.359	0.323	0.290	1.5	0.953	0.904	0.852	0.800	0.748	0.696	0.646	0.597	0.551	0.506
2.0	0.811	0.757	0.705	0.654	0.606	0.559	0.515	0.473	0.433	0.397	2.0	0.970	0.936	0.899	0.859	0.817	0.775	0.731	0.688	0.645	0.604
2.5	0.876	0.833	0.789	0.744	0.700	0.657	0.615	0.574	0.534	0.496	2.5	0.981	0.958	0.931	0.901	0.869	0.834	0.798	0.761	0.723	0.685
3.0	0.919	0.885	0.850	0.813	0.775	0.737	0.698	0.660	0.622	0.585	3.0	0.988	0.972	0.953	0.931	0.906	0.879	0.849	0.818	0.786	0.753
3.5	0.947	0.922	0.894	0.864	0.833	0.800	0.766	0.732	0.698	0.663	3.5	0.992	0.982	0.968	0.952	0.933	0.912	0.889	0.863	0.836	0.808
4.0	0.966	0.947	0.926	0.902	0.877	0.850	0.821	0.791	0.761	0.730	4.0	0.995	0.988	0.979	0.967	0.953	0.936	0.918	0.898	0.876	0.852
4.5	0.978	0.964	0.948	0.930	0.910	0.888	0.864	0.839	0.813	0.786	4.5	0.997	0.992	0.986	0.977	0.967	0.954	0.940	0.924	0.906	0.887
5.0	0.986	0.976	0.964	0.950	0.934	0.917	0.898	0.877	0.855	0.831	5.0	0.998	0.995	0.990	0.984	0.977	0.967	0.956	0.944	0.930	0.914
	1	2	3	4	5	6	7	8	9	10		1	2	3	4	5	6	7	8	9	10
1	0.346	0.183	0.094	0.047	0.023	0.011	0.005	0.003	0.001	0.001	0	0.368	0.135	0.050	0.018	0.007	0.002	0.001	0.000	0.000	0.000
2	0.606	0.397	0.247	0.148	0.086	0.049	0.027	0.015	0.008	0.004	1	0.654	0.394	0.225	0.123	0.066	0.034	0.017	0.009	0.004	0.002
3	0.775	0.585	0.417	0.283	0.185	0.118	0.072	0.043	0.026	0.015	2	0.817	0.604	0.415	0.270	0.169	0.102	0.060	0.034	0.019	0.011
4	0.877	0.730	0.573	0.428	0.307	0.213	0.142	0.093	0.059	0.037	3	0.906	0.753	0.583	0.427	0.298	0.201	0.131	0.082	0.051	0.031
5	0.934	0.831	0.702	0.565	0.436	0.325	0.234	0.163	0.112	0.074	4	0.953	0.852	0.717	0.572	0.435	0.319	0.225	0.153	0.103	0.066
6	0.966	0.898	0.800	0.682	0.559	0.442	0.338	0.251	0.181	0.128	5	0.977	0.914	0.815	0.693	0.564	0.441	0.333	0.243	0.174	0.120
7	0.983	0.940	0.870	0.776	0.667	0.555	0.447	0.349	0.266	0.197	6	0.989	0.951	0.883	0.788	0.676	0.558	0.446	0.344	0.260	0.190
8	0.991	0.966	0.918	0.847	0.757	0.656	0.551	0.450	0.358	0.278	7	0.995	0.973	0.928	0.857	0.766	0.662	0.554	0.449	0.355	0.273
9	0.996	0.981	0.949	0.898	0.827	0.741	0.646	0.548	0.453	0.366	8	0.997	0.985	0.957	0.907	0.837	0.749	0.652	0.550	0.453	0.363
10	0.998	0.989	0.969	0.934	0.880	0.811	0.728	0.638	0.545	0.455	9	0.999	0.992	0.974	0.941	0.888	0.819	0.735	0.643	0.548	0.455
11				0.958	0.919	0.865	0.796	0.717	0.631	0.543	10	0.999	0.996	0.985	0.963	0.926	0.872	0.804	0.723	0.635	0.546
12				0.974	0.946	0.905	0.851	0.784	0.708	0.625	11				0.977	0.951	0.912	0.858	0.790	0.713	0.629
13				0.984	0.965	0.935	0.893	0.838	0.773	0.699	12				0.986	0.969	0.940	0.899	0.844	0.778	0.704

14				0.990	0.978	0.956	0.925	0.881	0.827	0.763	13				0.992	0.980	0.960	0.930	0.887	0.833	0.768
15				0.994	0.986	0.971	0.948	0.915	0.871	0.817	14				0.995	0.988	0.974	0.952	0.919	0.876	0.822
16				0.997	0.991	0.981	0.965	0.940	0.905	0.861	15				0.997	0.993	0.983	0.968	0.943	0.910	0.866
17				0.998	0.995	0.988	0.976	0.958	0.932	0.896	16				0.998	0.996	0.989	0.979	0.961	0.935	0.901
18				0.999	0.997	0.992	0.984	0.971	0.951	0.924	17				0.999	0.997	0.993	0.986	0.973	0.954	0.927
19				0.999	0.998	0.995	0.990	0.980	0.966	0.945	18					0.998	0.996	0.991	0.982	0.968	0.948
											19					0.999	0.998	0.994	0.988	0.978	0.963
	12	14	16	18	20	22	24	26	28	30		12	14	16	18	20	22	24	26	28	30
2	0.001	0.000	0.000								0	0.000	0.000								
4	0.014	0.005	0.002	0.000	0.000	0.000					2	0.003	0.001	0.000	0.000						
6	0.060	0.026	0.011	0.004	0.002	0.001	0.000				4	0.026	0.010	0.003	0.001	0.000	0.000				
8	0.155	0.081	0.039	0.018	0.006	0.003	0.001	0.000	0.000		6	0.095	0.044	0.019	0.008	0.003	0.001	0.000	0.000		
10	0.297	0.178	0.099	0.052	0.026	0.012	0.005	0.002	0.001	0.000	8	0.216	0.118	0.060	0.029	0.013	0.006	0.002	0.001	0.000	0.000
12	0.459	0.312	0.196	0.116	0.065	0.034	0.017	0.008	0.004	0.002	10	0.375	0.237	0.139	0.076	0.039	0.019	0.009	0.004	0.002	0.001
14	0.616	0.462	0.324	0.212	0.131	0.077	0.043	0.023	0.012	0.006	12	0.541	0.384	0.254	0.156	0.091	0.050	0.026	0.013	0.006	0.003
16	0.746	0.608	0.464	0.334	0.226	0.145	0.088	0.051	0.028	0.015	14	0.689	0.538	0.392	0.268	0.172	0.104	0.060	0.033	0.017	0.009
18	0.844	0.733	0.602	0.465	0.341	0.237	0.157	0.099	0.059	0.034	16	0.804	0.677	0.535	0.398	0.279	0.185	0.117	0.070	0.040	0.022
20	0.909	0.828	0.721	0.597	0.467	0.350	0.248	0.169	0.109	0.067	18	0.884	0.788	0.667	0.535	0.403	0.289	0.197	0.128	0.079	0.047
22	0.950	0.896	0.815	0.709	0.591	0.469	0.355	0.257	0.178	0.118	20	0.935	0.869	0.774	0.659	0.533	0.408	0.298	0.208	0.138	0.088
24	0.974	0.940	0.883	0.803	0.700	0.587	0.473	0.382	0.265	0.187	22	0.966	0.923	0.855	0.762	0.651	0.531	0.412	0.306	0.218	0.148
26	0.987	0.967	0.930	0.872	0.792	0.694	0.583	0.472	0.367	0.273	24	0.983	0.957	0.912	0.843	0.752	0.645	0.531	0.415	0.313	0.226
28	0.994	0.983	0.960	0.921	0.862	0.782	0.687	0.582	0.473	0.372	26	0.992	0.977	0.949	0.901	0.832	0.743	0.639	0.529	0.418	0.319
30	0.997	0.991	0.978	0.953	0.912	0.852	0.774	0.681	0.579	0.474	28	0.996	0.988	0.972	0.941	0.891	0.822	0.735	0.635	0.529	0.420
32	0.999	0.996	0.988	0.973	0.946	0.903	0.843	0.766	0.675	0.577	30	0.998	0.994	0.985	0.966	0.932	0.882	0.813	0.727	0.630	0.527
34		0.998	0.994	0.985	0.968	0.939	0.895	0.834	0.758	0.670	32	0.999	0.997	0.992	0.981	0.960	0.925	0.873	0.804	0.721	0.626
36		0.999	0.997	0.992	0.982	0.963	0.932	0.887	0.827	0.752	34		0.999	0.996	0.990	0.977	0.954	0.917	0.865	0.796	0.714
38			0.999	0.996	0.990	0.978	0.958	0.926	0.880	0.819	36		0.999	0.998	0.995	0.987	0.973	0.948	0.910	0.857	0.789
40			0.999	0.998	0.995	0.988	0.975	0.953	0.919	0.873	38			0.999	0.997	0.993	0.984	0.968	0.942	0.903	0.849
42				0.999	0.997	0.993	0.985	0.971	0.948	0.913	40				0.999	0.996	0.991	0.981	0.964	0.936	0.896
											42				0.999	0.998	0.995	0.989	0.978	0.959	0.931

$\dagger \tau^* = (\theta^* - x^*)(N_{tu}/\bar{C}_w^*)$, $x^* = 1$ at exit section.

Table 3-4 Insulated Duct or Porous Matrix Solution

N_{tu}	Maximum slope	Time at max slope	N_{tu}	Maximum slope	Time at max slope
1.0	0.368	0.000	8.0	0.840	0.806
1.25	0.448	0.000	8.5	0.863	0.817
1.5	0.502	0.000	9.0	0.885	0.828
1.75	0.532	0.000	9.5	0.907	0.837
2.0	0.541	0.000	10.0	0.929	0.846
2.5	0.553	0.271	15.0	1.121	0.898
3.0	0.577	0.424	20.0	1.284	0.924
3.5	0.604	0.521	25.0	1.432	0.940
4.0	0.632	0.590	30.0	1.564	0.950
4.5	0.660	0.640	40.0	1.797	0.963
5.0	0.688	0.680	50.0	2.009	0.970
5.5	0.715	0.711	60.0	2.196	0.974
6.0	0.741	0.737	70.0	2.371	0.977
6.5	0.767	0.758	80.0	2.533	0.981
7.0	0.792	0.776	90.0	2.683	0.984
7.5	0.816	0.792			

$$N_{tu} = \frac{hA}{C} = St \frac{L}{r_h}$$

$$\text{Maximum slope} = \left[\frac{d\varepsilon_f^*}{d\left(\frac{\theta^* - 1}{\bar{C}_w^*}\right)} \right]_{\max}$$

$$\text{Time at max slope} = \left(\frac{\theta^* - 1}{\bar{C}_w^*} \right)_{\max \text{ slope}} \approx \left(\frac{\theta^*}{\bar{C}_w^*} \right)_{\max \text{ slope}} \quad \text{for } \theta^* \gg 1$$

Table 3-5 Illustrative Examples of Gas-Turbine-Plant Regenerators

	Direct-transfer	Periodic-flow
Typical design conditions:		
1. Steady-state effectiveness ε , %	75	90.9
2. Number of transfer units N_{tu} , $N_{tu,o}$	3.0	10.0
3. Rotor capacity-rate ratio C_r^*	—	10.0
4. Wall-capacitance ratio $\bar{C}_w^*$, $\bar{C}_r^*$	1,500	1,000
5. Resistance ratio R^* (cold-to-hot sides)	2/3	2
6. Rotor period θ_r , s	—	2
7. Dwell time for C_r , $\theta_{d,c}$, s	0.1	0.02
Response time calculation:		
1. $\theta^*/\bar{C}_w^*$ for $\varepsilon_{f,2}^* = 0.90$, from Fig. 3-3	1.2	
2. θ_R^*/C_r^* for $\varepsilon_{f,2}^* = 0.90$, from Fig. 3-4	—	0.92
3. $\frac{\theta^*}{\bar{C}_w^*} - 0.40 \left(\frac{R^* - 1}{R^* + 1} \right)$, from Fig. 3-3, for $\varepsilon_{f,1}^* = 0.70$	1.04	
4. $\frac{\theta_R^*}{C_r^* - 0.6} - 0.53 \left(\frac{R^* - 1}{R^* + 1} \right)$, from Fig. 3-4, for $\varepsilon_{f,1}^* = 0.70$	—	2.32
5. $0.40 \left(\frac{R^* - 1}{R^* + 1} \right)$	-0.08	
6. $0.53 \left(\frac{R^* - 1}{R^* + 1} \right)$	—	0.18
7. $\varepsilon_{f,2}^* = 0.90$ response times		
$\theta^* = \bar{C}_w^*(\theta^*/\bar{C}_w^*)$	1,800	
$\theta = \theta_{d,c}\theta^*$, s	180	
$\theta_R^* = C_r^*(\theta_R^*/C_r^*)$	—	9.2
$\theta = \theta_r\theta_R^*$, s	—	18.4
8. $\varepsilon_{f,1}^* = 0.70$ response times		
$\theta^*/\bar{C}_w^*$	0.96	
θ , s	144	
$\theta_R^*/(C_r^* - 0.6)$	—	2.50
θ , s	—	47

Fig. 3-3 Transient response solutions 1, 2, 5, and 6 of Table 3-1.

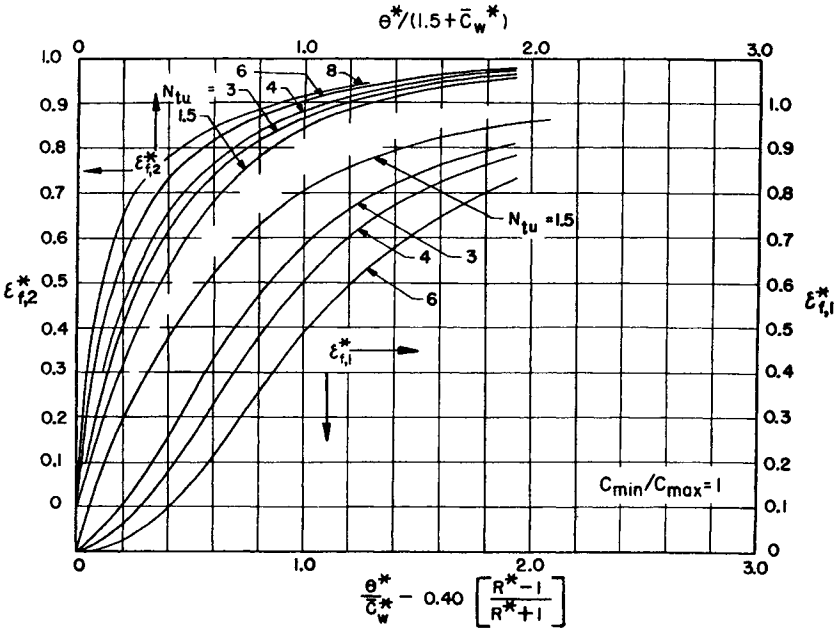


Fig. 3-4 Transient response solutions 3 and 4 of Table 3-1.

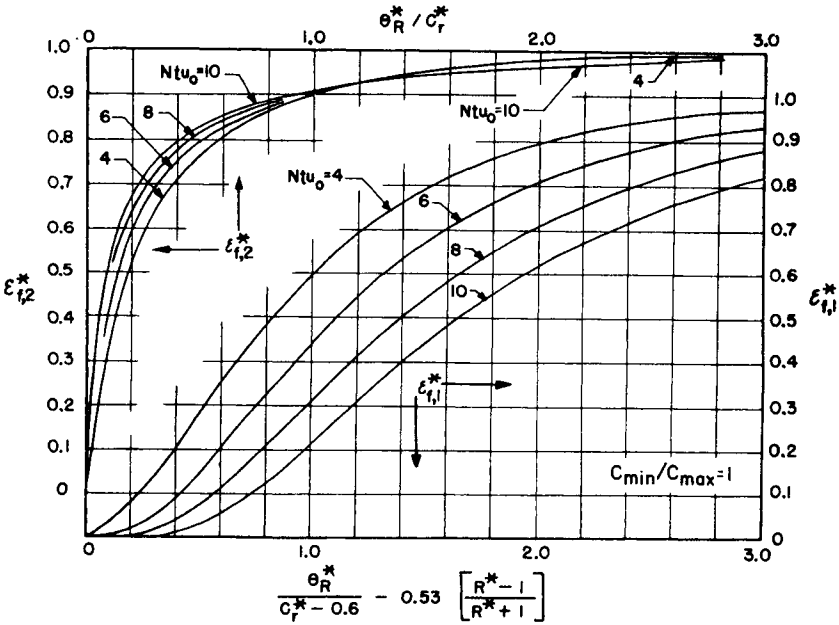


Fig. 3-5 Transient response solution 3 of Table 3-1. Enlarged abscissa scale for $\varepsilon_{f,2}^{\circ}$ of Fig. 3-4.

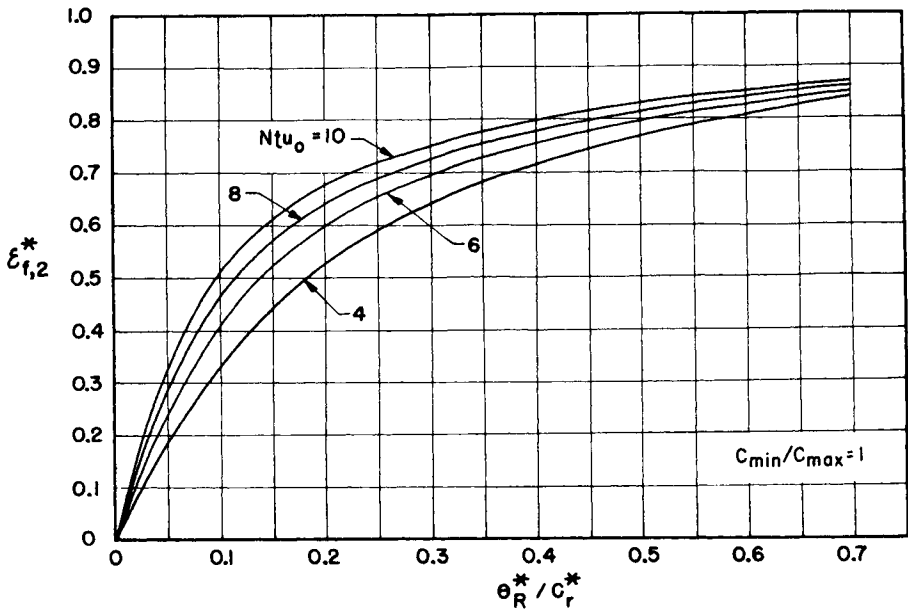


Fig. 3-6 Transient response solution 7 of Table 3-1: $Z\theta^* = \theta^* N_{tu} \left[\left(\frac{R^* + 1}{\bar{C}_w^*} \right) \left(\frac{R^* + 1 + \bar{C}_w^*}{2R^*} \right) \right] \varepsilon_{ss}$

$$\varepsilon_{ss} = 1 - e^{-N_{tu}}. \text{ Thus, } \varepsilon_f^* = \Phi(\theta^*, N_{tu}, R^*, \bar{C}_w^*)$$

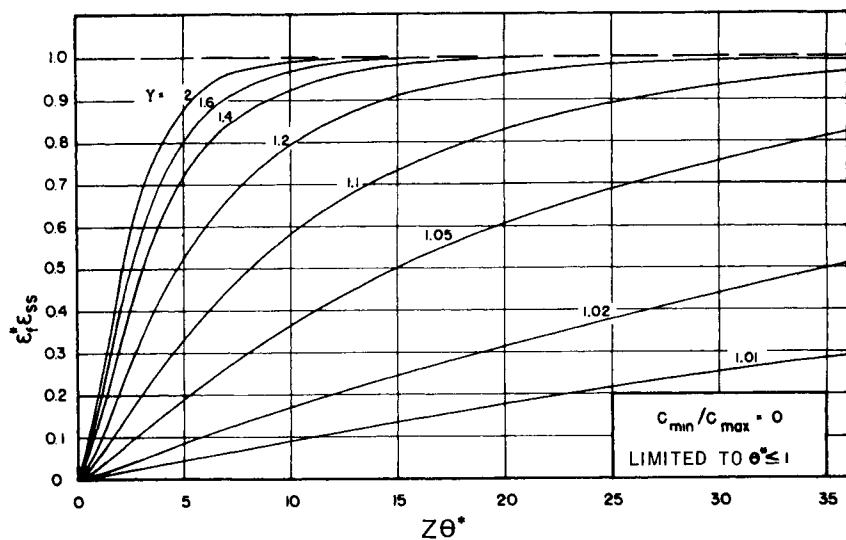
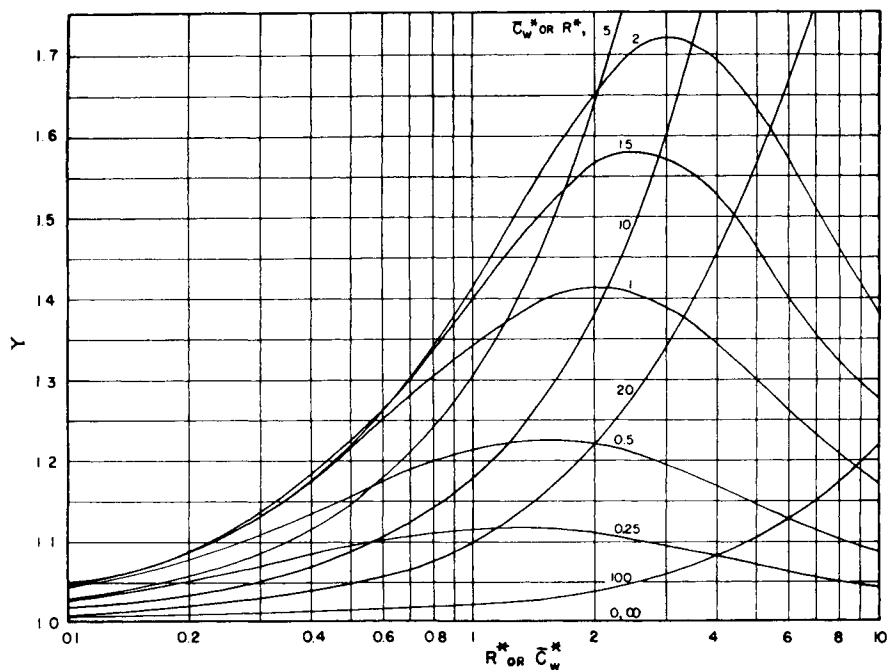


Fig. 3-7 Transient response solutions 8, 9, and 10 of Table 3-1.

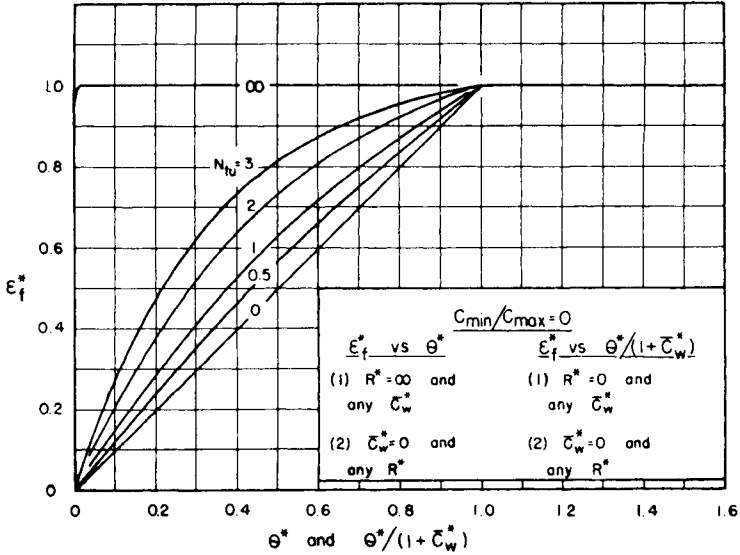


Fig. 3-8 Transient response solution 11 of Table 3-1, from the equation set of Table 3-2.

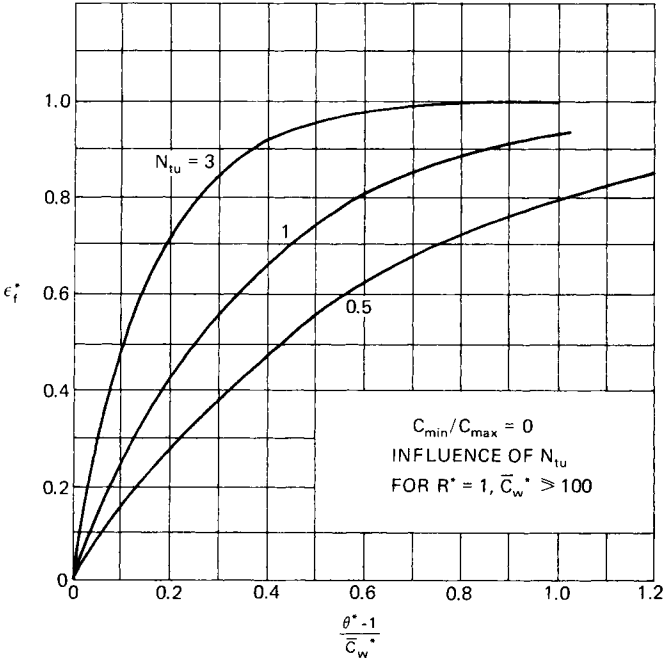


Fig. 3-9 Transient response solution 12 of Table 3-1, which demonstrates insensitivity to $\bar{C}_w^*$ for $\bar{C}_w^* \geq 100$.

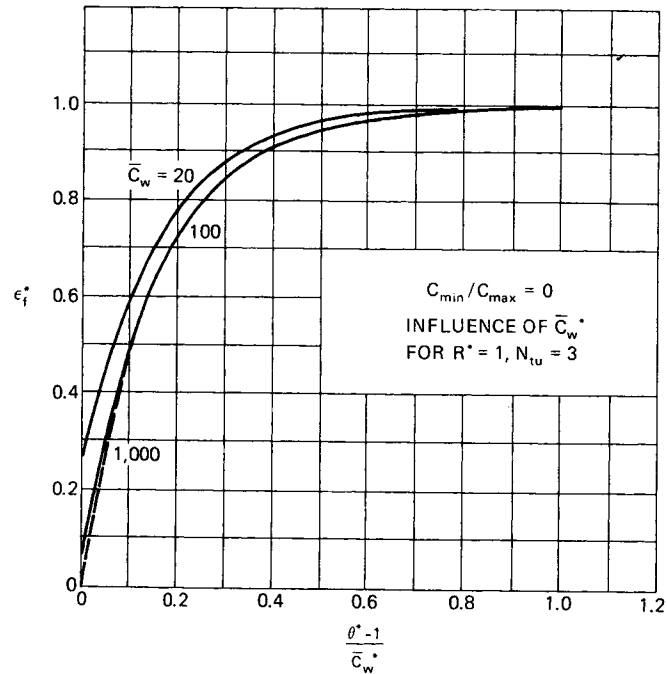


Fig. 3-10 Transient response solution 13 of Table 3-1. $R^* = 0$ solution applies for $\bar{C}_w^* > 0$ from Fig. 3-7.

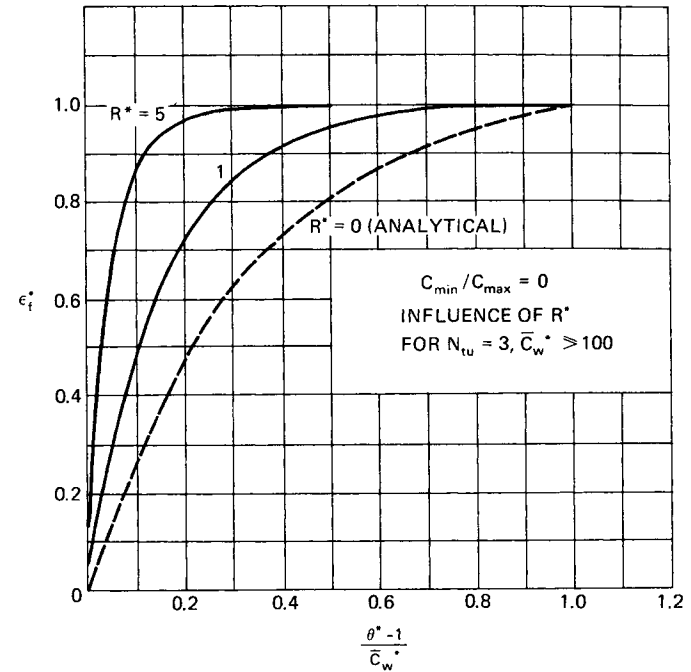


Fig. 3-11 Transient response solution 14 of Table 3-1.

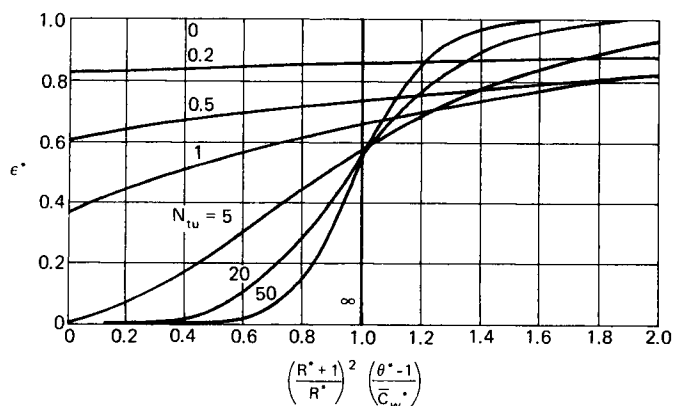


Fig. 3-12 Transient response solutions 15 and 16 of Table 3-1. See Table 3-3 for numerical values.

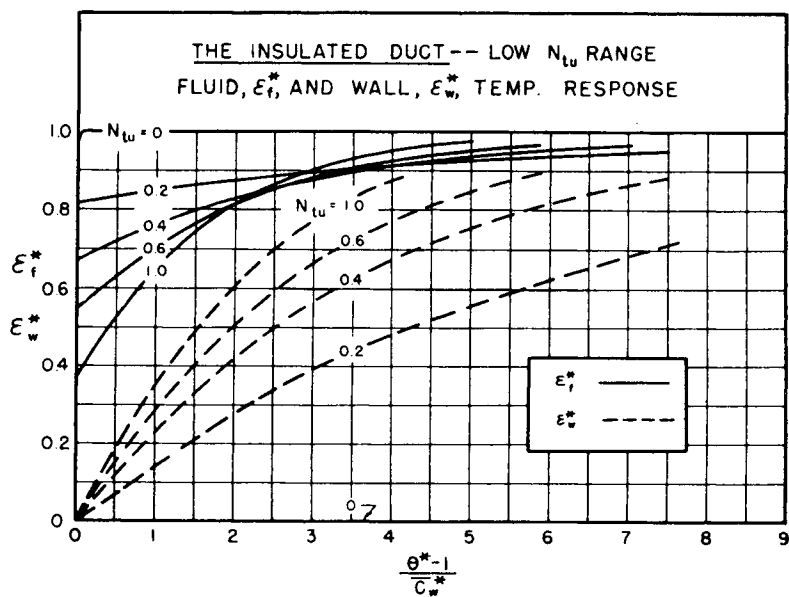


Fig. 3-13 Transient response solutions 15 and 16 of Table 3-1. See Table 3-3 for numerical values.

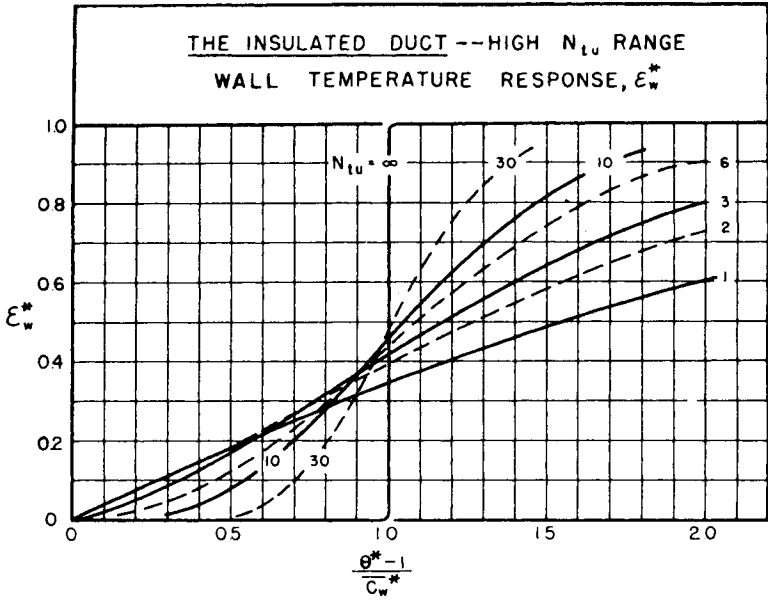


Fig. 3-14 Transient response solutions 15 and 16 of Table 3-1. See Table 3-3 for numerical values.

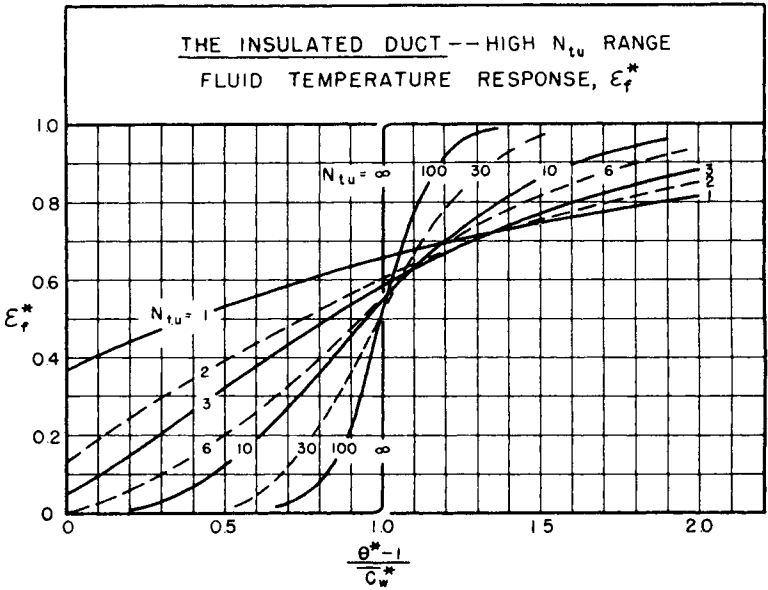
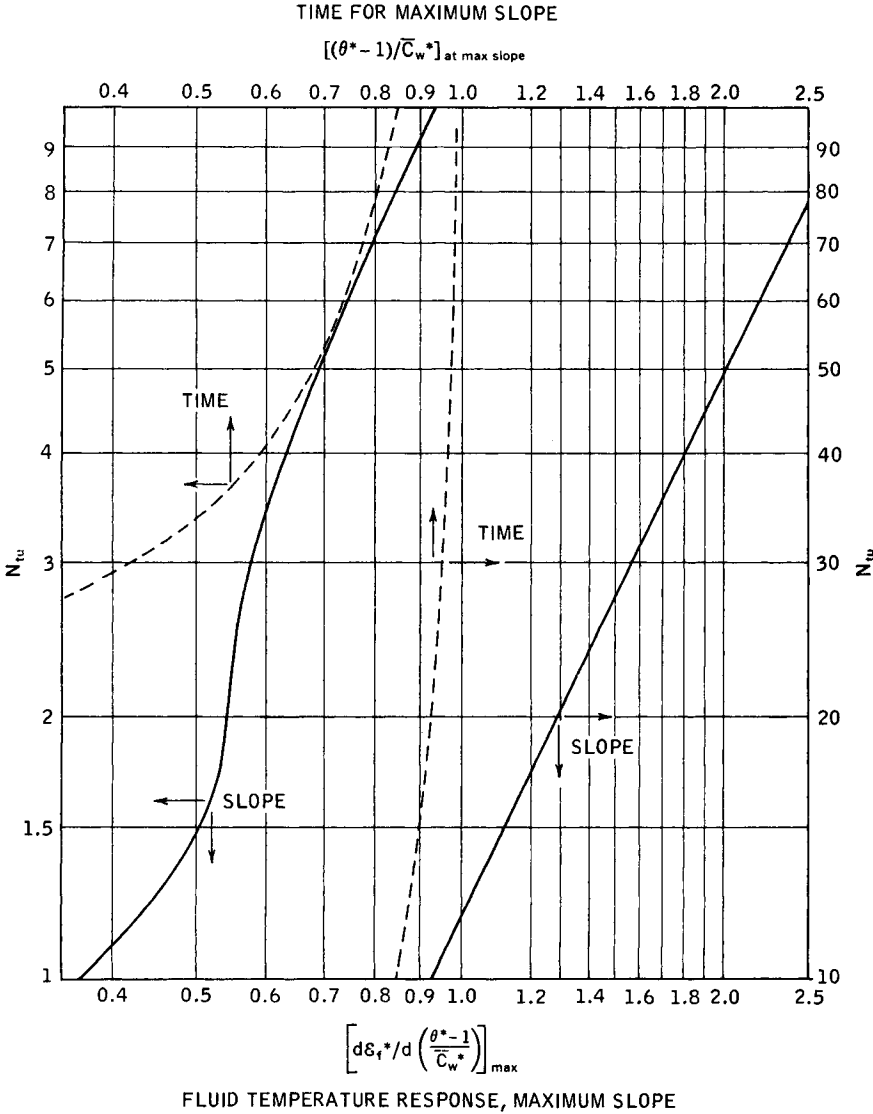


Fig. 3-15 Maximum slope of the transient response ϵ_f^2 of solution 16 of Table 3-1. See Table 3-3 for numerical values.



4

The Effects of Temperature-Dependent Fluid Properties

In the application of basic heat transfer and flow-friction data to heat exchanger design where temperature differences between the fluid and the surface are large, consideration must be given to the fact that fluid-transport properties, viscosity and thermal conductivity, and density may vary considerably with temperature. Such property variations distort both velocity and temperature profiles, and the question as to what temperature to use in evaluation of the fluid properties immediately arises.

The variable-properties problems for gases and liquids differ substantially. For gases, thermal conductivity, viscosity, and density all vary with temperature to a considerable degree. On the other hand, the only liquid property that is highly temperature dependent is viscosity, but viscosity typically varies more with temperature than any of the gas properties. Because of these differences, this chapter will be subdivided into a section on gases and a section on liquids. In both cases, the concern will be with the effect of the temperature profile at a particular flow cross section. In a final section of the chapter, the effect of the variation of temperature in the direction of flow will be considered.

EFFECT OF GAS-PROPERTY VARIATIONS AT A PARTICULAR FLOW SECTION

Property-variation effects at a particular flow cross section can usually be compensated for by evaluating certain or all of the properties at some specified reference temperature with respect to the wall and mixed mean fluid temperatures (or freestream temperature in the case of external flows). For gases, while such a scheme can generally be made to work, it often involves a certain awkwardness when applied to internal flows, such as in a heat exchanger. In an internal flow, one of the least ambiguous flow parameters is the mass velocity G . It is the product of the mean velocity and mean density, $V\rho$, but more importantly it is simply the mass flow rate divided by flow cross-sectional area W/A_c . In a gas flow where temperature is varying, the density varies everywhere and

the mean velocity then varies along the tube. But G is constant and unambiguously defined regardless of density variation, and it is thus very convenient to use in defining the Reynolds number, $4r_h G/\mu$, and the Stanton number, h/Gc_p . However, temperature-dependent property-compensation schemes based on a reference temperature generally require that G be broken up into its V and ρ components so that ρ may be evaluated at the reference temperature. For an internal flow, this really involves working with two different densities, for a mean density must also be evaluated in order to evaluate the mean velocity from the continuity equation. It is submitted that such a scheme is both awkward and subject to ambiguous interpretations. Of course, for an external flow (flow over an airfoil, for example), the possibility of ambiguity is greatly lessened because the significant velocity is the free-stream velocity which is usually unambiguously defined regardless of density variations in the boundary layer.

An alternative method of compensation, which is believed to be less subject to ambiguities, is based on the fact that the pertinent gas properties vary in a similar manner with absolute temperature for all gases. Thus all fluid properties can be evaluated at the convenient mixed mean temperature, and the effects of properties variation over the flow section can be expressed as a function of the absolute temperature ratio T_w/T_m . Fortunately, both analysis and experiment indicate that for most cases of interest the temperature-dependent-properties effect at a given Re can be expressed as a simple power of T_w/T_m :

$$\frac{Nu}{Nu_m} = \left(\frac{T_w}{T_m} \right)^n \quad (4-1)^\dagger$$

$$\frac{f}{f_m} = \left(\frac{T_w}{T_m} \right)^m \quad (4-2)$$

where it is understood that *all fluid properties* are to be evaluated at the *mixed mean fluid temperature*, and m and n are exponents that assume various magnitudes for different flow geometries and flow situations.

In Table 4-1, values of n and m are given for various situations, based on the authors' interpretation of the available experimental data and analytic solutions. These data are believed applicable for absolute temperature ratios up to about 3.0 or down to 0.33, and for absolute temperatures for which gas dissociation and chemical reaction do not occur. The experiments cited were carried out primarily with air, but enough included other gases to give some confidence in the generality of the results.

It will be noted in Table 4-1 that only flow through circular tubes is included. Information on flow through noncircular tubes is sketchy and incomplete, but it is relatively certain that, for turbulent flow, tube geometry has little effect. For laminar flow, tube shape probably has some effect, but in lieu of better data, it is recommended that the circular-tube results be used. For the turbulent bound-

[†]Subscript m refers to the constant-property solution evaluated at the mixed mean fluid temperature.

Table 4-1 Exponents for (T_w/T_m) , Gases

	n	m	Ref.	Basis
Laminar boundary layer on flat plate:				
Gas heating	-0.01	-0.1	1	Analytic solution
Gas cooling	0	-0.05	1	Analytic solution
Flow normal to circular tube or bank of circular tubes:				
Gas heating	0.0	0.0	2	Heat transfer experiments
Gas cooling	0.0	0.0	2	Heat transfer experiments
Fully developed laminar flow in circular tube:				
Gas heating	0.0	1.0	3	Analytic solution
Gas cooling	0.0	1.0	3	Analytic solution
Fully developed turbulent flow in circular tube:				
Gas heating†	-0.5	. . .	4	Experiments
		-0.1	4	Experiments
Gas cooling	0.0	-0.1	4	Experiments

† Sleicher and Rouse [5] examined a large amount of experimental heat transfer data for this case and recommended

$$n = -\left(\log_{10} \frac{T_w}{T_m}\right)^{1/4} + 0.3$$

any layer on an external surface, the behavior should differ very little from that for turbulent flow inside a circular tube.

All properties in the experimental data presented in Chap. 10 have been evaluated at the mixed mean temperature without a correction of the type indicated here. Since the wall-to-fluid temperature differences in the experiments were always moderately small, the indicated correction would in most cases be negligible, but this would not necessarily be the case in application of the data to large-temperature-difference heat exchangers.

It is to be noted that most of the test surfaces under consideration are of the interrupted-fin variety, where the predominant mechanism is heat transfer through a laminar boundary layer; for this geometry the temperature-dependent-properties effect is very small, since m and n are close to zero. It is therefore recommended that, for all the interrupted-boundary-layer surfaces, properties should be evaluated at the mixed mean fluid temperature with no additional correction for properties variation.

Property-variation effects become much more significant for flow through long uninterrupted tubes. The necessary corrections, from Table 4-1, have been incorporated in all the *summaries* of long-tube behavior presented in Chapter 7. If the analytic solutions in Chap. 6 are used for gas flow, these too should be corrected by reference to Table 4-1.

EFFECT OF LIQUID-PROPERTY VARIATIONS AT A PARTICULAR FLOW SECTION

The only property of importance that varies markedly with temperature for a liquid is the viscosity. The two common methods of compensation of viscosity-variation effects are the use of a reference temperature (the "film average" temperature scheme), and the method of lumping the effects into a ratio of wall to mean fluid viscosity (μ_w/μ_m), with the properties otherwise evaluated at the mixed mean fluid temperature. The latter method will be employed here, primarily because it is more consistent with the suggested procedure for handling the temperature-dependent-properties problem for gases. Thus the corrections take on the form

$$\frac{Nu}{Nu_m} = \left(\frac{\mu_w}{\mu_m} \right)^n \quad (4-3)$$

$$\frac{f}{f_m} = \left(\frac{\mu_w}{\mu_m} \right)^m \quad (4-4)$$

For *laminar flow in a circular tube*, calculations by Yang [6] for the thermal entry region for both constant heat rate and constant surface temperature may be very well correlated by $n = -0.11$ for a wide range of viscosity ratios. Deissler [7] suggests -0.14 on the basis of fully developed flow calculations, and this figure has been extensively used for many years to correlate experimental data both for laminar flow in tubes and for flow normal to tubes. There are evidently no definitive experiments on which to base a choice here, but in any case, the difference is small and either figure is probably adequate for design purposes. Deissler's analysis also yields data for the friction-factor exponent m . The effect is rather substantial; $m = 0.58$ for liquid heating and 0.50 for liquid cooling correlates the analysis satisfactorily. There is evidently little experimental confirmation of these results.

For *turbulent flow in a circular tube* Petukhov [8] has examined heat transfer data covering these ranges: Pr, 2 to 140; Re, 5,000 to 123,000; and μ_w/μ_m , 0.025 to 12.5. He finds that the data are quite well correlated by

$$\frac{\mu_w}{\mu_m} > 1 \quad n = -0.25 \text{ (cooling)}$$

$$\frac{\mu_w}{\mu_m} < 1 \quad n = -0.11 \text{ (heating)}$$

Allen and Eckert [9] conducted experiments with water at Pr = 8 for the Reynolds number range 13,000 to 111,000 for the heating case and found that the friction coefficient results are reasonably correlated by

$$\frac{\mu_w}{\mu_m} < 1 \quad m = 0.25 \text{ (heating)}$$

Friction data for the cooling case appear to be scarce, but the very early data of Rohonczy [10] for $Re = 33,000$ to $225,000$, $Pr = 1.3$ to 5.8 , and $\mu_w/\mu_m = 1$ to 2 are quite well correlated by the same value of m , i.e.,

$$\frac{\mu_w}{\mu_m} > m = 0.25 \text{ (cooling)}$$

EFFECT OF PROPERTY VARIATION IN THE FLOW DIRECTION

In many applications, temperature (and thus property) variations along the tube in the flow direction are more pronounced than those occurring over the flow cross section. Analysis of typical cases for gases indicates that if the absolute temperature variation is less than $2:1$ from one end of the flow tube to the other, it is adequate to evaluate properties at a *mean temperature with respect to flow-tube length*. This mean is then also the T_m used to correct for property variation at a flow section, as discussed above. For greater temperature variations in the flow direction, it is probably desirable to consider the exchanger in separate sections, in each of which the temperature variation is not excessive.

The evaluation of a correct mean temperature with respect to flow-tube length often leads to some difficulty if very precise results are desired. Analysis of the heat transfer performance of a heat exchanger is not so critical in this respect as is the pressure drop, which is directly proportional to the mean fluid density. In most cases, however, a reasonable estimate is satisfactory.

The simplest case is that of a counterflow heat exchanger with equal fluid capacity rates; the mean fluid temperatures are merely the arithmetic averages of inflow and outflow temperatures. Where one fluid is at constant temperature, as in an evaporator or condenser, or where one capacity rate is very much greater than the other, the temperature variation of the other fluid will be close to exponential, provided the heat transfer conductance does not vary markedly with tube length. Denoting the constant temperature, entering fluid, and leaving fluid temperatures as t_{const} , t_1 , and t_2 , respectively,

$$t_{\text{mean}} = t_{\text{const}} - \frac{t_2 - t_1}{N_{\text{tu}}} \quad (4-5)$$

Where the heat transfer conductance decreases with tube length, as occurs most importantly with laminar flow, the true mean temperature with respect to tube length is nearer the fluid exit temperature, and the designer must exercise judgment since exact analysis is very tedious.

The problem of the crossflow heat exchanger is much more complex, because each flow tube behaves differently. However, the recommendations cited above should be a satisfactory guide for most applications. At the very least, they will provide an estimate of the importance of the variable-properties effects.

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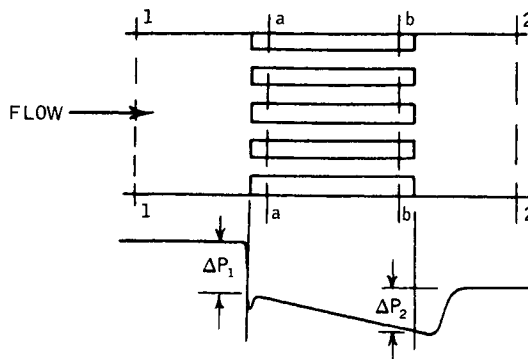
Abrupt Contraction and Expansion Pressure-Loss Coefficients

The typical installation of a compact heat exchanger usually involves a flow contraction at the heat exchanger core entrance and a flow expansion at the core exit. Often an abrupt (right-angle) contraction and expansion are used, and these introduce additional flow-stream pressure drop. The designer is generally interested in the overall core pressure drop, including any entrance and exit loss, rather than merely the core friction pressure drop. He or she thus must be able to predict these additional losses.

This problem has been investigated both analytically and experimentally by Kays [1]. The present chapter is a brief summary of the pertinent conclusions of this reference and includes graphs of core entrance and exit pressure-loss coefficients, K_c and K_e , that can be used for heat exchanger design.

Figure 5-1 shows the entrance pressure drop and the exit pressure rise characteristic of flow through a heat exchanger core. The entrance pressure

Fig. 5-1 Entrance pressure drop and exit pressure rise in a heat exchanger core.



drop is made up of two parts which have been arbitrarily separated as follows: The first is the pressure drop which would occur due to flow-area change alone, without friction. The second is the pressure loss due to the irreversible free expansion that always follows the abrupt contraction, which arises from boundary-layer separation (as characterized by the vena contracta), and the consequent pressure change due to change of momentum rate associated with changes in velocity profile downstream from the vena contracta. Although the principal applications under consideration here involve gas flow, density changes in the usual case are sufficiently small that a constant-density treatment is satisfactory. The entrance *pressure drop* can then be expressed as

$$\frac{\Delta P_1}{\rho} = \frac{V^2}{2g_c} (1 - \sigma^2) + K_c \frac{V^2}{2g_c} \quad (5-1)$$

where V is the velocity in the smaller tube (or tubes), i.e., inside the heat exchanger core, and σ is the core free-flow to frontal-area ratio. The irreversible component of the pressure drop is contained in the abrupt contraction, or entrance, coefficient K_c .

The exit *pressure rise* is similarly broken into two parts. The first is the pressure rise which would occur due to area change alone, without friction, and is identical to the corresponding term in the entrance pressure drop. The second is the pressure loss associated with the irreversible free expansion and momentum changes following an abrupt expansion, and this term in the present case subtracts from the other. Thus

$$\frac{\Delta P_2}{\rho} = \frac{V^2}{2g_c} (1 - \sigma^2) - K_e \frac{V^2}{2g_c} \quad (5-2)$$

where K_e is the abrupt expansion, or exit, coefficient.

K_c and K_e are functions of the contraction and expansion geometry and, in some cases, of the Reynolds number in the tubes. The latter dependence arises from the influence of the velocity profile in the tubes on the momentum rates and from the resulting effect on the change of momentum at the entrance and exit. These coefficients have been established analytically for a number of simple entrance and exit geometries [1] and are presented graphically in Figs. 5-2 to 5-5. Sufficient experimental confirmation is contained in Ref. 1 to establish the validity of the analysis at least to a precision adequate for the present application.

The coefficients have been evaluated on the assumption of essentially uniform velocity in the large duct leading to the core, of uniform velocity in the leaving duct, but of a fully established velocity profile in the small tubes of the core. This assumption is generally justified in heat exchanger core analysis because the Reynolds number in the core tubes is invariably many times smaller than that in the entering and leaving ducts. It is only at low Reynolds numbers that velocity profile has any appreciable influence. In applying the data, however, it should be remembered that the coefficients already include pressure change associated with change of velocity profile. *Thus the core friction factor used in Eq. (2-22) to evaluate the overall pressure drop should be defined on the*

basis of mean wall shear and not on the basis of overall pressure drop, as is sometimes done. All the friction factors contained in this book are defined in this manner, unless otherwise specified. This distinction becomes of particular importance when evaluating pressure drop with laminar flow in the core tubes.

Since Figs. 5-2 to 5-5 cover only a few of the simplest geometries, the designer must exercise judgment in the application of these results. In particular, the only case that is rigorously applicable to very short tubes is that of circular tubes. The others have been worked out only for the case of tubes of sufficient length so that there is an essentially fully established velocity profile at the tube exit. A less than fully established velocity profile will result in entrance coefficients K_e that are lower, and exit coefficients K_e that are higher, than those obtained for the fully established condition. Interrupted-fin surfaces, and this includes a large portion of the surfaces considered here, never have velocity profiles that approach the fully established profiles of long uninterrupted tubes, because the very purpose of the fin interruption is to break up the profile. For such surfaces with very frequent fin interruptions, the K_e and K_e curves for $Re = \infty$ should be used, and it will be noted that the $Re = \infty$ curves in Figs. 5-3 to 5-5 are all identical. There will, of course, be interrupted-fin surfaces where a partial profile is established, and the designer can only rely on his or her judgment. However, it should be noted that the entrance and exit pressure losses in heat exchangers using interrupted-fin surfaces do not contribute greatly to the overall exchanger pressure drop, as is illustrated by the examples of Appendix B. Thus the curves of Figs. 5-2 to 5-5 provide the designer with estimates of entrance and exit pressure-drop data which are usually adequate for design purposes.

For flow over banks of either bare or finned tubes, each tube row consists of a contraction and an expansion, and the friction behavior of the first and last tube rows in the core is not materially different from interior tube rows. Thus the entrance and exit behavior is in effect already taken into consideration in the core friction factors, and the entrance and exit loss coefficients K_e and K_e are then *zero*. This consideration also applies to the matrix surfaces—sphere-bed, crossed-rod, and woven-wire screens.

REFERENCES

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Fig. 5-2 Entrance and exit pressure-loss coefficients for a multiple-circular-tube heat exchanger core with abrupt-contraction entrance and abrupt-expansion exit.

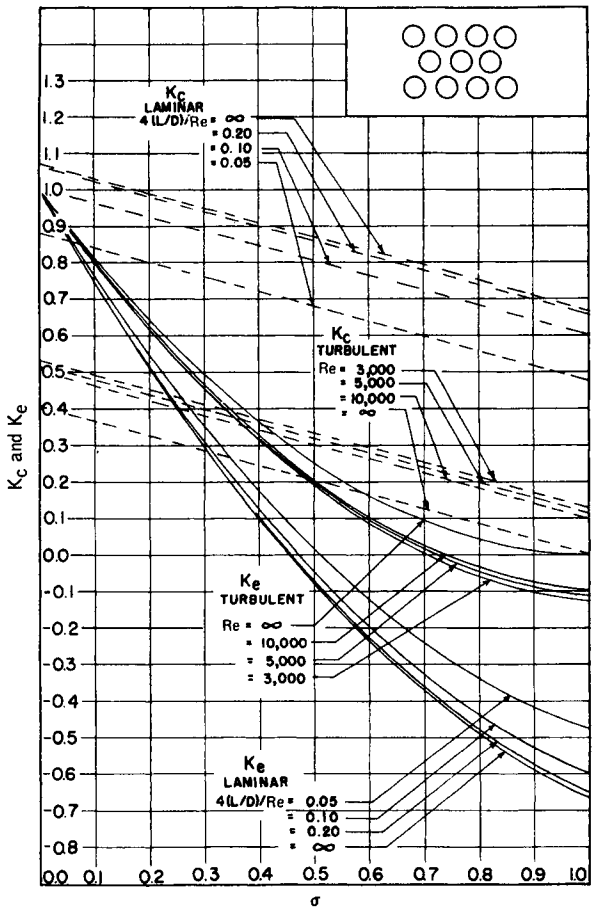


Fig. 5-3 Entrance and exit pressure-loss coefficients for a multiple-tube flat-duct heat exchanger core with abrupt-contraction entrance and abrupt-expansion exit.

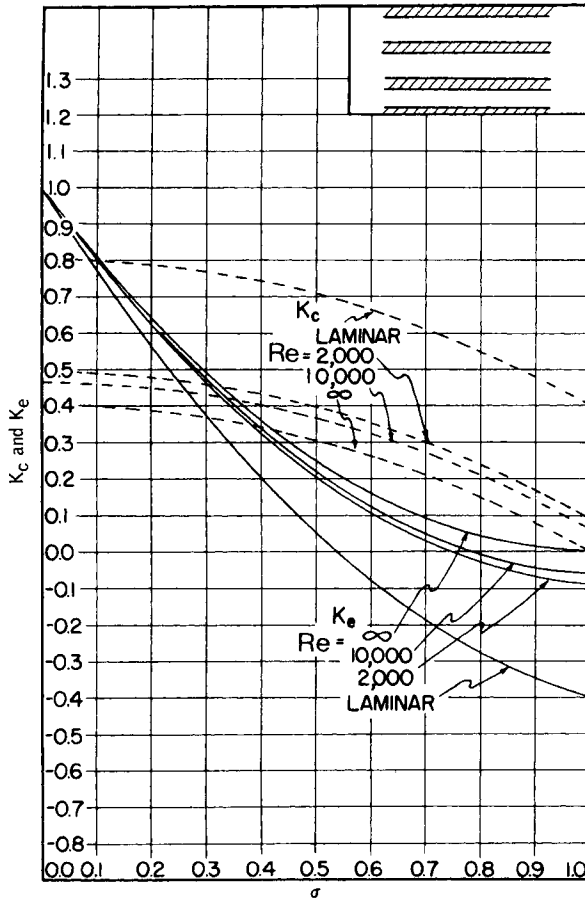


Fig. 5-4 Entrance and exit pressure-loss coefficients for a multiple-square-tube heat exchanger core with abrupt-contraction entrance and abrupt-expansion exit.

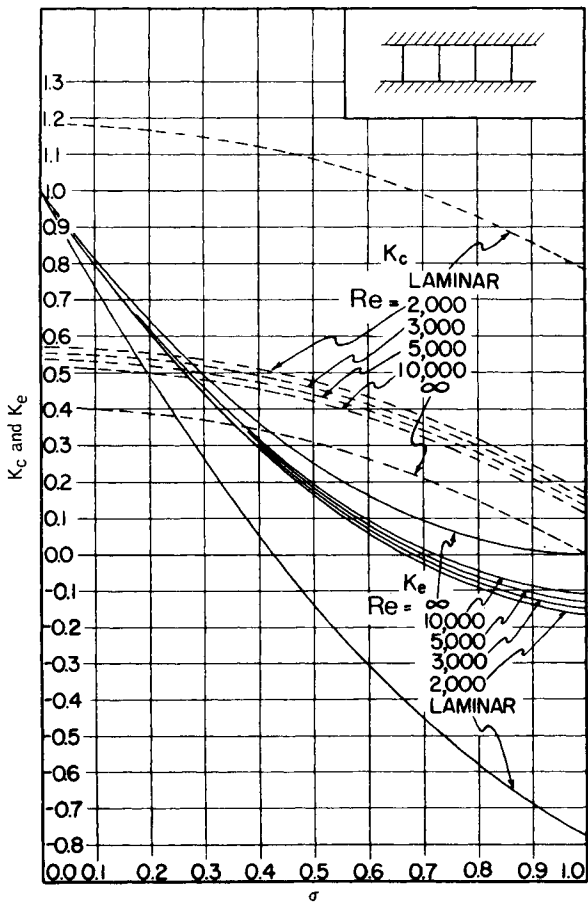
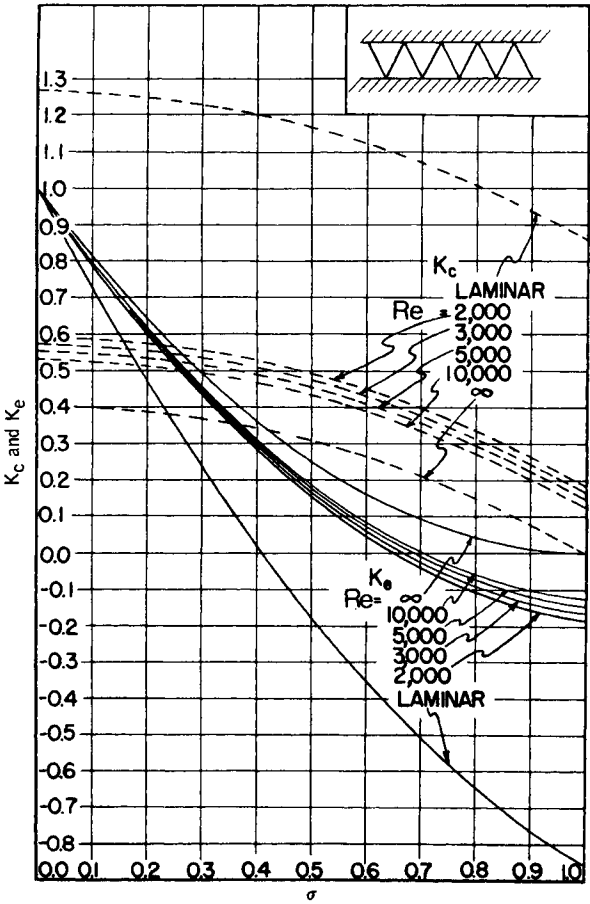


Fig. 5-5 Entrance and exit pressure-loss coefficients for a multiple-triangular-tube heat exchanger core with abrupt-contraction entrance and abrupt-expansion exit.



6

Analytic Solutions for Flow in Tubes

This chapter contains, in graphic and tabular form, a number of analytic solutions (and two experimental correlations) for both laminar and turbulent flow inside continuous tubes of various cross section—circular-, rectangular-, triangular-, and concentric-circular-tube annuli. The validity of most of these solutions has been adequately established experimentally, and thus where there is overlap, there is no serious discrepancy between these data and the experimental data in Chap. 10. The data of Chap. 10 cover primarily complex geometrical flow configurations for which a completely analytic treatment is impracticable. For applications where long noninterrupted tubes are employed, it is suggested that the reader investigate the solutions presented here before attempting to find appropriate test data in Chap. 10. It should also be noted that the test data presented in Chap. 10 are largely restricted to the Prandtl number range of gases, whereas the analytic solutions presented here cover, with few exceptions, the entire Prandtl number spectrum.

The solutions are presented in the following order:

1. Fully developed velocity and temperature profiles
 - a. Laminar flow, various geometries
 - b. Turbulent flow, circular-tube and concentric-circular-tube annuli
2. Thermal entry length, fully developed velocity profiles
 - a. Laminar flow, circular-tube, concentric-circular-tube annuli, rectangular tubes
 - b. Turbulent flow, circular-tube
3. Combined thermal and hydrodynamic entry length laminar flow, circular-tube

In all the solutions presented, the length dimension in the Nusselt number and the Reynolds number is the *hydraulic diameter*, defined by Eq. (1-5).

In Fig. 6-1, Nusselt numbers and friction factors are presented for hydrodynamically and thermally fully developed flow in a number of tubes of simple geometry. The Nusselt numbers are presented for two heating conditions for

which a fully developed temperature profile is possible: constant heat rate per unit of tube length ($\bar{H}$), and constant surface temperature ($\bar{T}$).† The triangular- and rectangular-tube results are also restricted to a constant surface temperature around the heated tube periphery. Additionally, the solution of Reynolds [1] is presented for a circular tube with constant heat rate per unit of tube length, but with a cosine heat-flux variation around the periphery.

Figures 6-2 and 6-3 give the corresponding solutions for the complete family of rectangular tubes [2]. Again, temperature is always constant around the tube periphery.

Figure 6-4 gives the fully developed laminar-flow friction factors for the concentric-circular-tube annulus, where r^* is the annulus radius ratio r_i/r_o , inner radius over outer radius. In Fig. 6-5 the fully developed laminar-flow constant-heat-rate Nusselt numbers for the annulus are presented for the inner tube when the inner tube alone is heated, Nu_{ii} , and for the outer tube when the outer tube alone is heated, Nu_{oo} [3]. Also presented are influence coefficients Z_i and Z_o , which may be used together with the one-surface-heated Nusselt numbers to calculate Nusselt numbers on both surfaces for any ratio of heat flux on the two surfaces:

$$Nu_i = \frac{Nu_{ii}}{1 - (q''_o/q''_i)Z_i} \quad (6-1)$$

$$Nu_o = \frac{Nu_{oo}}{1 - (q''_i/q''_o)Z_o} \quad (6-2)$$

The inner- and outer-tube Nusselt numbers are then related to the heat flux on the two surfaces and the surface and fluid mixed mean temperatures as follows:

$$q''_i = h_i(t_i - t_m) \quad (6-3)$$

$$q''_o = h_o(t_o - t_m) \quad (6-4)$$

$$Nu_i = \frac{h_i D_h}{k} \quad Nu_o = \frac{h_o D_h}{k} \quad (6-5)$$

t_i is the temperature on the inner tube surface, t_o is the temperature on the outer tube surface, and t_m is the mixed mean fluid temperature. Heat flux is here defined as positive inward to the fluid from the surfaces. Heat flux on either surface may be either inward or outward, positive or negative. The relations and results given hold whether the heat-flux ratio be positive or negative.

In Fig. 6-6 is plotted the Kármán-Nikuradse equation (an experimental correlation) for the friction factor for fully developed turbulent flow in a smooth circular tube. Also shown are two simplified equations which may be used where simplicity of an algebraic equation is desired. Turbulent-flow friction factors in noncircular tubes that do not involve sharp corners with very acute angles differ very little from the circular tube.

† These are not the only cases for which a fully developed temperature profile is possible.

The complete, fully developed, turbulent flow heat transfer behavior for a circular tube with constant heat rate per unit of tube length is shown in Fig. 6-7. These results are based on experimental correlations suggested by Sleicher and Rouse [4].

The ratio of the Nusselt number for constant heat rate to the Nusselt number for constant surface temperature is plotted in Fig. 6-8 from the calculations of Sleicher and Tribus [5]. Note that for turbulent flow it is only at very low Prandtl numbers that there is a significant difference.

Figure 6-9 is a plot of the fully developed, turbulent-flow, constant-heat-rate Nusselt numbers for flow between parallel planes with one wall insulated [6]. The corresponding influence coefficients are given in Fig. 6-10. These data can be used directly in Eq. (6-1) or (6-2) to calculate asymmetric heating.

Figures 6-11 to 6-13, from the calculations of Leung [6], give the complete, fully developed, turbulent-flow, constant-heat-rate performance for a concentric-circular-tube annulus with radius ratio 0.20. These results may be used in Eqs. (6-1) and (6-2) to calculate asymmetric heating in the annulus for any heat-flux ratio. Since the parallel planes and the circular tube (outer tube only) are also members of the annulus family, sufficient results are presented to make reasonable estimates by interpolation for any annulus radius ratio. To further assist in interpolation, Figs. 6-14 and 6-15 are presented.

Figures 6-9 through 6-15 are the results of analytic solutions based on an internally consistent set of turbulence model parameters. However, in the low Prandtl number range (< 0.03) a considerably higher value of turbulent Prandtl number was employed than now seems reasonable. As a result the Nusselt numbers in the low Prandtl number range are generally a few percent lower than they in fact should be, and some inconsistency will be noted when comparing these results with Fig. 6-7.

The solutions for thermal entry length, but with hydrodynamically fully developed flow, start with Fig. 6-16. These solutions are all based on fluid temperature uniform over the flow cross section at $x = 0$, the point where heat transfer starts. Figure 6-16 gives the constant-surface-temperature results of Sellars et al. [7]. Both the local Nusselt number Nu_x and the mean Nusselt number with respect to tube length Nu_m are plotted for laminar flow in a circular tube.

In Fig. 6-17 the constant-heat-rate laminar-flow solutions for circular tubes [8] and flow between parallel planes [3] are presented. The Nusselt number is the local Nusselt number, which is the only Nusselt number of any useful significance in constant-heat-rate applications. The parallel planes results include the influence coefficients so that asymmetric heating problems can be handled. Again, Eqs. (6-1) and (6-2) are applicable.

The laminar-flow constant-heat-rate solutions for two annulus radius ratios, from the analysis of Lundberg [3], are given in Fig. 6-18. Again, Eqs. (6-1) and (6-2) are applicable.

Figure 6-19 gives the laminar-flow thermal-entry-length local Nusselt numbers for rectangular tubes with constant surface temperature, both axially and peripherally. These results are based on numerical solutions discussed by Shah

and London [9] and are actually a composite of more than one analysis. These are not "exact" solutions as the precision of the numerical methods used is not high.

Figures 6-20 to 6-22 show the calculated thermal-entry-length performance for turbulent flow in a circular tube with constant heat rate per unit of tube length. The $Pr = 0.7, 10.0$ results are based on the analysis of Sparrow et al. [10], while the $Pr = 0.01$ results are based on Kays and Nicoll [11]. The corresponding results for constant surface temperature differ little from these.

Figure 6-23 shows the friction factors for the hydrodynamic entry length of a circular tube, based on the analysis of Langhaar [12]. Velocity is assumed uniform over the flow cross section at $x = 0$. Three friction factors are plotted: f_x is the local friction factor, defined on the basis of local wall shear stress; f is the mean friction factor from 0 to x , defined on the basis of mean shear stress over the same length; $\bar{f}_{app}$ is the apparent mean friction factor, defined on the basis of total pressure drop from 0 to x , as follows:

$$\Delta P = \bar{f}_{app} \frac{G^2}{2g_c\rho} \frac{x}{r_h} \quad (6-6)$$

This friction factor takes into consideration pressure drop due to increased axial momentum flux accompanying the development of the velocity profile. f and $\bar{f}_{app}$ can be evaluated from one another if the change of momentum flux can be evaluated. A momentum-flux correction factor K_d is plotted in Fig. 6-23 and can be used to evaluate total momentum flux at any flow section; K_d relates the actual momentum flux to that which would be evaluated from the mean velocity.

$$\text{Momentum flow rate} = K_d(A_c G) \frac{G}{g_c\rho} \quad (6-7)$$

In Figs. 6-24 to 6-26 are plotted three laminar-flow heat transfer solutions for combined hydrodynamic and thermal entry length for a circular tube, i.e., both velocity and temperature uniform at $x = 0$. The results in Figs. 6-24 and 6-25 are for a constant surface temperature and are based on numerical solutions by Hornbeck [13, 14]. Fig. 6-26 is for constant heat rate and is based on the results of Heaton [15].

Calculations have been made for the combined hydrodynamic and thermal entry length for turbulent flow in a tube. However, it is the opinion of the authors that such solutions are of very limited utility and can be positively misleading. If a tube has a nozzle entrance, a laminar boundary layer will tend to develop, followed by a transition to turbulent flow with entirely different heat transfer characteristics than would be the case when a turbulent boundary layer develops from the tube entrance, as is assumed in all such analyses. If the tube has a square or other sharp-cornered entrance which causes boundary-layer separation at the entrance, the turbulence generated causes much greater heat transfer rates in the entry region than would be calculated on the basis of a developing turbulent boundary layer. In Chap. 7, performance curves based on experimental data are presented for several types of tubes with sharp-cornered entrances, and these are believed to be much more useful and accurate for heat exchanger design than the available analytic solutions.

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Fig. 6-1 Laminar-flow heat transfer and friction solutions for fully developed velocity and temperature profiles.

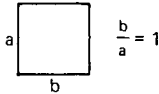
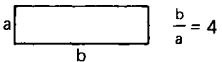
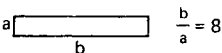
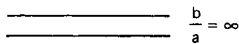
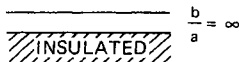
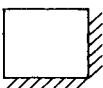
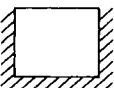
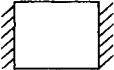
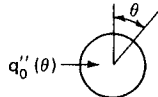
GEOMETRY ($L/4r_h > 100$)	$Nu_{(H)}$	$Nu_{(T)}$	$f Re$	$\frac{Nu_{(H)}}{Nu_{(T)}}$	$\frac{St Pr^{2/3}}{f}$
	3.11	2.47	13.33	1.26	0.263
	3.61	2.98	14.2	1.21	0.286
	4.364	3.66	16	1.19	0.307
	5.33	4.44	18.3	1.20	0.328
	6.49	5.60	20.6	1.16	0.355
	8.235	7.54	24	1.09	0.386
	5.385	4.86	24	1.11	0.253
	3.556	3.018	14.2		
	2.836	2.375	14.2		
	2.712	2.437	14.2		
	4.094	3.703	14.2		
	If $q''_0(\theta) = q''_a(1 + b \cos \theta)$ $Nu_{(H)}(\theta) = \frac{(1 + b \cos \theta)}{(\frac{11}{48} + \frac{b}{2} \cos \theta)}$				

Fig. 6-2 Friction factors for fully developed laminar flow in rectangular tubes.

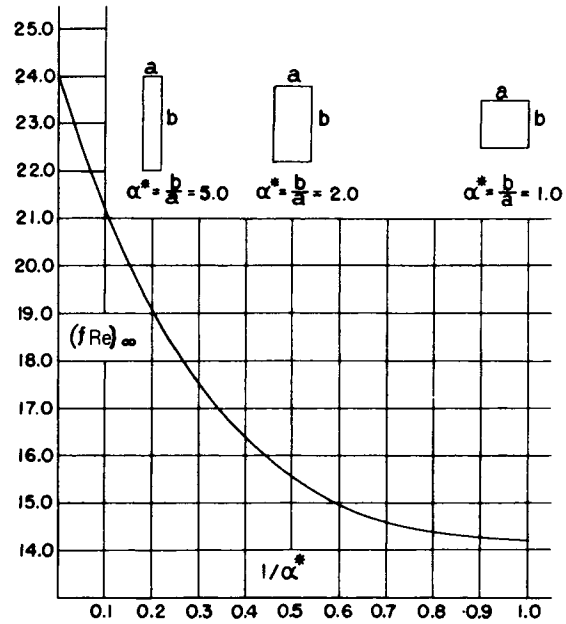


Fig. 6-3 Nusselt numbers for laminar flow in rectangular tubes with fully developed velocity and temperature profiles.

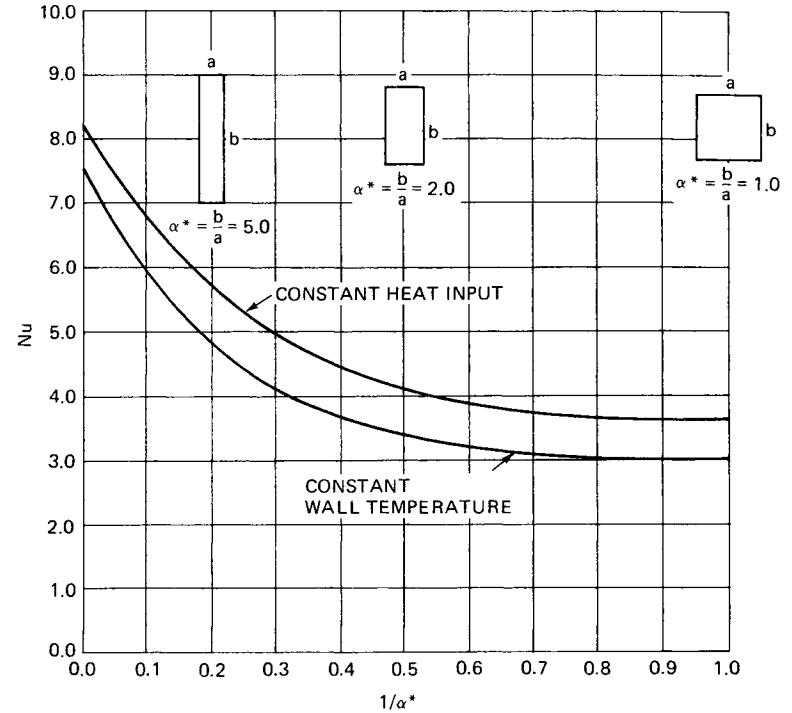


Fig. 6-4 Friction factors for fully developed laminar flow in concentric-circular-tube annuli.

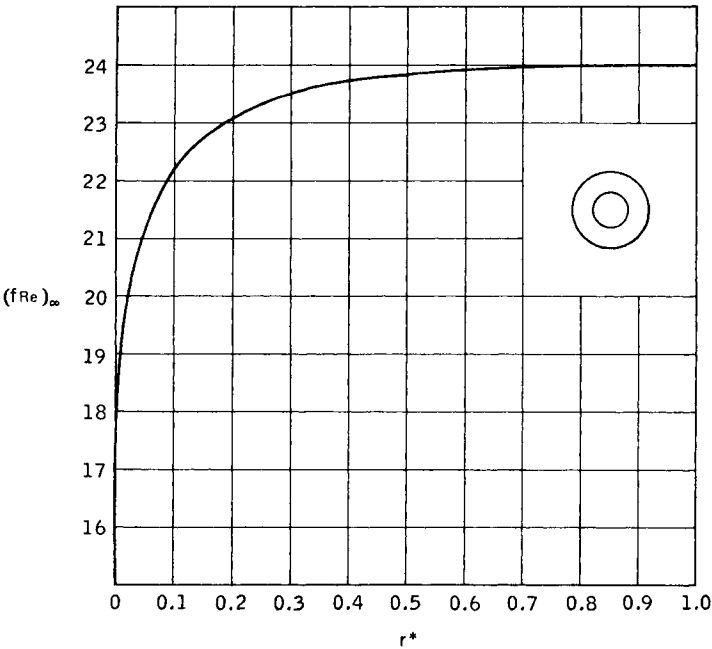


Fig. 6-5 Nusselt numbers and influence coefficients for laminar flow in concentric-circular-tube annuli for constant heat rate per unit of tube length and fully developed velocity and temperature profiles.

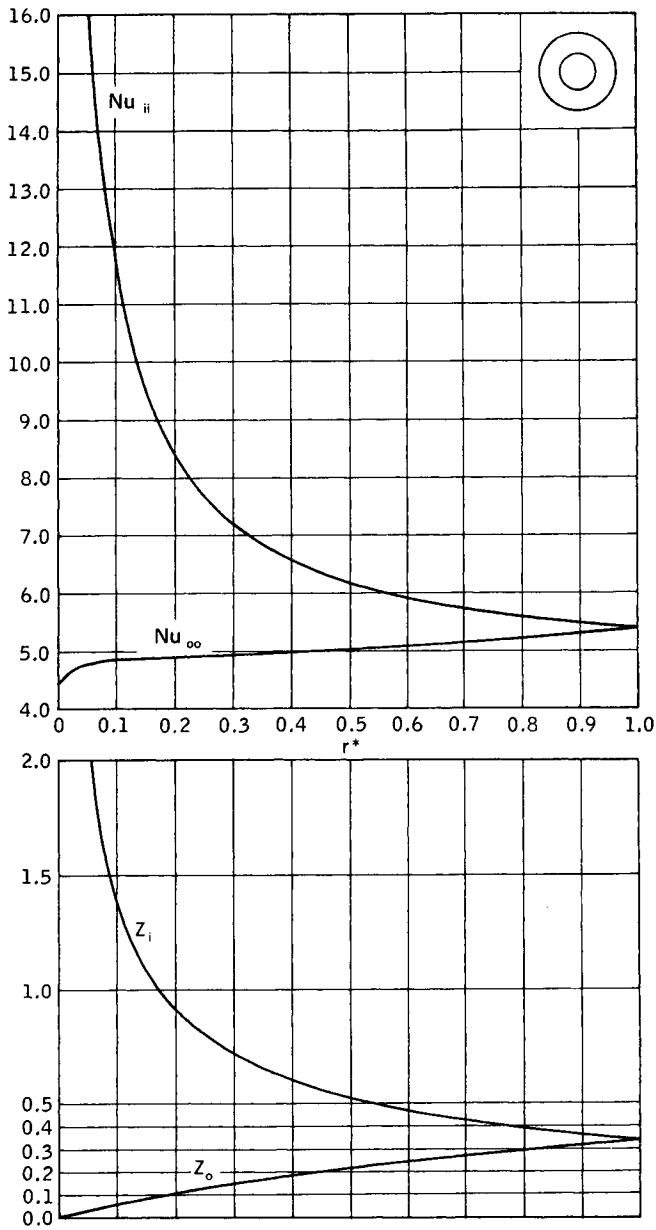


Fig. 6-6 Friction factors for fully developed turbulent flow in a smooth-walled circular tube. Turbulent flow friction factors for rectangular and annular tubes do not differ significantly from these.

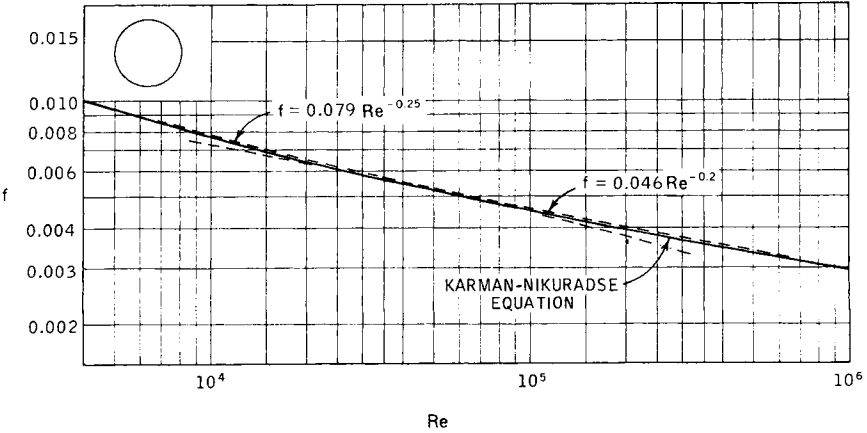


Fig. 6-7 Nusselt numbers for turbulent flow in a circular tube with constant heat rate per unit of tube length and fully developed velocity and temperature profiles.

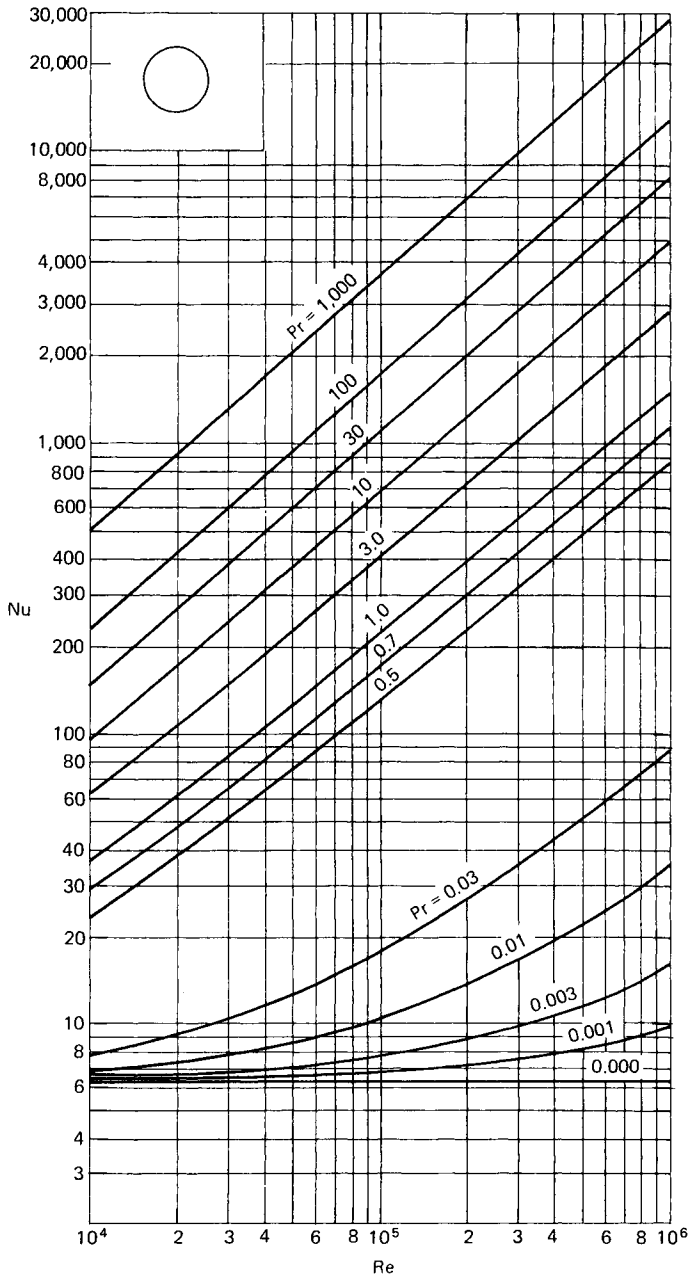


Fig. 6-8 Ratio of Nusselt number for constant heat rate to Nusselt number for constant surface temperature for fully developed turbulent flow in a circular tube.

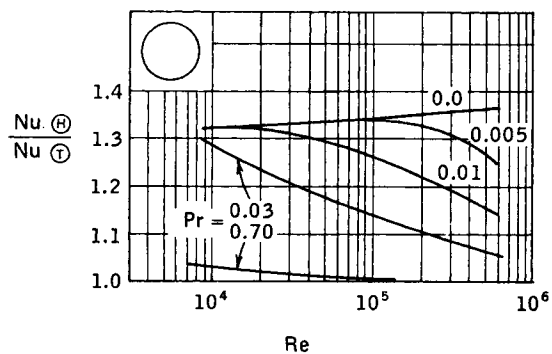


Fig. 6-9 Nusselt numbers for turbulent flow between parallel planes with one side insulated, constant heat rate per unit of tube length, and fully developed velocity and temperature profiles.

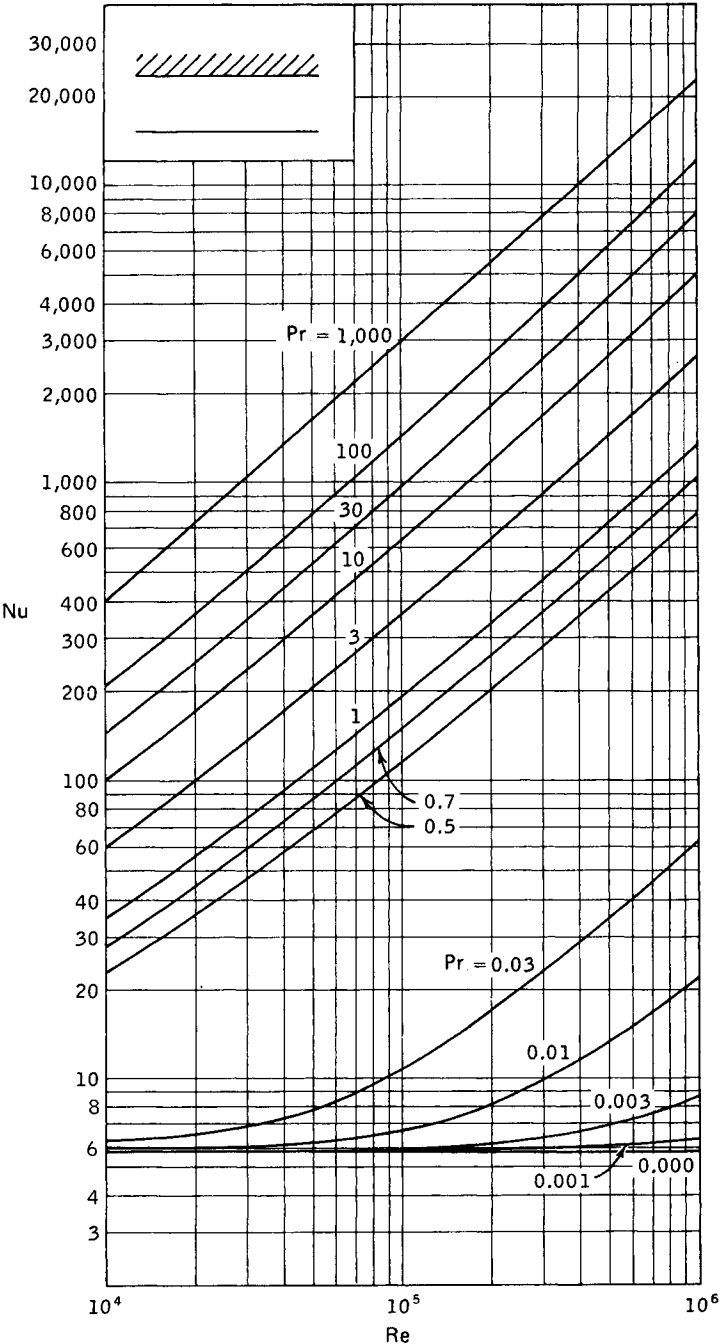


Fig. 6-10 Influence coefficients for computation of asymmetric heating for turbulent flow between parallel planes, constant heat rate per unit of tube length, and fully developed velocity and temperature profiles.

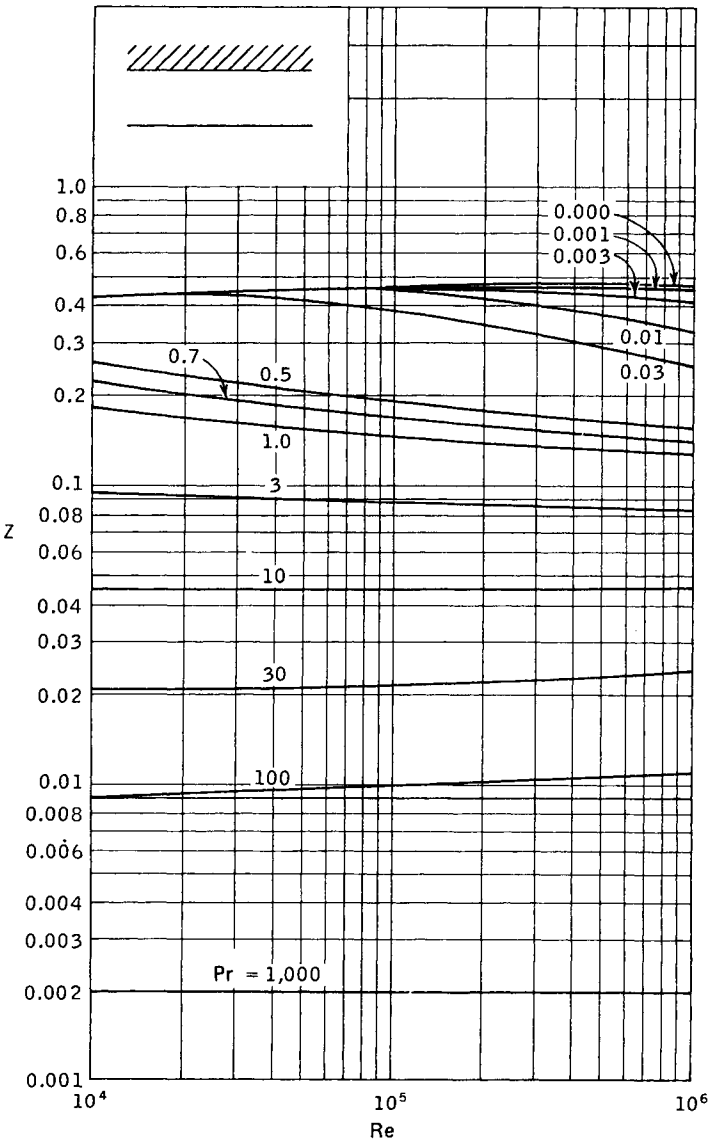


Fig. 6-11 Nusselt numbers for turbulent flow in a concentric-circular-tube annulus of radius ratio 0.20, outer surface insulated, constant heat rate per unit of tube length, and fully developed velocity and temperature profiles.

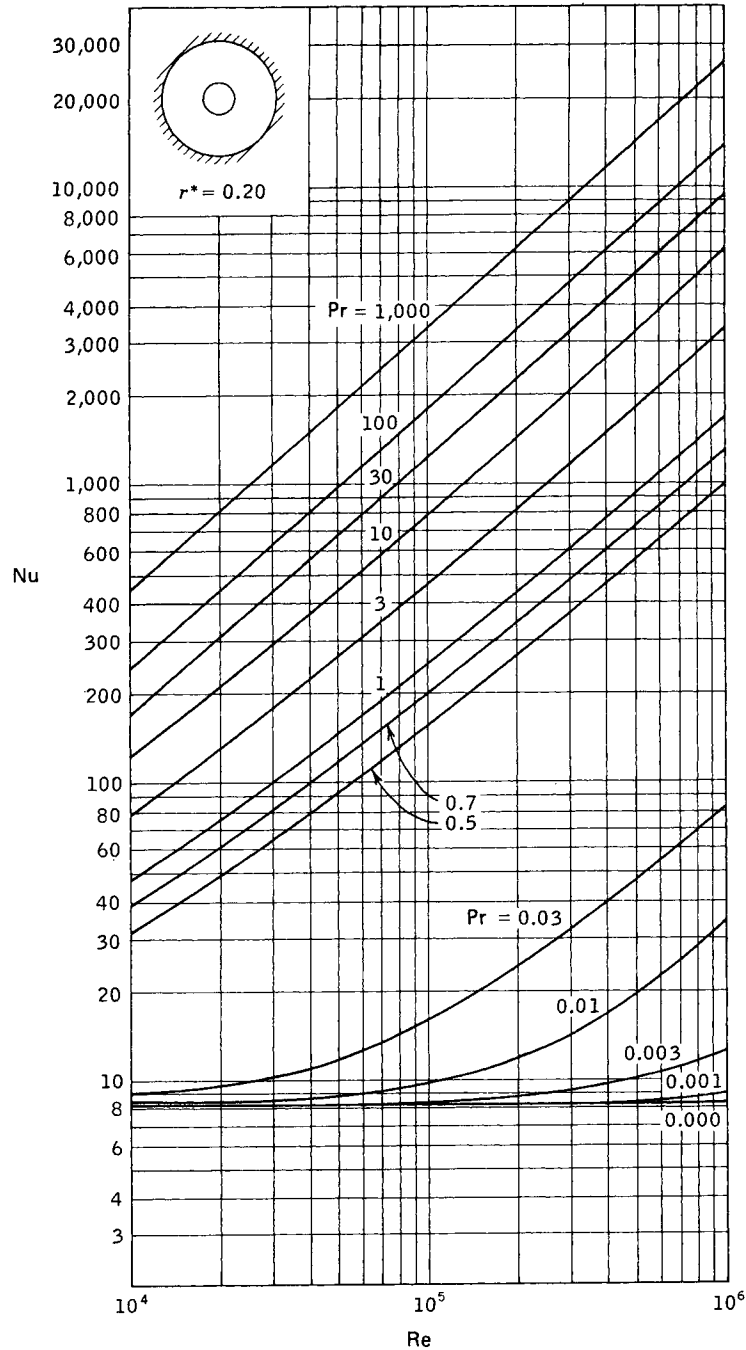


Fig. 6-12 Nusselt numbers for turbulent flow in a concentric-circular-tube annulus of radius ratio 0.20, inner surface insulated, constant heat rate per unit of tube length, and fully developed velocity and temperature profiles.

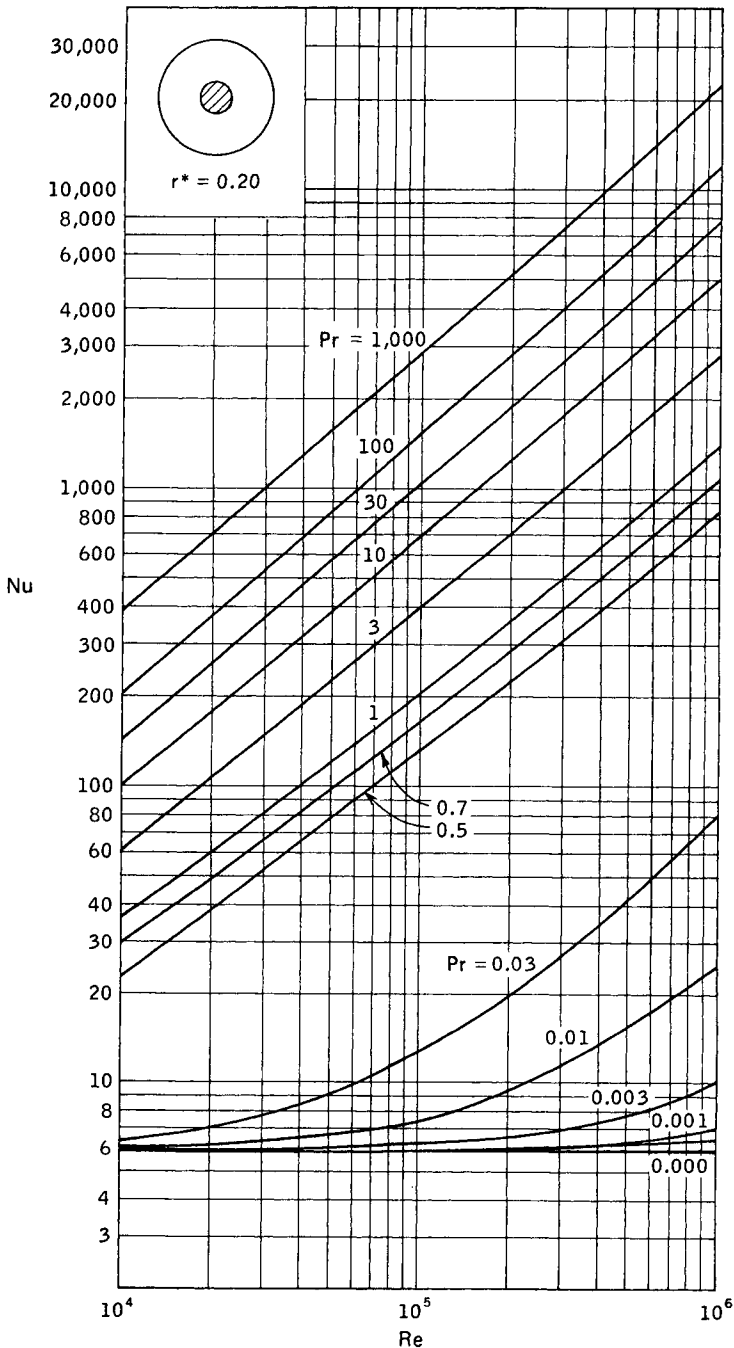


Fig. 6-13 Influence coefficients for computation of asymmetric heating for turbulent flow in a concentric-circular-tube annulus of radius ratio 0.20, constant heat rate per unit of tube length, and fully developed velocity and temperature profiles.

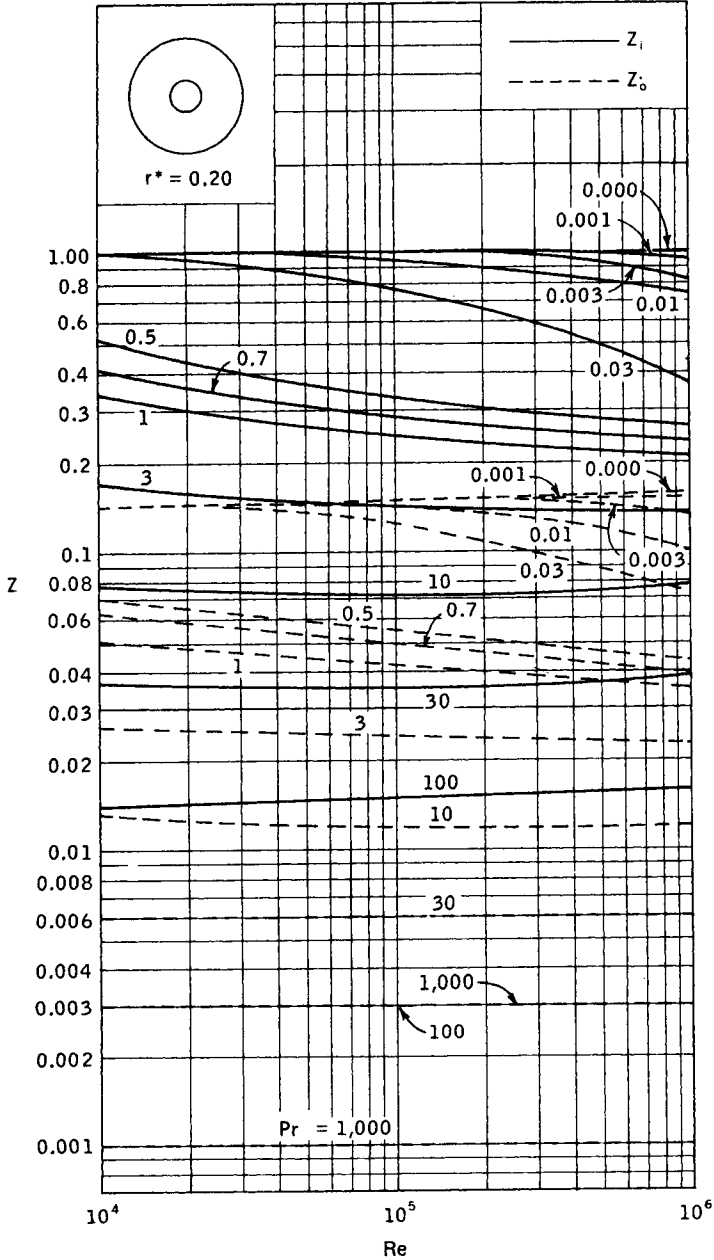


Fig. 6-14 Effect of radius ratio on Nusselt numbers and influence coefficients for turbulent flow in concentric-circular-tube annuli with constant heat rate and fully developed velocity and temperature profiles; $Re = 100,000$, $Pr = 0.7$.

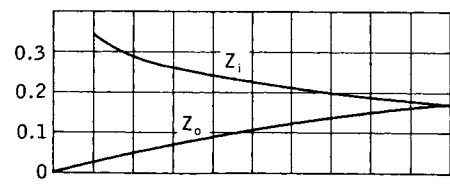
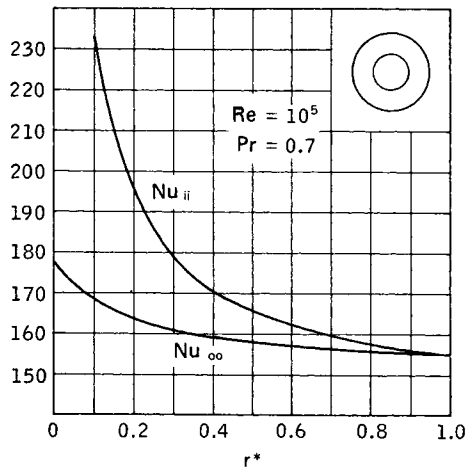


Fig. 6-15 Effect of radius ratio on Nusselt numbers and influence coefficients for turbulent flow in concentric-circular-tube annuli with constant heat rate and fully developed velocity and temperature profiles; $Re = 100,000$, $Pr = 0.01$.

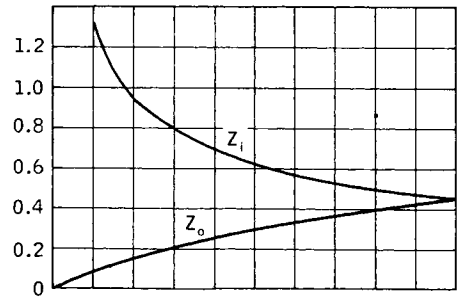
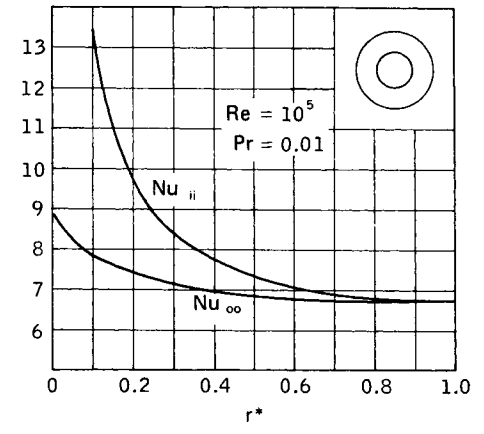


Fig. 6-16 Thermal-entry-length Nusselt numbers for laminar flow in a circular tube with constant surface temperature and fully developed velocity profile.

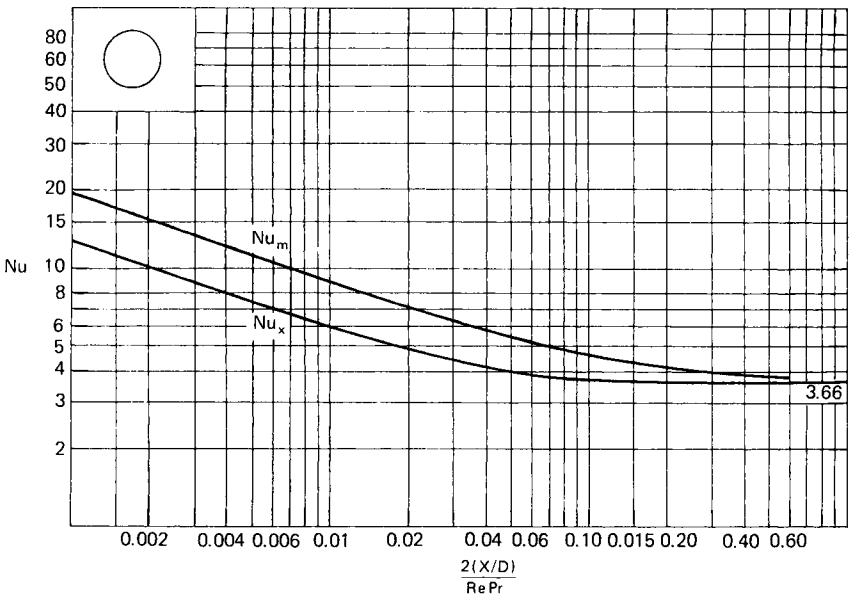


Fig. 6-17 Thermal-entry-length Nusselt numbers and influence coefficients for laminar flow in a circular tube and flow between parallel planes with one side insulated, constant heat rate, and fully developed velocity profiles.

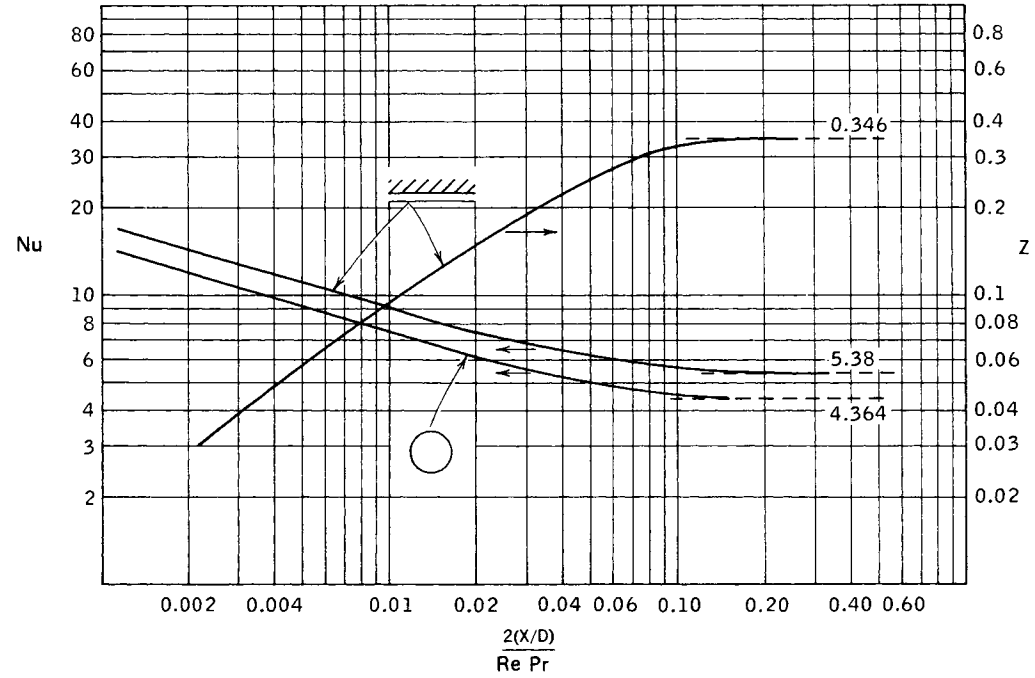
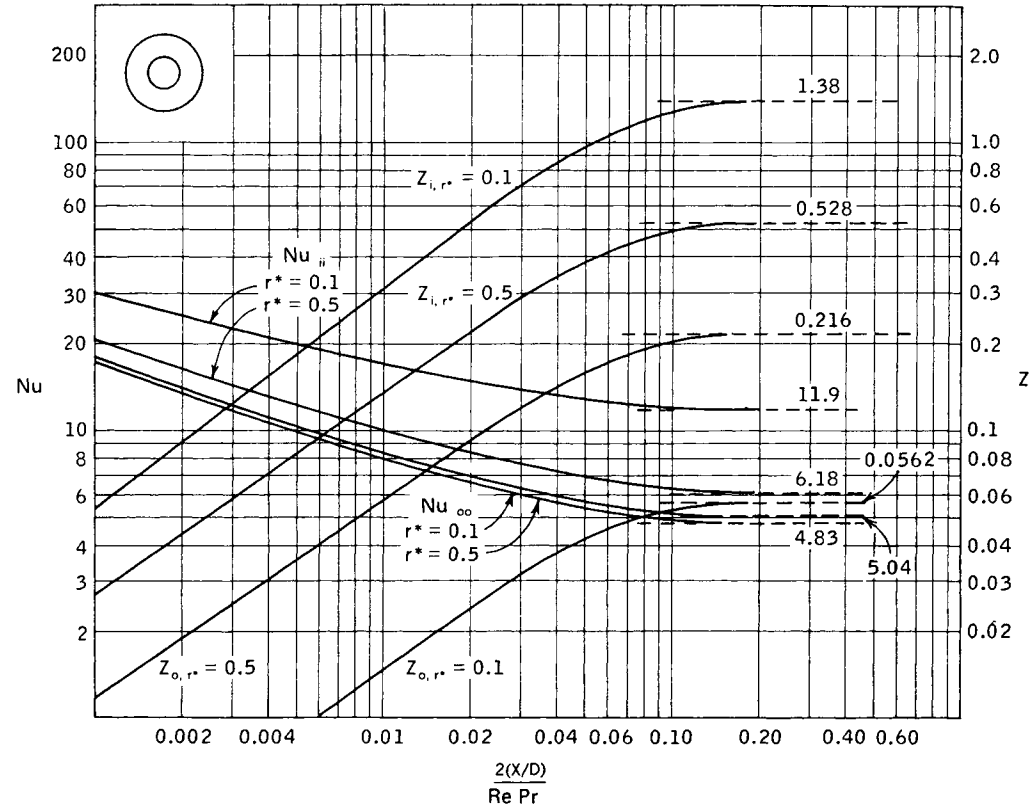


Fig. 6-18 Thermal-entry-length local Nusselt numbers and influence coefficients for laminar flow in concentric-circular-tube annuli with constant heat rate and fully developed velocity profiles.



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Fig. 6-19 Thermal-entry-length local Nusselt numbers for laminar flow in rectangular ducts with constant surface temperature and fully developed velocity profiles.

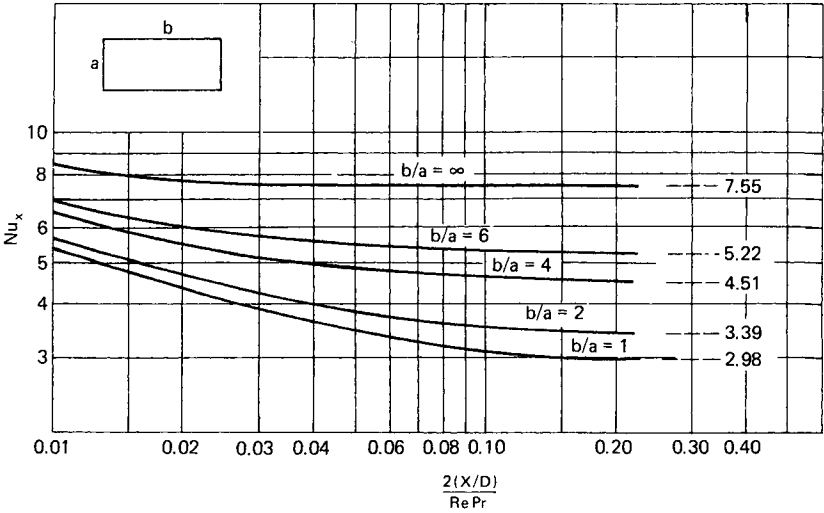


Fig. 6-20 Thermal-entry-length local Nusselt numbers for turbulent flow in a circular tube with constant heat rate and fully developed velocity profiles, showing influence of Reynolds number at $Pr = 0.7$.

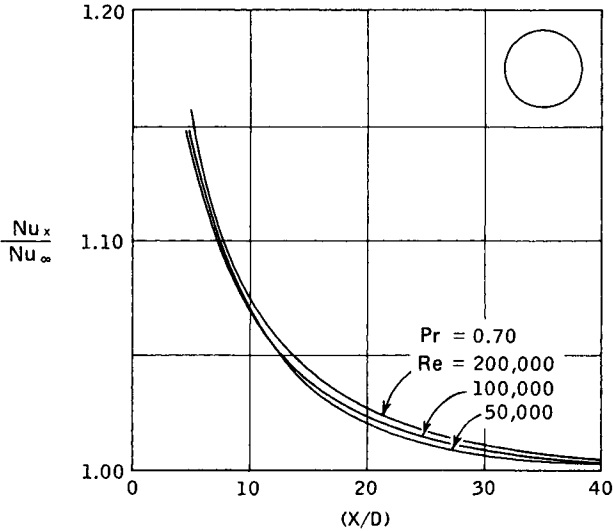


Fig. 6-21 Thermal-entry-length local Nusselt numbers for turbulent flow in a circular tube with constant heat rate and fully developed velocity profiles, showing influence of Reynolds number at $Pr = 0.01$.

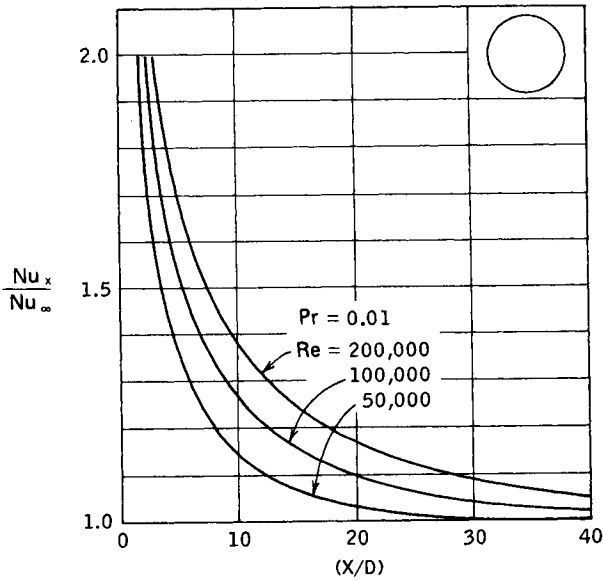


Fig. 6-22 Thermal-entry-length local Nusselt numbers for turbulent flow in a circular tube with constant heat rate and fully developed velocity profiles, showing influence of Prandtl number at $Re = 100,000$.

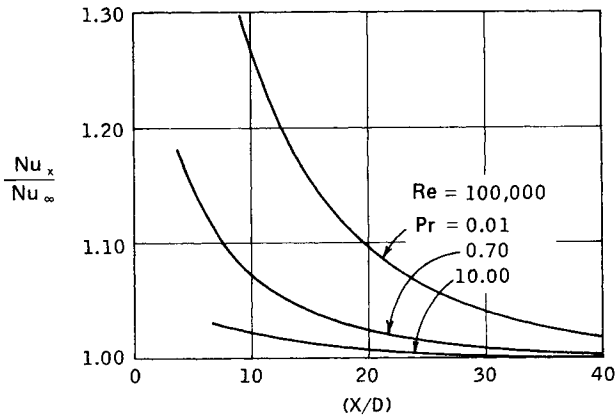


Fig. 6-23 Local, mean, and total apparent friction factors for laminar flow in the hydrodynamic entry length of a circular tube. The momentum flux correction factor for the hydrodynamic entry length is also plotted.

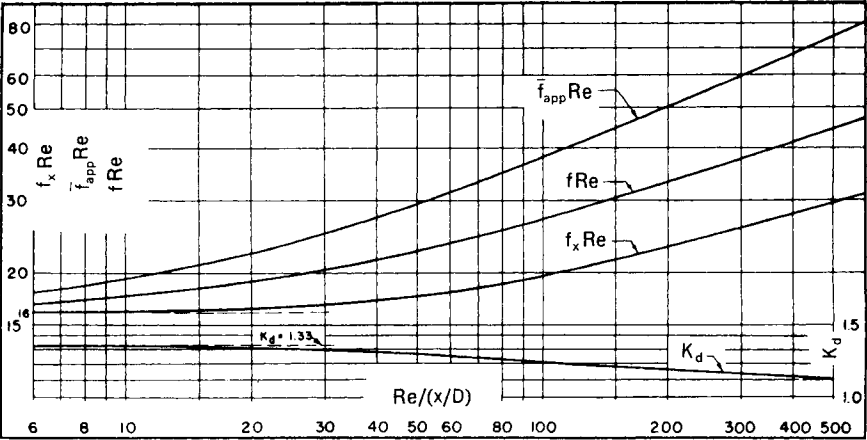


Fig. 6-24 Local Nusselt numbers for the combined hydrodynamic and thermal entry length of a circular tube, laminar flow.

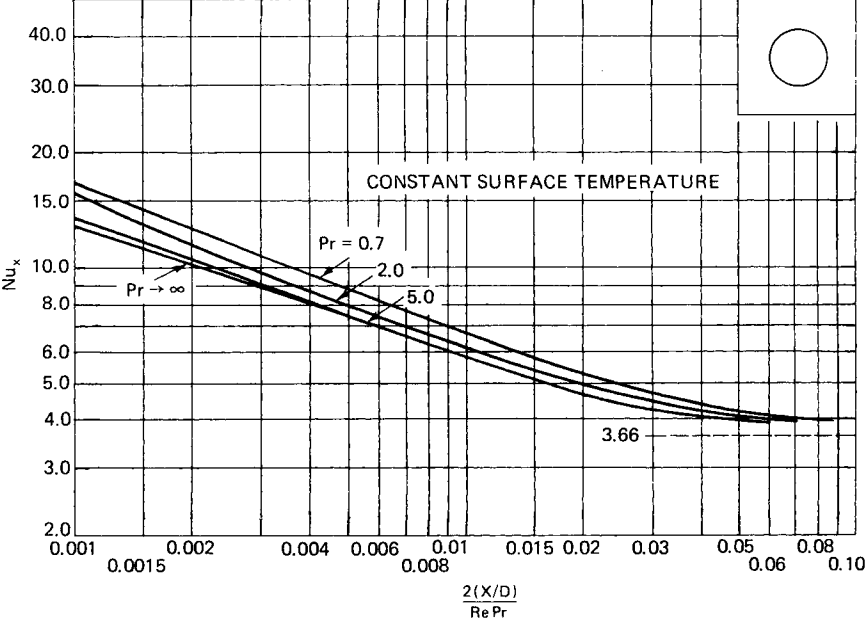


Fig. 6-25 Mean Nusselt numbers for the combined hydrodynamic and thermal entry length of a circular tube, laminar flow.

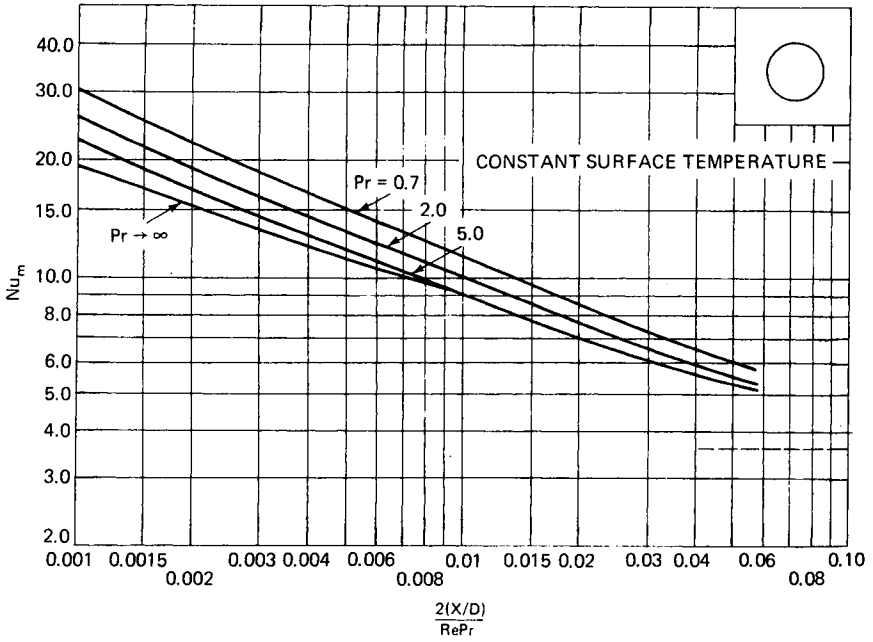
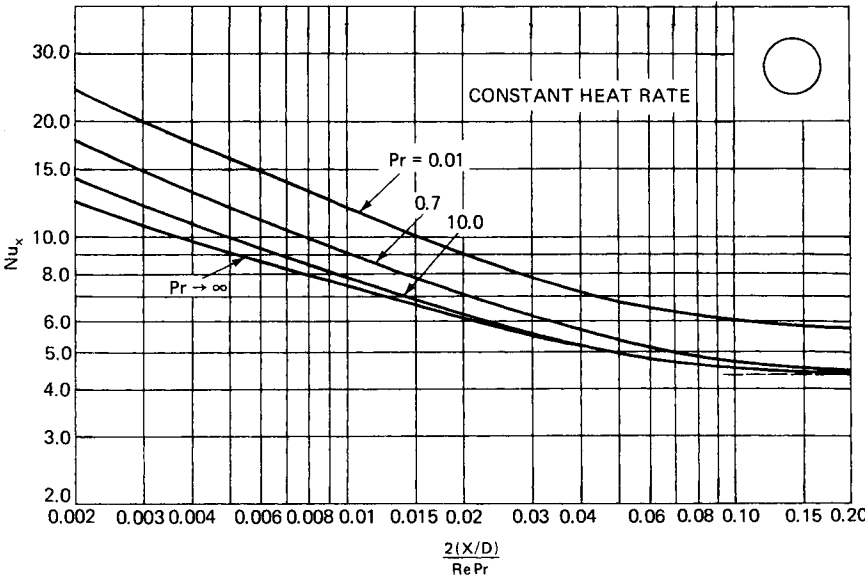


Fig. 6-26 Local Nusselt numbers for the combined hydrodynamic and thermal entry length of a circular tube, laminar flow.



7

Experimental Correlations for Simple Geometries

In the course of the experimental work leading to the compilation of test data in Chap. 10, sufficient data have been obtained for certain of the simple flow geometries to permit a generalized presentation of basic heat transfer and flow-friction performance. In addition, the simple geometries, such as flow through circular and rectangular tubes, have been at least partially handled by analytic means. Thus, by a judicious combination of analytical and experimental data, it is possible to present a comprehensive treatment of gas flow through circular and rectangular tubes with abrupt contraction entrances, including effects of tube length, type of heating, and temperature-dependent fluid properties. Similarly, as a result of the large amount of experimental data obtained for flow normal to staggered banks of circular tubes, a generalized treatment of this particular geometry is possible, which allows interpolation to staggered patterns that have not been tested. Extensive data have been obtained by various investigators for three common types of matrix surfaces that can be used in periodic-flow heat exchangers. Correlations of experimental data on randomly stacked screen matrices, sphere matrices, and triangular-tube matrices are presented in this chapter, rather than in Chap. 10, because each graph represents a range of geometry, and the curves are based on an interpretation of the work of various experimenters. The experimental data presented in Chap. 10 are the direct test results of a particular geometry and a single set of tests, and with a few exceptions the actual test points are presented.

It is recommended that the generalized curves in this chapter, rather than the direct test results, be used for heat exchanger design for gas flow in circular and rectangular tubes, except for those cases where the surface to be used is identical with the test surface; this includes length-to-diameter ratio and the type of entrance. Even in this case, it should be noted that if the direct test data are used for flow in long tubes, a separate correction for temperature-dependent properties should be made. Also, the test data were practically all obtained with constant wall temperature, and if applied to heat exchangers with other types of wall-temperature variation, slightly different performance will be obtained (see

Fig. 10-22). These considerations are not important for the interrupted type of surface but can be significant for flow in long uninterrupted tubes.

For flow normal to banks of staggered circular tubes, the test data in Chap. 10 are recommended for design for the particular layout patterns tested, and the generalized data in the present chapter are recommended for interpolation to patterns that have not been tested. However, even if the direct data of Chap. 10 are employed, they should be corrected for number of tube rows by Fig. 7-7.

The only direct test data on matrix surfaces presented in Chap. 10 are for flow through crossed-rod matrices. There is no counterpart in Chap. 10 for the matrix-surface summary curves presented in this chapter.

Figure 7-1 presents a summary of both analysis and experiment for gas flow through circular tubes in the moderate Reynolds number range. The laminar-flow curves are based on the data in Chap. 6 for combined hydrodynamic and thermal entry length, and have been found experimentally to be applicable even when an abrupt-contraction entrance is used. The turbulent-flow data are the result of a correlation of a large amount of experimental data, including rather importantly the entry-length data of Boelter et al. [1] for a tube with an abrupt-contraction entrance. The Stanton numbers are all averages with respect to tube length, and this can be used directly in the heat exchanger theory of Chap. 2. The transition data, in the Reynolds number range 2,500 to 10,000, are subject to considerable uncertainty because of the stability nature of the transition problem. For a single tube these data may be meaningless, but the curves presented represent what is typically obtained for a large number of tubes in parallel, all with abrupt-contraction entrances, as is typical of most circular-tube heat exchangers. Some typical direct test data are shown in Fig. 10-1, but the transition region of Fig. 7-1 is based not only on Fig. 10-1 but also on the results of several other investigators [2]. As a general rule, one would prefer to design outside this region, but in compact heat exchanger applications, the Reynolds number range of interest is usually about 500 to 15,000; thus the transition region can hardly be avoided. Even if the design point falls above 10,000, the heat exchanger will operate in the transition region at part load. These curves should not be used outside the gas Prandtl number range.

Figures 7-2 to 7-4 are based on much the same kinds of considerations as Fig. 7-1, although the analytic basis for the laminar combined hydrodynamic and thermal-entry-length data is less complete. The rectangular tubes are assumed to be formed by fins extending between plates as in a plate-fin heat exchanger, and thus the abrupt contraction is essentially a two-dimensional one rather than three-dimensionally axially symmetric as for the circular tubes. This results in a less pronounced entry-length effect in the turbulent flow region. An interesting feature of high-aspect-ratio rectangular tubes is that the transition "dip" is less pronounced than for other shapes; there is thus less uncertainty in designing in this region. Another more bothersome feature is that if the rectangular tubes are formed by thin fins placed between plates, slight irregularities in the fin spacing can cause a substantial drop in the apparent heat transfer performance of the overall heat exchanger core. This is due to the unequal flow distribution between fins that results from the higher flow resistance in the smaller passages.

Buckling of fins has the same effect. Direct test data for a great many plain plate-fin surfaces in which the flow passages are close to rectangular will be found in Chap. 10, and comparison of these data with the curves of Figs. 7-2 to 7-4 will give an idea of the kind of performance that can be expected from commercially available plain-fin surfaces. These curves should not be used outside of the gas Prandtl number range.

Correlations of heat transfer and friction data for flow normal to banks of circular tubes are presented in Figs. 7-5 to 7-7. These correlations are based entirely on the direct experimental data in Chap. 10 and should be used for interpolation, since the direct data are probably more accurate for the particular tube patterns tested. The tube-bank data can probably be employed with confidence at moderately high Prandtl numbers, although the test data on which the curves are based were obtained with air. Figure 7-7 shows the effect of a finite number of tube rows in a tube bank. The correlations in Figs. 7-5 and 7-6 are based on a bank of infinite extent, and this is also true of most of the direct test data in Chap. 10. Thus Fig. 7-7 should be used in conjunction with both the correlations and the direct data of Chap. 10. Figure 7-7 should also be used in conjunction with the crossed-rod matrices in Chap. 10, Figs. 10-90 to 10-92.

The remaining graphs in this chapter are for surfaces of potential use as matrices in periodic-flow heat exchangers. Figures 7-8 and 7-9 give correlations of experimental data for flow through randomly stacked woven-wire screens with various porosities. These correlations have not been included with the direct test data in Chap. 10 because each curve has been established from tests of wire screens at the low Reynolds number end but crossed-rod matrices at the high Reynolds number end [3]. The correlation is believed to be valid, but the curves do not represent the direct performance of any one screen matrix. The applicability of these data outside the gas Prandtl number range is uncertain.

Figure 7-10 shows a correlation of heat transfer and friction performance for flow through a matrix formed of randomly stacked spheres, based on the work of Coppage and London [4] and Denton et al. [5]. Again the applicability of these results to Prandtl numbers outside the gas range is uncertain.

Matrix surfaces can be constructed in much the same manner as plate-fin surfaces, and Fig. 7-11 shows a correlation of the test results by Mondt [6] for matrices with triangular flow passages. The bulk of the data are in the laminar-flow region, and the dashed lines in the laminar region are from Fig. 6-1 for an equilateral triangle with fully established flow and constant surface temperature. Single curves represent equally well the performance of 50 to 90° triangles, and the curves also demonstrate the applicability of the analytic solutions to a gas in the low Reynolds number range (<400).

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Fig. 7-1 Gas flow inside circular tubes with abrupt-contraction entrances; a summary of experimental and analytical data.

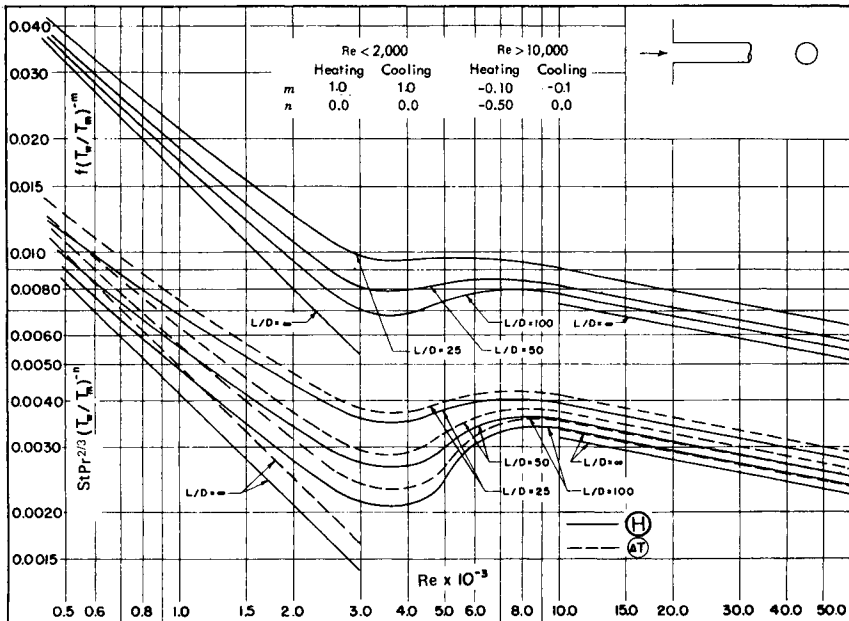


Fig. 7-2 Gas flow inside rectangular tubes with abrupt-contraction entrances; a summary of experimental and analytical data.

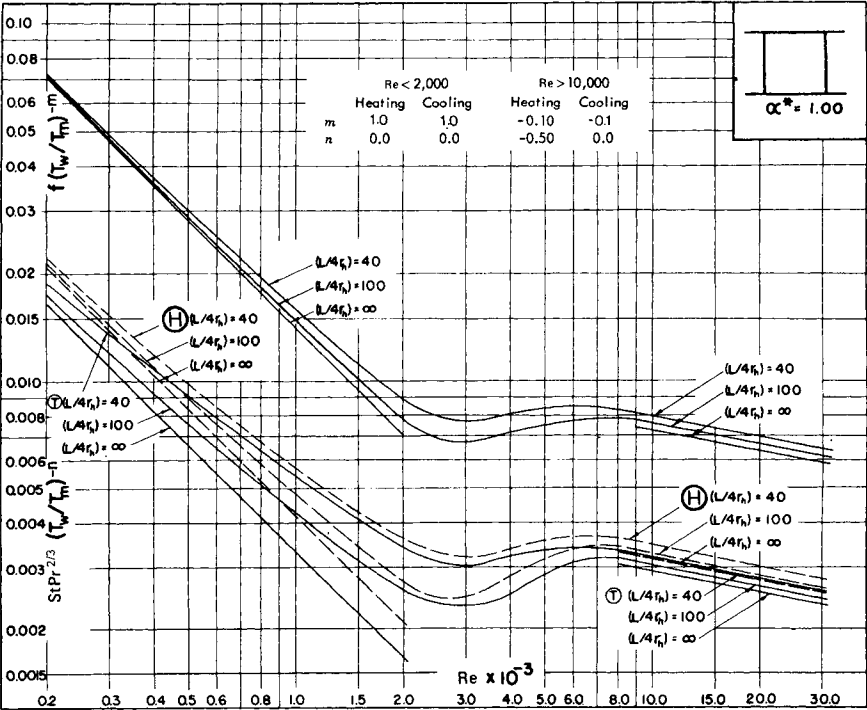


Fig. 7-3 Gas flow inside rectangular tubes with abrupt-contraction entrances; a summary of experimental and analytical data.

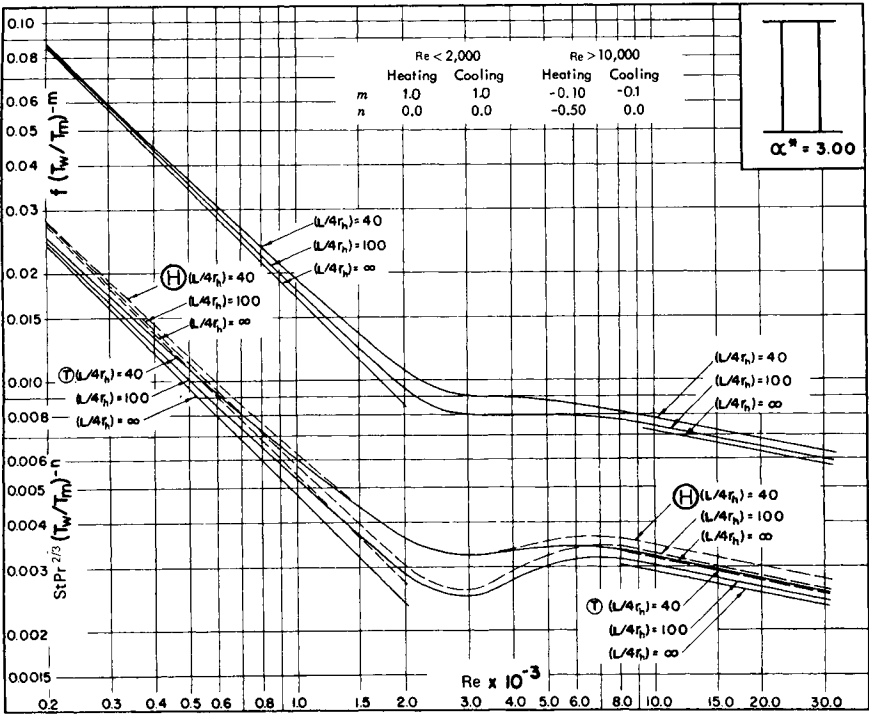


Fig. 7-4 Gas flow inside rectangular tubes with abrupt-contraction entrances; a summary of experimental and analytical data.

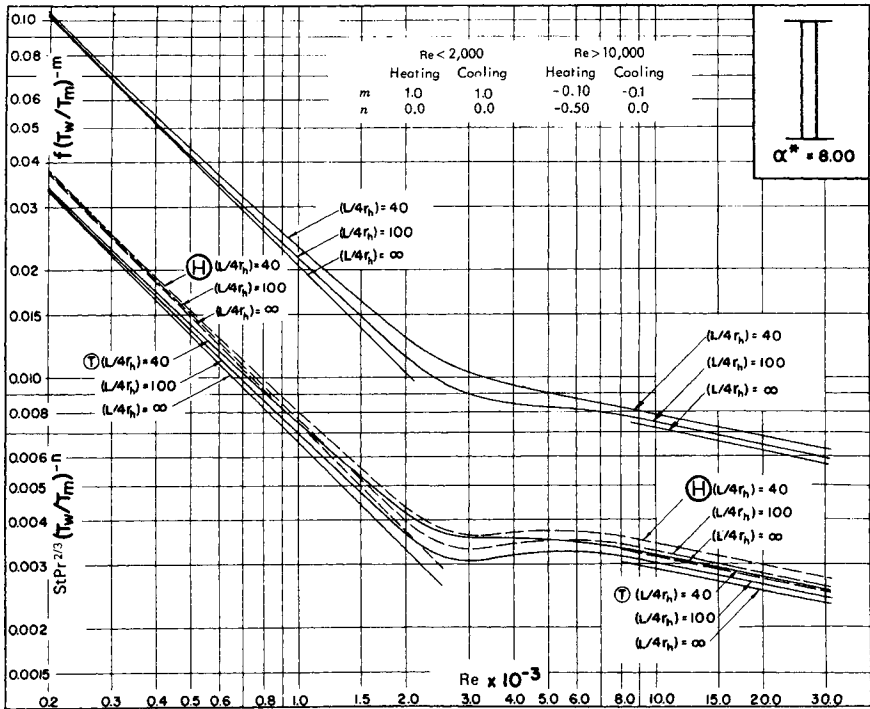


Fig. 7-5 Gas flow normal to an infinite bank of staggered circular tubes, heat transfer characteristics; a correlation of experimental data, average heat transfer coefficient around tube periphery. $StPr^{2/3} = C_h Re^{-0.4}$ $300 < Re < 15,000$.

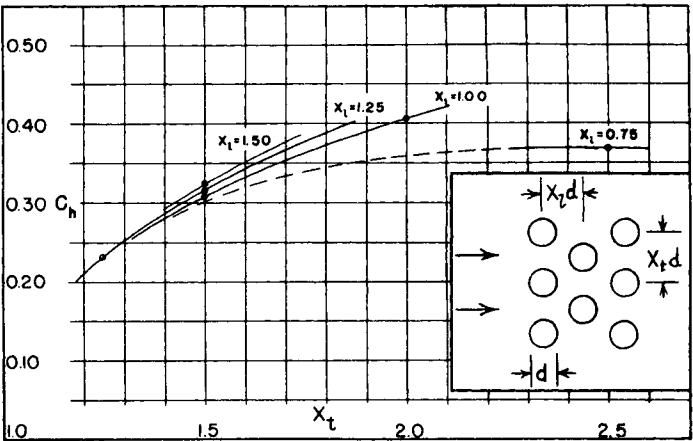


Fig. 7-6 Flow normal to an infinite bank of staggered circular tubes, flow friction characteristics; a correlation of experimental data. $f = C_f Re^{-0.18}$ $300 < Re < 15,000$.

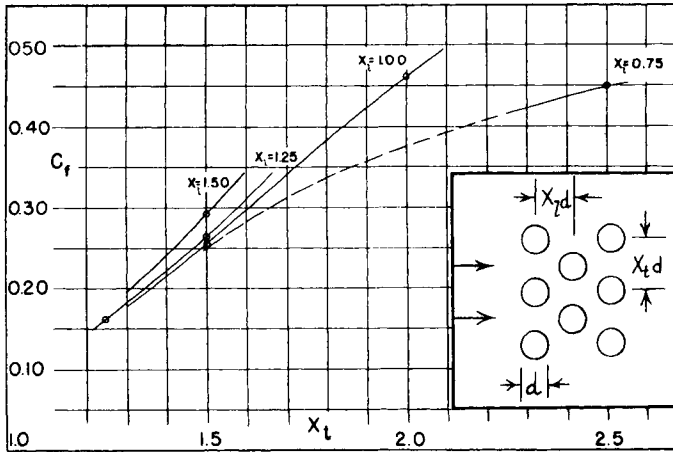


Fig. 7-7 Overall influence of row-to-row variations in the heat transfer coefficient in tube banks. Data are based on staggered tube banks but apply equally well to in-line tube banks and are a good approximation for crossed-rod and woven-screen matrices.

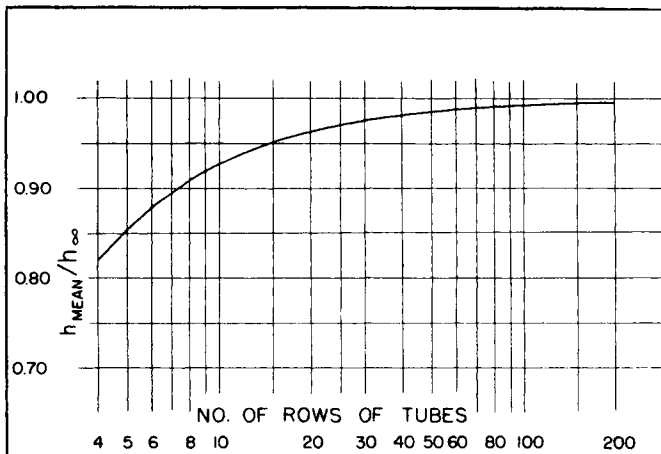


Fig. 7-8 Gas flow through an infinite randomly stacked woven-screen matrix, heat transfer characteristics; a correlation of experimental data from wire screens and crossed rods simulating wire screens. Perfect stacking, i.e., screens touching, is assumed. $G = W/pA_{fr}$, $r_h = p/\alpha$.

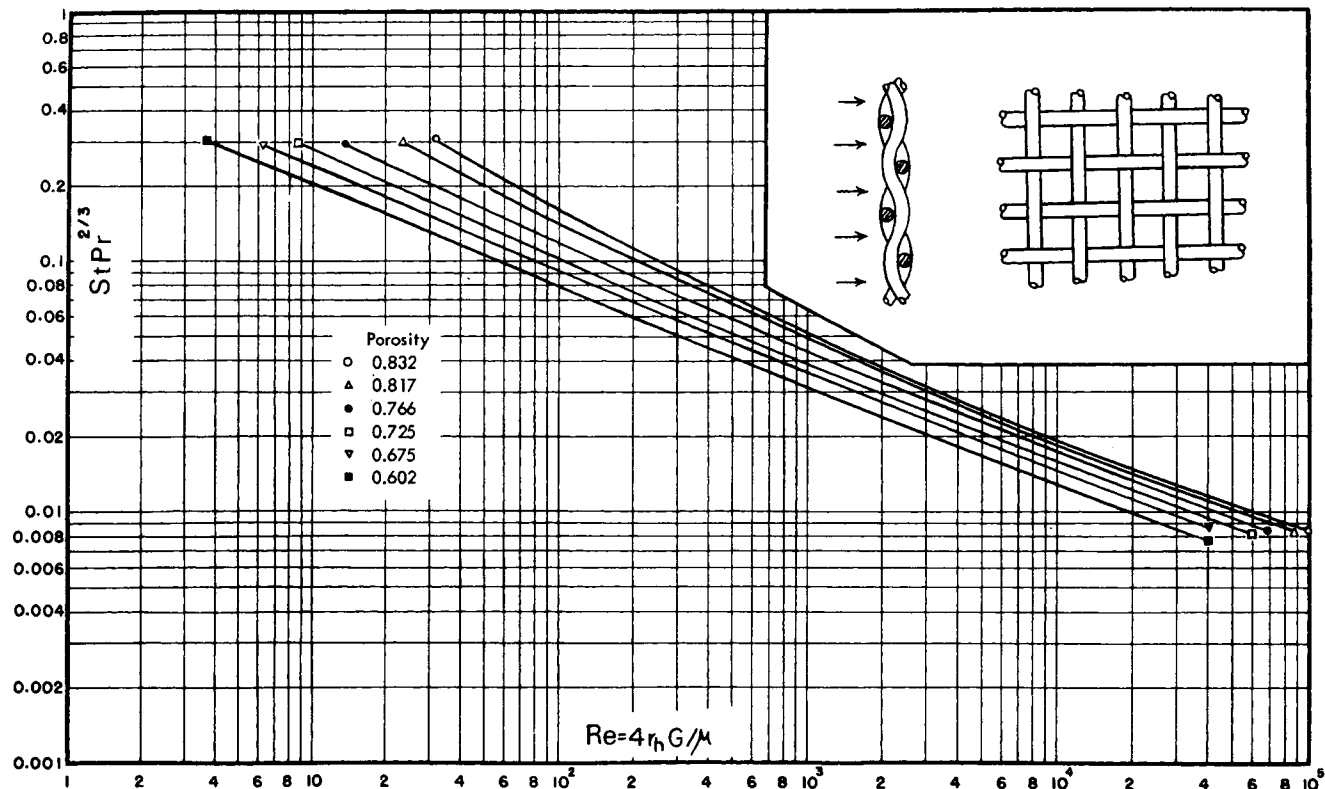


Fig. 7-9 Flow through an infinite randomly stacked woven-screen matrix, flow-friction characteristics; a correlation of experimental data from wire screens and crossed rods simulating wire screens. Perfect stacking, i.e., screens touching, is assumed. $G = W/pA_{fr}$, $r_h = p/\alpha$.

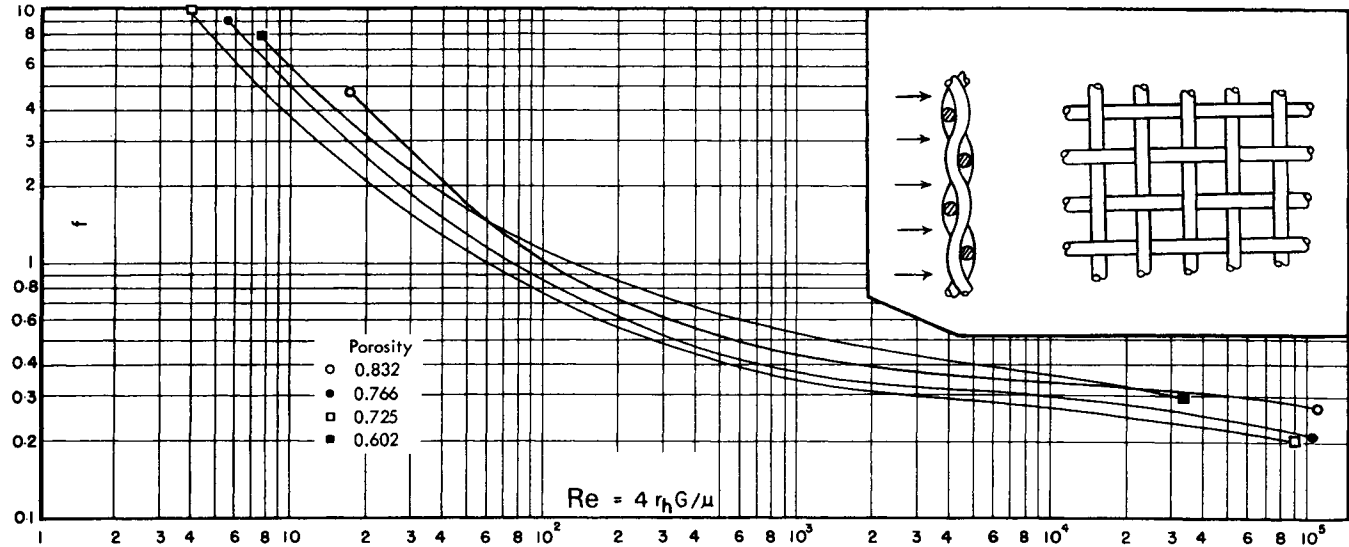


Fig. 7-10 Gas flow through an infinite randomly stacked sphere matrix. A correlation of experimental data with porosity varying from 0.37 to 0.39. See tabulation of data in Chap. 10.
 $G = W/pA_{fr}$, $r_h = p/\alpha$.

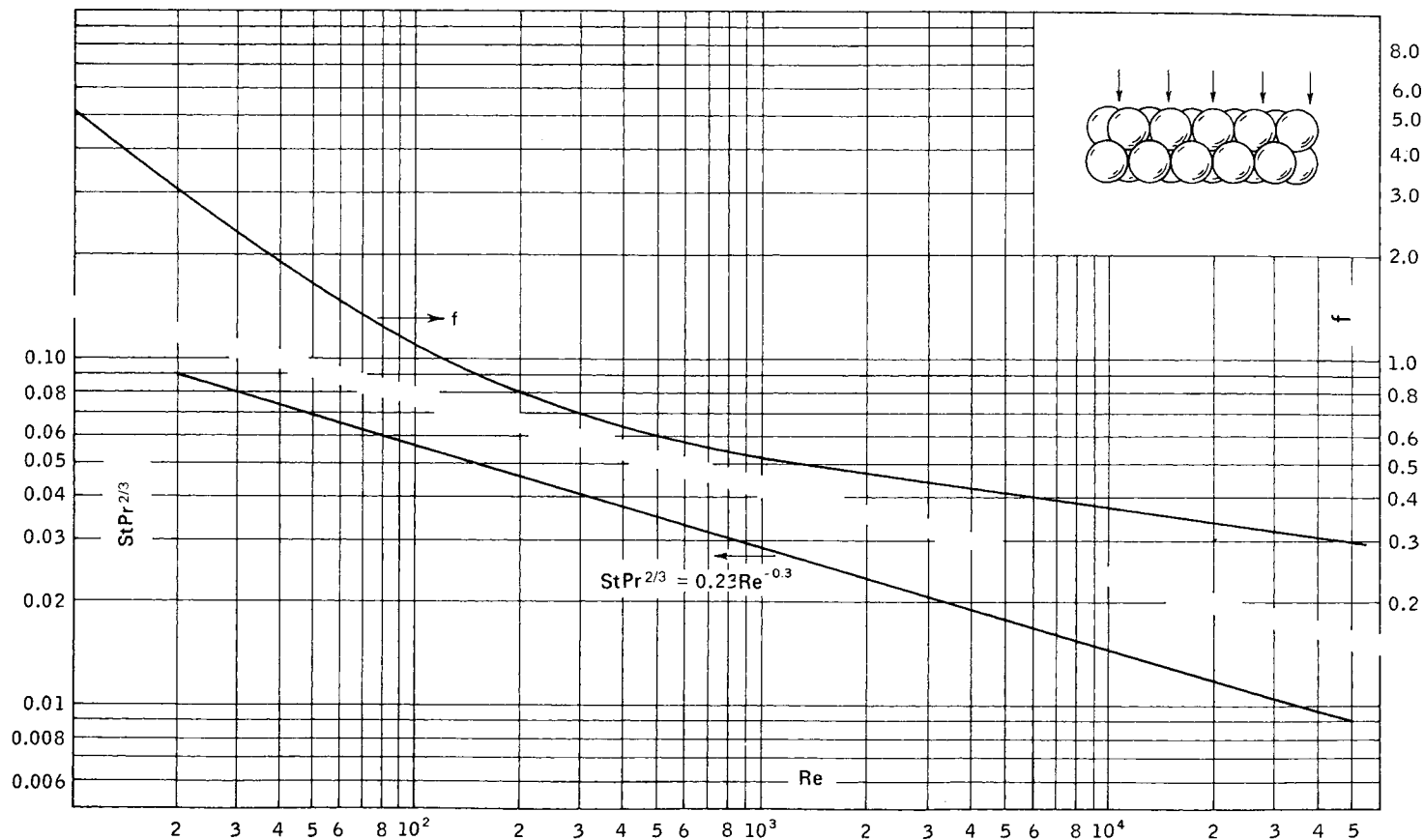
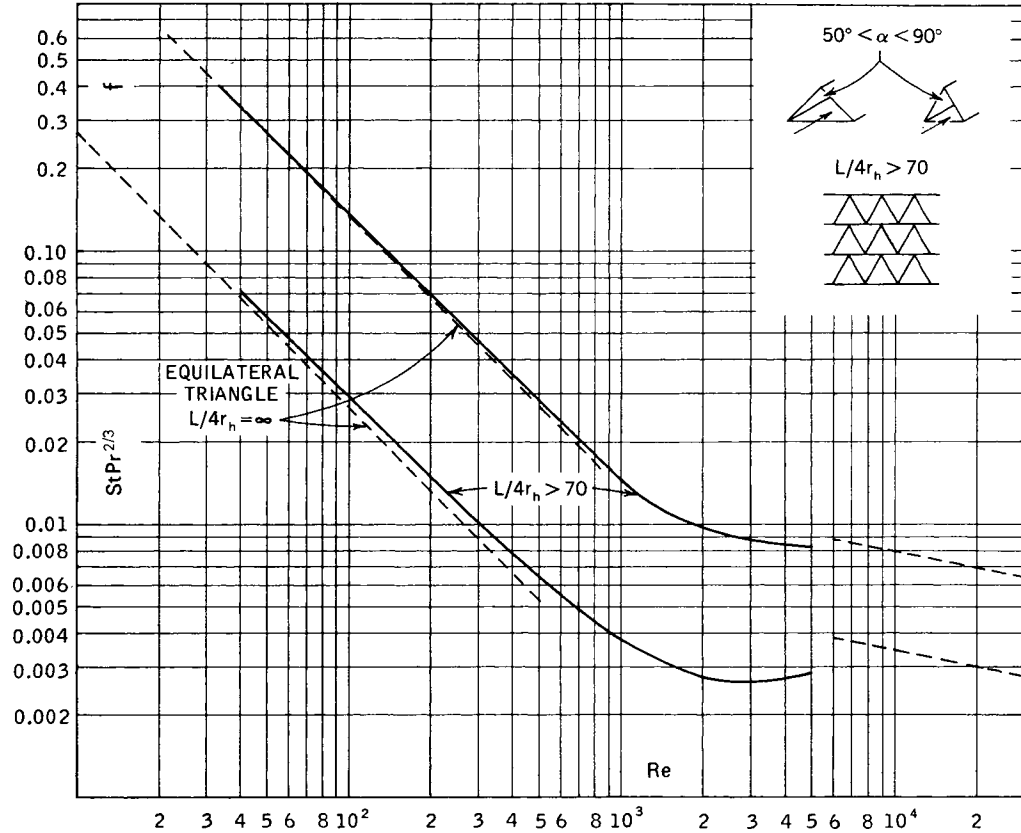


Fig. 7-11 Gas flow through a matrix composed of triangular flow passages. A correlation of experimental data for isosceles triangle matrices, apex angle varying from 50 to 90°. Dashed lines in the laminar-flow region are the analytic solutions for constant surface temperature for an equilateral triangle, $L/r_h = \infty$; in the turbulent-flow region the dashed lines are the circular-tube characteristics.



8

Experimental Methods

The bulk of the experimental data reported in Chap. 10 was obtained at Stanford University and at the U.S. Navy Engineering Experiment Station, Annapolis, Maryland. At both laboratories a steam-to-air test system was employed for all tests, with the exception of the circular-tube banks, one plate-fin test core, and the matrix surfaces. For the circular-tube banks and the crossed-rod matrices tested at Stanford, a transient test technique was employed, although the same air duct was used as for the steam-to-air tests. One plain plate-fin core tested at Stanford was hot-water-heated rather than steam-heated, but again the same air duct was used. The other matrix surfaces were tested using a transient technique different from that employed for the tube-bank tests.

Detailed descriptions of the experimental apparatus and method of test data analysis employed at the U.S. Navy Engineering Experiment Station and at Stanford are contained in Refs. 1 to 6. Essentially the same test technique, and similar equipment, were used at the Engineering Experiment Station and at Stanford for the steam-to-air tests; therefore, only the Stanford system will be described. The steam-to-air apparatus and methods will be described first, followed by the transient systems.

STEAM-TO-AIR TESTS

The principal element of the test apparatus was a controllable-flow air duct with a rectangular test section, $8\frac{3}{4}$ by $9\frac{3}{4}$ in., into which cores containing the heat transfer surfaces under investigation were inserted. The test cores were cross-flow heat exchangers allowing for air heating on one side at the expense of condensing steam on the other side. The air-flow rate was metered by standard flat-plate orifices installed in a 12-in.-diameter circular portion of the test duct downstream from the test section. The orifices were constructed and installed in accordance with ASME specifications, Ref. 7. Orifice coefficients from this reference were used.

Air temperature measurements upstream and downstream of the test section and at the orifice were made by calibrated, radiation-shielded, single-junction,

iron-constantan thermocouples. For the duct section downstream from the test core, 27-point traverses were made in order to obtain an accurate bulk mean temperature. For the other positions, 9-point traverses were taken. Steam temperatures were measured by iron-constantan thermocouples welded into brass tubes.

Air pressure differentials and gauge pressures at both the test section and the orifice were measured by a system of piezometer rings with inclined draft gauges and vertical single-leg water manometers. Steam-gauge pressures were measured with mercury manometers.

A steam system supplied slightly superheated steam at an accurately controlled pressure to the test core. Roughly four times as much steam was blown through the core as was condensed, in order to prevent the build-up of a substantial condensate-film thermal resistance. Both condensate and excess steam were separately metered so that an energy-balance check could be made on the accuracy of the instrumentation by comparing the energy loss of the steam to the energy gain of the air. The average deviation from balance was consistently less than 3 percent.

Measurements of air temperature upstream and downstream of the test core, air-flow rate, steam pressures and temperatures, and air pressure drop through the test core, together with a consideration of the steam-side condensate-film heat transfer resistance (less than 10 percent of the total), the pressure losses at the entrance and exit to the test core, and the temperature ineffectiveness of the extended surface, provided sufficient information for the determination of the Stanton number (h/Gc_p) and friction factor f . Together with the Reynolds number $(4r_t G/\mu)$, also established from the above measurements, these comprise the nondimensional heat transfer and flow-friction characteristics of the surface under investigation.

The thermocouple systems were designed to measure all temperatures to an accuracy of $\pm 0.2^\circ\text{F}$. Straightening vanes and trim tabs were provided and adjusted to ensure a very nearly uniform air velocity in the duct at the test section. Considerable effort was expended to provide accurately controlled conditions for steady-state operation. Various cross-checks were provided so that a continuous verification was made of the accuracy of the instrumentation.

An analysis of experimental uncertainty was made considering the following possible sources of error:

1. Uncertainties inherent in the design and performance of the test system
2. Instrument uncertainty
3. Uncertainties introduced by virtue of the idealizations and the accuracy of the transport properties, viscosity, and thermal conductivity
4. Uncertainties in the determination of the test-core dimensions

The probable uncertainty in the final results of the steam-to-air tests varies somewhat for different test cores and for different air-flow rates. A conservative estimate is given below from the analysis of a number of different tests, and this estimate is believed to be approximately applicable to all of the test results reported.

St	± 3.5 percent
Pr	± 5.0 percent
StPr ^{2/3}	± 5.0 percent
f	± 5.0 percent
Re	± 2.0 percent

The principal difference between the test ducts at the U.S. Navy Engineering Experiment Station and at Stanford University is that the former had a forced draft while the latter had an induced-draft blower. With induced draft, uniform flow conditions across the core could be established with relative ease.

TRANSIENT TESTS OF TUBE BANKS AND CROSSED-ROD MATRICES

The transient tests of tube banks and crossed-rod matrices were carried out in the same air duct as described for the steady-state steam-to-air tests. The tube banks under test were built into test cores, with a $8\frac{3}{8}$ by $9\frac{3}{4}$ in frontal area, using $\frac{3}{8}$ -in.-diameter aluminum tubes fitted into a pair of plastic tube sheets at the top and bottom of the test core. The same core frame and tubes were used for all patterns, the only differences being in the removable tube sheets. For the friction-factor tests the procedure was identical to that described for the steam-to-air cores. However, heat transfer performance was established by a novel transient-test scheme whereby one of the aluminum tubes was replaced by an externally identical copper thermal-capacitor rod. The copper rod contained a thermocouple, and the test procedure involved heating the rod to about 30°F above air-flow temperature, inserting it at the desired location in the core, and recording its temperature-time history on cooling. From these data the heat transfer conductance can be readily deduced. The validity of this method of testing was established by comparing the results for two circular-tube patterns with the results obtained for the same patterns with steam-heated steady-state test cores. It was found that the method was excellent for staggered tubes, but somewhat limited in usefulness for in-line banks. The virtue of the transient method lies in its simplicity and accuracy and the speed with which data for a large variety of tube patterns can be obtained. Experimental uncertainty has been analyzed and is found to be about the same for both f and StPr^{2/3} as found for the steady-state tests.

The procedure for the crossed-rod matrices was virtually identical to that for the tube banks, except that the nonactive tubes were plastic and the test duct was circular rather than rectangular.

TRANSIENT TESTS OF SCREEN AND SPHERE MATRICES

The matrix surfaces were tested in an entirely different test apparatus. Because of the character of the matrix surfaces, a transient test technique is virtually mandatory. For the small flow passages used, it was also not practicable to measure directly the matrix surface temperature. The system used involved heating the matrix uniformly with a stream of hot air, changing the entering air

temperature suddenly to a lower uniform temperature by a quick flow-switching method, and recording the temperature-time history of the air leaving the test core. The surface heat transfer conductance could then be deduced from the maximum slope of the temperature-time plot by an analysis that is fully described in Refs. 4 and 5. The theoretical solution used for this purpose is presented in Fig. 3-15.

The test duct and test cores for the stacked screen and sphere matrices were circular in section with diameters of 2.93 in. The stacked-screen test cores were constructed by stacking together 2.93-in-diameter sections of screen to the desired core depth. The sphere matrix was similar, with the small spheres packed in a 2.93-in-diameter can. The glass ceramic matrix test cores were $3\frac{1}{4}$ in square in cross section. Air-flow rates were measured by both ASME standard orifices and a Fischer-Porter flowrater. All temperatures were measured by iron-constantan thermocouples. Eight thermocouples in series were used to measure the critical core upstream and downstream temperatures. Pressures were measured by inclined draft gauges and vertical water manometers.

A detailed analysis of the experimental uncertainty of this system yielded the following results:

St	± 13 percent
StPr ^{2/3}	± 14 percent
f	± 3 percent
Re	± 2 percent

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9

Heat Transfer Surface Geometry

The purpose of this chapter is to bring together in one place the complete geometrical descriptions of the compact heat transfer surfaces for which the direct performance data are given in Chap. 10. Actually, all the data presented here are also given in the captions to the direct-performance-data graphs in Chap. 10 and on the graphs themselves. However, by assembling this information in one place, it is much more convenient to compare one surface with another when it is only the geometry that is being studied.

The data are assembled in Tables 9-1 to 9-7, and scale diagrams of the surfaces are given in Figs. 9-1 to 9-18. The various types of compact surfaces are grouped together in the following order:

1. Tubular surfaces (Figs. 9-1 to 9-2)
 - a.* Flow inside circular and flattened circular tubes
 - b.* Flow normal to banks of tubes
2. Plate-fin surfaces (Figs. 9-3 to 9-9)
 - a.* Plain fins
 - b.* Louvered fins
 - c.* Strip fins
 - d.* Wavy fins
 - e.* Pin fins
 - f.* Perforated fins
3. Flow normal to banks of finned tubes (Figs. 9-10 to 9-13)
 - a.* Circular fins and tubes
 - b.* Circular tubes, continuous fins
 - c.* Flat tubes, continuous fins
4. Matrix surfaces (Figs. 9-14 to 9-18)

Each of these categories will now be described briefly.

The tubular surfaces, illustrated in Figs. 9-1 and 9-2, are the simplest form of compact heat transfer surface. Test data have been obtained for flow both inside and normal to tube banks, using circular tubes and tubes that have been flat-

tened to an oval shape. A further variation of the flattened tube is the dimpled tube; dimpling is sometimes used as a method of boundary-layer interruption to increase the heat transfer conductance without increasing flow velocity.

A semidescriptive method of surface designation is employed. Thus ST indicates flow inside straight tubes, while FT refers to flattened tubes and FTD to flattened and dimpled tubes. For flow normal to banks of circular tubes, the tube arrangement is designated according to whether the arrangement is staggered or in-line, and then by the transverse and longitudinal pitch ratios. Thus surface S 1.50–1.00 is a staggered arrangement with a transverse pitch ratio of 1.50 diameters and a longitudinal (flow direction) pitch ratio of 1.00 diameters. The letter I before the numerals indicates an in-line arrangement. Part of the circular-tube-bank surfaces were tested by a steady-state steam-to-air technique, and the remainder by a transient technique. The type of test is indicated in Fig. 9-2 and also in the captions to the direct test data in Chap. 10.

The dimensional information for the tubular surfaces in Tables 9-1 and 9-2 includes the hydraulic diameter, free-flow to frontal-area ratio, and heat transfer area per unit of volume for each of the surfaces. These are needed in addition to the basic heat transfer and flow-friction performance data in order to complete the heat exchanger analysis. The given dimensions apply to the particular surfaces as tested. Since the basic heat transfer and flow-friction design data in Chap. 10 are presented in nondimensional form, they should apply equally well to surfaces of different size, i.e., tubes of different diameter, providing only that complete geometrical similarity with the test surface is maintained. This is also true of all the other heat transfer surfaces for which direct performance data are presented in Chap. 10. However, it should be kept in mind that complete geometrical similarity also applies to such secondary features as surface roughness, and therefore extrapolation to very great changes in surface size may introduce some additional uncertainty into the performance data. Note that the dimensional data are given in both English and SI units.

In addition to the tube banks shown in Fig. 9-2, data are also presented in Chap. 10 (Figs. 10-11, 10-15, and 10-16) for other staggered and in-line arrangements from the extensive data of Grimison [1].

In Figs. 9-3 to 9-9 are shown diagrams of a great many surfaces of the plate-fin type. The plate-fin surfaces are especially useful where both fluids in the heat exchanger are gases, since extended surface can be effectively employed on both fluid sides. With the plate-fin construction it is possible to achieve a very large area compactness or heat transfer area per unit of volume. Flexibility in heat exchanger design is possible because the two fluid sides are independent of one another and the most suitable type of extended surface can be chosen for each of the fluids. This is not possible with the circular-tube-bank surfaces where, for example, the choice of flow normal to a bank of tubes for one fluid automatically fixes the flow geometry for the other fluid, with nearly the same heat transfer area on both sides.

The surface configurations shown in Figs. 9-3 to 9-9 represent most of the common types in present use, any two of which could be built together into a heat exchanger core using the sandwich type of assembly shown. The plate-fin

surfaces have been further subdivided into plain-fin, louvered-fin, strip-fin, pin-fin, wavy-fin, and perforated-fin types.

The plain-fin surfaces are characterized by long uninterrupted flow passages, with performance similar to that obtained inside long circular tubes. There is, however, a decided tube-length effect on both the heat transfer and friction performance, and the length to hydraulic-diameter ratio $L/4r_h$ is indicated on each of the surface diagrams. The length L is not necessarily the flow length of the total heat exchanger but is rather the flow length of uninterrupted fin; the actual heat exchanger flow length may involve several flow lengths of fin material placed end to end. Unless absolutely perfect alignment is obtained, the surface will behave as if there were a new hydrodynamic and thermal entry length, and in commercial heat exchangers no attempt is usually made to obtain perfect alignment. Note that the plain-fin surfaces include rectangular passages, triangular passages, and passages with rounded and reentry corners, so that there is considerably more variation in geometry than in the summary correlation curves in Chap. 7 where only plain-fin surfaces with rectangular flow passages are considered. Note that some of the triangular-passage surfaces are of a double-passage construction, which is a scheme that provides the heat exchanger designer with a method for obtaining flexibility in relative flow area, hot and cold sides, while still using a common fin dimension.

The semidescriptive method of designating plain-fin surfaces refers to the number of fins per inch transverse to the flow direction. Thus surface 14.77 has 14.77 fins per inch. The additional letter T indicates the surfaces with definite triangular passages.

The louvered-fin surfaces are characterized by fins that have been cut and bent out into the flow stream at frequent intervals. The purpose of louvering is to break up the boundary layers so as to yield higher heat transfer coefficients than are possible with plain fins under the same flow conditions. As a general rule, the more frequent the interruption, the higher the conductance, although the friction factor is also increased. The resulting improvement in heat transfer versus friction-power behavior is discussed in Chap. 1.

The louvered-fin surfaces are designated by two figures. The first refers to the length of the louvered fin in the flow direction, the second to the fins per inch transverse to the flow. Thus surface $\frac{3}{8}$ -11.1 has $\frac{3}{8}$ -in louvers and 11.1 fins per inch.

The strip-fin surfaces are similar in principle to the louvered-fin surfaces, the only difference being that the short sections of fins are aligned entirely with the flow direction. With the strip-fin configuration it is feasible to have very short flow-length fins and thus very high heat transfer coefficients. The experimental uncertainty for the strip-fin surfaces is probably somewhat greater than for any of the other surfaces because the performance, and most especially the friction performance, is materially affected by the thickness and character of the fin leading edge. Fins of this type are generally constructed by a machine-cutting process that inevitably leaves a slightly bent and scarfed fin edge that differs depending upon the fin material and the character of the cutting tool. Since a few ten-thousandths of an inch of scarfing can have a considerable effect, it is difficult to either dimensionally describe this effect or to exactly duplicate one of

the test surfaces. In the captions to the direct-test-data curves in Chap. 10 for the strip-fin surfaces, the fin material used for the test surfaces is indicated, because a soft material, such as aluminum, may be expected to yield fin edges somewhat different than would be obtained from a hard material, such as stainless steel. Most of the fins of the test surfaces were of aluminum construction, and although this by no means precludes the use of the performance data for steel fins, it does suggest some conservatism in design. Briggs and London [4 or 5] touch on this point.

The designation scheme for the strip-fin surfaces is essentially the same as that used for the louvered surfaces.

The wavy-fin surfaces are also high-performance surfaces with performance quite similar to the louvered and strip-fin surfaces. The change in flow direction induced by the fins caused boundary-layer separation with effects similar to complete fin interruption. The wavy-fin surfaces are designated by two figures, giving the number of fins per inch and the wavelength, followed by the letter W. Thus surface 11.5- $\frac{3}{8}$ W has 11.5 fins per inch and a complete wave every $\frac{3}{8}$ in.

Pin-fin surfaces are another example of the plate-fin system, where the purpose is to achieve very high heat transfer coefficients by maintaining thin boundary layers on the fins. By constructing the fins from small-diameter wire, the effective flow length of the fins can be very small indeed. The pin-fin surfaces are, however, characterized by quite high friction factors attributable primarily to form drag associated with the boundary-layer separation that occurs on the pins. Nevertheless, the very high heat transfer coefficients attainable often more than offset the high friction factors when the final heat exchanger design is considered. The designation scheme for the pin-fin surfaces is not descriptive.

One perforated-fin surface is described, designated simply by the number of fins per inch transverse to the flow and the letter P. Holes cut out of the fins again provide boundary-layer interruption. The friction factors for this surface are quite low, suggesting that there is very little form drag induced, but there are unfortunately insufficient data on this kind of surface to draw much in the way of general conclusions.

The dimensional information given in Table 9-3 provides all that is necessary, in addition to the basic heat transfer and flow-friction data in Chap. 10, to apply the plate-fin surfaces to heat exchanger design. It will be noted that the heat transfer area density is given as β , the area per unit of volume between the plates on one flow side. This is not area per unit of total heat exchanger volume, because the latter depends also upon which particular surface is incorporated on the other side. In Chap. 2 are given some useful geometrical relations which can be used with the information of Table 9-3 to determine the total area density, as well as the free-flow to frontal-area ratio of the heat exchanger.

Extrapolation of the plate-fin performance data to surfaces possessing a superficial geometrical similarity but different hydraulic diameter can probably be accomplished without introducing serious uncertainty for moderate changes in hydraulic diameter. For the plain-fin surfaces, this has already in effect been suggested in the generalized performance curves of Chap. 7. For the various interrupted-fin surfaces, the uncertainty is greater because of the proportionally greater influence of such factors as the cleanliness of the fin leading and trailing

edges and fin root connections. An aerodynamically cleaner surface is generally obtained in the larger-scale than in the smaller-scale surfaces.

If the original scale is maintained but fins of different thicknesses are used, there will be little effect on the nondimensional performance of the plain-fin surfaces but an increasingly important effect on louvered- and strip-fin performances. It should also be noted that changing fin thickness for any of the surfaces affects free-flow area and hydraulic diameter. Thus the dimensional information in the tables must be altered accordingly.

Figures 9-10 to 9-13 show diagrams of 21 surfaces of the finned-tube type. The finned-tube surfaces are commonly employed where one of the fluids in the heat exchanger is a gas and the other is a liquid. Since gases characteristically tend to operate with lower heat transfer coefficients than liquids, more area is required on the gas side of the surface for a balanced design. The finned-tube surfaces satisfy this requirement.

Three fundamental types of finned-tube surfaces are considered: (1) circular tubes with integral circular spiraled fins, (2) circular tubes with continuous fins, and (3) finned flat tubes. In order to present as complete a picture as possible, some of what are believed to be the more reliable data in the literature have been included, in addition to the data obtained on this project. In this category are the data of Jameson [2] on finned circular tubes and data obtained by the Trane Company [3] on circular tubes with continuous fins.

Again, a semidescriptive surface designation scheme is employed. The letters CF refer to circular fins; otherwise, the fins may be assumed to be continuous. The first numerals refer to the number of fins per inch; the second (if any) refer either to the nominal outside diameter of the tube in the case of circular tubes, or the tube dimension in the flow direction in the case of flat tubes.

The dimensional data on all the finned-tube surfaces are contained in Table 9-4. Since the basic heat transfer and flow-friction design data in Chap. 10 are in generalized nondimensional form, they are presumably applicable to surfaces of different sizes than the original test surfaces, but again complete geometrical similarity must be maintained if the surfaces are to be scaled up or down. No attempt has been made to generalize the correlations to the point where variations in the nondimensional geometry can be handled.

All the fins on the finned-circular-tube surfaces are of the continuous-spiraled-fin type. However, the basic performance data presented will probably not differ markedly for separate disk fins.

Figures 9-14 to 9-18 show six types of matrix surfaces. Figure 9-14 shows a series of matrices formed by stacked crossed rods. Staggered, in-line, and randomly stacked rods with perfect stacking (each row touching) have been tested, and the geometrical data on seven patterns considered are tabulated in Table 9-5. The corresponding direct test data are given in Figs. 10-98 to 10-100 in Chap. 10 and in Table 10-10.

Performance data on the randomly stacked woven-screen matrix (Fig. 9-15) are not tabulated or graphed in Chap. 10, and thus detailed geometrical data are not given in this chapter. However, correlations of tests of woven-screen matrices are presented in Chap. 7, based on a number of sources described in Chap. 7.

Matrices can also be constructed using the plate-fin scheme. Figure 9-16 shows an example which is essentially a very compact strip-fin surface. The only difference is that the fins are attached to the base plates by mechanical pressure rather than by brazing, since there is no necessity that the joints be heat-conducting. The result is that there is no fillet in the flow passage corners, and this has some effect on performance. The surface designation for this matrix is nondescriptive.

Figure 9-17 shows two examples of glass ceramic matrices [6, 7]. Besides being made of a material that is usable at quite high temperatures, these matrices can be constructed with extremely high area compactness. They have the further virtue that the thermal conductivity of the ceramic material can be very low, which is important in high-performance counterflow heat exchangers where longitudinal heat conduction can be of critical importance (see Chap. 2). (Note that the strip-fin construction described in the previous paragraph also has the virtue of an interrupted longitudinal conduction path.) Again the surface designation is nondescriptive.

Another matrix surface is that formed by randomly stacked spheres (Fig. 9-18). Again, a correlation of data from a number of sources is presented in Chap. 7, and no geometrical data are given in this chapter because the performance data presented are not the direct measured performance of a particular defined surface. However, since only a single pair of performance curves is proposed for a randomly stacked bed of spheres, the coordinates of these curves are tabulated in Chap. 10.

The matrix surfaces are characterized by a potential for very high area compactness and by high heat transfer coefficients. Typical uses of such surfaces might be in periodic-flow heat exchangers (see Chap. 2), as fuel elements in nuclear reactors, and as energy-storage devices in gas blow-down systems.

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Table 9-1 Surface Geometry, Flow Inside Circular and Flattened Circular Tubes

Surface designation	Type	Hydraulic diameter $4r_h$			Test core ($L/4r_h$)	Minimum free-flow area per tube	
		ft	in	10^{-3}m		ft^2	10^{-6}m^2
ST-1	Circular	0.01925	0.231	5.867	79.2	0.0002908	27.016
FT-1	Flattened	0.01433	0.1719	4.368	—	0.0002195	20.392
FTD-1	Flattened, dimpled	0.01116	0.1338	3.399	—	0.000169	15.701

Table 9-2 Surface Geometry, Flow Normal to Banks of Bare Tubes

Surface designation	Pattern	Type of tests	Tube diameter		Transverse spacing		Longitudinal spacing		Hydraulic diameter $4r_h$		Free-flow/frontal area σ	Heat transfer area/total volume α	
			ft	10^{-3}m	ft	10^{-3}m	ft	10^{-3}m	ft	10^{-3}m		ft^2/ft^3	m^2/m^3
S 1.50–1.25(s)	Staggered	Steady state	0.02085	6.350	0.03125	9.525	0.02605	7.938	0.0166	5.029	0.333	80.3	263.5
S 1.50–1.25	Staggered	Transient	0.03125	9.525	0.04688	14.288	0.0391	11.908	0.0248	7.569	0.333	53.4	175.2
S 1.25–1.25	Staggered	Transient	0.03125	9.525	0.0391	11.908	0.0391	11.908	0.0125	3.810	0.200	64.4	211.3
S 1.50–1.00	Staggered	Transient	0.03125	9.525	0.04688	14.288	0.03125	9.525	0.0199	6.071	0.333	67.1	220.1
S 1.50–1.50	Staggered	Transient	0.03125	9.525	0.04688	14.288	0.04688	14.288	0.0298	9.093	0.333	44.8	147.0
S 2.00–1.00	Staggered	Transient	0.03125	9.525	0.0625	19.050	0.03125	9.525	0.0327	9.957	0.414	50.3	165.0
S 2.50–0.75	Staggered	Transient	0.03125	9.525	0.0782	23.813	0.02344	7.145	0.0271	8.255	0.366	53.6	175.9
I 1.50–1.25(s)	In line	Steady state	0.02085	6.350	0.03125	9.525	0.02605	7.938	0.0166	5.029	0.338	80.4	263.8
I 1.50–1.25	In line	Transient	0.03125	9.525	0.04688	14.288	0.0391	11.908	0.0248	7.569	0.333	53.6	175.9
I 1.25–1.25	In line	Transient	0.03125	9.525	0.0391	11.908	0.0391	11.908	0.01237	3.767	0.200	64.4	211.3
FT-2	Staggered	Steady state	Flattened		See surface diagrams				0.01433	4.374	0.386	108	354.3
FTD-2	Staggered	Steady state	Flattened		See surface diagrams				0.0160	4.877	0.423	108	354.3

Table 9-3 Surface Geometry, Plate-Fin Surfaces

(a) Plain fins												
Surface designation	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Flow length of uninterrupted fin		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
2.0	0.0625	19.05	2.0	0.0474	14.45	0.032	0.813	12.0	304.8	76.1	249.7	0.606
3.01	0.0625	19.05	3.01	0.03546	10.82	0.032	0.813	12.0	304.8	98.3	322.5	0.706
3.97	0.0625	19.05	3.97	0.0282	8.59	0.032	0.813	12.0	304.8	119.4	391.7	0.766
4.00	0.015	4.57	See surface diagrams	0.0150	4.57	See surface diagrams		18.0	457.2	See surface diagrams		0.500
5.3	0.0392	11.94	5.3	0.02016	6.15	0.006	0.152	2.49	63.2	188	616.8	0.719
6.2	0.0337	10.29	6.2	0.0182	5.54	0.010	0.254	1.20	30.5	204	669.3	0.728
9.03	0.0686	20.90	9.03	0.01522	4.64	0.008	0.203	1.19	30.2	244	800.5	0.888
11.1	0.0208	6.35	11.1	0.01012	3.08	0.006	0.152	2.50	63.5	367	1204.0	0.756
11.11(a)	0.0400	12.19	11.11	0.01153	3.52	0.008	0.203	8.00	203.2	312	1023.6	0.854
14.77	0.0275	8.38	14.77	0.00848	2.59	0.006	0.152	2.51	63.8	420	1378.0	0.844
15.08	0.0348	10.62	15.08	0.00876	2.67	0.006	0.152	6.84	173.7	414	1358.3	0.870
19.86	0.0208	6.35	19.86	0.00615	1.875	0.006	0.152	2.51	63.8	561	1840.6	0.849
10.27T	0.0453	13.82	10.27	0.01259	3.84	0.010	0.254	5.0	127.0	289.9	951.1	0.863
11.94T	0.0207	6.33	11.94	0.009396	2.86	0.006	0.152	5.0	127.0	393.0	1289.4	0.769
12.00T	0.0208	6.35	12.00	0.009412	2.87	0.006	0.152	2.5	63.5	392.7	1288.4	0.773
16.96T	0.0213	6.50	16.96	0.005652	1.722	0.006	0.152	5.0	127.0	607.8	1994.1	0.861
25.79T	0.0170	5.18	25.79	0.003771	1.151	0.006	0.152	2.5	63.5	855.6	2807.1	0.884
30.33T	0.0287	8.76	30.33	0.004009	1.222	0.006	0.152	2.5	63.5	812.5	2665.7	0.928
46.45T	0.00833	2.54	46.45	0.002643	0.805	0.002	0.051	2.63	66.8	1332.5	4371.7	0.837

(b) Louvered fins														
Surface designation	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Louver spacing		Louver gap		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
$\frac{3}{8}-6.06$	0.0208	6.35	6.06	0.01460	4.453	0.006	0.152	0.375	9.525	0.055	1.397	256	840	0.640
$\frac{3}{8}(a)-6.06$	0.0208	6.35	6.06	0.01460	4.453	0.006	0.152	0.375	9.525	0.130	3.302	256	840	0.640
$\frac{1}{2}-6.06$	0.0208	6.35	6.06	0.01460	4.453	0.006	0.152	0.500	12.70	0.055	1.397	256	840	0.640
$\frac{1}{2}(a)-6.06$	0.0208	6.35	6.06	0.01460	4.453	0.006	0.152	0.500	12.70	0.130	3.302	256	840	0.640
$\frac{3}{8}-8.7$	0.0208	6.35	8.7	0.01196	3.650	0.006	0.152	0.375	9.525	0.055	1.397	307	1007	0.705
$\frac{3}{8}(a)-8.7$	0.0208	6.35	8.7	0.01196	3.650	0.006	0.152	0.375	9.525	0.080	2.032	307	1007	0.705
$\frac{1}{16}-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.1875	4.763	0.055	1.397	367	1204	0.756
$\frac{1}{4}-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.250	6.350	0.035	1.397	367	1204	0.756
$\frac{1}{4}(b)-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.250	6.350	0.055	1.397	367	1204	0.756
$\frac{3}{8}-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.375	9.525	0.055	1.397	367	1204	0.756
$\frac{3}{8}(b)-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.375	9.525	0.055	1.397	367	1204	0.756
$\frac{1}{2}-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.500	12.70	0.055	1.397	367	1204	0.756
$\frac{3}{4}-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.750	19.05	0.040	1.016	367	1204	0.756
$\frac{3}{4}(b)-11.1$	0.0208	6.35	11.1	0.01012	3.084	0.006	0.152	0.750	19.05	0.040	1.016	367	1204	0.756

(continued)

Table 9-3 Surface Geometry, Plate-Fin Surfaces (continued)

(c) Strip fins												
Surface designation	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Flow length of uninterrupted fin		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
$\frac{1}{4}(s)-11.1$	0.0208	6.35	11.1	0.01012	3.08	0.006	0.152	0.25	6.35	367	1204	0.756
$\frac{1}{8}-15.2$	0.0346	10.55	15.2	0.00868	2.65	0.006	0.152	0.125	3.175	417	1368	0.873
$\frac{1}{8}-13.95$	0.0313	9.54	13.95	0.00879	2.68	0.010	0.254	0.125	3.175	381	1250	0.840
$\frac{1}{8}-15.61$	0.0208	6.35	15.61	0.007812	2.38	0.004	0.102	0.125	3.175	472	1548	0.809
$\frac{1}{8}-19.86$	0.00817	2.49	19.86	0.00506	1.54	0.004	0.102	0.125	3.175	687	2254	0.785
$\frac{1}{8}-22.68$	0.0251	7.65	22.68	0.00569	1.74	0.004	0.102	0.111	2.80	631	2069	0.885
$\frac{1}{8}-25.01$	0.0167	5.08	25.01	0.00491	1.50	0.004	0.102	0.111	2.80	719	2360	0.850
$\frac{1}{8}-24.12$	0.00625	1.91	24.12	0.00397	1.21	0.004	0.102	0.111	2.80	863	2830	0.665
$\frac{1}{10}-27.03$	0.0208	6.35	27.03	0.00467	1.42	0.004	0.102	0.100	2.54	752	2466	0.887
$\frac{1}{10}-19.35$	0.00625	1.91	19.35	0.00460	1.40	0.004	0.102	0.100	2.54	759	2490	0.611
$\frac{1}{10}-19.74$	0.00423	1.29	19.74	0.00400	1.22	0.002	0.051	0.100	2.54	923	3028	0.508
$\frac{3}{32}-12.2$	0.0404	12.3	12.2	0.01120	3.41	0.004	0.102	0.094	2.40	340	1115	0.862
$\frac{1}{2}-11.94(\text{D})$	0.0198	6.02	11.94	0.007436	2.27	0.006	0.152	0.500	12.7	461	1512	0.796
$\frac{1}{4}-15.4(\text{D})$	0.0172	5.23	15.4	0.00527	1.61	0.006	0.152	0.250	6.35	642	2106	0.816
$\frac{1}{8}-12.18(\text{D})$	0.0294	8.97	12.18	0.008848	2.70	0.004	0.102	0.178	4.52	422	1385	0.847
$\frac{1}{7}-15.75(\text{D})$	0.0253	7.72	15.75	0.006790	2.07	0.004	0.102	0.143	3.63	526	1726	0.859
$\frac{1}{8}-16.00(\text{D})$	0.0212	6.48	16.00	0.006112	1.86	0.006	0.152	0.125	3.175	550	1804	0.845
$\frac{1}{8}-16.12(\text{D})$	0.0172	5.23	16.12	0.00509	1.55	0.006	0.152	0.125	3.175	660	2165	0.823
$\frac{1}{8}-19.82(\text{D})$	0.0171	5.21	19.82	0.005049	1.54	0.004	0.102	0.125	3.175	680	2231	0.841
$\frac{1}{8}-20.06(\text{D})$	0.0167	5.11	20.06	0.004892	1.49	0.004	0.102	0.125	3.175	698	2290	0.843
$\frac{1}{8}-16.12(\text{T})$	0.0261	7.98	16.12	0.00514	1.57	0.006	0.152	0.125	3.175	650	2133	0.882

(d) Wavy fins

Surface designation	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Wavelength		Double wave amplitude		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
11.44- $\frac{3}{8}\text{W}$	0.0345	10.49	11.44	0.01060	3.23	0.006	0.152	0.375	9.53	0.0775	1.97	351	1152	0.847
11.5- $\frac{3}{8}\text{W}$	0.0313	9.53	11.5	0.00993	3.02	0.010	0.254	0.375	9.53	0.078	1.98	345	1138	0.822
17.8- $\frac{3}{8}\text{W}$	0.0345	10.49	17.8	0.00696	2.12	0.006	0.152	0.375	9.53	0.0775	1.97	514	1686	0.892

(e) Pin fins

Surface designation	Plate spacing b		Pin pattern	Hydraulic diameter $4r_h$		Pin diameter.		Transverse pin spacing		Longitudinal pin spacing		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
AP-1	0.020	6.10	In-line	0.01444	4.40	0.040	1.02	0.125	3.18	0.125	3.18	188	617	0.512
AP-2	0.0332	10.1	In-line	0.01172	3.58	0.040	1.02	0.12	3.05	0.096	2.44	204	669	0.686
PF-3	0.0625	19.1	In-line	0.00536	1.636	0.031	0.79	0.062	1.53	0.062	1.53	339	1112	0.834
PF-4(F)	0.0418	12.75	Staggered	0.0186	5.66	0.065	1.65	0.199	5.06	0.125	3.18	140	459	0.704
PF-9(F)	0.0425	12.95	In-line	0.0297	9.04	0.065	1.65	0.238	6.05	0.196	4.98	96.2	316	0.546
PF-10(F)	0.0367	11.18	In-line	0.01426	4.34	0.063×0.021	1.6×0.53	0.090	2.29	0.110	2.79	214	702	0.693

(f) Perforated fins

Surface designation	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Hole diameter		Hole spacing		Heat transfer area/volume between plates β		Fin area/total area
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3	
13.95(P)	0.0167	5.08	12.95	0.00822	2.50	0.012	0.305	0.079	2.0	32	4.96×10^6	381	1250	0.705

Table 9-4 Surface Geometry, Flow Normal to Banks of Finned Tubes

(a) Circular tubes, circular fins																		
Surface designation	Tube arrangement	Tube diameter		Fin outside diameter		Transverse tube spacing		Longitudinal tube spacing		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Free-flow/ frontal area σ	Heat transfer area/ total volume α		Fin area/ total area
		in	10^{-3}m	in	10^{-3}m	in	10^{-3}m	in	10^{-3}m		ft	10^{-3}m	in	10^{-3}m		ft^2/ft^3	m^2/m^3	
CF-7.34	Staggered	0.38	9.65	0.92	23.4	0.975	24.8	0.800	20.3	7.34	0.0154	4.75	0.018	0.46	0.538	140	459	0.892
CF-8.72	Staggered	0.38	9.65	0.92	23.4	0.975	24.8	0.800	20.3	8.72	0.01288	3.93	0.018	0.46	0.524	163	535	0.910
CF-8.72(c)	Staggered	0.42	10.67	0.861	21.9	0.975	24.8	0.800	20.3	8.72	0.01452	4.43	0.019	0.48	0.494	136	446	0.876
CF-11.46	Staggered	0.38	9.65	0.92	23.4	0.975	24.8	0.800	20.3	11.46	0.00976	2.98	0.016	0.41	0.510	209	686	0.931
CF-7.0- $\frac{1}{2}$ J	Staggered	0.645	16.38	1.121	28.5	1.232	31.3	1.35	34.3	7.0	0.0219	6.68	0.010	0.25	0.449	82	269	0.830
CF-8.7- $\frac{1}{2}$ J(a)	Staggered	0.645	16.38	1.121	28.5	1.232	31.3	1.35	34.3	8.7	0.01797	5.48	0.010	0.25	0.443	98.7	324	0.862
CF-8.7- $\frac{1}{2}$ J(b)	Staggered	0.645	16.38	1.121	28.5	1.848	46.9	1.35	34.3	8.7	0.383	11.67	0.010	0.25	0.628	65.7	216	0.862
CF-9.05- $\frac{1}{2}$ J(a)	Staggered	0.774	19.66	1.463	37.2	1.557	39.5	1.75	44.5	9.05	0.01681	5.13	0.012	0.31	0.455	108	354	0.917
CF-9.05- $\frac{1}{2}$ J(b)	Staggered	0.774	19.66	1.463	37.2	1.982	50.3	1.75	44.5	9.05	0.02685	8.18	0.012	0.31	0.572	85.1	279	0.917
CF-9.05- $\frac{1}{2}$ J(c)	Staggered	0.774	19.66	1.463	37.2	2.725	69.2	1.75	44.5	9.05	0.0445	13.59	0.012	0.31	0.688	61.9	203	0.917
CF-9.05- $\frac{1}{2}$ J(d)	Staggered	0.774	19.66	1.463	37.2	2.725	62.9	0.80	20.3	9.05	0.01587	4.85	0.012	0.31	0.537	135	443	0.917
CF-9.05- $\frac{1}{2}$ J(e)	Staggered	0.774	19.66	1.463	37.2	1.982	50.3	1.375	34.9	9.05	0.02108	6.43	0.012	0.31	0.572	108	354	0.917
CF-8.8-1.0-J(a)	Staggered	1.024	26.0	1.737	44.1	1.959	49.8	2.063	52.4	8.8	0.01927	5.89	0.012	0.31	0.439	91.2	299	0.825
CF-8.8-1.0-J(b)	Staggered	1.024	26.0	1.737	44.1	3.079	78.2	2.063	52.4	8.8	0.0443	13.21	0.012	0.31	0.642	58.1	191	0.825

(b) Circular tubes, continuous fins

Surface designation	Tube arrangement	Tube diameter		Fin type	Fins/in	Hydraulic diameter, $4r_h$		Fin thickness δ		Free-flow/ frontal area σ	Heat transfer area/ total volume α		Fin area/ total area
		in	10^{-3}m			ft	10^{-3}m	in	10^{-3}		ft^2/ft^3	m^2/m^3	
8.0- $\frac{3}{8}$ T	Staggered	0.402	10.21	Plain	8.0	0.01192	3.63	0.013	0.33	0.534	179	587	0.913
7.75- $\frac{3}{8}$ T	Staggered	0.676	17.17	Plain	7.75	0.0114	3.48	0.016	0.41	0.481	169	554	0.950

(c) Flat tubes, continuous fins

Surface designation	Tube arrangement	Fin type	Tube length (parallel to flow)		Tube width (normal to flow)		Fins/in	Hydraulic diameter $4r_h$		Fin thickness δ		Free flow/ frontal area σ	Heat transfer area/ total volume α		Fin area/ total area
			in	10^{-3}m	in	10^{-3}m		ft	10^{-3}m	in	10^{-3}m		ft^2/ft^3	m^2/m^3	
9.68-0.87	In-line	Plain	0.870	22.1	0.120	3.0	9.68	0.01180	3.60	0.004	0.102	0.697	229	751	0.795
9.1-0.737S	Staggered	Plain	0.737	18.7	0.100	2.5	9.1	0.01380	4.21	0.004	0.102	0.788	224	735	0.813
9.68-0.87R	In-line	Ruffled	0.870	22.1	0.120	3.0	9.68	0.01180	3.60	0.004	0.102	0.697	229	751	0.795
9.21-0.737SR	Staggered	Ruffled	0.737	18.7	0.100	2.5	9.29	0.01352	4.12	0.004	0.102	0.788	228	748	0.814
11.32-0.737SR	Staggered	Ruffled	0.737	18.7	0.100	2.5	11.32	0.01152	3.51	0.004	0.102	0.780	270	886	0.845

Table 9-5 Surface Geometry, Flow Through Crossed-Rod Matrices†

Matrix no.	Porosity p	Rod diameter d		Transverse pitch X_t	Hydraulic radius r_h		Free-flow/ frontal area σ	Heat transfer area/ total volume α	
		in	10^{-3}		ft	10^{-3}m		ft^2/ft^3	m^2/m^3
1	0.832	0.375	9.53	4.675	0.0387	11.80	0.618	21.5	70.5
2	0.817	0.375	9.53	4.292	0.0349	10.64	0.588	23.4	76.8
3	0.766	0.375	9.53	3.356	0.0256	7.80	0.493	30.0	98.4
4	0.725	0.375	9.53	2.856	0.0206	6.28	0.422	35.2	115.5
5	0.675	0.375	9.53	2.417	0.0162	4.94	0.344	41.6	136.5
6	0.602	0.375	9.53	1.974	0.0118	3.60	0.243	50.9	167.0
7	0.500	0.375	9.53	1.571	0.00781	2.30	0.132	64.0	210.0

† Applicable to in-line, staggered, or random stacking.

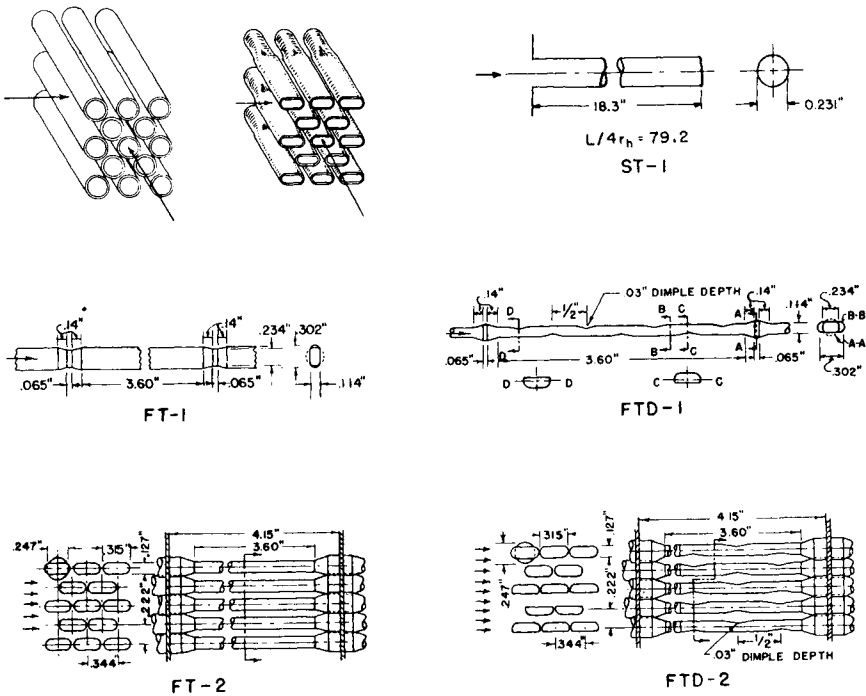
Table 9-6 Surface Geometry, Plate-Fin Matrix

Matrix no.	Plate spacing b		Fins/in	Hydraulic diameter $4r_h$		Fin thickness, δ		Flow length of uninterrupted fin		Heat transfer area/volume between plates β		Porosity p	Plate thickness		Total heat transfer area/total volume α	
	ft	10^{-3}m		ft	10^{-3}m	in	10^{-3}m	in	10^{-3}m	ft^2/ft^3	m^2/m^3		in	10^{-3}m	ft^2/ft^3	m^2/m^3
501MOD	0.00218	0.665	35.74	0.001885	0.575	0.001	0.0254	0.05	1.27	1976	6,483	0.896	0.001	0.0254	1,902	6,240

Table 9-7 Glass Ceramic Matrices

Matrix no.	Porosity p	Passage count		Hydraulic diameter $4r_h$		$L/4r_h$	Heat transfer area/ total volume α	
		1/in ²	1/m ²	ft	10 ⁻³ m		ft ² /ft ³	m ² /m ³
505A	0.794	526	815,300	0.00247	0.753	101	1,285	4,216
519	0.811	1,683	2,608,655	0.00197	0.600	127	1,646	5,400

Fig. 9-1 Tubular surfaces.



S-150-1.25 (s)

I-150-1.25 (s)

Fig. 9-3 Plate-fin surfaces, plain fins.

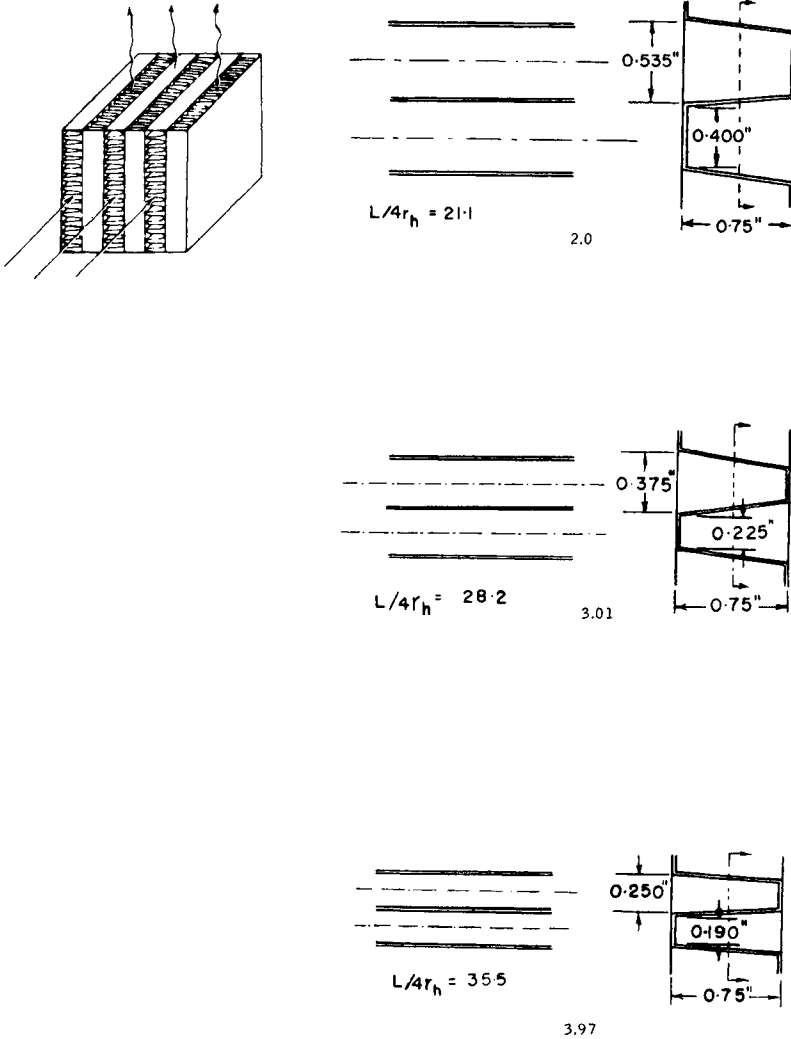


Fig. 9-4 Plate-fin surfaces, plain fins.

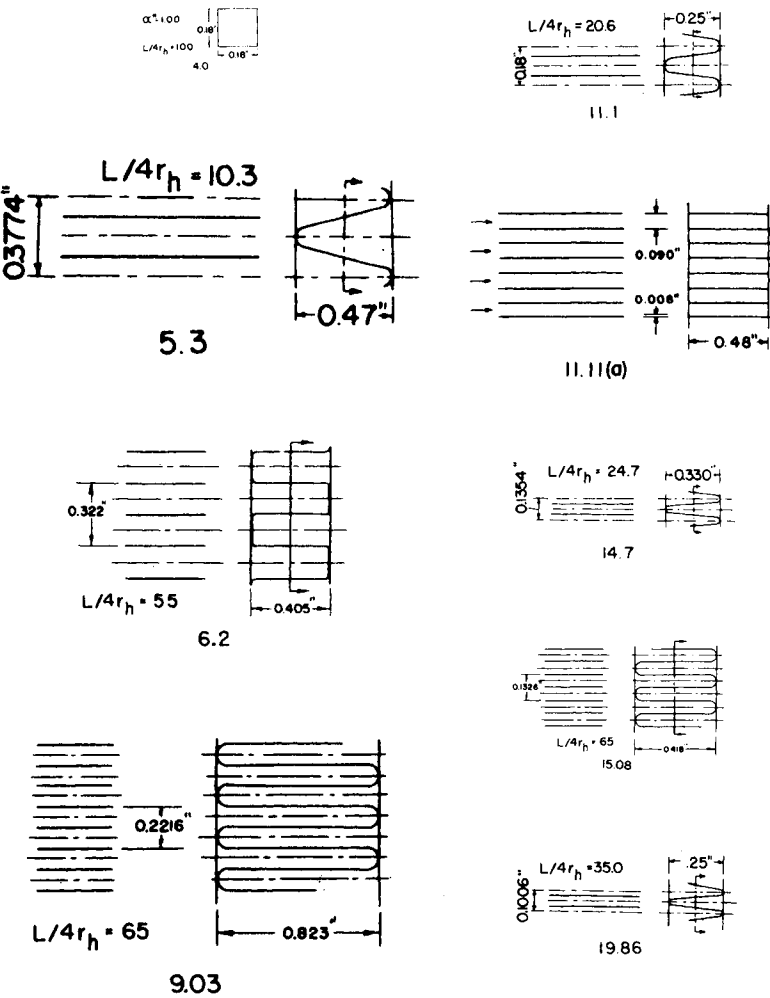


Fig. 9-4 (continued)

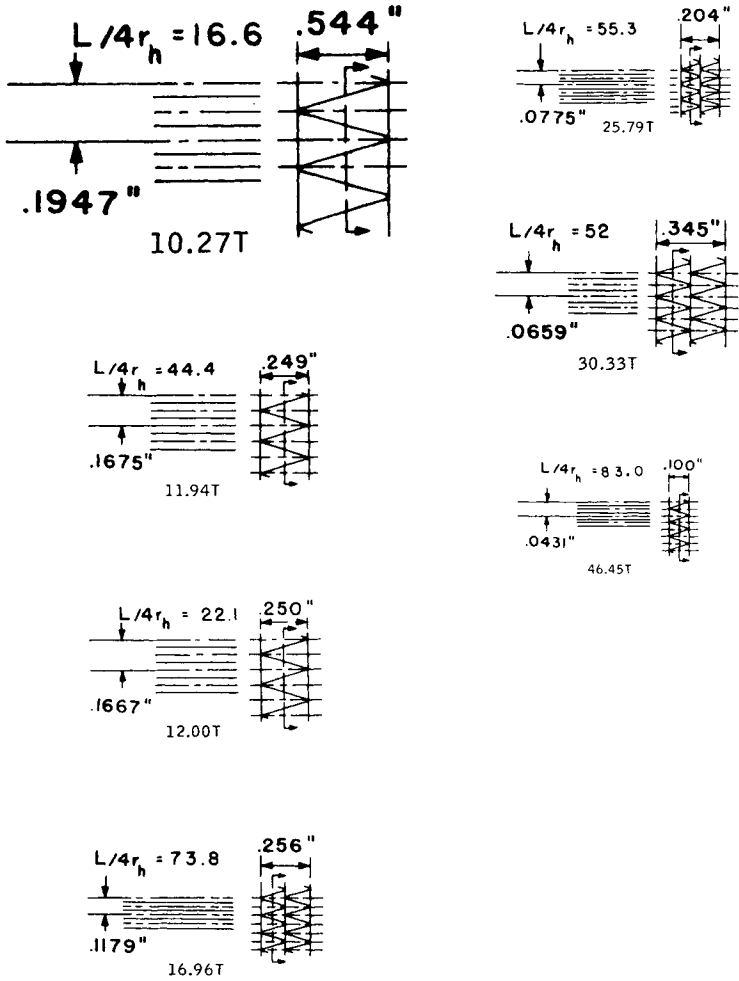


Fig. 9-5 Plate-fin surfaces, louvered fins.

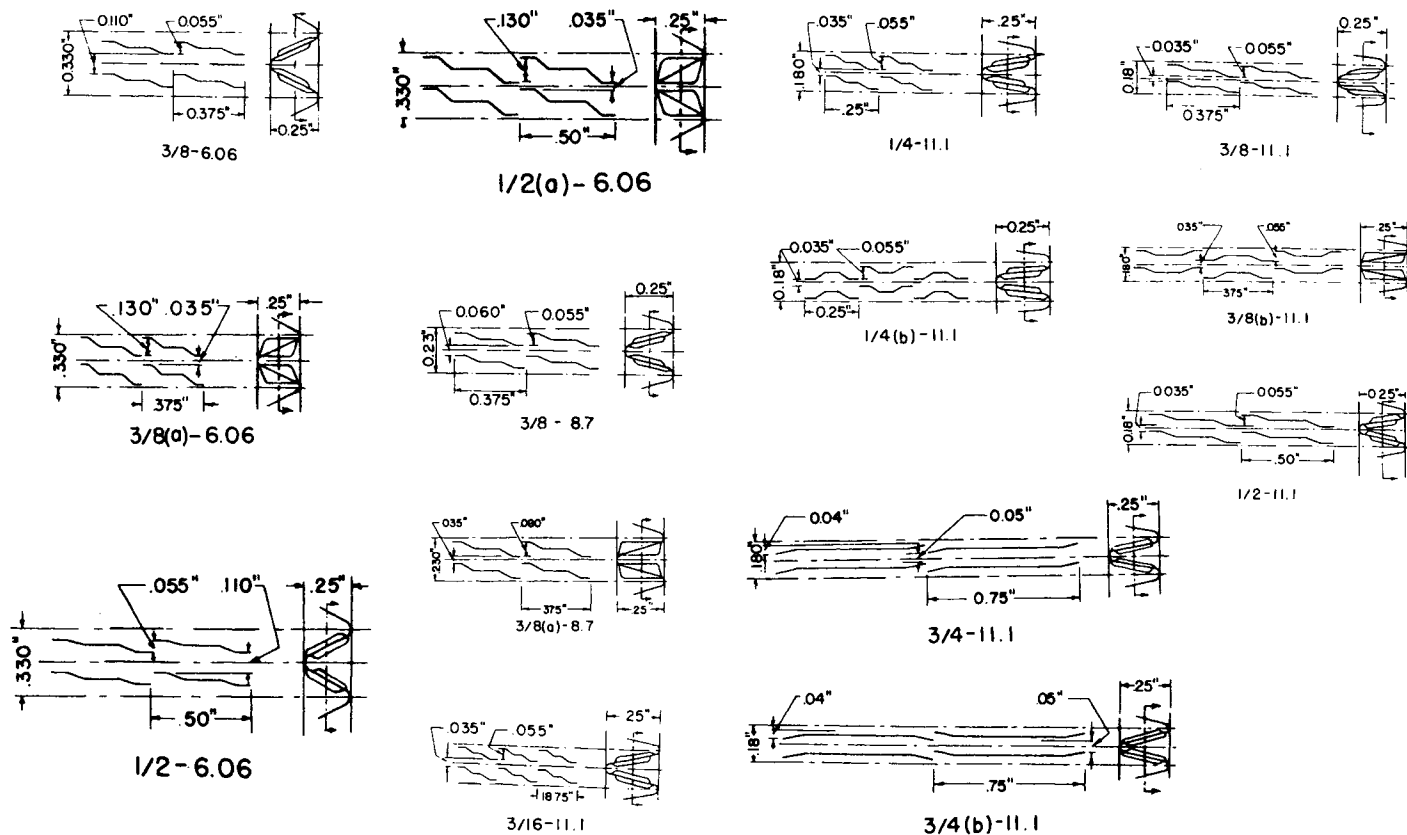


Fig. 9-6 Plate-fin surfaces, strip fins.

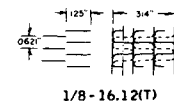
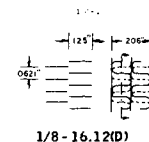
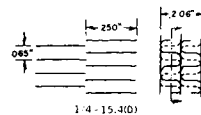
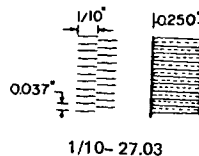
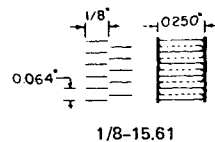
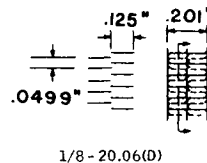
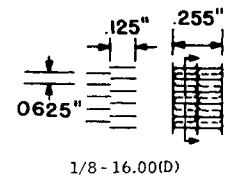
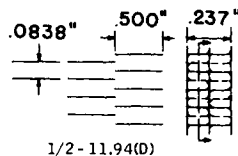
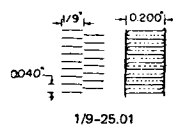
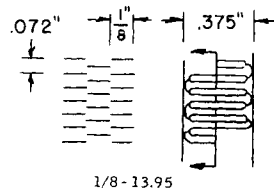
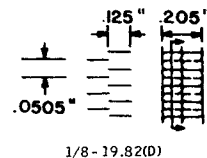
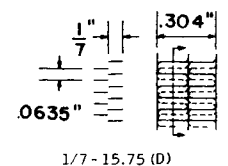
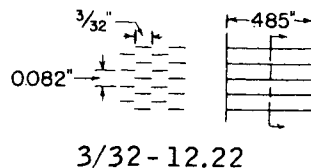
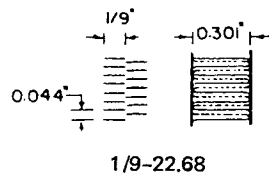
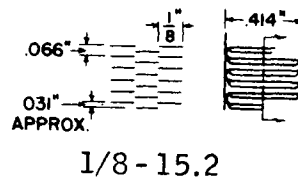
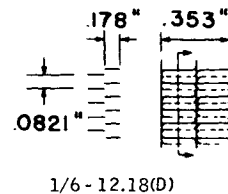
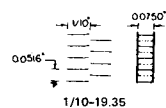
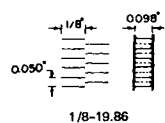
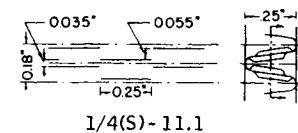


Fig. 9-7 Plate-fin surfaces, wavy fins.

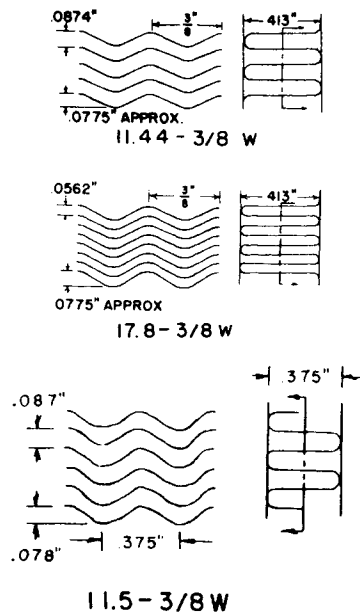


Fig. 9-9 Plate-fin surfaces, perforated fins.

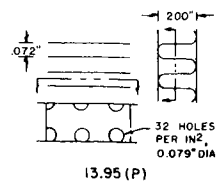


Fig. 9-8 Plate-fin surfaces, pin fins.

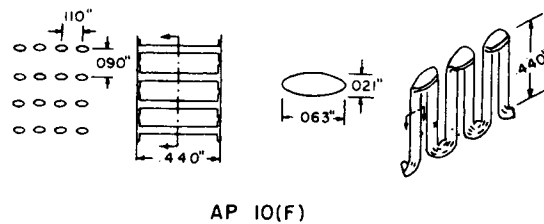
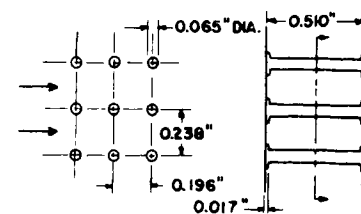
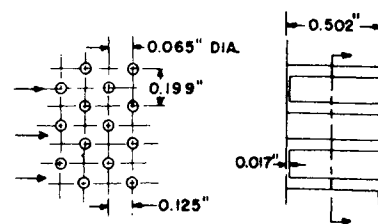
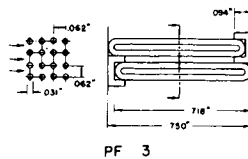
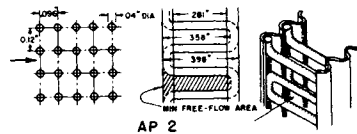
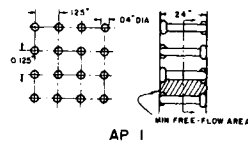
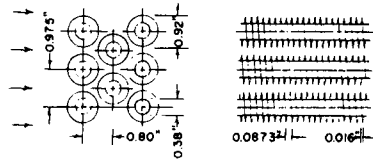
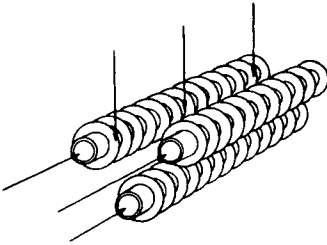
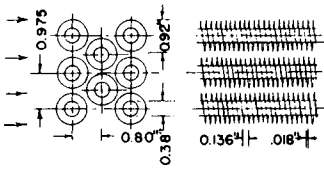


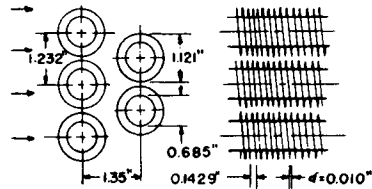
Fig. 9-10 Finned-tube surfaces, circular tubes, circular fins.



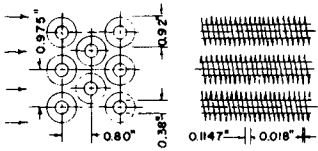
CF-11.46



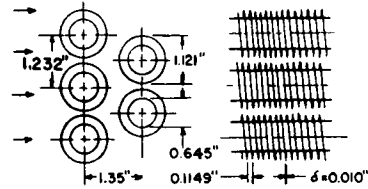
CF-7.34



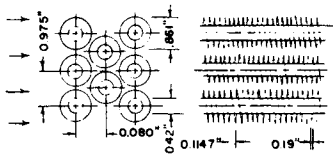
CF 7.0-5/8J



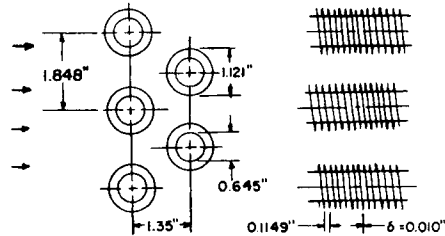
CF-8.72



CF 8.7-5/8J(a)



CF-8.72 (c)



CF 8.7-5/8J(b)

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Fig. 9-11 Finned-tube surfaces, circular tubes, circular fins.

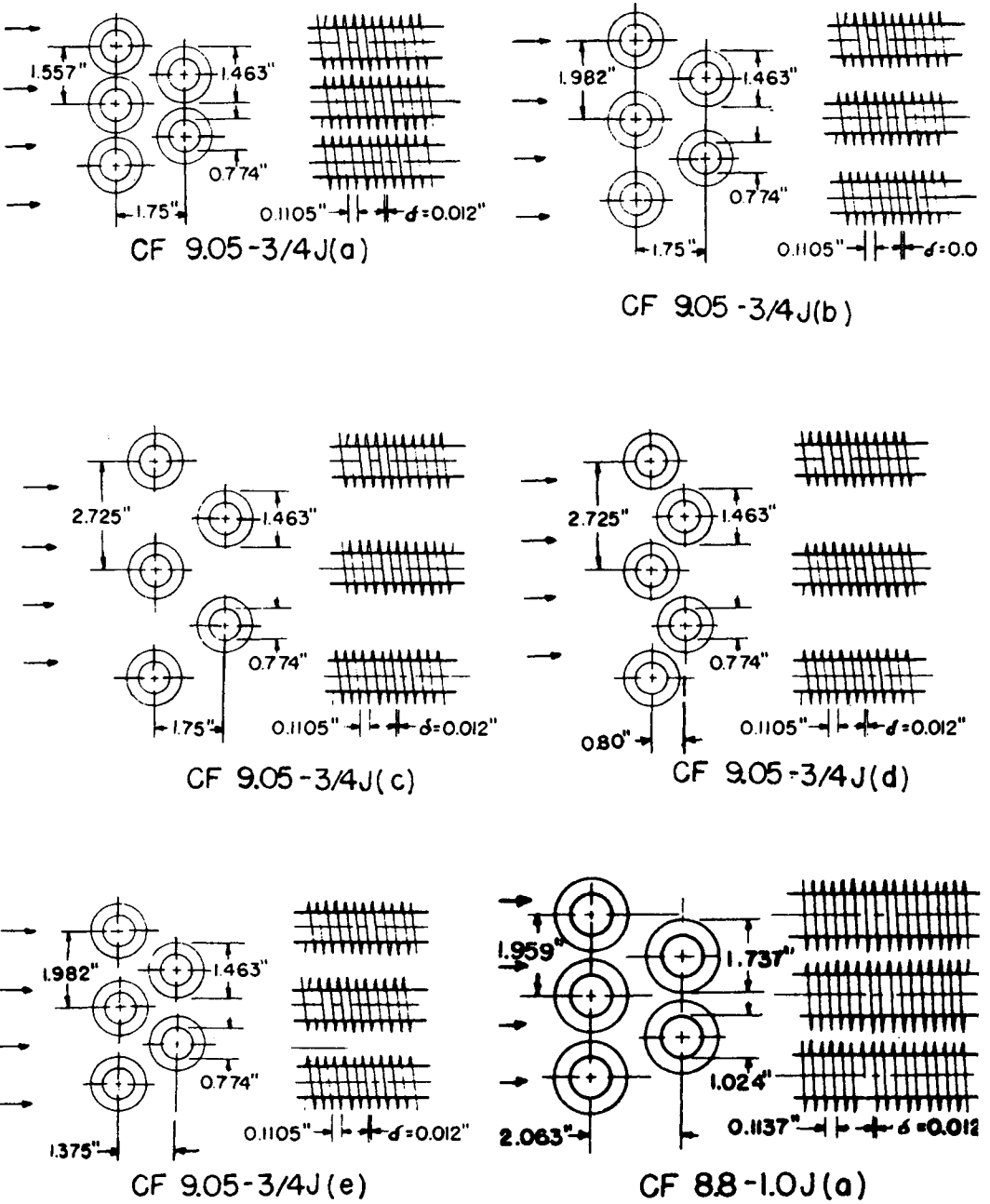
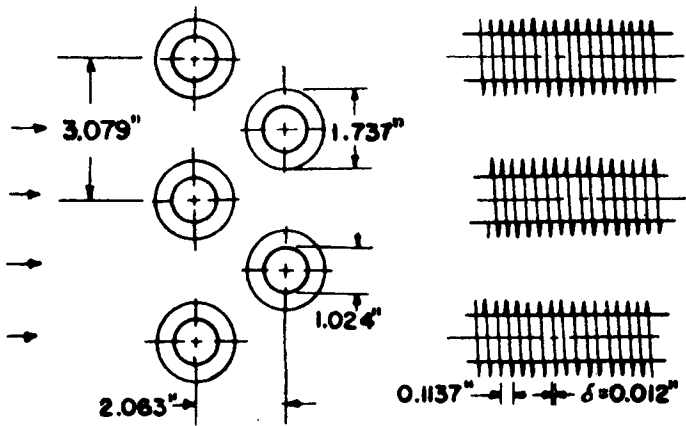


Fig. 9-11 (continued)



CF 88-1.0J (b)

Fig. 9-12 Finned-tube surfaces, circular tubes, continuous fins.

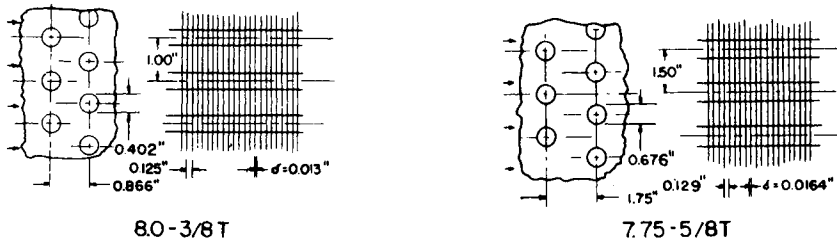


Fig. 9-13 Finned-tube surfaces, flat tubes, continuous fins.

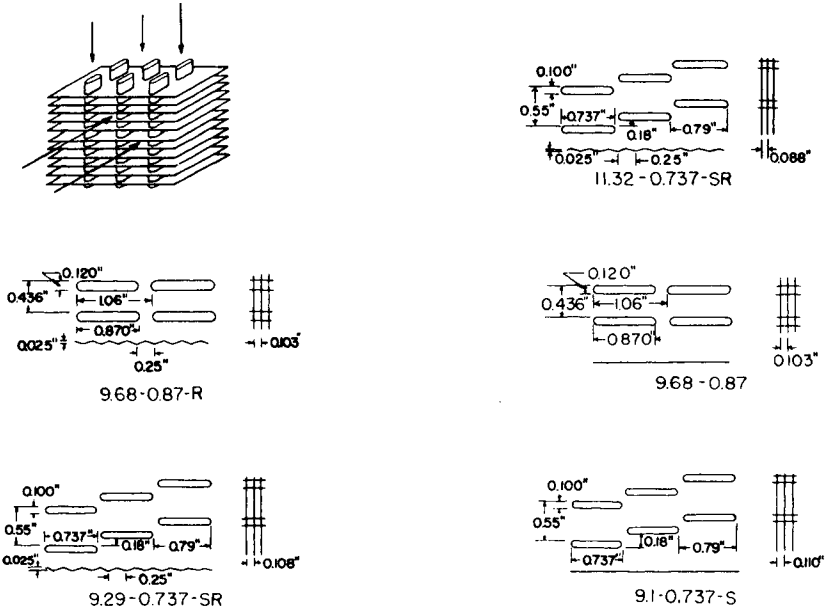


Fig. 9-14 Matrix surfaces, crossed-rod matrices.

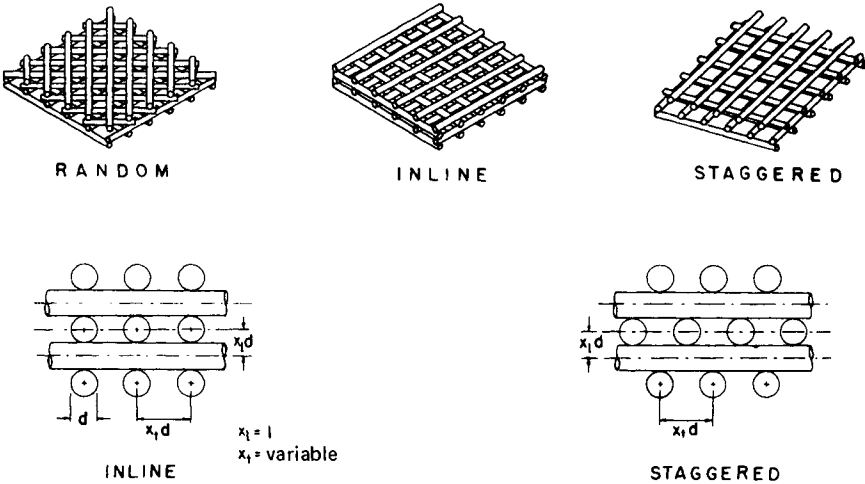


Fig. 9-15 Matrix surfaces, woven-screen matrices. See summary curves (Figs. 7-8 and 7-9) for performance characteristics.

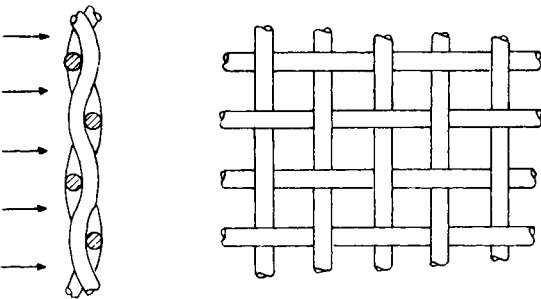


Fig. 9-16 Matrix surfaces, plate-fin surfaces, stacked.

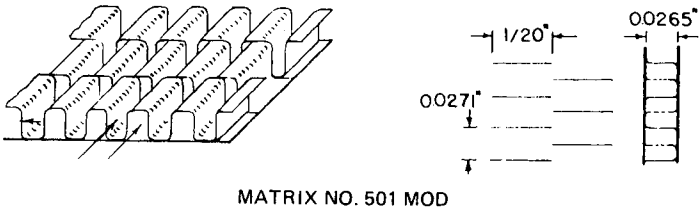


Fig. 9-17 Matrix surfaces, glass ceramic.

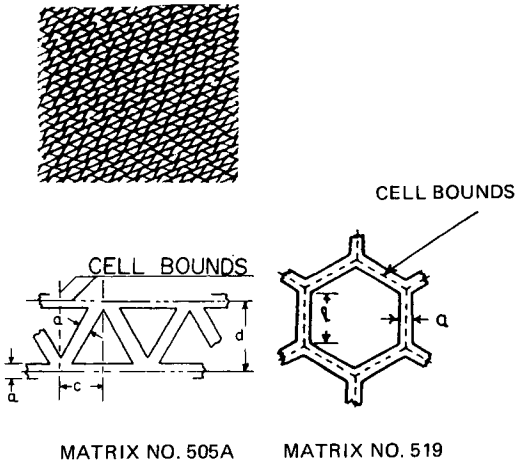
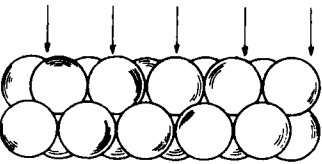


Fig. 9-18 Matrix surfaces, sphere matrices. See summary curves (Fig. 7-10) for performance characteristics.



10

Heat Transfer and Flow-Friction Design Data

Direct test data are presented in this chapter for 132 compact heat transfer surfaces. The geometries of the surfaces are completely described in the preceding chapter. The sources of the data and the experimental methods are discussed in Chap. 8. The present chapter will consist of a brief description of the tables and graphs, followed by some comments on certain of the test results which require more explanation.

The test data for the various configurations are presented in the same order as the surface diagrams in Figs. 9-1 to 9-18. The surfaces may be identified by both the designation number and the surface diagrams which appear on each of the graphs of basic heat transfer and flow-friction design data. The pertinent dimensional data for each of the surfaces are given in the captions to the figures. These are the same dimensional data that are given in Tables 9-1 to 9-7.

The abscissa on each figure is the Reynolds number, $Re = 4r_h G/\mu$. In *all cases*, except for the matrix surfaces, G is based on minimum free-flow area A_e , and $4r_h$ is as defined by Eq. (1-4). For the matrix surfaces, G is based on pA_{fr} , an "effective" free-flow area. The ordinate is used for two parameters: (1) the friction factor f , in *all cases* defined by Eq. (1-5); (2) the heat transfer parameter, Stanton number times Prandtl number to the two-thirds power, $StPr^{2/3} = (h/Gc_p)(\mu c_p/k)^{2/3}$.

The individual test points are indicated on the graphs for all the test results obtained under the supervision of the authors. Test points shown as small circles (o) were obtained during heat transfer tests (hot-core tests). Test points shown as crosses (x) are from friction-factor tests conducted with no core heating (cold-core tests). The discrepancy noted in some cases between the hot-core and cold-core friction factors (see, for example, Fig. 10-1) is due to temperature-dependent fluid-property effects discussed in Chap. 4

The solid lines drawn through the data points represent a best interpretation of the test data. It will be noted that the $StPr^{2/3}$ best-interpretation curve for the steam-to-air tests lies somewhat above the test points, especially at high Reynolds numbers. This arises from the fact that the test points were calculated from the raw test data with no correction made for the thermal resistance of the condensate layer on the steam side of the heat transfer surface. Before drawing

in the best-interpretation curves, a correction was made for an estimate of the steam-side resistance, and the difference between the test points and the best-interpretation curves represents the magnitude of this correction. Since the steam-side resistance was generally less than 10 percent of the overall resistance, only a moderately accurate estimate of this resistance was necessary. No correction for a steam-side resistance was necessary for the test results obtained by transient techniques; therefore, in these cases, the best-interpretation curve is faired through the test points.

Where both hot- and cold-core friction data were obtained, the best-interpretation curves are in all cases faired through the cold-core friction-factor test points, there being no uncertainty about temperature-dependent property effects.

Figures 10-11, 10-15, 10-16, and 10-87 to 10-92 do not show any test data points. The curves are plots of equations derived from the indicated sources and do not represent data obtained under the authors' supervision. These results have been included in this compilation in order to provide a more complete treatment of compact heat exchanger surfaces; they represent what are believed to be the most reliable information in the literature on the surfaces indicated.

Tables 10-1 to 10-13 are summaries of the basic heat transfer and flow-friction design data for all the surfaces tested under the authors' supervision. The tabular summaries are taken directly from the best-interpretation curves and do not represent test points. They are presented for the reader who may wish to plot his or her own curves.

Certain features of the individual graphs will now be pointed out in the order in which they appear.

In Fig. 10-12, the high Reynolds number end of the friction-factor curve is dashed. In-line tube banks often show a marked tendency to whistle, and if the sound is sufficiently intense, it may be reflected in increased friction factors. The surface shown in Fig. 10-12 whistled above $Re = 9,000$ with an increase in friction factor. The dashed line represents the probable course of the friction-factor curve without whistling. Since the whistling is an acoustic phenomenon and is affected by other factors besides the Reynolds number, those test points markedly affected by the whistle are not considered to be significant. The tendency to whistle may in some applications disallow the use of such surfaces unless suitable acoustic treatments are developed [1, 2]. Other surfaces with serious whistle problems will be noted below.

In Figs. 10-13 and 10-14, no best-interpretation curve is shown for the heat transfer test points at the low Reynolds number end of the test range. These surfaces were tested by a transient technique, which is satisfactory in the turbulent-flow range but does not yield valid results in laminar flow. Thus the best-interpretation curve is shown only in the region of fully established turbulent flow.

In Fig. 10-17, it may be noted that severe whistle difficulty was encountered for Reynolds numbers above 5,000, with both the heat transfer and friction results affected.

Figure 10-22 shows test results for a core made up of cylindrical tubes of

square cross section. A hot-water-to-air test system was used, and wall temperatures were measured directly so that no correction for water-side heat transfer resistance was necessary. Two curves are shown for the heat transfer results, one with constant wall temperature and the other with an approximately constant heat rate along the tubes. In all other tests the surfaces were at constant wall temperature.

Some of the pin-fin surfaces (Figs. 10-76 to 10-81) whistled at the high Reynolds number end of the test range. Especially severe whistle trouble was encountered with surface PF 9(F) (Fig. 10-80), where the whistle was of great intensity at Reynolds numbers above 4,000. Surface PF (10F) (Fig. 10-81) gave whistle trouble also at low Reynolds numbers.

The circular-fin-surface data in Figs. 10-83, 10-84, and 10-86 were obtained from test cores constructed from bimetal tubes. The original test results showed definite evidence of considerable contact heat transfer resistance between the inner copper liner and the outer aluminum fin muff. The data in Fig. 10-85 were obtained from a core with copper tubes and integral copper spiraled fins; thus no contact resistance existed. On the basis of the test results in Figs. 10-84 and 10-85, the magnitude of the contact resistance could be estimated; this estimation was used to correct the heat transfer curves in Figs. 10-83 and 10-84, so that the data are then representative only of the outer finned-tube-surface performance. Since this estimation was only an approximation, the correction introduced a considerable additional experimental uncertainty into the data. For this reason, the $StPr^{2/3}$ curves in Figs. 10-83 and 10-84 are shown as dashed rather than full lines, and the ± 5 percent experimental uncertainty quoted for the other test data does not apply. The effect of the contact resistance on the data from surface CF 11.46 (Fig. 10-86) was so great that no correction was attempted; thus only the friction factors are shown for this surface.

The figures for the glass ceramic matrices, Figs. 10-103 and 10-104, show by dashed lines some theoretical curves for fully developed flow (constant heat rate) for these particular shapes of flow passage. The differences between these lines and the experimental data are probably primarily due to slight variations in the sizes of the very large number of parallel-flow passages. It is difficult to quantify these differences, but it can be shown theoretically that small differences can have substantial effects of the type shown [3].

REFERENCES

1. Putnam, A. A.: Flow Induced Noise in Heat Exchangers, *Trans. ASME*, vol. 81, p. 417, 1959.
2. Grotz, B. J., and F. R. Arnold: Flow-induced Vibrations in Heat Exchangers, *Tech. Rept. 31*, prepared under contract Nonr 225(23) (NR-065-104) for the Office of Naval Research, Department of Mechanical Engineering, Stanford University, August, 1956.
3. Shah, R. K., and A. L. London: Effects of Non-Uniform Passages on Compact Heat Exchanger Performance, *Trans. ASME*, vol. 102, p. 653, 1980.

Table 10-1 Heat Transfer and Friction Data for Flow Inside Circular and Flattened Circular Tubes

ST-1			FT-1			FTD-1		
Re	$StPr^{2/3}$	f	Re	$StPr^{2/3}$	f	Re	$StPr^{2/3}$	f
15,000	0.00288	0.00668	15,000			15,000		
12,000	0.00300	0.00697	12,000			12,000		
10,000	0.00310	0.00725	10,000	0.00292	0.00687	10,000	0.00356	0.00660
8,000	0.00314	0.00760	8,000	0.00298	0.00718	8,000	0.00378	0.00701
6,000	0.00294	0.00770	6,000	0.00305	0.00770	6,000	0.00409	0.00764
5,000	0.00254	0.00725	5,000	0.00310	0.00820	5,000	0.00430	0.00807
4,000	0.00193	0.00635	4,000	0.00317	0.00895	4,000	0.00457	0.00878
3,000	0.00202	0.00672	3,000	0.00291	0.00939	3,000	0.00494	0.00982
2,500	0.00229	0.00750	2,500	0.00272	0.00939	2,500	0.00520	0.01070
2,000	0.00269	0.00902	2,000	0.00310	0.0107	2,000	0.00540	0.0116
1,500	0.00334	0.01170	1,500	0.00381	0.0135	1,500	0.00503	0.0119
1,200	0.00399	0.01425	1,200	0.00453	0.0163	1,200	0.00475	0.0125
1,000	0.00463	0.01680	1,000	0.00522	0.0191	1,000	0.00518	0.0136
800	0.00558	0.0206	800	0.00621	0.0232	800	0.00605	0.0158
600	0.00707	0.0266	600	0.00780	0.0296	600	0.00738	0.0191
500			500	0.00900	0.0346	500		
400			400			400		
300			300			300		

Table 10-2 Heat Transfer and Friction Data for Flow Normal to Banks of Bare Tubes

Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f
S 1.50–1.25(s)			S 1.50–1.25			S 1.25–1.25			S 1.50–1.00		
15,000	0.00632	0.0508	15,000	0.00690	0.0503	15,000			15,000		
12,000	0.00698	0.0530	12,000	0.00753	0.0525	12,000			12,000		
10,000	0.00755	0.0550	10,000	0.00808	0.0543	10,000			10,000		
8,000	0.00832	0.0578	8,000	0.00883	0.0566	8,000			8,000		
6,000	0.00941	0.0614	6,000	0.00987	0.0596	6,000			6,000		
5,000	0.0102	0.0640	5,000	0.0106	0.0615	5,000	0.00778	0.0375	5,000		
4,000	0.0112	0.0670	4,000	0.0116	0.0641	4,000	0.00850	0.0391	4,000	0.0111	0.0613
3,000	0.0127	0.0702	3,000	0.0129	0.0675	3,000	0.00950	0.0415	3,000	0.0124	0.0652
2,500	0.0137	0.0725	2,500	0.0138	0.0700	2,500	0.0102	0.0430	2,500	0.0134	0.0680
2,000	0.0149	0.0750	2,000	0.0151	0.0728	2,000	0.0111	0.0450	2,000	0.0147	0.0713
1,500	0.0166	0.0780	1,500	0.0169	0.0768	1,500	0.0124	0.0477	1,500	0.0164	0.0761
1,200	0.0178	0.0800	1,200	0.0184	0.0800	1,200	0.0136	0.0498	1,200	0.0180	0.0799
1,000	0.0189	0.0812	1,000	0.0198	0.0828	1,000	0.0146	0.0518	1,000	0.0193	0.0833
800	0.0201	0.0827	800	0.0216	0.0860	800	0.0159	0.0540	800	0.0213	0.0874
600			600	0.0241	0.0907	600	0.0178	0.0572	600	0.0239	0.0932
500			500	0.0258	0.0940	500	0.0191	0.0593	500	0.0257	0.0972
400			400			400	0.0208	0.0620	400		
300			300			300	0.0233	0.0658	300		
S 1.50–1.50			S 2.00–1.00			S 2.50–0.75			I 1.50–1.25(s)		
15,000			15,000			15,000			15,000		
12,000			12,000			12,000			12,000		
10,000	0.00809	0.0593	10,000			10,000			10,000	0.00752	0.0505
8,000	0.00888	0.0619	8,000	0.0108	0.0982	8,000	0.01060	0.1000	8,000	0.00820	0.0525
6,000	0.00997	0.0653	6,000	0.0122	0.104	6,000	0.01175	0.1036	6,000	0.00900	0.0549
5,000	0.0107	0.0674	5,000	0.0132	0.108	5,000	0.0126	0.1065	5,000	0.00958	0.0558
4,000	0.0117	0.0701	4,000	0.0145	0.112	4,000	0.0136	0.1100	4,000	0.01020	0.0562

3,000	0.0132	0.0740	3,000	0.0164	0.118	3,000	0.0152	0.1145	3,000	0.01080	0.0554
2,500	0.0142	0.0755	2,500	0.0177	0.122	2,500	0.0162	0.1175	2,500	0.01095	0.0535
2,000	0.0155	0.0769	2,000	0.0194	0.128	2,000	0.0176	0.1180	2,000	0.01075	0.0497
1,500	0.0174	0.0790	1,500	0.0219	0.135	1,500	0.0195	0.1200	1,500	0.00960	0.0410
1,200	0.0190	0.0803	1,200	0.0241	0.141	1,200	0.0212	0.1213	1,200	0.00778	0.0331
1,000	0.0204	0.0818	1,000	0.0261	0.146	1,000	0.0227	0.1220	1,000	0.00750	0.0281
800	0.0224	0.0834	800	0.0286	0.152	800	0.0246	0.1230	800	0.00790	0.0265
600			600	0.0323	0.160	600	0.0273	0.1250	600		
500			500			500			500		
400			400			400			400		
300			300			300			300		
I 1.50-1.25			I 1.25-1.25			FT-2			FTD-2		
15,000			15,000			15,000			15,000		
12,000			12,000			12,000			12,000		
10,000	0.00715	0.0483	10,000			10,000			10,000	0.00580	0.0248
8,000	0.00790	0.0501	8,000	0.00580	0.0364	8,000			8,000	0.00611	0.0248
6,000	0.00892	0.0511	6,000	0.00654	0.0373	6,000			6,000	0.00646	0.0248
5,000	0.00960	0.0515	5,000	0.00706	0.0380	5,000			5,000	0.00660	0.0248
4,000	0.01050	0.0519	4,000	0.00777	0.0387	4,000	0.00490	0.0190	4,000	0.00661	0.0248
3,000		0.0520	3,000		0.0397	3,000	0.00481	0.0184	3,000	0.00655	0.0248
2,500		0.0516	2,500		0.0401	2,500	0.00450	0.0175	2,500	0.00640	0.0248
2,000		0.0483	2,000		0.0410	2,000	0.00447	0.0177	2,000	0.00611	0.0249
1,500		0.0401	1,500		0.0420	1,500	0.00500	0.0195	1,500	0.00588	0.0259
1,200		0.0328	1,200		0.0429	1,200	0.00590	0.0221	1,200	0.00639	0.0274
1,000		0.0301	1,000		0.0436	1,000	0.00678	0.0245	1,000	0.00714	0.0297
800			800		0.0444	800	0.00809	0.0278	800	0.00836	0.0337
600			600		0.0456	600	0.0101	0.0329	600	0.01020	0.0395
500			500		0.0459	500	0.0117	0.0368	500	0.01160	0.0439
400			400		0.0450	400			400		
300			300		0.0421	300			300		

Table 10-3 Heat Transfer and Friction Data for Plain Plate-Fin Surfaces

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
2.0			3.01			3.97		
60,000	0.00228	0.00549	45,000	0.00233	0.00602	35,000	0.00246	0.00595
50,000	0.00237	0.00562	40,000	0.00239	0.00608	30,000	0.00254	0.00605
40,000	0.00248	0.00579	30,000	0.00254	0.00630	25,000	0.00263	0.00620
30,000	0.00264	0.00601	25,000	0.00264	0.00645	20,000	0.00276	0.00638
25,000	0.00274	0.00616	20,000	0.00277	0.00667	15,000	0.00291	0.00667
20,000	0.00288	0.00638	15,000	0.00295	0.00700	12,000	0.00302	0.00695
15,000	0.00305	0.00672	12,000	0.00310	0.00732	10,000	0.00316	0.00720
12,000	0.00320	0.00703	10,000	0.00322	0.00762	8,000	0.00330	0.00761
10,000	0.00333	0.00734	8,000	0.00336	0.00808	6,000	0.00348	0.00826
8,000	0.00347	0.00778	6,000	0.00355	0.00886	5,000	0.00357	0.00880
6,000	0.00363	0.00847	5,000	0.00366	0.00950	4,000	0.00367	0.00963
5,000	0.00373	0.00904	4,000	0.00373	0.01045	3,000	0.00367	0.01110
4,000	0.00379	0.01023	3,000	0.00368	0.01190	2,500	0.00357	0.01230

4.0			
Re	StPr ^{2/3}		<i>f</i>
	(<i>T</i>)	(ΔT)	
15,000			
12,000	0.00287	0.00314	0.00631
10,000	0.00298	0.00326	0.00658
8,000	0.00312	0.00339	0.00689
6,000	0.00313	0.00334	0.00700
5,000		0.00321	0.00670
4,000		0.00290	0.00594
3,000		0.00244	0.00563
2,500		0.00238	0.00633

2,000	0.00237	0.00255	0.00772
1,500	0.00296	0.00325	0.01010
1,200	0.00353	0.00394	0.01244
1,000	0.00407	0.00461	0.0148
800	0.00488	0.00562	0.0183
600	0.00614	0.00727	0.0241
500	0.00712	0.00858	0.0288
400			
300			

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
5.3			6.2			9.03		
15,000			15,000			15,000	0.00255	0.00708
12,000			12,000	0.00303	0.00708	12,000	0.00265	0.00740
10,000	0.00373	0.00764	10,000	0.00310	0.00735	10,000	0.00273	0.00763
8,000	0.00397	0.00806	8,000	0.00317	0.00768	8,000	0.00283	0.00799
6,000	0.00427	0.00870	6,000	0.00325	0.00807	6,000	0.00296	0.00842
5,000	0.00448	0.00913	5,000	0.00330	0.00838	5,000	0.00304	0.00870
4,000	0.00477	0.00978	4,000	0.00333	0.00875	4,000	0.00310	0.00903
3,000	0.00515	0.0108	3,000	0.00326	0.00923	3,000	0.00310	0.00980
2,500	0.00535	0.0115	2,500	0.00301	0.00958	2,500	0.00318	0.0106
2,000	0.00554	0.0127	2,000	0.00312	0.0103	2,000	0.00347	0.0122
1,500	0.00571	0.0146	1,500	0.00371	0.0127	1,500	0.00421	0.0152
1,200	0.00606	0.0167	1,200	0.00435	0.0152	1,200	0.00499	0.0182
1,000	0.00654	0.0189	1,000	0.00496	0.0176	1,000	0.00575	0.0214
800	0.00728	0.0228	800	0.00581	0.0211	800	0.00692	0.0262
600	0.00851	0.0299	600			600		

(continued)

Table 10-3 Heat Transfer and Friction Data for Plain Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
11.1			11.11(a)			14.77		
10,000	0.00314	0.00878	10,000	0.00288	0.00768	10,000	0.00310	0.00920
8,000	0.00333	0.00923	8,000	0.00303	0.00807	8,000	0.00326	0.00955
6,000	0.00356	0.00971	6,000	0.00324	0.00862	6,000	0.00352	0.0101
5,000	0.00372	0.00991	5,000	0.00338	0.00900	5,000	0.00367	0.0106
4,000	0.00390	0.0103	4,000	0.00353	0.00958	4,000	0.00389	0.0112
3,000	0.00412	0.0112	3,000	0.00368	0.0105	3,000	0.00417	0.0123
2,500	0.00424	0.0119	2,500	0.00373	0.0112	2,500	0.00435	0.0133
2,000	0.00436	0.0129	2,000	0.00375	0.0119	2,000	0.00456	0.0147
1,500	0.00444	0.0149	1,500	0.00420	0.0137	1,500	0.00495	0.0173
1,200	0.00471	0.0169	1,200	0.00505	0.0166	1,200	0.00538	0.0202
1,000	0.00515	0.0190	1,000	0.00586	0.0198	1,000	0.00585	0.0231
800	0.00599	0.0228	800	0.00704	0.0243	800	0.00663	0.0274
600	0.00733	0.0294	600	0.00890	0.0319	600	0.00791	0.0346
500	0.00840	0.0350	500	0.0103	0.0380	500	0.00898	0.0403
400			400			400		
300			300			300		
15.08			19.86			10.27T		
15,000			15,000			10,000	0.00295	0.00723
12,000			12,000			9,000	0.00299	0.00740
10,000			10,000			8,000	0.00303	0.00763
8,000			8,000	0.00320	0.00851	7,000	0.00310	0.00790
6,000	0.00308	0.00882	6,000	0.00337	0.00900	6,000	0.00318	0.00826
5,000	0.00310	0.00900	5,000	0.00348	0.00931	5,000	0.00328	0.00871
4,000	0.00309	0.00925	4,000	0.00363	0.00972	4,000	0.00341	0.00945
3,000	0.00309	0.00970	3,000	0.00382	0.0104	3,000	0.00372	0.01085
2,500	0.00322	0.01040	2,500	0.00395	0.0112	2,000	0.00445	0.01370

2,000	0.00352	0.01205	2,000	0.00410	0.0123	1,500	0.00523	0.01645
1,500	0.00420	0.0151	1,500	0.00443	0.0142	1,200	0.00608	0.0195
1,200	0.00491	0.0182	1,200	0.00497	0.0167	1,000	0.00682	0.0228
1,000	0.00562	0.0215	1,000	0.00567	0.0197	800	0.00797	0.0278
800	0.00662	0.0264	800	0.00672	0.0242	600	0.00889	0.0357
600	0.00815	0.0343	600	0.00834	0.0314	500	0.01101	0.0419
500	0.00930	0.0405	500	0.00960	0.0372	400	0.0129	0.0511
400			400	0.0113	0.0457	300		0.0662
300			300			200		
11.94T			12.00T			16.96T		
10,000	0.00294	0.00716	10,000			10,000		
9,000	0.00302	0.00730	9,000		0.00820	9,000		
8,000	0.00309	0.00755	8,000	0.00302	0.00851	8,000		
7,000	0.00317	0.00782	7,000	0.00312	0.00881	7,000		
6,000	0.00322	0.00819	6,000	0.00322	0.00928	6,000		0.00790
5,000	0.00323	0.00856	5,000	0.00333	0.00980	5,000	0.00281	0.00809
4,000	0.00330	0.00885	4,000	0.00344	0.01045	4,000	0.00281	0.00835
3,000	0.00317	0.00956	3,000	0.00350	0.01128	3,000	0.00263	0.00875
2,000	0.00329	0.01145	2,000	0.00346	0.01285	2,000	0.00268	0.00962
1,500	0.00379	0.01350	1,500	0.00388	0.01475	1,500	0.00294	0.01088
1,200	0.00437	0.0159	1,200	0.00441	0.0170	1,200	0.00338	0.0125
1,000	0.00498	0.0181	1,000	0.00493	0.0195	1,000	0.00379	0.0144
800	0.00589	0.0220	800	0.00555	0.0238	800	0.00448	0.0178
600	0.00729	0.0285	600	0.00713	0.0306	600	0.00561	0.0232
500	0.00833	0.0336	500	0.00815	0.0359	500	0.00658	0.0275
400	0.00980	0.0411	400	0.00955	0.0437	400	0.00796	0.0339
300	0.01215	0.0535	300	0.01185	0.0566	300	0.01020	0.0442
200			200	0.01600	0.0811	200		0.0650

(continued)

Table 10-3 Heat Transfer and Friction Data for Plain Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
25.79T			30.33T			46.45T		
7,000			7,000			7,000		
6,000			6,000			6,000		
5,000			5,000			5,000		
4,000		0.00785	4,000		0.00891	4,000		
3,000	0.00277	0.00831	3,000	0.00293	0.00981	3,000		0.0109
2,000	0.00312	0.00981	2,000	0.00356	0.01185	2,000	0.00294	0.0118
1,500	0.00354	0.01165	1,500	0.00418	0.01395	1,500	0.00349	0.0135
1,200	0.00401	0.0134	1,200	0.00481	0.0162	1,200	0.00418	0.0157
1,000	0.00450	0.0153	1,000	0.00545	0.0189	1,000	0.00482	0.0183
800	0.00529	0.0185	800	0.00643	0.0230	800	0.00581	0.0228
600	0.00670	0.0240	600	0.00802	0.0302	600	0.00735	0.0301
500	0.00782	0.0286	500	0.00922	0.0361	500	0.00856	0.0359
400	0.00942	0.0351	400	0.0110	0.0448	400		0.0446
300	0.01193	0.0460	300	0.0138	0.0595	300		0.0592
200		0.0673	200		0.0888	200		0.0875

Table 10-4 Heat Transfer and Friction Data for Louvered Plate-Fin Surfaces

Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f
$\frac{3}{8}-6.06$			$\frac{3}{8}(a)-6.06$			$\frac{1}{2}-6.06$			$\frac{1}{2}(a)-6.06$			$\frac{3}{8}-8.7$		
10,000	0.00551	0.0331	10,000	0.00638	0.0494	10,000	0.00568	0.0300	10,000	0.00598	0.0400	10,000	0.00542	0.0297
8,000	0.00593	0.0340	8,000	0.00688	0.0510	8,000	0.00605	0.0310	8,000	0.00645	0.0413	8,000	0.00583	0.0306
6,000	0.00651	0.0354	6,000	0.00760	0.0531	6,000	0.00655	0.0322	6,000	0.00714	0.0432	6,000	0.00640	0.0319
5,000	0.00690	0.0363	5,000	0.00810	0.0547	5,000	0.00690	0.0332	5,000	0.00760	0.0447	5,000	0.00678	0.0328
4,000	0.00738	0.0375	4,000	0.00878	0.0568	4,000	0.00734	0.0347	4,000	0.00809	0.0463	4,000	0.00737	0.0340
3,000	0.00805	0.0394	3,000	0.00970	0.0596	3,000	0.00791	0.0366	3,000	0.00895	0.0491	3,000	0.00794	0.0359
2,500	0.00849	0.0406	2,500	0.0102	0.0620	2,500	0.00829	0.0381	2,500	0.00941	0.0511	2,500	0.00835	0.0374
2,000	0.00900	0.0426	2,000	0.0110	0.0646	2,000	0.00875	0.0402	2,000	0.0100	0.0540	2,000	0.00885	0.0394
1,500	0.00970	0.0461	1,500	0.0119	0.0696	1,500	0.00948	0.0438	1,500	0.0108	0.0588	1,500	0.00951	0.0430
1,200	0.0104	0.0496	1,200	0.0127	0.0745	1,200	0.0102	0.0474	1,200	0.0113	0.0634	1,200	0.0103	0.0472
1,000	0.0112	0.0532	1,000	0.0132	0.0795	1,000	0.0109	0.0512	1,000	0.0118	0.0680	1,000	0.0112	0.0515
800	0.0124	0.0587	800	0.0140	0.0860	800	0.0118	0.0571	800	0.0122	0.0752	800	0.0126	0.0585
600	0.0144	0.0682	600	0.0149	0.0962	600	0.0133	0.0667	600	0.0128	0.0880	600	0.0149	0.0700
500	0.0160	0.0755	500			500			500			500	0.0169	0.0793
400			400			400			400			400		
300			300			300			300			300		
$\frac{3}{8}(a)-8.7$			$\frac{1}{16}-11.1$			$\frac{1}{4}-11.1$			$\frac{1}{4}(b)-11.1$			$\frac{3}{8}-11.1$		
10,000			10,000			10,000			10,000			10,000	0.00548	0.0242
8,000	0.00630	0.0372	8,000			8,000	0.00666	0.0309	8,000	0.00701	0.0349	8,000	0.00588	0.0253
6,000	0.00690	0.0395	6,000	0.00690	0.0350	6,000	0.00728	0.0333	6,000	0.00761	0.0364	6,000	0.00645	0.0271
5,000	0.00730	0.0410	5,000	0.00740	0.0367	5,000	0.00771	0.0351	5,000	0.00800	0.0375	5,000	0.00684	0.0283
4,000	0.00790	0.0428	4,000	0.00802	0.0390	4,000	0.00825	0.0374	4,000	0.00853	0.0390	4,000	0.00735	0.0300
3,000	0.00870	0.0450	3,000	0.00899	0.0426	3,000	0.00901	0.0408	3,000	0.00922	0.0412	3,000	0.00811	0.0326
2,500	0.00920	0.0470	2,500	0.00960	0.0452	2,500	0.00954	0.0431	2,500	0.00972	0.0430	2,500	0.00861	0.0346
2,000	0.00980	0.0497	2,000	0.0103	0.0491	2,000	0.0102	0.0464	2,000	0.0103	0.0456	2,000	0.00930	0.0375
1,500	0.0106	0.0550	1,500	0.0113	0.0553	1,500	0.0112	0.0512	1,500	0.0112	0.0502	1,500	0.0102	0.0423
1,200	0.0113	0.0580	1,200	0.0122	0.0610	1,200	0.0119	0.0558	1,200	0.0120	0.0550	1,200	0.0111	0.0469
1,000	0.0121	0.0620	1,000	0.0130	0.0662	1,000	0.0125	0.0600	1,000	0.0128	0.0595	1,000	0.0121	0.0513
800	0.0131	0.0680	800	0.0142	0.0738	800	0.0137	0.0670	800	0.0139	0.0662	800	0.0135	0.0582
600	0.0145	0.0790	600	0.0161	0.0848	600	0.0155	0.0772	600	0.0157	0.0780	600	0.0156	0.0700
500	0.0154	0.0890	500	0.0177	0.0925	500	0.0168	0.0850	500	0.0170	0.0870	500	0.0170	0.0796
400			400			400			400			400		
300			300			300			300			300		

(continued)

Table 10-4 Heat Transfer and Friction Data for Louvered Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
$\frac{3}{8}(b) - 11.1$			$\frac{1}{2} - 11.1$			$\frac{3}{4} - 11.1$			$\frac{3}{4}(b) - 11.1$		
10,000			10,000			10,000	0.00432	0.0151	10,000	0.00440	0.0156
8,000	0.00590	0.0257	8,000	0.00557	0.0220	8,000	0.00462	0.0158	8,000	0.00469	0.0163
6,000	0.00650	0.0271	6,000	0.00604	0.0233	6,000	0.00508	0.0170	6,000	0.00510	0.0175
5,000	0.00694	0.0281	5,000	0.00640	0.0242	5,000	0.00537	0.0178	5,000	0.00537	0.0183
4,000	0.00752	0.0296	4,000	0.00680	0.0255	4,000	0.00576	0.0190	4,000	0.00572	0.0194
3,000	0.00835	0.0319	3,000	0.00739	0.0271	3,000	0.00630	0.0208	3,000	0.00621	0.0213
2,500	0.00889	0.0336	2,500	0.00777	0.0283	2,500	0.00663	0.0222	2,500	0.00655	0.0227
2,000	0.00960	0.0363	2,000	0.00825	0.0299	2,000	0.00711	0.0244	2,000	0.00699	0.0248
1,500	0.0105	0.0406	1,500	0.00888	0.0332	1,500	0.00787	0.0289	1,500	0.00762	0.0288
1,200	0.0112	0.0442	1,200	0.00950	0.0368	1,200	0.00859	0.0330	1,200	0.00831	0.0323
1,000	0.0119	0.0483	1,000	0.0104	0.0410	1,000	0.00928	0.0370	1,000	0.00894	0.0362
800	0.0130	0.0550	800	0.0117	0.0474	800	0.0103	0.0427	800	0.00981	0.0416
600	0.0148	0.0659	600	0.0137	0.0570	600	0.0119	0.0516	600	0.0112	0.0500
500	0.0161	0.0741	500	0.0150	0.0641	500	0.0132	0.0580	500	0.0122	0.0565
400			400			400			400		
300			300			300			300		

Table 10-5 Heat Transfer and Friction Data for Strip-Fin Plate-Fin Surfaces

Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f
$\frac{1}{8}(s) - 11.1$			$\frac{1}{8} - 15.2$			$\frac{1}{8} - 13.95$		
10,000			10,000			8,000		0.0628
8,000	0.00525	0.0197	8,000			6,000	0.01110	0.0650
6,000	0.00580	0.0209	6,000	0.00850	0.0487	5,000	0.01170	0.0664
5,000	0.00620	0.0218	5,000	0.00896	0.0498	4,000	0.01250	0.0684
4,000	0.00669	0.0231	4,000	0.00959	0.0516	3,000	0.0137	0.0712
3,000	0.00740	0.0253	3,000	0.01040	0.0540	2,500	0.0144	0.0733
2,500	0.00789	0.0272	2,500	0.01110	0.0558	2,000	0.0155	0.0765
2,000	0.00850	0.0298	2,000	0.01177	0.0584	1,500	0.0168	0.0817
1,500	0.00940	0.0348	1,500	0.01267	0.0628	1,200	0.0181	0.0870
1,200	0.0102	0.0394	1,200	0.01327	0.0676	1,000	0.0192	0.0927
1,000	0.0109	0.0438	1,000	0.01373	0.0726	800	0.0204	0.1020
800	0.0122	0.0500	800	0.01427	0.0800	600	0.0223	0.1170
600	0.0139	0.0595	600	0.01520	0.0913	500	0.0233	0.131
500	0.0155	0.0665	500	0.01580	0.1010	400	0.0247	0.154
400			400	0.01675	0.1145			
300			300	0.01810	0.1390			
$\frac{1}{8} - 15.61$			$\frac{1}{8} - 19.86$			$\frac{1}{8} - 22.68$		
7,000	—	0.0330	5,000	—	0.0275	5,000	—	0.0298
6,000	0.00758	0.0336	4,000	0.00770	0.0284	4,000	0.00721	0.0310
5,000	0.00802	0.0343	3,000	0.00850	0.0306	3,000	0.00810	0.0332
4,000	0.00865	0.0355	2,000	0.00988	0.0350	2,000	0.00960	0.0382
3,000	0.00952	0.0376	1,500	0.0110	0.0396	1,500	0.0108	0.0430
2,000	0.0111	0.0420	1,200	0.0119	0.0440	1,200	0.0119	0.0476
1,500	0.0125	0.0467	1,000	0.0128	0.0484	1,000	0.0130	0.0523
1,200	0.0138	0.0519						
1,000	0.0150	0.0569						

(continued)

Table 10-5 Heat Transfer and Friction Data for Strip-Fin Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f
$\frac{1}{8} - 15.61$			$\frac{1}{8} - 19.86$			$\frac{1}{8} - 22.68$		
800	0.0167	0.0645	800	0.0141	0.0550	800	0.0147	0.0600
600	0.0195	0.0772	600	0.0161	0.0660	600	0.0176	0.0736
500	0.0216	0.0880	500	0.0176	0.0750	500	0.0199	0.0845
400	0.0246	0.104	400	0.0198	0.0876	400	0.0234	0.0997
300	0.0295	0.131	300	0.0233	0.1090	300	0.0290	0.125
						200	—	0.172
						150	—	0.217
						120	—	0.260
$\frac{1}{8} - 25.01$			$\frac{1}{8} - 24.12$			$\frac{1}{16} - 27.03$		
5,000	—	—				5,000	—	—
4,000	—	0.0385				4,000	—	0.0388
3,000	0.00950	0.0398	3,000	—	0.0362	3,000	0.00960	0.0401
2,000	0.0107	0.0437	2,000	0.00965	0.0398	2,000	0.0113	0.0449
1,500	0.0121	0.0483	1,500	0.0103	0.0432	1,500	0.0129	0.0497
1,200	0.0133	0.0528	1,200	0.0110	0.0467	1,200	0.0143	0.0540
1,000	0.0145	0.0573	1,000	0.0117	0.0500	1,000	0.0156	0.0586
800	0.0162	0.0647	800	0.0125	0.0551	800	0.0176	0.0663
600	0.0187	0.0772	600	0.0138	0.0639	600	0.0210	0.0810
500	0.0207	0.0880	500	0.0152	0.0713	500	0.0238	0.0922
400	—	0.103	400	0.0170	0.0828	400	—	0.108
300	—	0.128	300	0.0202	0.102	300	—	0.135
200	—	0.175	200	—	0.137	200	—	0.185
150	—	0.219	150	—	0.171	150	—	0.231
120	—	0.261				120	—	0.276
$\frac{1}{16} - 19.35$			$\frac{1}{16} - 19.74$			$\frac{1}{32} - 12.22$		
5,000	—	—	5,000	—	—	10,000	0.00629	0.0394
4,000	—	0.0246	4,000	—	—	8,000	0.00688	0.0413

3,000	0.00830	0.0263	3,000	0.00764	0.0339	6,000	0.00770	0.0440
2,000	0.00940	0.0303	2,000	0.00843	0.0369	5,000	0.00828	0.0458
1,500	0.0105	0.0342	1,500	0.00910	0.0398	4,000	0.00903	0.0487
1,200	0.0115	0.0381	1,200	0.00974	0.0430	3,000	0.0101	0.0530
1,000	0.0125	0.0418	1,000	0.0104	0.0461	2,500	0.0108	0.0560
800	0.0138	0.0471	800	0.0114	0.0510	2,000	0.0119	0.0607
600	0.0158	0.0561	600	0.0129	0.0599	1,500	0.0133	0.0680
500	0.0174	0.0635	500	0.0141	0.0672	1,200	0.0146	0.0752
400	0.0195	0.0750	400	0.0158	0.0783	1,000	0.0156	0.0826
300	—	0.0937	300	0.0185	0.0962	800	0.0171	0.0942
200	—	0.130	200	—	0.131	600	0.0192	0.113
						500	0.0205	0.130
						400		
						300		
$\frac{1}{2} - 11.94(D)$			$\frac{1}{4} - 15.4(D)$			$\frac{1}{8} - 12.18(D)$		
9,000		0.0122				9,000	0.00512	0.0183
8,000		0.0123				8,000	0.00530	0.0184
7,000	0.00452	0.0126				7,000	0.00557	0.0189
6,000	0.00471	0.0131	6,000	0.00510	0.0167	6,000	0.00591	0.0196
5,000	0.00492	0.0137	5,000	0.00537	0.0175	5,000	0.00635	0.0203
4,000	0.00522	0.0146	4,000	0.00570	0.0186	4,000	0.00692	0.0218
3,000	0.00575	0.0162	3,000	0.00617	0.0202	3,000	0.00782	0.0241
			2,500	0.00650	0.0213	2,000		
2,000	0.00682	0.0198	2,000	0.00692	0.0228	1,500	0.00933	0.0290
1,500	0.00744	0.0231	1,500	0.00756	0.0255		0.01065	0.0341
						1,200		0.0388
1,200	0.00830	0.0265	1,200	0.00809	0.0283	1,000	0.0119	0.0438
1,000	0.00911	0.0306	1,000	0.00864	0.0314	800	0.0129	0.0490
800	0.01045	0.0347	800	0.00952	0.0362	600	0.0141	0.0592
600	0.01255	0.0429	600	0.01107	0.0443	500	0.0169	0.0695
500	0.01415	0.0493	500	0.01227	0.0507	400	0.0191	0.0808
400	0.0166	0.0592	400	0.01407	0.0601	300	0.0230	0.1025
300	0.0205	0.0758	300	0.0169	0.0757	200	0.0278	0.1480

(continued)

Table 10-5 Heat Transfer and Friction Data for Strip-Fin Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
$\frac{1}{4} - 15.75(D)$			$\frac{1}{4} - 16.00(D)$			$\frac{1}{4} - 16.12(D)$		
9,000								
8,000								
7,000		0.0199						
6,000	0.00619	0.0203						
5,000	0.00649	0.0211	5,000	0.00778	0.0295			
4,000	0.00713	0.0227	4,000	0.00838	0.0307			
3,000	0.00813	0.0248	3,000	0.00925	0.0328			
2,000	0.00992	0.0294	2,000	0.01085	0.0373			
1,500	0.01125	0.0339	1,500	0.01205	0.0418	1,500	0.00987	0.0400
1,200	0.0124	0.0386	1,200	0.0132	0.0460	1,200	0.01060	0.0429
1,000	0.0136	0.0440	1,000	0.0142	0.0502	1,000	0.01123	0.0464
800	0.0154	0.0499	800	0.0159	0.0568	800	0.01205	0.0517
600	0.0185	0.0608	600	0.0188	0.0675	600	0.01352	0.0607
500	0.0209	0.0690	500	0.0209	0.0765	500	0.0150	0.0679
400		0.0819	400		0.0892	400	0.0176	0.0781
300		0.1025	300		0.1115	300	0.0226	0.0937
200		0.1425	200					

‡-19.82(D)			‡-20.06(D)			‡-16.12(T)		
			9,000					
			8,000					
			7,000					
			6,000					
			5,000			5,000		0.0358
4,000		0.0390	4,000		0.0289	4,000	0.00705	0.0370
3,000	0.00880	0.0420	3,000	0.00855	0.0309	3,000	0.00773	0.0386
2,000	0.01015	0.0450	2,000	0.00995	0.0349	2,500	0.00820	0.0397
1,500	0.01155	0.0492	1,500	0.01115	0.0387	2,000	0.00880	0.0416
1,200	0.0128	0.0535	1,200	0.0120	0.0422	1,500	0.00967	0.0440
1,000	0.0139	0.0577	1,000	0.0129	0.0459	1,200	0.01045	0.0464
800	0.0154	0.0640	800	0.0144	0.0520	1,000	0.01122	0.0496
600	0.0180	0.0747	600	0.0173	0.0621	800	0.01225	0.0552
500	0.0202	0.0832	500	0.0197	0.0699	600	0.01395	0.0644
						500	0.0155	0.0721
400		0.0955	400		0.0810			
300		0.1175	300		0.0995	400	0.0179	0.0830
200		0.1585	200			300	0.0219	0.1003

Table 10-6 Heat Transfer and Friction Data for Wavy-Fin Plate-Fin Surfaces

11.44— $\frac{1}{8}$ W			11.5— $\frac{1}{8}$ W			17.8— $\frac{1}{8}$ W		
Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
8,000	0.00712	0.0359	10,000	0.00686	0.0331			
6,000	0.00794	0.0401	8,000	0.00746	0.0357			
5,000	0.00846	0.0430	6,000	0.00831	0.0398			
4,000	0.00920	0.0469	5,000	0.00890	0.0427	5,000	0.00675	0.0293
			4,000	0.00970	0.0467	4,000	0.00740	0.0320
3,000	0.01025	0.0524	3,000	0.01077	0.0525	3,000	0.00835	0.0358
2,500	0.0110	0.0563	2,500	0.01155	0.0567	2,500	0.00900	0.0385
2,000	0.0119	0.0615	2,000	0.0126	0.0625	2,000	0.00982	0.0421
1,500	0.0132	0.0691	1,500	0.0140	0.0704	1,500	0.0110	0.0478
1,200	0.0144	0.0758	1,200	0.0150	0.0779	1,200	0.0120	0.0530
1,000	0.0153	0.0819	1,000	0.0158	0.0845	1,000	0.0129	0.0579
800	0.0165	0.0888	800	0.0167	0.0926	800	0.0142	0.0643
600	0.0175	0.0984	600	0.0178	0.1035	600	0.0158	0.0738
500	0.0179	0.1045	500	0.0185	0.111	500		
400			400	0.0194	0.118	400		
300						300		

Table 10-7 Heat Transfer and Friction Data for Pin-Fin Plate-Fin Surfaces

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
AP-1			AP-2			PF-3		
8,000	0.00808	0.0815						
6,000	0.00905	0.0793						
5,000	0.00970	0.0780						
4,000	0.0105	0.0762	4,000	0.0130	0.166			
3,000	0.0116	0.0741	3,000	0.0145	0.165	2,000	0.00645	
2,500	0.0123	0.0725	2,500	0.0155	0.164	1,500	0.00720	
2,000	0.0134	0.0707	2,000	0.0168	0.158	1,200	0.00784	0.0373
1,500	0.0152	0.0708	1,500	0.0183	0.151	1,000	0.00840	0.0383
1,200	0.0161	0.0718	1,200	0.0190	0.150	800	0.00912	0.0399
1,000	0.0175	0.0755	1,000	0.0200	0.151	600	0.01020	0.0430
800	0.0183	0.0800	800	0.0209	0.156	500	0.01085	0.0454
600			600	0.0218	0.170	400	0.0118	0.0491
500			500	0.0222	0.180	300	0.0132	0.0551
400			400			250		0.0600
300			300			200		0.0673
PF-4(F)			PF-9(F)			PF-10(F)		
			25,000	0.00740		15,000	0.00757	0.0488
			20,000	0.00800		12,000	0.00826	0.0493
			17,000	0.00842		10,000	0.00890	0.0503
8,000	0.0128	0.153	15,000	0.00880		8,000	0.00973	0.0530
6,000	0.0144	0.153	12,000	0.00951		6,000	0.01085	0.0573
5,000	0.0156	0.153	10,000	0.0101		5,000	0.01172	0.0589
4,000	0.0171	0.153	8,000	0.0108		4,000	0.0128	0.0603
3,000	0.0192	0.156	6,000	0.0120		3,000	0.0142	0.0626
2,500	0.0207	0.160	5,000	0.0128		2,500	0.0157	0.0640
2,000	0.0227	0.167	4,000	0.0137	0.0692	2,000	0.0170	0.0640

(continued)

Table 10-7 Heat Transfer and Friction Data for Pin-Fin Plate-Fin Surfaces (*continued*)

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
PF-4(F)			PF-9(F)			PF-10(F)		
1,500	0.0255	0.175	3,000	0.0152	0.0718	1,500	0.0170	0.0605
1,200	0.0279	0.182	2,500	0.0161	0.0737	1,200	0.0174	0.0561
1,000	0.0301	0.189	2,000	0.0173	0.0760	1,000	0.0182	0.0577
800	0.0330	0.198	1,500	0.0191	0.0800	800	0.0198	0.0670
600			1,200					
500			1,000					
400			900					
300			800					

Table 10-8 Perforated-Fin Plate-Fin Surfaces

Re	StPr ^{2/3}	<i>f</i>
13.95(P)		
12,000		0.01168
10,000	0.00503	0.01220
8,000	0.00547	0.01285
6,000	0.00600	0.01386
5,000	0.00631	0.0146
4,000	0.00660	0.0155
3,000	0.00695	0.0169
2,500	0.00708	0.0178
2,000	0.00700	0.0187
1,500	0.00736	0.0207
1,200	0.00817	0.0232
1,000	0.00893	0.0257
800	0.01010	0.0295
600	0.01180	0.0357
500	0.0132	0.0407

Table 10-9 Heat Transfer and Friction Data for Flow Normal to Banks of Finned Tubes

Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f	Re	StPr ^{2/3}	f
CF-7.34			CF-8.72			CF-8.72C		
10,000	0.00388	0.0304	10,000		0.0306	10,000		
8,000	0.00465	0.0313	8,000		0.0314	8,000	0.00638	0.0347
6,000	0.00500	0.0326	6,000	0.00645	0.0326	6,000	0.00685	0.0359
5,000	0.00547	0.0334	5,000	0.00685	0.0333	5,000	0.00727	0.0368
4,000	0.00609	0.0348	4,000	0.00740	0.0344	4,000	0.00785	0.0381
3,000	0.00701	0.0365	3,000	0.00820	0.0360	3,000	0.00879	0.0401
2,500	0.00770	0.0379	2,500	0.00880	0.0371	2,500	0.00945	0.0418
2,000	0.00860	0.0395	2,000	0.00960	0.0388	2,000	0.01035	0.0440
1,500	0.00990	0.0420	1,500	0.0108	0.0412	1,500	0.01170	0.0470
1,200	0.01105	0.0441	1,200	0.0119	0.0433	1,200	0.01285	0.0495
1,000	0.01210	0.0461	1,000	0.0129	0.0453	1,000	0.01390	0.0520
800	0.01350	0.0490	800	0.0143	0.0483	800	0.01504	0.0551
600	0.01506	0.0530	600	0.0165	0.0532	600	0.01760	0.0597
500			500	0.0180	0.0572	500	0.01915	0.0630
400			400			400		
300			300			300		
CF-11.46			9.68—.87			9.1—.737-S		
10,000		0.0255	10,000	0.00326	0.0106	10,000	0.00389	0.0133
8,000		0.0260	8,000	0.00339	0.0108	8,000	0.00417	0.0142
6,000		0.0267	6,000	0.00359	0.0113	6,000	0.00460	0.0156
5,000		0.0272	5,000	0.00369	0.0116	5,000	0.00493	0.0167
4,000		0.0278	4,000	0.00385	0.0122	4,000	0.00539	0.0183
3,000		0.0290	3,000	0.00405	0.0133	3,000	0.00608	0.0206
2,500		0.0300	2,500	0.00411	0.0142	2,500	0.00656	0.0223
2,000		0.0314	2,000	0.00427	0.0155	2,000	0.00726	0.0247
1,500		0.0337	1,500	0.00477	0.0174	1,500	0.00831	0.0284
1,200		0.0358	1,200	0.00536	0.0196	1,200	0.00934	0.0320

1,000		0.0379	1,000	0.00600	0.0219	1,000	0.01030	0.0354
800		0.0408	800	0.00701	0.0256	800	0.01175	0.0402
600		0.0450	600	0.00861	0.0321	600	0.01377	0.0476
500		0.0481	500	0.00982	0.0376	500	0.01526	0.0531
400		0.0522	400	0.01150	0.0463	400		
300			300			300		
9.68—.87-R			9.29—.737-SR			11.32—.737-SR		
10,000	0.00390	0.0183	10,000	0.00514	0.0206	10,000	0.00453	0.0180
8,000	0.00412	0.0191	8,000	0.00546	0.0219	8,000	0.00468	0.0192
6,000	0.00445	0.0201	6,000	0.00590	0.0236	6,000	0.00533	0.0207
5,000	0.00468	0.0208	5,000	0.00620	0.0247	5,000	0.00563	0.0218
4,000	0.00498	0.0217	4,000	0.00660	0.0261	4,000	0.00602	0.0232
3,000	0.00543	0.0228	3,000	0.00712	0.0280	3,000	0.00655	0.0253
2,500	0.00573	0.0238	2,500	0.00751	0.0294	2,500	0.00690	0.0269
2,000	0.00615	0.0255	2,000	0.00802	0.0315	2,000	0.00739	0.0290
1,500	0.00676	0.0284	1,500	0.00885	0.0346	1,500	0.00816	0.0323
1,200	0.00727	0.0309	1,200	0.00964	0.0380	1,200	0.00892	0.0353
1,000	0.00772	0.0333	1,000	0.01045	0.0416	1,000	0.00968	0.0384
800	0.00832	0.0366	800	0.01176	0.0472	800	0.0108	0.0429
600	0.00920	0.0422	600	0.01390	0.0569	600	0.0127	0.0505
500	0.01000	0.0470	500			500	0.0142	0.0565
400	0.01128	0.0550	400			400	0.0163	0.0652
300			300			300		

Table 10-10 Heat Transfer and Friction Data for Crossed-Rod Matrices

Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>	Re	StPr ^{2/3}	<i>f</i>
In-line stacking														
1I			3I			5I			6I			7I		
120,000	0.00784	0.255	80,000	0.00804	0.223	60,000	0.00820	0.213	40,000	0.00865	0.185	30,000	0.00860	0.153
100,000	0.00815	0.227	60,000	0.00872	0.230	40,000	0.00940	0.226	30,000	0.00930	0.190	20,000	0.00995	0.163
80,000	0.00858	0.228	40,000	0.00990	0.239	30,000	0.0103	0.235	20,000	0.0107	0.202	15,000	0.0111	0.171
60,000	0.00922	0.230	30,000	0.0110	0.243	20,000	0.0119	0.248	15,000	0.0120	0.213	10,000	0.0129	0.183
40,000	0.0105	0.232	20,000	0.0126	0.252	15,000	0.0132	0.257	10,000	0.0140	0.228	8,000	0.0140	0.190
30,000	0.0115	0.234	15,000	0.0142	0.257	10,000	0.0154	0.267	8,000	0.0155	0.234	6,000	0.0156	0.200
20,000	0.0133	0.236	10,000	0.0166	0.264	8,000	0.0167	0.273	6,000	0.0174	0.241	4,000	0.0183	0.212
15,000	0.0147	0.240	8,000	0.0182	0.268	6,000	0.0188	0.280	4,000	0.0206	0.251	3,000	0.0204	0.222
10,000	0.0170	0.247	6,000	0.0203	0.273	4,000	0.0221	0.290	3,000	0.0233	0.259	2,000	0.0239	0.233
8,000	0.0185	0.250	4,000	0.0236	0.278	3,000	0.0247	0.298	2,000	0.0276	0.267	1,500	0.0270	0.242
6,000	0.0206	0.256	3,000	0.0264	0.283	2,000	0.0294	0.310	1,500	0.0311	0.274	1,000	0.0324	0.252
4,000	0.0242	0.267	2,000	0.0304	0.295	1,500	0.0332	0.321	1,000	0.0370	0.291			
3,000	0.0268		1,500	0.0337	0.308	1,000	0.0397	0.347						
2,000	0.0306		1,000	0.0388	0.300	800	0.0436	0.361						
1,500	0.0337					600	0.0492	0.386						
Staggered stacking														
1S			3S			5S			6S			7S		
120,000	0.00840	0.356	80,000	0.00798	0.253	60,000	0.00704	0.173	40,000	0.00745	0.142	30,000	0.00806	0.132
100,000	0.00880	0.370	60,000	0.00888	0.267	40,000	0.00832	0.189	30,000	0.00825	0.150	20,000	0.00932	0.142
80,000	0.00942	0.383	40,000	0.0101	0.292	30,000	0.00935	0.200	20,000	0.00973	0.164	15,000	0.0104	0.151
60,000	0.0103	0.401	30,000	0.0111	0.308	20,000	0.0110	0.217	15,000	0.0110	0.175	10,000	0.0121	0.165
40,000	0.0119	0.429	20,000	0.0130	0.330	15,000	0.0124	0.230	10,000	0.0130	0.019	8,000	0.0132	0.174
30,000	0.0133	0.443	15,000	0.0145	0.342	10,000	0.0148	0.249	8,000	0.0142	0.198	6,000	0.0147	0.185
20,000	0.0155	0.463	10,000	0.0173	0.361	8,000	0.0162	0.258	6,000	0.0160	0.209	4,000	0.0172	0.200
15,000	0.0175	0.478	8,000	0.0190	0.371	6,000	0.0185	0.271	4,000	0.0189	0.223	3,000	0.0192	0.210
10,000	0.0208	0.498	6,000	0.0215	0.383	4,000	0.0222	0.290	3,000	0.0212	0.235	2,000	0.0225	0.222
8,000	0.0227	0.510	4,000	0.0256	0.401	3,000	0.0255	0.307	2,000	0.0250	0.251	1,500	0.0254	0.231
6,000	0.0253	0.530	3,000	0.0293	0.419	2,000	0.0310	0.331	1,500	0.0286	0.263	1,000	0.0305	0.243
4,000	0.0292	0.561	2,000	0.0351	0.452	1,500	0.0356	0.351	1,000	0.0351	0.280			
3,000	0.0328		1,500	0.0401	0.480	1,000	0.0431	0.383						
2,000	0.0390		1,000	0.0488	0.527	800	0.0480	0.401						
1,500	0.0451					600	0.0550	0.430						

Random stacking

1R			2R			3R			4R			5R		
100,000	0.0086	0.280	80,000	0.0089	0.277	80,000	0.0081	0.232	80,000	0.0074	0.215	80,000	0.0070	
80,000	0.0093	0.290	60,000	0.0098	0.290	60,000	0.0090	0.243	60,000	0.0083	0.230	60,000	0.0079	0.190
60,000	0.0103	0.301	40,000	0.0114	0.308	40,000	0.0105	0.267	40,000	0.0099	0.248	40,000	0.0093	0.207
40,000	0.0117	0.315	30,000	0.0126	0.313	30,000	0.0118	0.280	30,000	0.0110	0.260	30,000	0.0103	0.218
30,000	0.0130	0.323	20,000	0.0145	0.330	20,000	0.0137	0.297	20,000	0.0120	0.274	20,000	0.0120	0.230
20,000	0.0150	0.334	10,000	0.0190	0.346	10,000	0.0178	0.318	10,000	0.0165	0.296	10,000	0.0155	0.254
10,000	0.0196	0.350	8,000	0.0208	0.350	8,000	0.0194	0.320	8,000	0.0180	0.300	8,000	0.0170	0.262
8,000	0.0215	0.354	6,000	0.0232	0.356	6,000	0.0215	0.329	6,000	0.0200	0.307	6,000	0.0187	0.270
6,000	0.0241	0.360	4,000	0.0272	0.362	4,000	0.0251	0.339	4,000	0.0231	0.317	4,000	0.0220	0.283
4,000	0.0283	0.370	3,000	0.0308	0.371	3,000	0.0281	0.343	3,000	0.0258	0.322	3,000	0.0247	0.290
3,000	0.0319	0.379	2,000	0.0364	0.381	2,000	0.0330	0.355	2,000	0.0300	0.349	2,000	0.0289	0.299
2,000	0.0379		1,000	0.0500		1,000	0.0443	0.375	1,000	0.0400		1,000	0.0382	0.330
1,000	0.0520		800	0.0560		800	0.0490	0.381	800	0.0442		800	0.0420	0.350
800	0.0580		600	0.0640		600	0.0560		600	0.0509		600	0.0479	0.384
600	0.0670								400	0.0620		400	0.0580	
									300	0.0720		300	0.0670	
6R						7R								
			60,000		0.154	30,000	0.00838	0.157						
			40,000	0.0086	0.169	20,000	0.00980	0.170						
			30,000	0.0094	0.180	15,000	0.0110	0.179						
			20,000	0.0108	0.195	10,000	0.0129	0.194						
			10,000	0.0138	0.218	8,000	0.0140	0.200						
			8,000	0.0150	0.227	6,000	0.0156	0.210						
			6,000	0.0167	0.237	4,000	0.0183	0.222						
			4,000	0.0195	0.250	3,000	0.0204	0.229						
			3,000	0.0219	0.257	2,000	0.0239	0.238						
			2,000	0.0255	0.263	1,500	0.0270	0.244						
			1,000	0.0340	0.286	1,000	0.0324	0.252						
			800	0.0375	0.303									
			600	0.0422	0.331									
			400	0.0510										
			300	0.0590										
			200	0.0730										

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Table 10-11 Plate-Fin Matrices

Re	$StPr^{2/3}$	f
50IMOD		
800	0.0116	0.0505
600	0.0130	0.0568
500	0.0142	0.0622
400	0.0161	0.0700
300	0.0197	0.0845
200	0.0272	0.113
150	0.0347	0.142
100	0.0490	0.196
80	0.0595	0.237
60	0.077	0.301
50	0.091	0.352

Table 10-12 Glass Ceramic Matrices

Re	$StPr^{2/3}$	f
505A		
700	0.0044	0.0210
500	0.0063	0.0290
400	0.0078	0.0357
300	0.0104	0.047
200	0.0155	0.069
150	0.0207	0.091
100	0.0310	0.132
80	0.0385	0.163
519		
800	0.00505	0.0229
600	0.00675	0.0298
500	0.00810	0.0353
400	0.0102	0.0438
300	0.0136	0.0577
200	0.0205	0.0856
150	0.0270	0.113
120	0.0340	0.142
100	0.0405	0.170
80	0.0510	0.212

Table 10-13 Heat Transfer and Friction Data for Sphere-Bed Matrices†

Random packing, $p = 0.37$ to 0.39		
Re	$StPr^{2/3}$	f
50,000	0.0089	0.30
20,000	0.0118	0.34
10,000	0.0144	0.37
5,000	0.0178	0.41
2,000	0.023	0.47
1,000	0.029	0.52
500	0.0355	0.59
200	0.046	0.80
100	0.056	1.10
50	0.069	1.65
20	0.091	0.30
10	0.112	5.2

† See Fig. 7-10 for graph of these data.

Fig. 10-1 Flow inside circular tubes, surface ST-1.

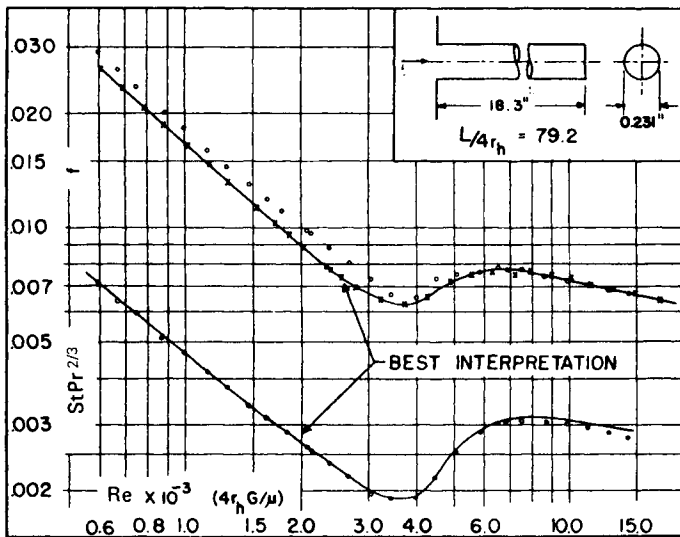
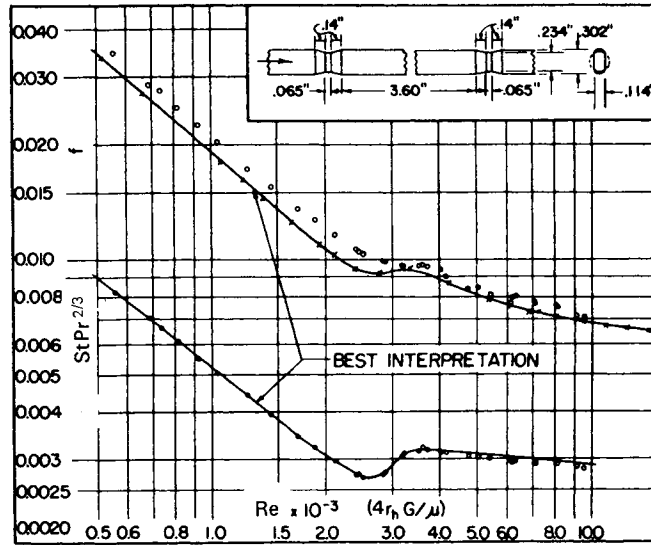
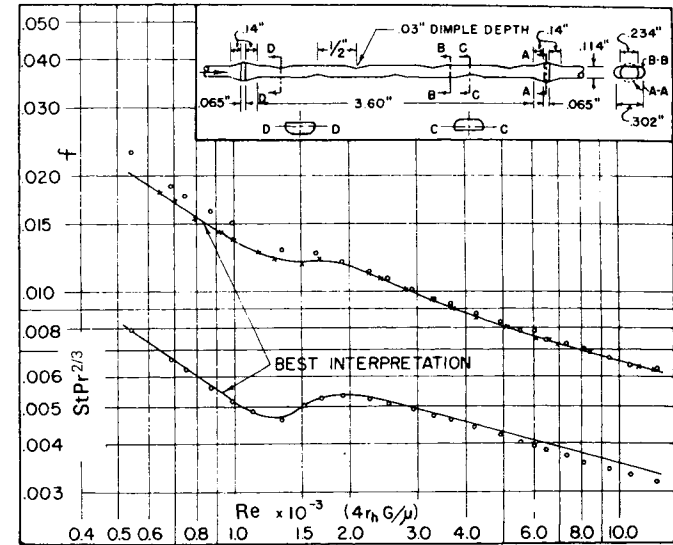
Tube inside diameter = 0.231 in = 5.876×10^{-3} mHydraulic diameter = 0.231 in, 0.01925 ft = 5.876×10^{-3} mFlow length/hydraulic diameter, $L/4r_h = 79.2$ Free-flow area per tube = 0.0002908 ft² = 27.016×10^{-6} m²

Fig. 10-2 Flow inside flattened circular tubes, surface FT-1.



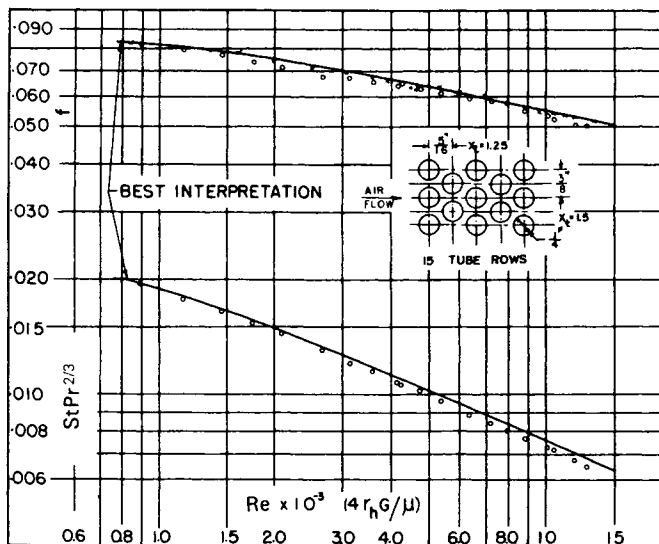
Tube ID before flattening = 0.234 in = 5.94×10^{-3} m
 Tube inside dimension perpendicular to flats = 0.114 in = 2.90×10^{-3} m
 Tube inside dimension parallel to flats = 0.302 in = 7.67×10^{-3} m
 Length of flat along tube = 3.60 in = 91.44×10^{-3} m
 Length of section from flat-to-circular-to-flat cross section
 = 0.345 in = 8.77×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01433$ ft = 4.368×10^{-3} m
 Minimum free-flow area per tube = 0.0002195 ft² = 20.392×10^{-6} m²

Fig. 10-3 Flow inside dimpled flattened tubes, surface FTD-1.



Tube ID before flattening = 0.234 in = 5.94×10^{-3} m
 Tube inside dimension perpendicular to flats = 0.114 in = 2.90×10^{-3} m
 Tube inside dimension parallel to flats = 0.302 in = 7.67×10^{-3} m
 Length of flat along tube = 3.60 in = 91.4×10^{-3} m
 Length of section from flat-to-circular-to-flat cross section
 = 0.345 in = 8.77×10^{-3} m
 Dimple depth = 0.03 in = 0.76×10^{-3} m
 Minimum distance between dimples = 0.5 in = 12.5×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01116$ ft = 3.339×10^{-3} m
 Minimum free-flow area per tube = 0.000169 ft² = 15.501×10^{-6} m²

Fig. 10-4 Flow normal to a staggered tube bank (steady-state tests), surface S 1.50–1.25(a).



Tube outside diameter = 0.250 in = 6.35×10^{-3} m

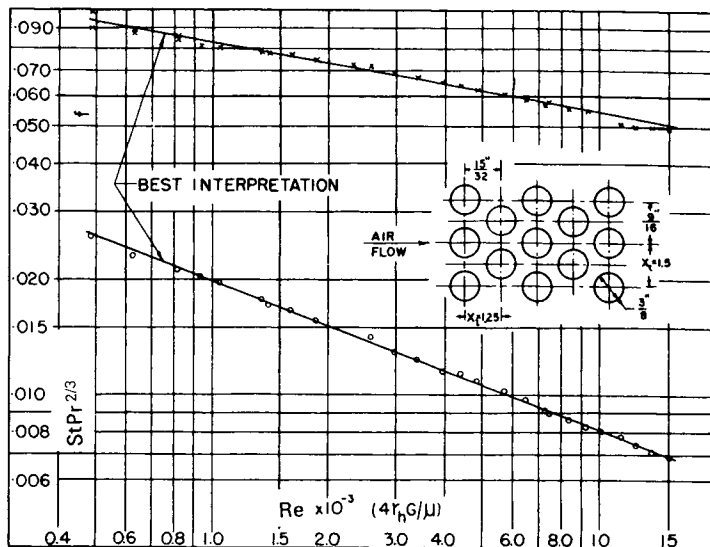
Hydraulic diameter, $4 r_h = 0.0166$ ft = 5.029×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.333$

Heat transfer area/total volume, $\alpha = 80.3$ ft²/ft³ = 263.451 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-5 Flow normal to a staggered tube bank (transient tests), surface S 1.50–1.25.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m

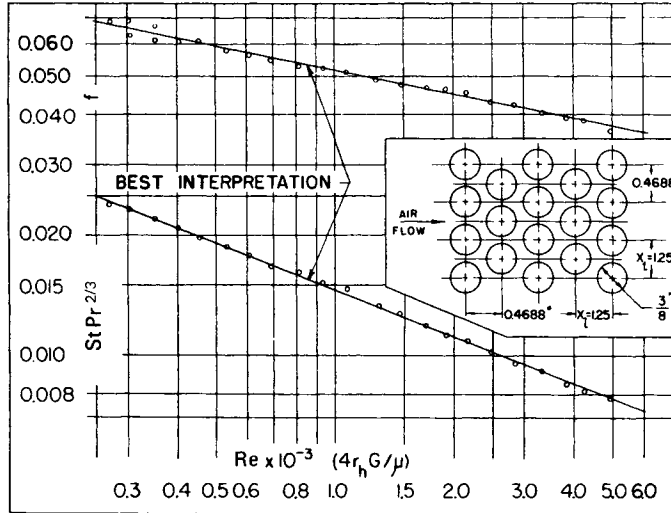
Hydraulic diameter, $4 r_h = 0.0249$ ft = 7.569×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.333$

Heat transfer area/total volume, $\alpha = 53.6$ ft²/ft³ = 175.197 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-6 Flow normal to a staggered tube bank (transient tests), surface S 1.25-1.25.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m

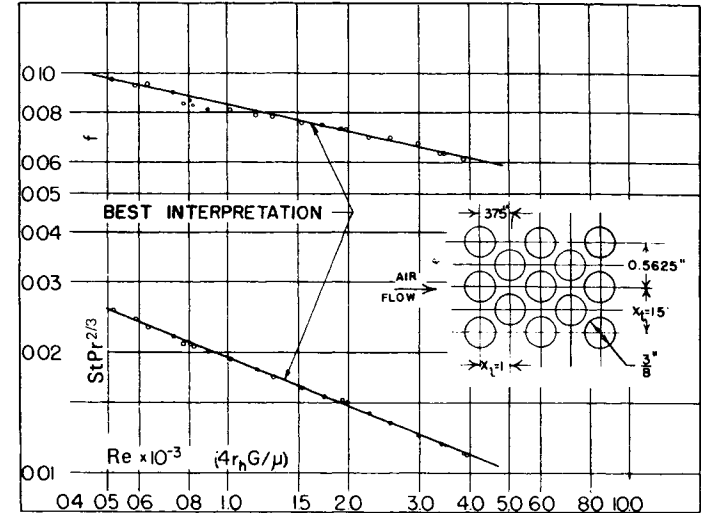
Hydraulic diameter, $4r_h = 0.0125$ ft = 3.810×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.200$

Heat transfer area/total volume, $\alpha = 64.4$ ft²/ft³ = 211.286 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-7 Flow normal to a staggered tube bank (transient tests), surface S 1.50-1.00.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m

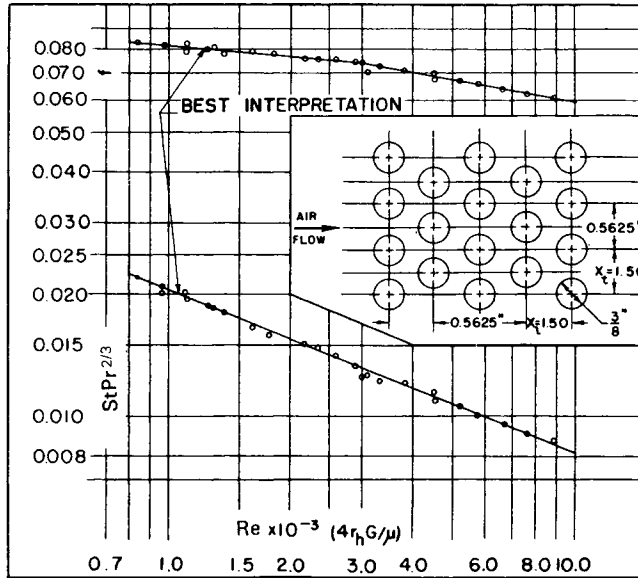
Hydraulic diameter, $4r_h = 0.0196$ ft = 6.071×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.333$

Heat transfer area/total volume, $\alpha = 67.1$ ft²/ft³ = 220.144 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-8 Flow normal to a staggered tube bank (transient tests), surface S 1.50–1.50.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m

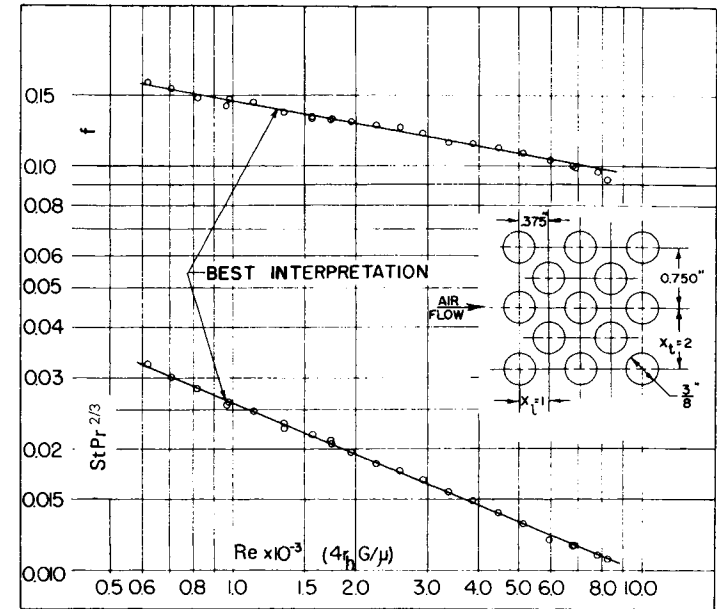
Hydraulic diameter, $4r_h = 0.0298$ ft = 9.093×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.333$

Heat transfer area/total volume, $\alpha = 44.8$ ft²/ft³ = 146.982 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-9 Flow normal to a staggered tube bank (transient tests), surface S 2.00–1.00.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m

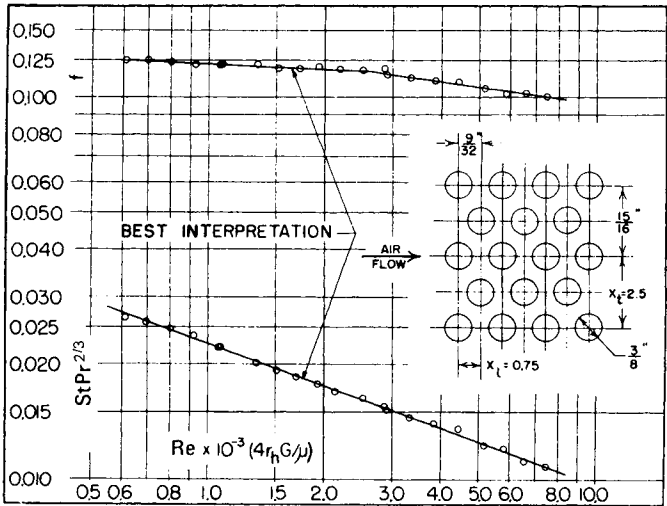
Hydraulic diameter, $4r_h = 0.0327$ ft = 9.957×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.414$

Heat transfer area/total volume, $\alpha = 50.3$ ft²/ft³ = 165.026 m²/m³

Note: Minimum free-flow area is in diagonal spaces.

Fig. 10-10 Flow normal to a staggered tube bank (transient tests), surface S 2.50-0.75.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m
Hydraulic diameter, $4r_h = 0.0271$ ft = 8.255×10^{-3} m
Free-flow area/frontal area, $\sigma = 0.366$
Heat transfer area/total volume, $\alpha = 53.6$ ft²/ft³ = 175.853 m²/m³
Note: Minimum free-flow area is in diagonal spaces.

Fig. 10-11 Flow normal to staggered tube banks. (Data of Grimison.)

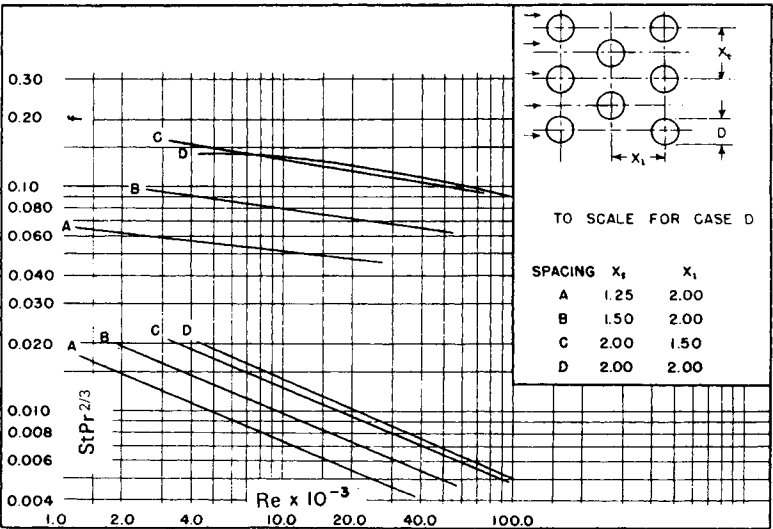
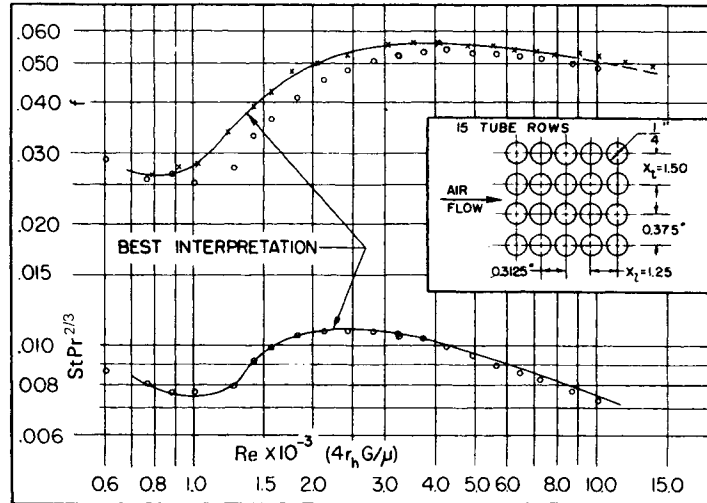


Fig. 10-12 Flow normal to an in-line tube bank (steady-state tests), surface I 1.50-1.25(a).



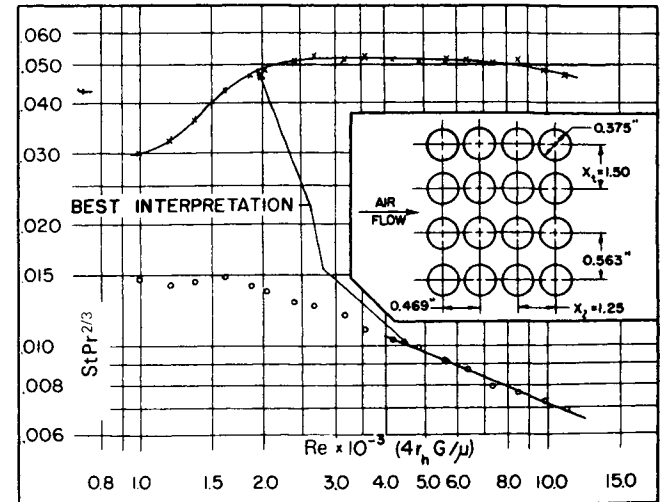
Tube outside diameter = 0.250 in = 6.35×10^{-3} m

Hydraulic diameter, $4r_h = 0.166 \text{ ft} = 5.029 \times 10^{-3} \text{ m}$

Free-flow area/frontal area, $\sigma = 0.338$

Heat transfer area/total volume, $\alpha = 80.4 \text{ ft}^2/\text{ft}^3 = 263.78 \text{ m}^2/\text{m}^3$

Fig. 10-13 Flow normal to an in-line tube bank (transient tests), surface I 1.50-1.25.



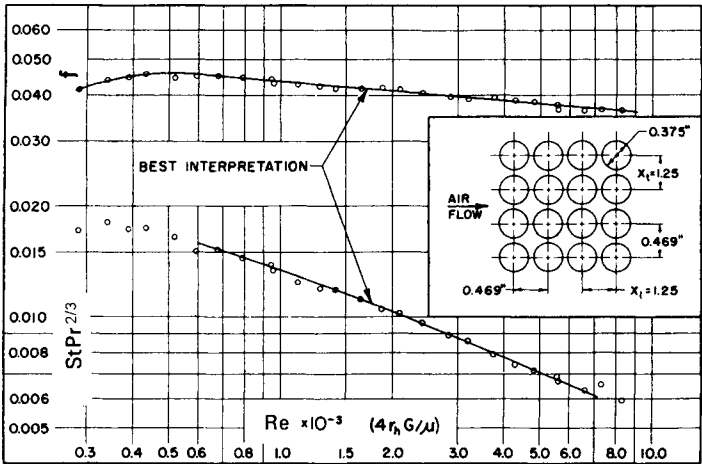
Tube outside diameter = 0.375 in = 9.525×10^{-3} m

Hydraulic diameter, $4r_h = 0.0248 \text{ ft} = 7.569 \times 10^{-3} \text{ m}$

Free-flow area/frontal area, $\sigma = 0.333$

Heat transfer area/total volume, $\alpha = 53.6 \text{ ft}^2/\text{ft}^3 = 175.853 \text{ m}^2/\text{m}^3$

Fig. 10-14 Flow normal to an in-line tube bank (transient tests), surface I 1.25 – 1.25.



Tube outside diameter = 0.375 in = 9.525×10^{-3} m
Hydraulic diameter, $4r_h = 0.01237$ ft = 3.767×10^{-3} m
Free-flow area/frontal area, $\sigma = 0.200$
Heat transfer area/total volume, $\alpha = 64.4$ ft²/ft³ = 211.286 m²/m³

Fig. 10-15 Flow normal to in-line tube banks. (Data of Grimison.)

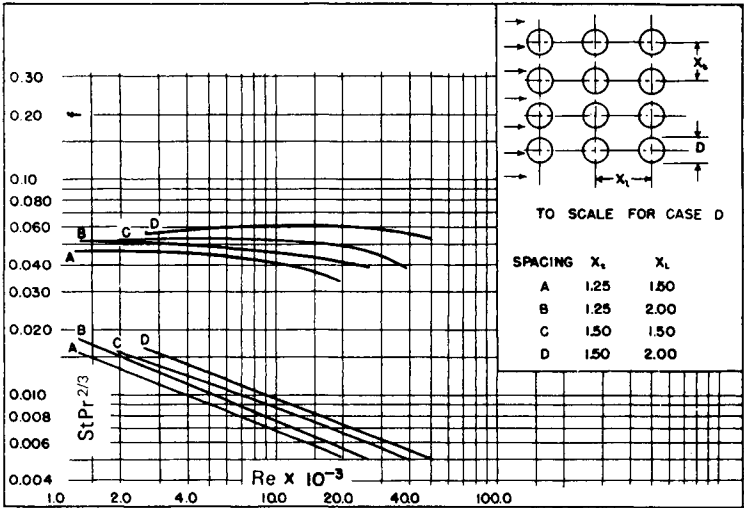


Fig. 10-16 Flow normal to in-line tube banks. (Data of Grimson.)

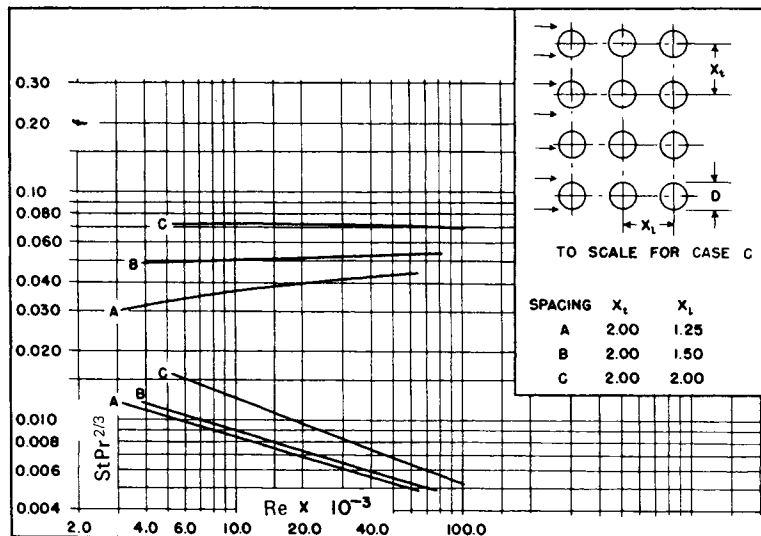
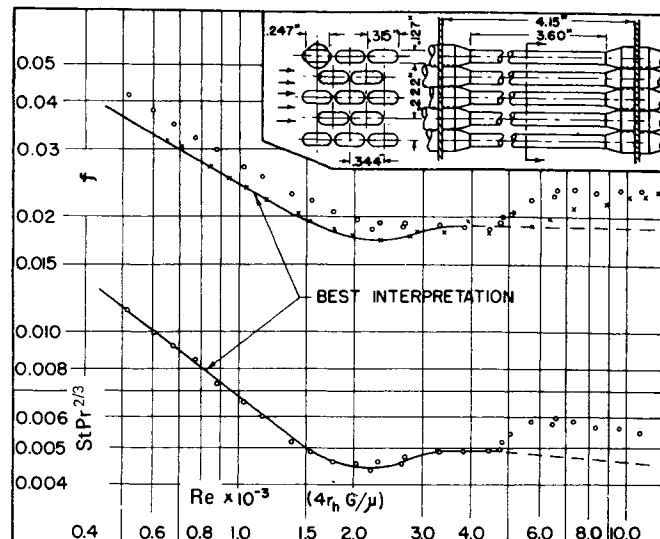
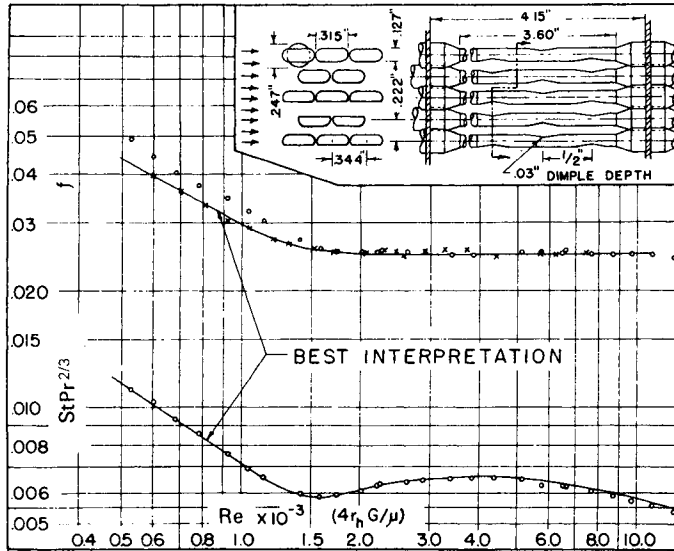


Fig. 10-17 Flow normal to flattened tubes, surface FT-2.



Tube outside diameter before flattening = 0.247 in = 6.27×10^{-3} m
 Distance between centers parallel to flow = 0.344 in = 8.74×10^{-3} m
 Distance between centers perpendicular to flow = 0.222 in = 5.64×10^{-3} m
 Tube dimension parallel to flow = 0.315 in = 8.00×10^{-3} m
 Tube dimension perpendicular to flow = 0.127 in = 3.23×10^{-3} m
 Distance between spacing plates = 4.15 in = 105.4×10^{-3} m
 Length of flat along tube = 3.60 in = 91.44×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01433$ ft = 4.374×10^{-3} m
 Total transfer area/total volume, $\alpha = 108$ ft²/ft³ = 354.331×10^{-3} m²/m³
 Free-flow area/frontal area, $\sigma = 0.386$

Fig. 10-18 Flow normal to dimpled flattened tubes, surface FTD-2.



Tube outside diameter before flattening = 0.247 in = 6.27×10^{-3} m

Distance between centers parallel to flow = 0.344 in = 8.74×10^{-3} m

Distance between centers perpendicular to flow = 0.222 in = 5.64×10^{-3} m

Tube dimension parallel to flow = 0.315 in = 8.00×10^{-3} m

Tube dimension perpendicular to flow = 0.127 in = 3.23×10^{-3} m

Distance between spacing plates = 4.15 in = 105.4×10^{-3} m

Length of flat along tube = 3.60 in = 91.44×10^{-3} m

Dimple depth = 0.03 in = 0.76×10^{-3} m

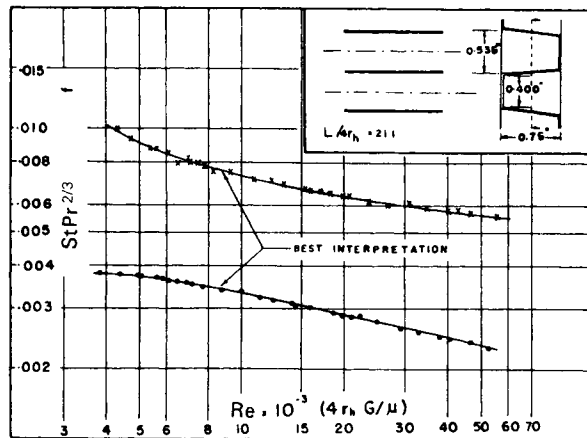
Minimum distance between dimples along tube = 0.5 in = 12.7×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.016$ ft = 4.877×10^{-3} m

Total transfer area/total volume, $\alpha = 108$ ft²/ft³ = 354.331 m²/m³

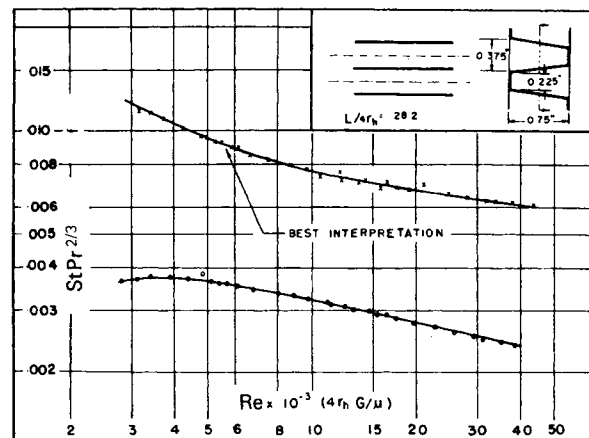
Free-flow area/frontal area, $\sigma = 0.423$

Fig. 10-19 Plain plate-fin surface 2.0.



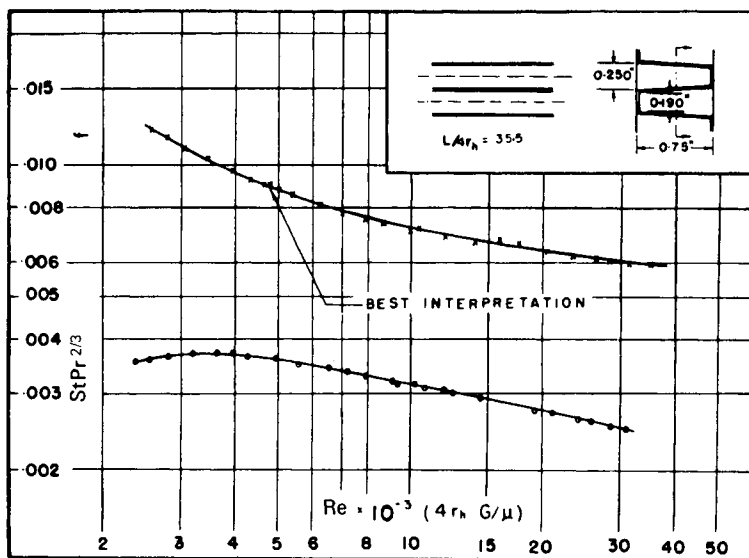
Fin pitch = 2.0 per in = 78.74 per m
 Plate spacing, $b = 0.750$ in = 19.05×10^{-3} m
 Fin length = 12.0 in = 304.8×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.0474$ ft = 14.453×10^{-3} m
 Fin metal thickness = 0.032 in, aluminum = 0.813×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 76.1$ ft²/ft³ = 249.672 m²/m³
 Fin area/total area = 0.606

Fig. 10-20 Plain plate-fin surface 3.01.



Fin pitch = 3.01 per in = 118.5 per m
 Plate spacing, $b = 0.750$ in = 19.05×10^{-3} m
 Fin length = 12.0 in = 304.8×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.03546$ ft = 10.82×10^{-3} m
 Fin metal thickness = 0.032 in, aluminum = 0.813×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 98.3$ ft²/ft³ = 322.507 m²/m³
 Fin area/total area = 0.706

Fig. 10-21 Plain plate-fin surface 3.97.



Fin pitch = 3.97 per in = 156.3 per m

Plate spacing, $b = 0.750$ in = 19.05×10^{-3} m

Fin length = 12.0 in = 304.8×10^{-3} m

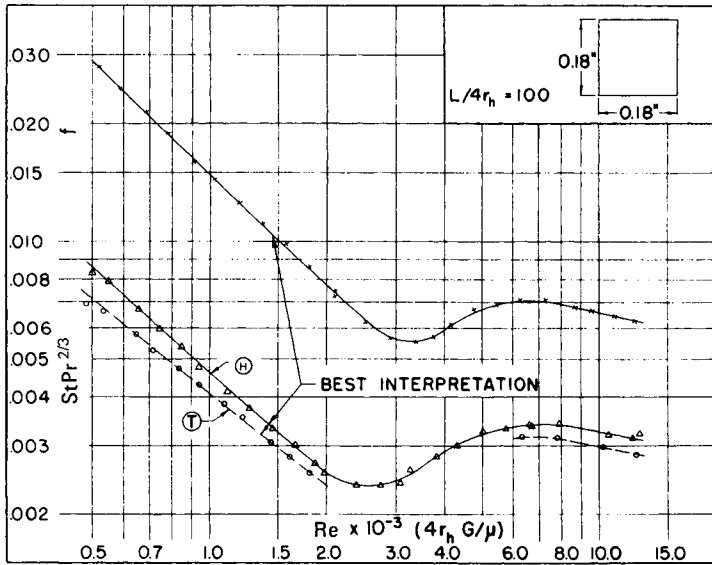
Flow passage hydraulic diameter, $4r_h = 0.0282$ ft = 8.585×10^{-3} m

Fin metal thickness = 0.032 in, aluminum = 0.813×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 119.4 \text{ ft}^2/\text{ft}^3 = 391.7 \text{ m}^2/\text{m}^3$

Fin area/total area = 0.766

Fig. 10-22 Plain plate-fin surface 4.00.



Square tube dimensions = 0.18 in = $4.57 \times 10^{-3} \times 4.57 \times 10^{-3}$ m

Flow passage hydraulic diameter, $4r_h = 0.015$ ft = 4.57×10^{-3} m

Fin area/total area = 0.500

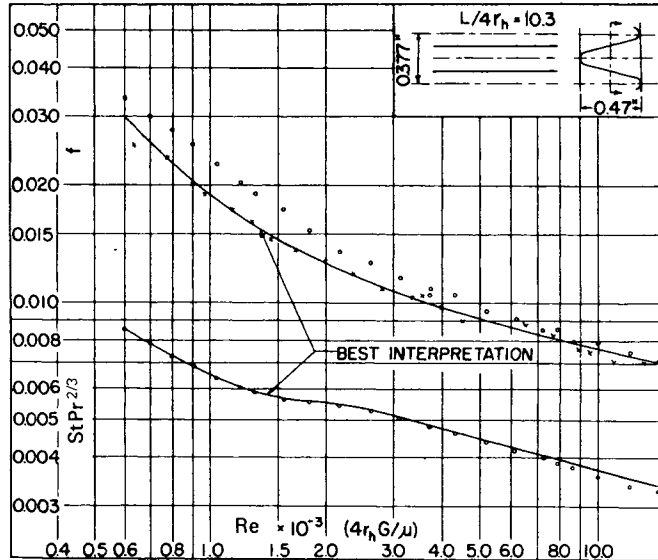
Δ = test points for constant wall-to-fluid temperature difference tests

$\circ$ = test points for constant wall temperature tests

$\times$ = test points for isothermal friction factor tests

Note: The test core for this surface was built up from square tubes rather than from plates and fins. Therefore, the data on fin thickness and area per unit of volume is not pertinent.

Fig. 10-23 Plain plate-fin surface 5.3.



Fin pitch = 5.3 in = 208.7 per m

Plate spacing, $b = 0.470$ in = 11.94×10^{-3} m

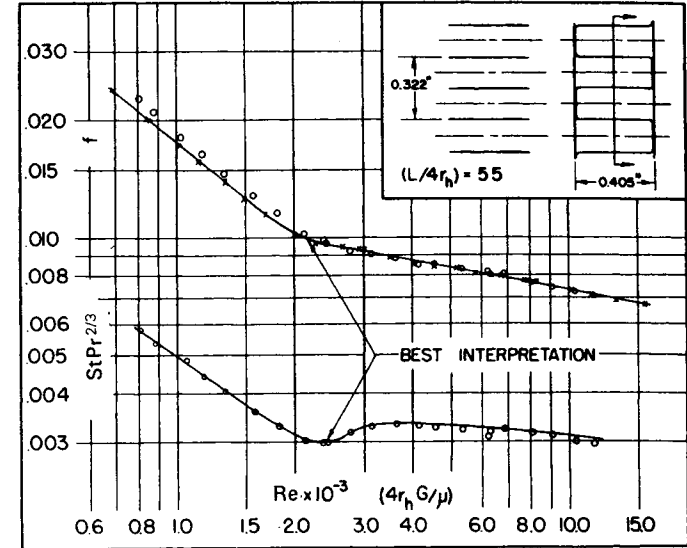
Flow passage hydraulic diameter, $4r_h = 0.02016$ ft = 6.147×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total transfer area/volume between plates, $\beta = 188$ ft²/ft³ = 616.8 m²/m³

Fin area/total area = 0.719

Fig. 10-24 Plain plate-fin surface 6.2.



Fin pitch = 6.2 per in = 244.1 per m

Plate spacing, $b = 0.405$ in = 10.29×10^{-3} m

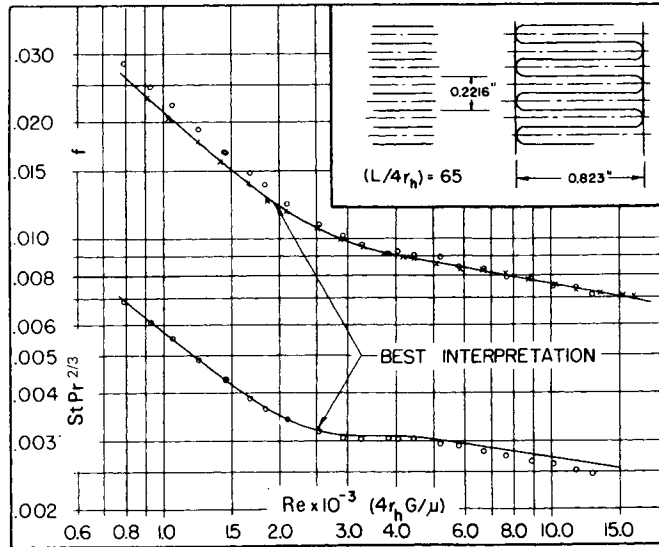
Flow passage hydraulic diameter, $4r_h = 0.0182$ ft = 5.54×10^{-3} m

Fin metal thickness = 0.010 in, aluminum = 0.254×10^{-3} m

Total transfer area/volume between plates, $\beta = 204$ ft²/ft³ = 669.3 m²/m³

Fin area/total area = 0.728

Fig. 10-25 Plain plate-fin surface 9.03.



Fin pitch = 9.03 per in = 355.5 per m

Plate spacing, $b = 0.823$ in = 20.9×10^{-3} m

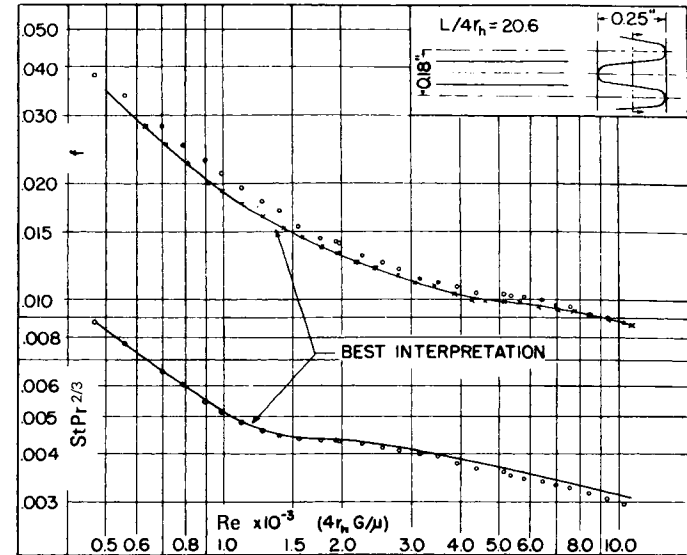
Flow passage hydraulic diameter, $4r_h = 0.01522$ ft = 4.643×10^{-3} m

Fin metal thickness = 0.008 in, aluminum = 0.203×10^{-3} m

Total transfer area/volume between plates, $\beta = 244$ ft²/ft³ = 800.5 m²/m³

Fin area/total area = 0.888

Fig. 10-26 Plain plate-fin surface 11.11.



Fin pitch = 11.1 per in = 437 per m

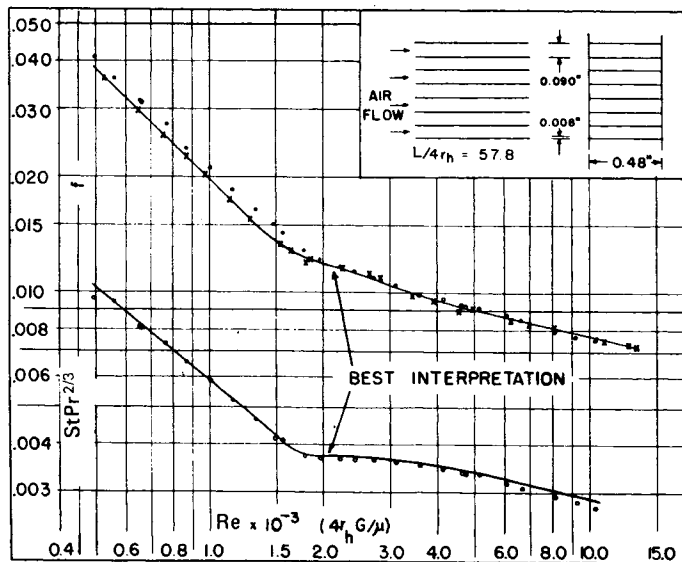
Plate spacing, $b = 0.250$ in = 6.350×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.081×10^{-3} m

Fin metal thickness = 0.006 in aluminum = 0.152×10^{-3} m

Total transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³

Fin area/total area = 0.756



Fin pitch = 11.11 per in = 437 per m

Plate spacing, $b = 0.48$ in = 12.19 mm

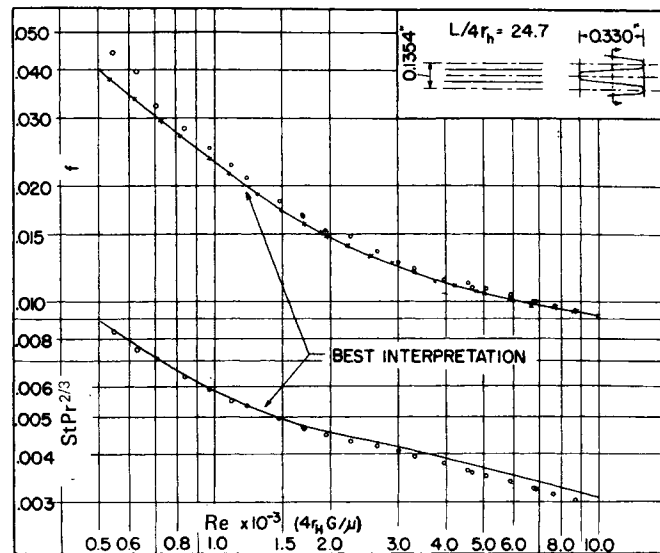
Flow passage hydraulic diameter, $4r_h = 0.01153$ ft = 3.518×10^{-3} m

Fin metal thickness = 0.008 in, copper = 0.203×10^{-3} m

Total transfer area/volume between plates, $\beta = 312$ ft²/ft³ = $1,024$ m²/m³

Fin area/total area = 0.854

Fig. 10-28 Plain plate-fin surface 14.77.



Fin pitch = 14.77 per in = 582 per m

Plate spacing, $b = 0.330$ in = 8.38×10^{-3} m

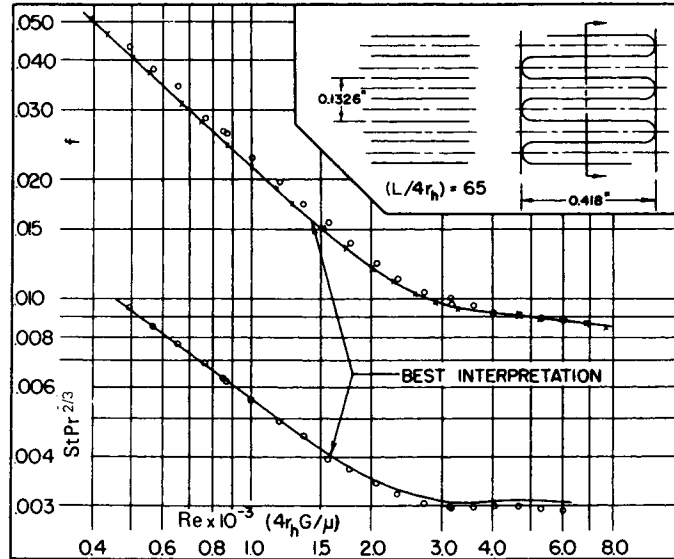
Flow passage hydraulic diameter, $4r_h = 0.00848$ ft = 2.59×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total transfer area/volume between plates, $\beta = 420$ ft²/ft³ = $1,378$ m²/m³

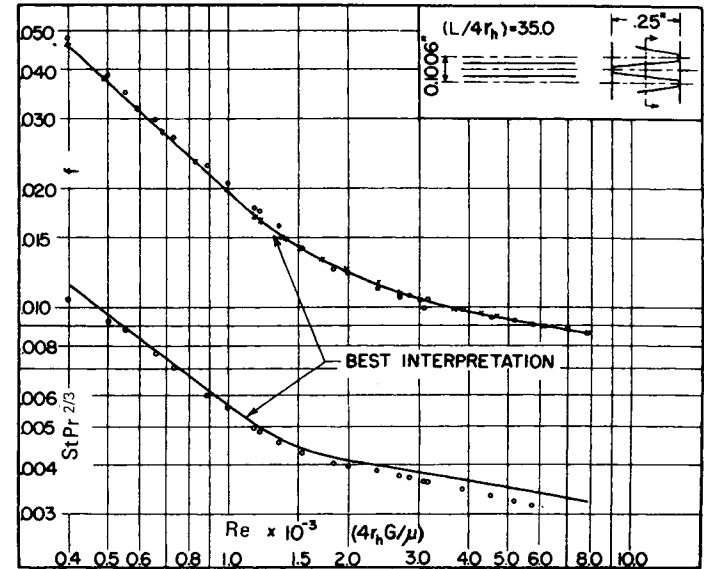
Fin area/total area = 0.844

Fig. 10-29 Plain plate-fin surface 15.08.



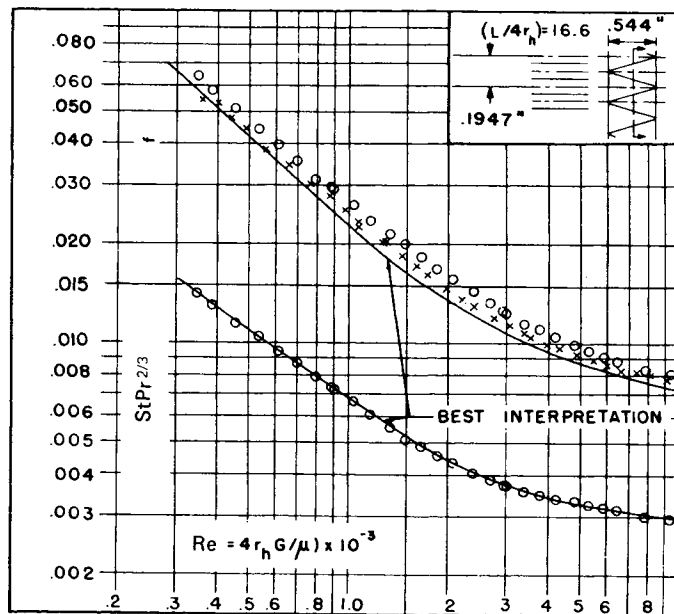
Fin pitch = 15.08 per in = 594 per m
 Plate spacing, $b = 0.418$ in = 10.6×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.00876$ ft = 2.67×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total transfer area/volume between plates, $\beta = 414$ ft²/ft³ = $1,358$ m²/m³
 Fin area/total area = 0.870

Fig. 10-30 Plain plate-fin surface 19.86.



Fin pitch = 19.86 per in = 782 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.00615$ ft = 1.875×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152 m
 Total transfer area/volume between plates, $\beta = 561$ ft²/ft³ = $1,841$ m²/m³
 Fin area/total area = 0.849

Fig. 10-31 Plain plate-fin surface 10.27T.†



Fin pitch = 10.27 per in = 404 per m

Plate spacing, $b = 0.544$ in = 13.8×10^{-3} m

Fin length flow direction = 5.00 in = 127×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.01259$ ft = 3.835×10^{-3} m

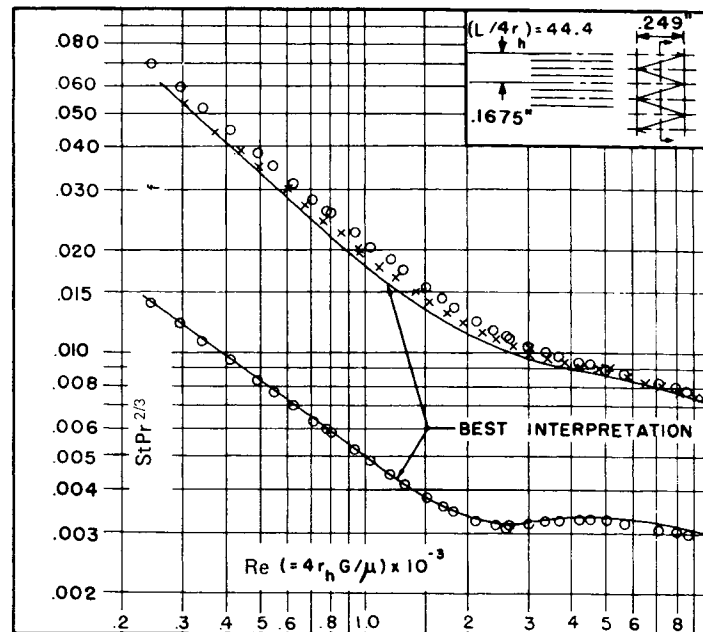
Fin metal thickness = 0.010 in, aluminum = 0.254×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 289.93$ ft²/ft³ = 951 m²/m³

Fin area/total area = 0.863

†Best interpretation to the friction data does not pass through the cold-core test points because of a separate correction for K_c , K_e as functions of Re .

Fig. 10-32 Plain plate-fin surface 11.94T.†



Fin pitch = 11.94 per in = 470 per m

Plate spacing, $b = 0.249$ in = 6.325×10^{-3} m

Fin length flow direction = 5.00 in = 127×10^{-3} m

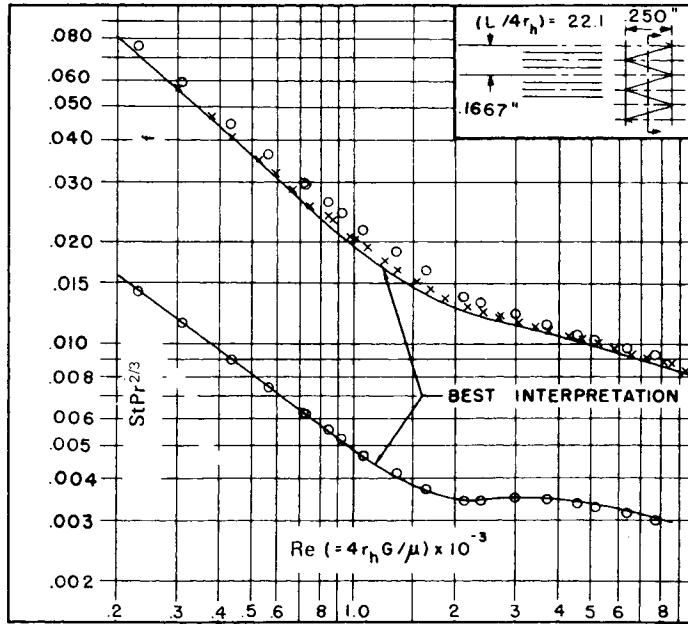
Flow passage hydraulic diameter, $4r_h = 0.009396$ ft = 2.870×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 393.0$ ft²/ft³ = 1289 m²/m³

Fin area/total area = 0.769

Fig. 10-33 Plain plate-fin surface 12.00T.†



Fin pitch = 12.00 per in = 472 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Fin length flow direction = 2.50 in strips (two = 5.00 in) = 63.5×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.009412$ ft = 2.870×10^{-3} m

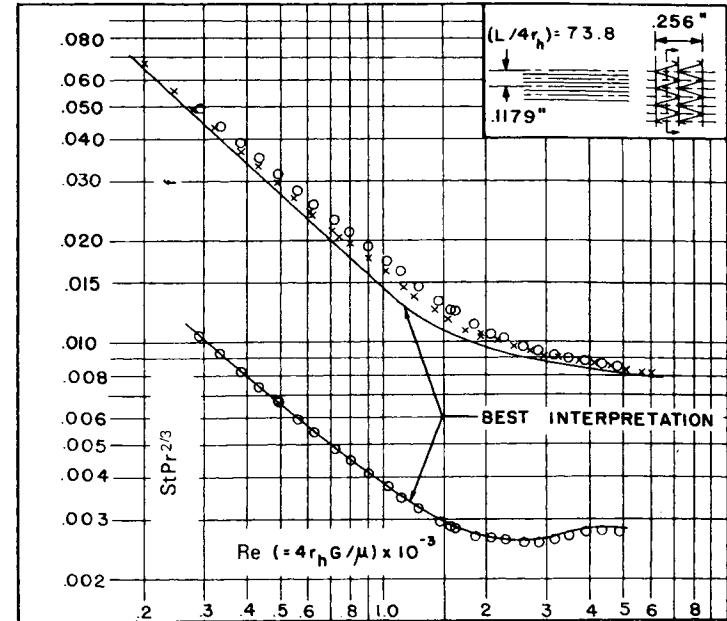
Fin metal thickness = 0.006 in, nickel = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 392.7$ ft²/ft³ = $1,288$ m²/m³

Fin area/total area = 0.773

†See footnote, Fig. 10-31.

Fig. 10-34 Plain plate-fin surface 16.96T.†



Fin pitch = 16.96 per in = 676.7 per m

Plate spacing, $b = 0.256$ in = 6.50×10^{-3} m

Fin length flow direction = 5.00 in = 127×10^{-3} m

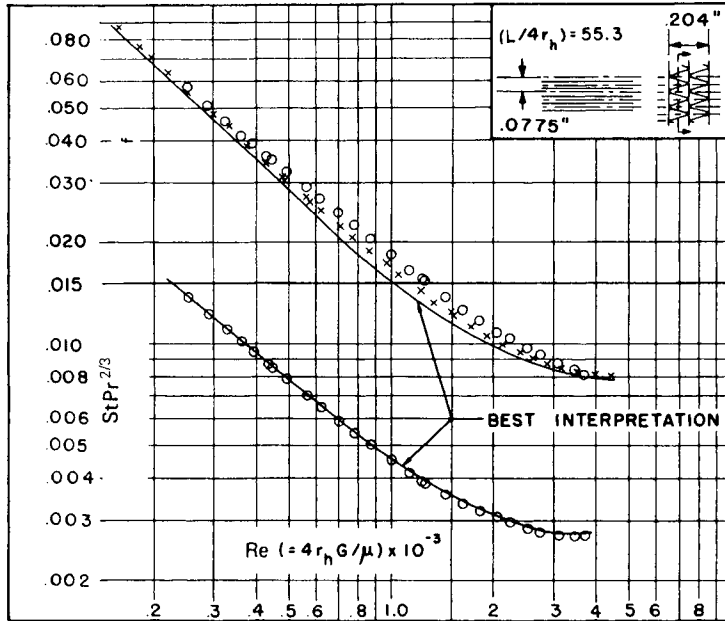
Flow passage hydraulic diameter, $4r_h = 0.0056524$ ft = 1.722×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 607.81$ ft²/ft³ = 1994 m²/m³

Fin area/total area = 0.861

Fig. 10-35 Plain plate-fin surface 25.79T.†



Fin pitch = 25.79 per in = 1015 per m

Plate spacing, $b = 0.204$ in = 5.182×10^{-3} m

Splitter symmetrically located

Fin length flow direction = 2.50 in = 63.5×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.003771$ ft = 1.151×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

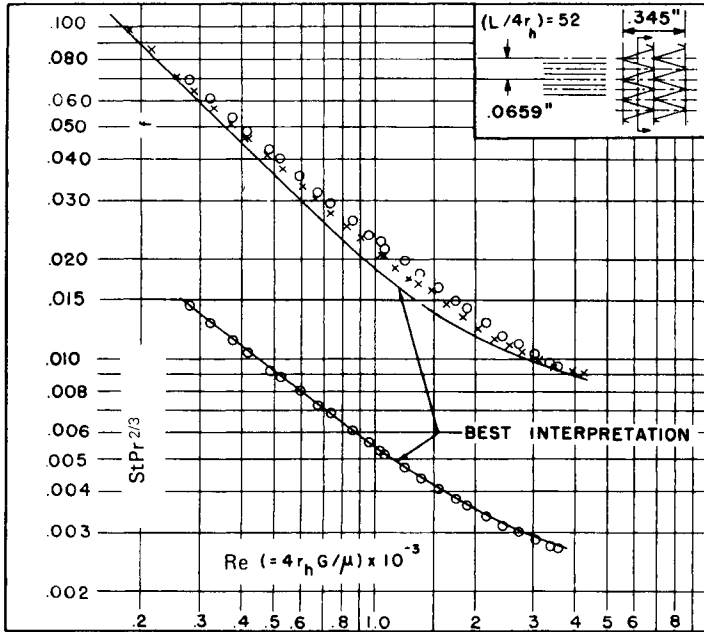
Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 855.58$ ft²/ft³ = 2807 m²/m³

Fin area (including splitter)/total area = 0.884

†See footnote, Fig. 10-31.

Fig. 10-36 Plain plate-fin surface 30.33T.†



Fin pitch = 30.33 per in = 1194 per m

Plate spacing, $b = 0.345$ in = 8.763×10^{-3} m

Splitter symmetrically located

Fin length flow direction = 2.50 in = 63.5×10^{-3} m

Flow passage hydraulic diameter, $4r_h$, 0.004009 ft = 1.222×10^{-3} m

Fin metal thickness = 0.004 in, aluminum = 0.101×10^{-3} m

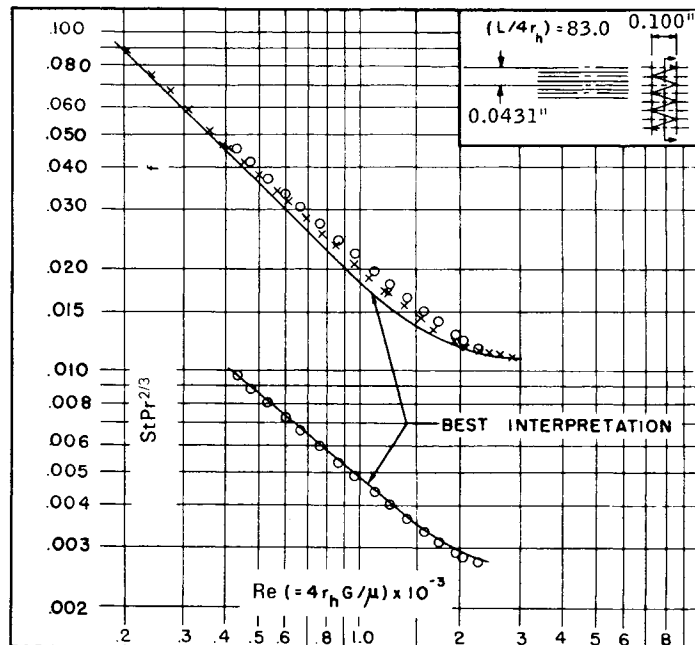
Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 812.51$ ft²/ft³ = $2,666$ m²/m³

Fin area (including splitter)/total area = 0.928

†See footnote, Fig. 10-31.

Fig. 10-37 Plain plate-fin surface 46.45T.†



Fin pitch = 46.45 per in = 1829 per m

Plate spacing, $b = 0.100$ in = 2.54×10^{-3} m

Fin length flow direction = 2.63 in = 66.8×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.002643$ ft = 0.805×10^{-3} m

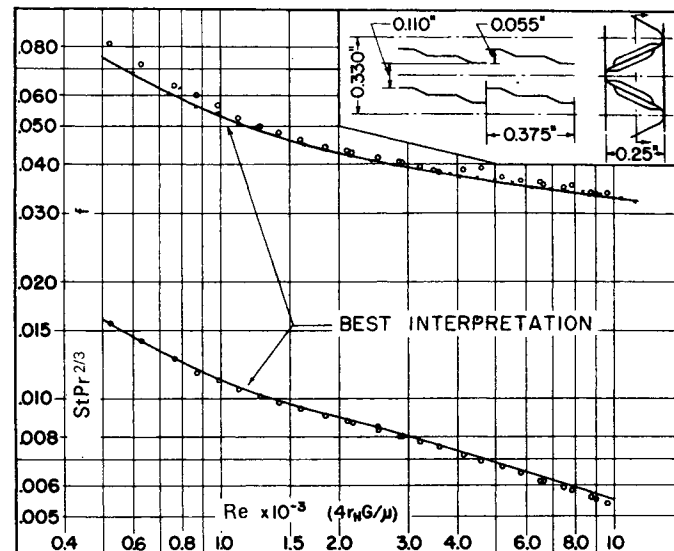
Fin metal thickness = 0.002 in, stainless steel = 0.051×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 1332.45$ ft²/ft³ = 4372 m²/m³

Fin area/total area = 0.837

†See footnote, Fig. 10-31,

Fig. 10-38 Louvered plate-fin surface 3/8-6.06.†



Fin pitch = 6.06 per in = 238.6 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.375 in = 9.525×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

Fin gap = 0.110 in = 2.79×10^{-3} m

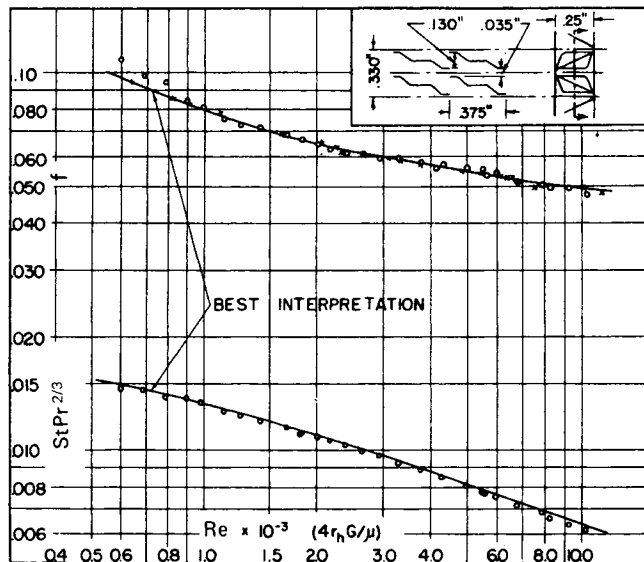
Flow passage hydraulic diameter, $4r_h = 0.01460$ ft = 4.453×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.153×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 256$ ft²/ft³ = 840 m²/m³

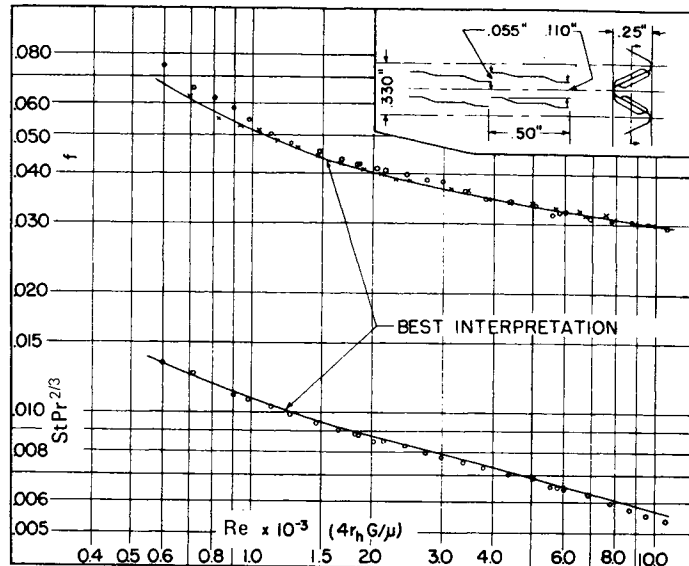
Fin area/total area = 0.640

Fig. 10-39 Louvered plate-fin surface 3/8(a)–6.06).



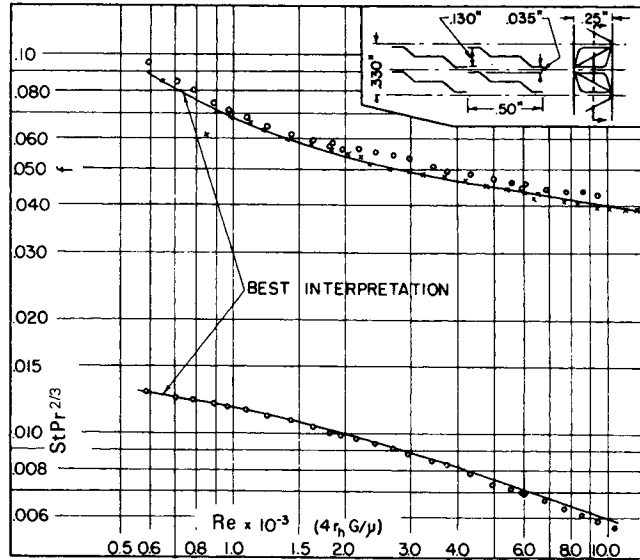
Fin pitch = 6.06 per in = 238 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Louver spacing = 0.375 in = 9.53×10^{-3} m
 Fin gap = 0.035 in = 0.89×10^{-3} m
 Louver gap = 0.130 in = 3.30×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01460$ ft = 4.453×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total transfer area/volume between plates, $\beta = 256$ ft²/ft³ = 840 m²/m³
 Fin area/total area = 0.640

Fig. 10-40 Louvered plate-fin surface 1/2–6.06.



Fin pitch = 6.06 per in = 239 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Louver spacing = 0.50 in = 12.7×10^{-3} m
 Fin gap = 0.110 in = 2.79×10^{-3} m
 Louver gap = 0.055 in = 1.4×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01460$ ft = 4.453×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 256$ ft²/ft³ = 840 m²/m³
 Fin area/total area = 0.640

Fig. 10-41 Louvered plate-fin surface 1/2(a)-6.06.



Fin pitch = 6.06 per in = 239 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.50 in = 12.7×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.130 in = 3.30×10^{-3} m

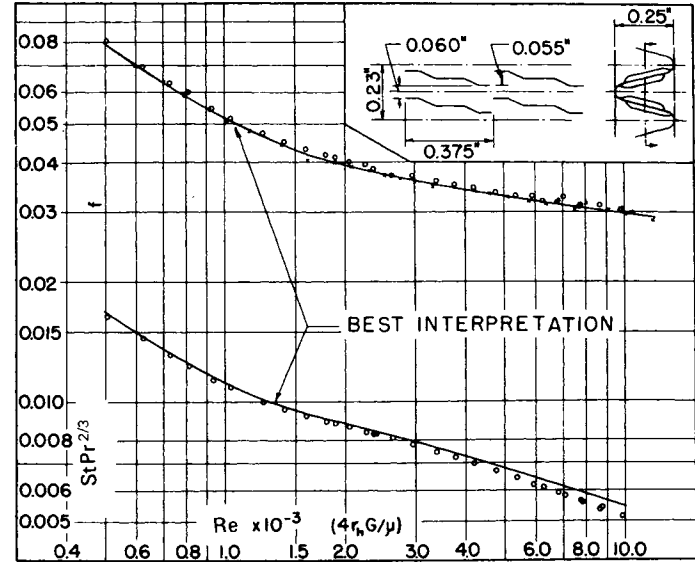
Flow passage hydraulic diameter, $4r_h = 0.01460$ ft = 4.453×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 256$ ft²/ft³ = 840 m²/m³

Fin area/total area = 0.640

Fig. 10-42 Louvered plate-fin surface 3/8-8.7.



Fin pitch = 8.7 per in = 343 per meter

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.375 in = 9.53×10^{-3} m

Fin gap = 0.060 in = 1.52×10^{-3} m

Louver gap = 0.055 in = 1.40×10^{-3} m

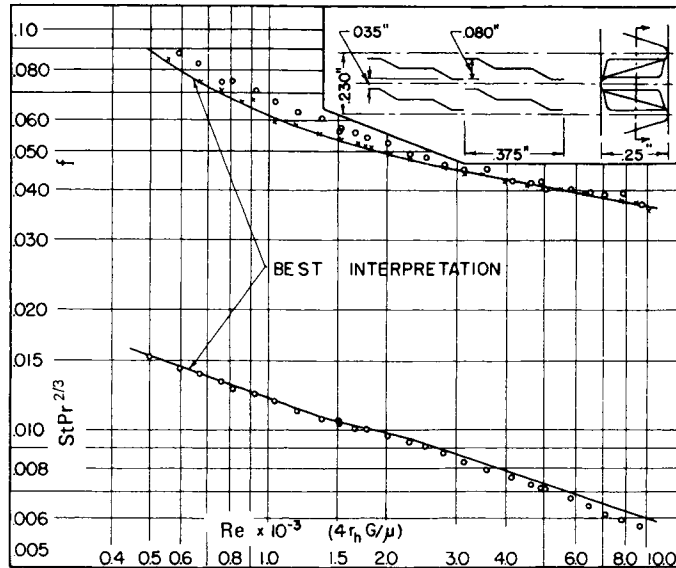
Flow passage hydraulic diameter, $4r_h = 0.01196$ ft = 3.650×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 307$ ft²/ft³ = 1007 m²/m³

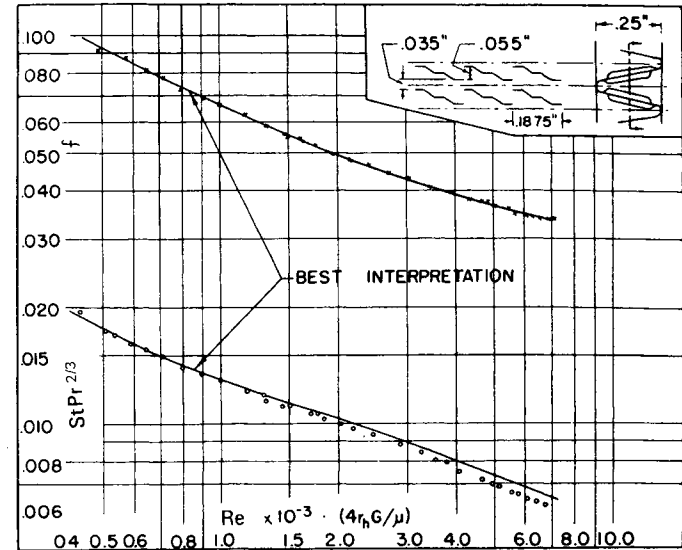
Fin area/total area = 0.705

Fig. 10-43 Louvered plate-fin surface 3/8(a)-8.7.



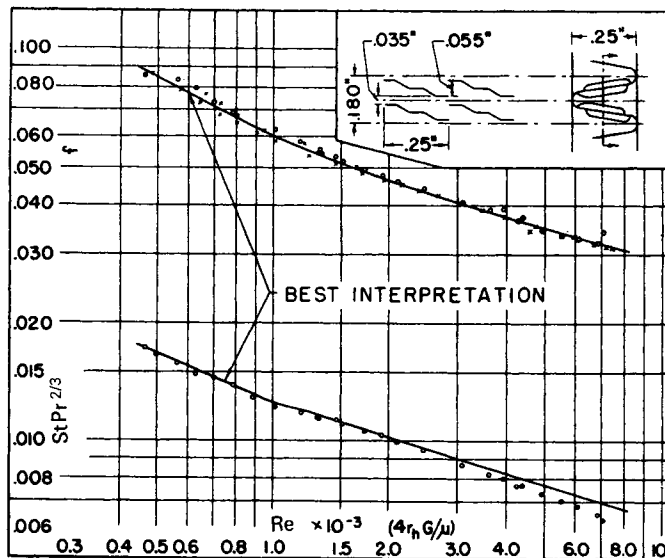
Fin pitch = 8.70 per in = 343 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Louver spacing = 0.375 in = 9.35×10^{-3} m
 Fin gap = 0.035 in = 0.89×10^{-3} m
 Louver gap = 0.080 in = 2.03×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01196$ ft = 3.650×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 307$ ft²/ft³ = 1007 m²/m³
 Fin area/total area = 0.705

Fig. 10-44 Louvered plate-fin surface 3/16-11.1.



Fin pitch = 11.1 per in = 437 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Louver spacing = 0.1875 in = 4.763×10^{-3} m
 Fin gap = 0.035 in = 0.89×10^{-3} m
 Louver gap = 0.055 in = 1.4×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³
 Fin area/total area = 0.756

Fig. 10-45 Louvered plate-fin surface 1/4-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.250 in = 6.35×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

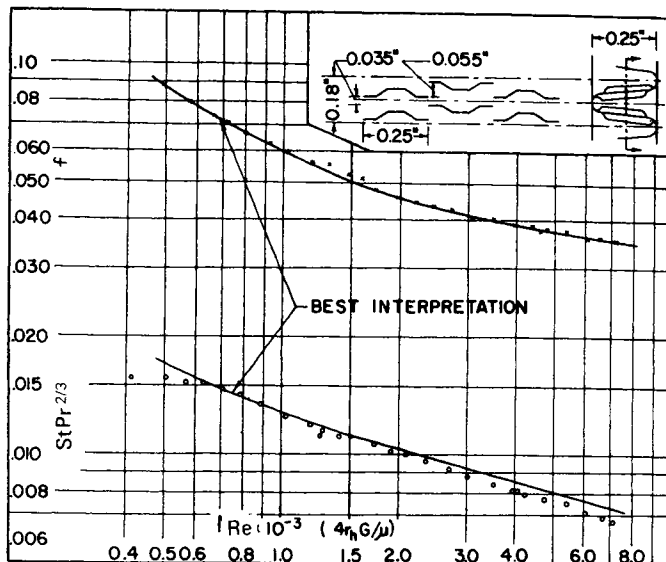
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³

Fin area/total area = 0.756

Fig. 10-46 Louvered plate-fin surface 1/4(b)-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.250 in = 6.35×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

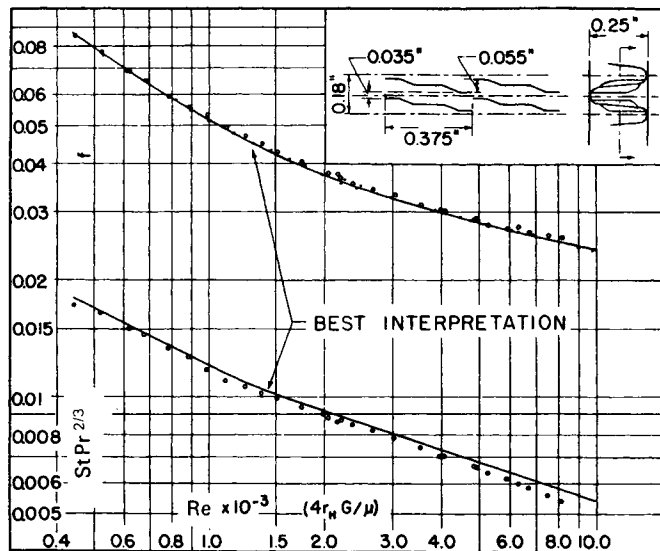
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³

Fin area/total area = 0.756

Fig. 10-47 Louvered plate-fin surface 3/8-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.375 in = 9.53×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

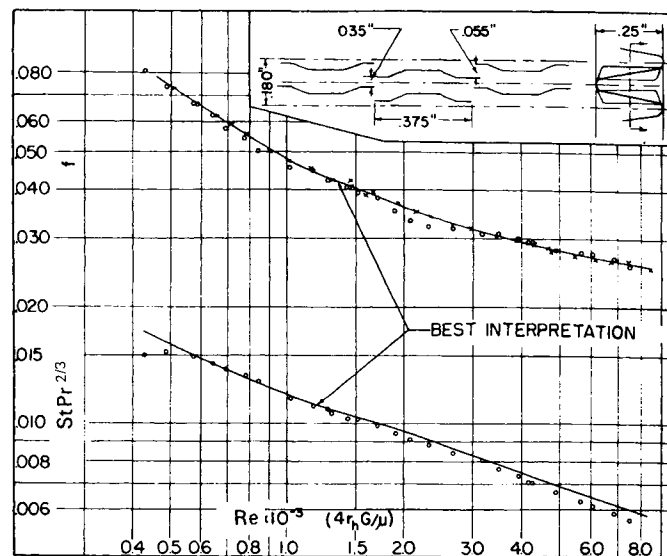
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = 1.204 m²/m³

Fin area/total area = 0.756

Fig. 10-48 Louvered plate-fin surface 3/8(b)-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.375 in = 9.53×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

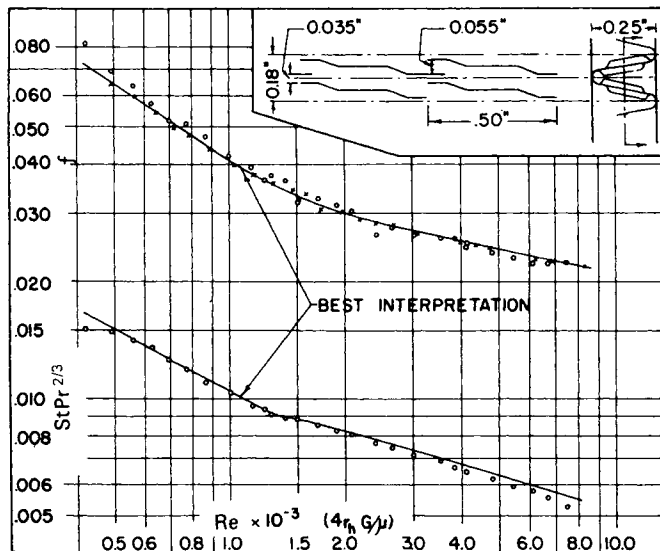
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = 1.204 m²/m³

Fin area/total area = 0.756

Fig. 10-49 Louvered plate-fin surface 1/2-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.50 in = 12.7×10^{-3} m

Fin gap = 0.035 in = 0.89×10^{-3} m

Louver gap = 0.055 in = 1.4×10^{-3} m

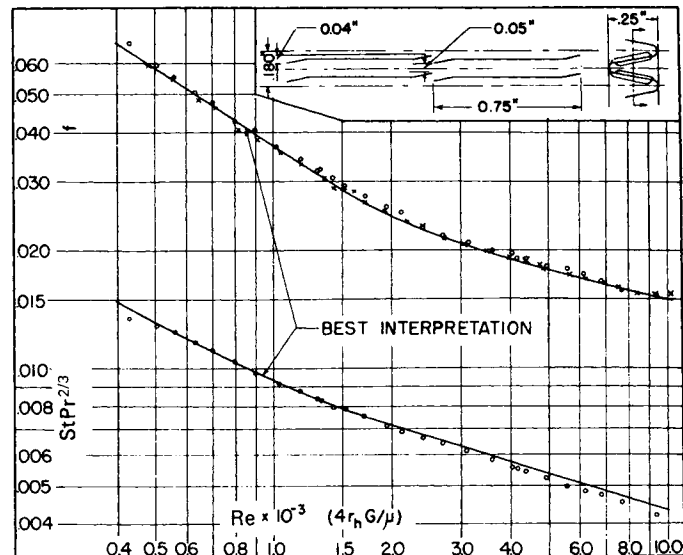
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³

Fin area/total area = 0.756

Fig. 10-50 Louvered plate-fin surface 3/4-11.1.



Fin pitch = 11.1 per in = 437 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Louver spacing = 0.75 in = 19.05×10^{-3} m

Fin gap = 0.05 in = 1.27×10^{-3} m

Louver gap = 0.04 in = 1.02×10^{-3} m

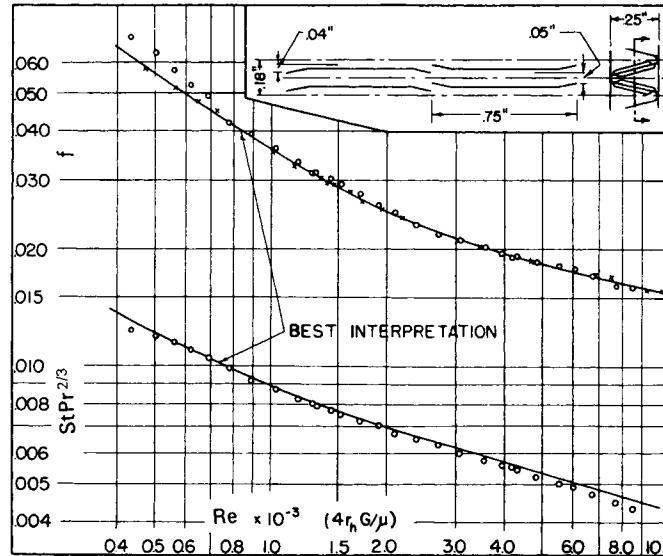
Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³

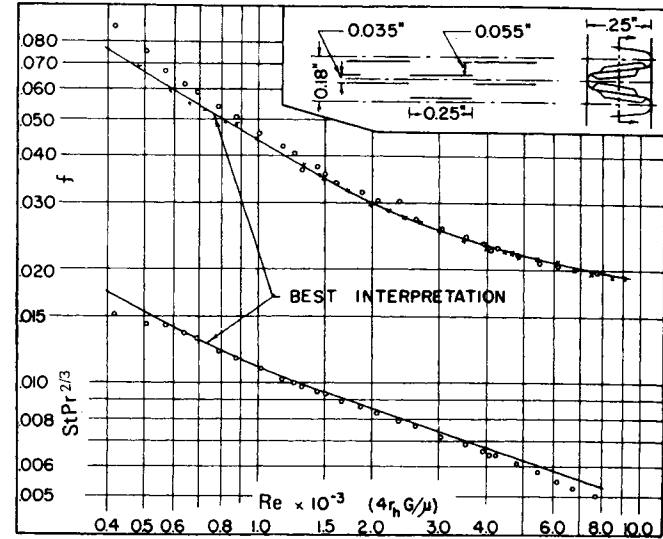
Fin area/total area = 0.756

Fig. 10-51 Louvered plate-fin surface 3/4(b)-11.1.



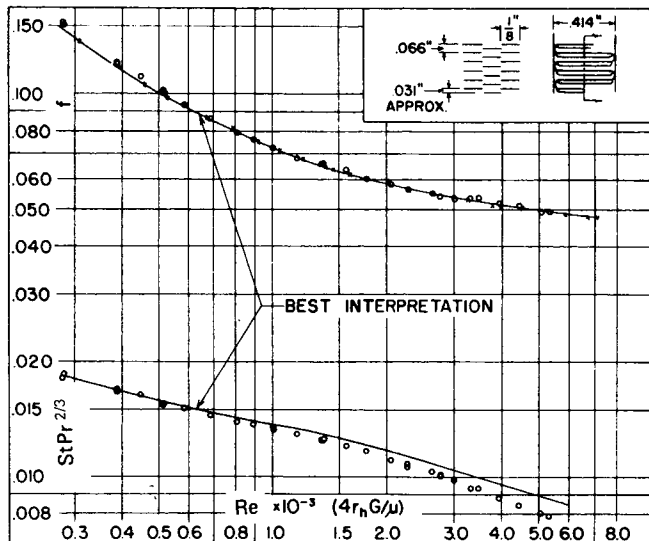
Fin pitch = 11.1 per in = 437 per m
 Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m
 Louver spacing = 0.55 in = 19.05×10^{-3} m
 Fin gap = 0.05 in = 1.27×10^{-3} m
 Louver gap = 0.04 in = 1.02×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³
 Fin area/total area = 0.756

Fig. 10-52 Strip-fin plate-fin surface 1/4(s)-11.1.



Fin pitch = 11.1 per in = 437 per m
 Plate spacing, $b = 0.25$ in = 6.35×10^{-3} m
 Fin length = 0.25 in = 6.35×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.01012$ ft = 3.084×10^{-3} m
 Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 367$ ft²/ft³ = $1,204$ m²/m³
 Fin area/total area = 0.756

Fig. 10-53 Strip-fin plate-fin surface 1/8-15.2.



Fin pitch = 15.2 per in = 598 per m

Plate spacing, $b = 0.414$ in = 10.5×10^{-3} m

Fin length = 0.125 in = 3.18×10^{-3} m

Fins staggered symmetrically

Flow passage hydraulic diameter, $4r_h = 0.00868$ ft = 2.647×10^{-3} m

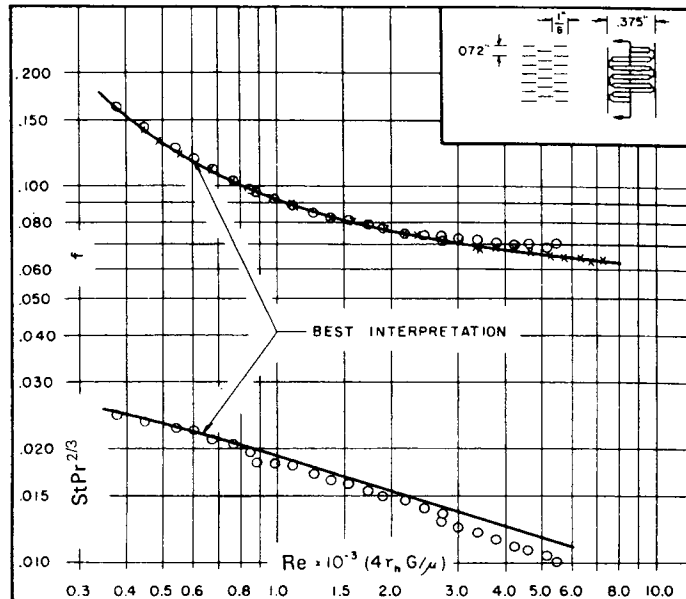
Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 417$ ft²/ft³ = 1368 m²/m³

Fin area/total area = 0.873

Note: Fin leading and trailing edges slightly scarfed from fin-cutting operation. Friction factors may be lower with clean fins.

Fig. 10-54 Strip-fin plate-fin surface 1/8-13.95.



Fin pitch = 13.95 per in = 549 per m

Plate spacing, $b = 0.375$ in = 9.525×10^{-3} m

Fin length = 0.125 in = 3.175×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.00879$ ft = 2.68×10^{-3} m

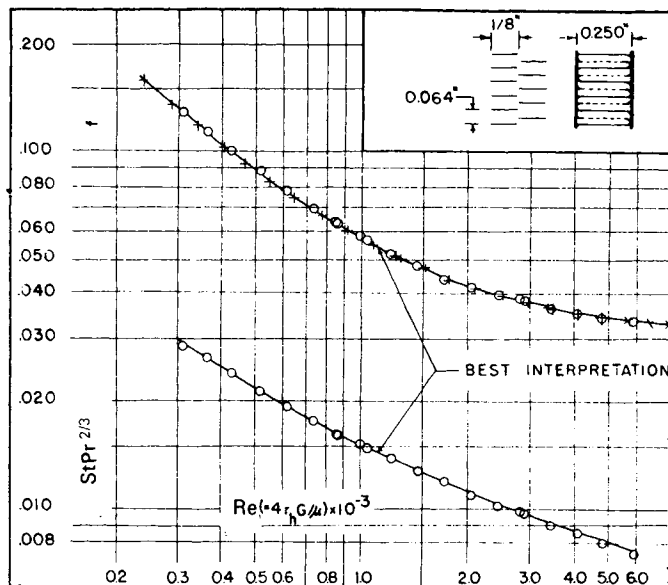
Fin metal thickness = 0.010 in, aluminum = 0.254×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 381$ ft²/ft³ = $1,250$ m²/m³

Fin area/total area = 0.840

Note: The fin surface area on the leading and trailing edges of the fins have not been included in area computation.

Fig. 10-55 Strip-fin plate-fin surface 1/8-15.61.



Fin pitch = 15.61 per in = 615 per m

Plate spacing, $b = 0.250$ in = 6.35×10^{-3} m

Fin length = 0.125 in = 3.175×10^{-3} m

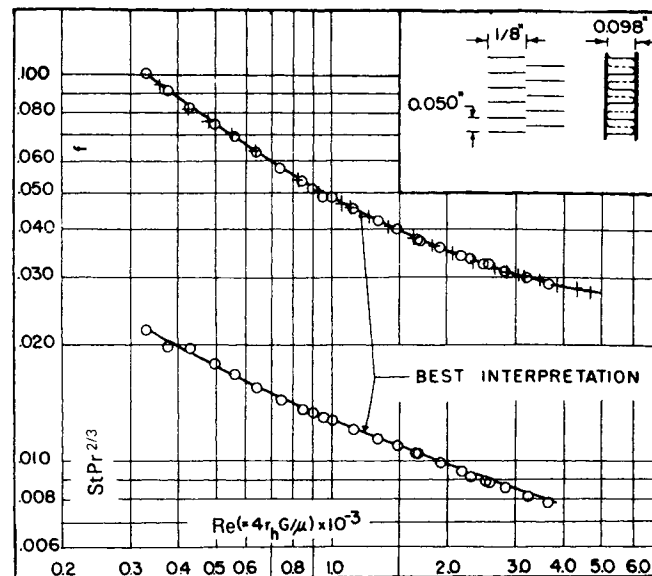
Flow passage hydraulic diameter, $4r_h = 0.007817$ ft = 2.38×10^{-3} m

Fin metal thickness = 0.004 in = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 471.9$ ft²/ft³ = $1,548$ m²/m³

Fin area/total area = 0.809

Fig. 10-56 Strip-fin plate-fin surface 1/8-19.86.



Fin pitch = 19.86 per in = 782 per m

Plate spacing, $b = 0.098$ in = 2.49×10^{-3} m

Fin length = 0.125 in = 3.175×10^{-3} m

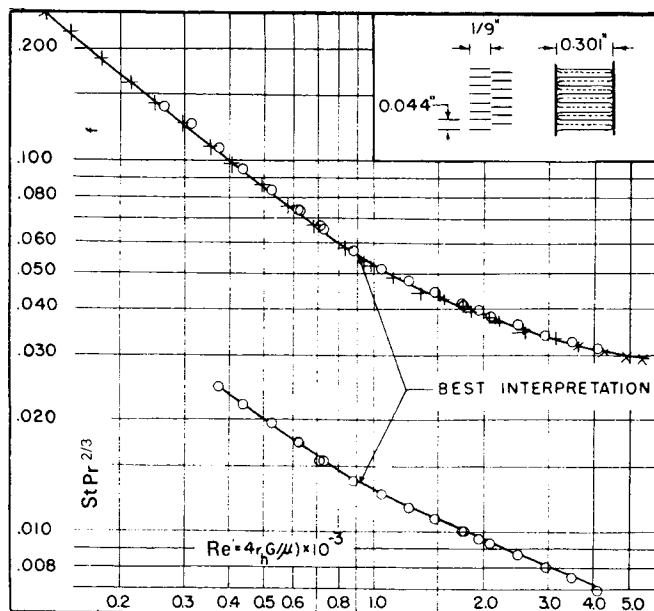
Flow passage hydraulic diameter, $4r_h = 0.00506$ ft = 1.54×10^{-3} m

Fin metal thickness = 0.004 in = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 687$ ft²/ft³ = 2254 m²/m³

Fin area/total area = 0.785

Fig. 10-57 Strip-fin plate-fin surface 1/9-22.68.



Fin pitch = 22.68 per in = 893 per m

Plate spacing, $b = 0.301$ in = 7.65×10^{-3} m

Fin length = 0.111 in = 2.8×10^{-3} m

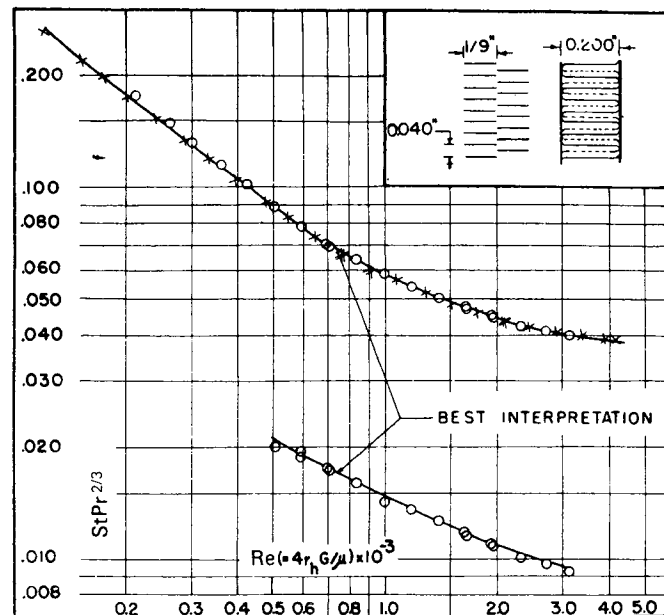
Flow passage hydraulic diameter, $4r_h = 0.005692$ ft = 1.735×10^{-3} m

Fin metal thickness = 0.0004 in = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 630.5$ ft²/ft³ = $2,069$ m²/m³

Fin area/total area = 0.885

Fig. 10-58 Strip-fin plate-fin surface 1/9-25.01.



Fin pitch = 25.01 per in = 985 per m

Plate spacing, $b = 0.200$ in = 5.08×10^{-3} m

Fin length = 0.111 in = 2.8×10^{-3} m

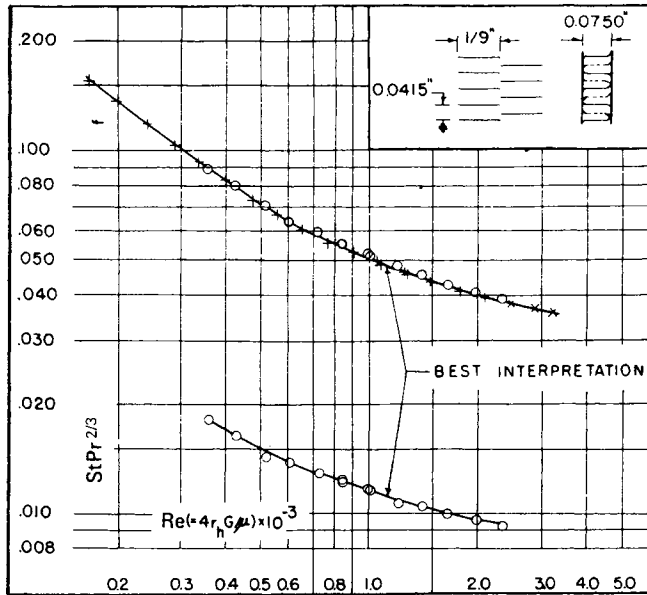
Flow passage hydraulic diameter, $4r_h = 0.004905$ ft = 1.50×10^{-3} m

Fin metal thickness = 0.004 in = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 719.4$ ft²/ft³ = $2,360$ m²/m³

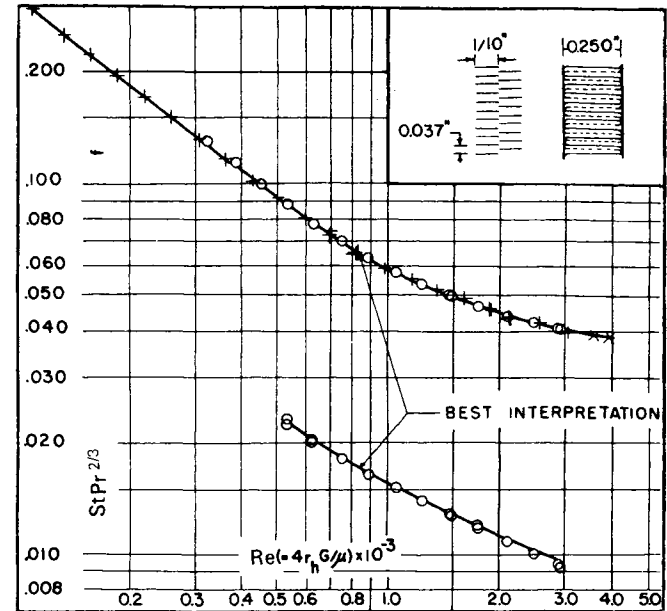
Fin area/total area = 0.850

Fig. 10-59 Strip-fin plate-fin surface 1/9-24.12.



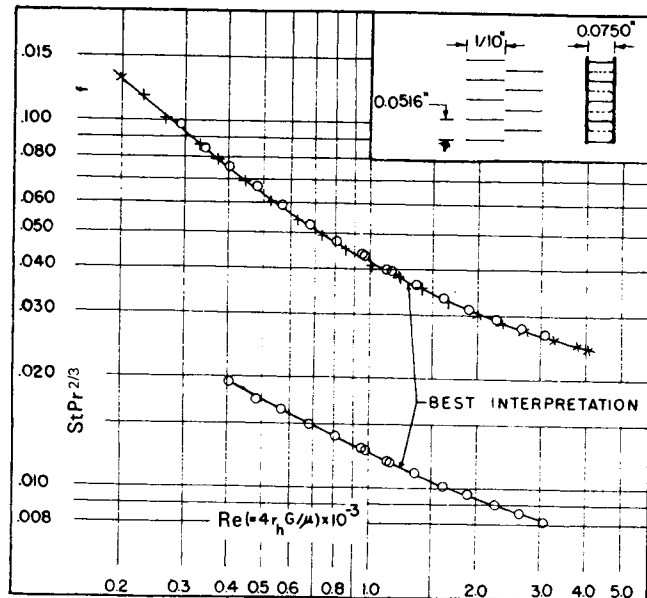
Fin pitch = 24.12 per in = 950 per m
 Plate spacing, $b = 0.075$ in = 1.91×10^{-3} m
 Fin length = 0.111 in = 2.8×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.003966$ ft = 1.209×10^{-3} m
 Fin metal thickness = 0.004 in = 0.102×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 862.7$ ft²/ft³ = $2,830$ m²/m³
 Fin area/total area = 0.665

Fig. 10-60 Strip-fin plate-fin surface 1/10-27.03.



Fin pitch = 27.03 per in = 1064 per m
 Plate spacing, $b = 0.250$ in = 6.38×10^{-3} m
 Fin length = 0.10 in = 2.54×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.004669$ ft = 1.423×10^{-3} m
 Fin metal thickness = 0.004 in = 0.102×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 751.7$ ft²/m³ = $2,466$ m²/m³
 Fin area/total area = 0.887

Fig. 10-61 Strip-fin surface 1/10-19.35.



Fin pitch = 19.35 per in = 762 per m

Plate spacing, $b = 0.075$ in = 1.91×10^{-3} m

Fin length = 0.10 in = 2.54×10^{-3} m

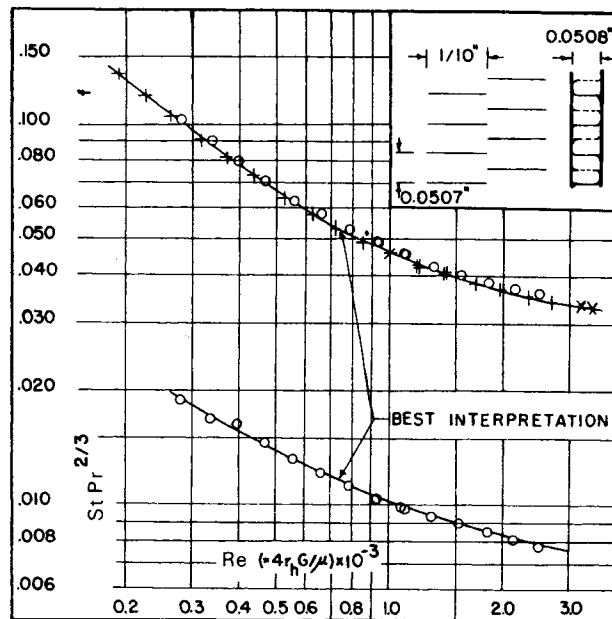
Flow passage hydraulic diameter, $4r_h = 0.004602$ ft = 1.403×10^{-3} m

Fin metal thickness = 0.004 in = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 758.9$ ft²/ft³ = $2,490$ m²/m³

Fin area/total area = 0.611

Fig. 10-62 Strip-fin plate-fin surface 1/10-19.74.



Fin pitch = 19.74 per in = 777 per m

Plate spacing, $b = 0.0508$ in = 1.29×10^{-3} m

Fin length = 0.10 in = 2.54×10^{-3} m

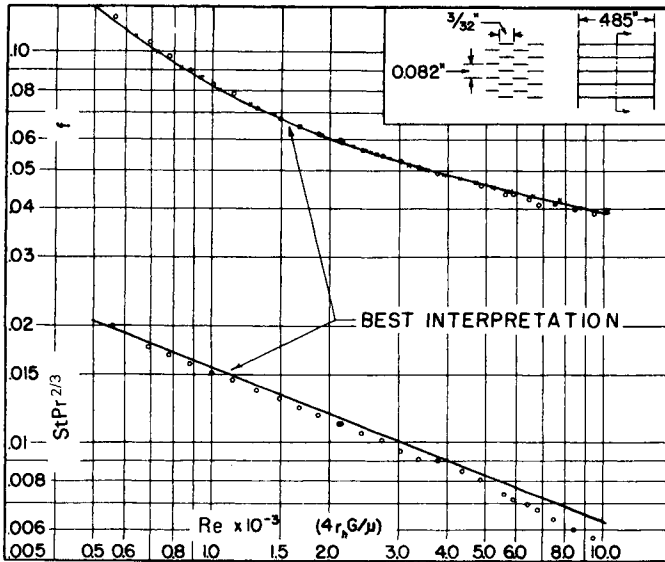
Flow passage hydraulic diameter, $4r_h = 0.0040$ ft = 1.219×10^{-3} m

Fin metal thickness = 0.002 in = 0.051×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 923$ ft²/ft³ = 3028 m²/m³

Fin area/total area = 0.508

Fig. 10-63 Strip-fin plate-fin surface 3/32-12.22.



Fin pitch = 12.22 per in = 480 per m

Plate spacing, $b = 0.485$ in = 12.3×10^{-3} m

Fin length = 0.094 in = 2.4×10^{-3} m

Fins staggered symmetrically

Flow passage hydraulic diameter, $4r_h = 0.01120$ ft = 3.41×10^{-3} m

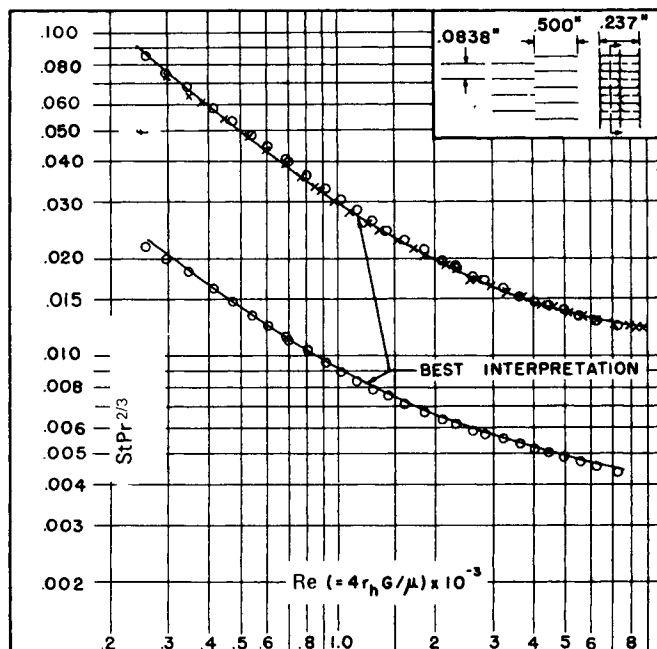
Fin metal thickness = 0.004 in, copper = 0.102×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 340$ ft²/ft³ = $1,115$ m²/m³

Fin area/total area = 0.862

Note: Fin leading and trailing edges slightly scarfed from fin-cutting operation. Friction factors may be lower with clean fins.

Fig. 10-64 Strip-fin plate-fin surface 1/2-11.94(D).



Fin pitch = 11.94 per in = 470 per m

Plate spacing, $b = 0.237$ in = 6.02×10^{-3} m

Splitter symmetrically located

Fin length flow direction = 0.500 in = 12.7×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.007436$ ft = 2.266×10^{-3} m

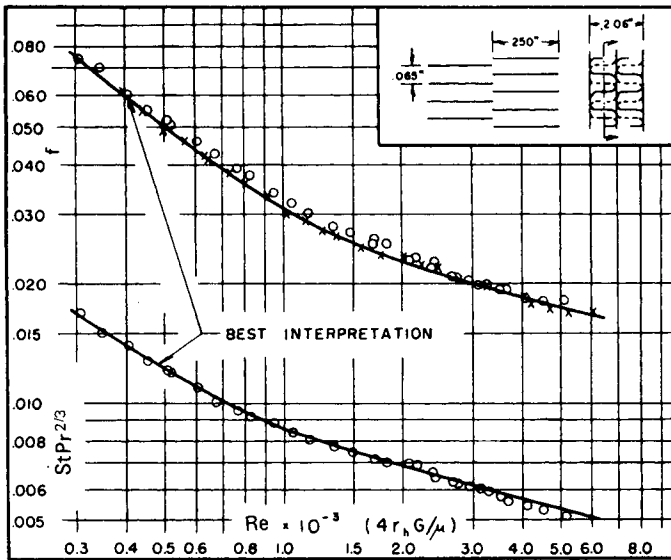
Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/column between plates, $\beta = 461.0$ ft²/ft³ = 1512 m²/m³

Fin area (including splitter)/total area = 0.796

Fig. 10-65 Strip-fin plate-fin surface 1/4-15.4(D).



Fin pitch = 15.4 per in = 606 per m

Plate spacing, $b = 0.206$ in = 5.23×10^{-3} m

Splitter symmetrically located

Fin length = 0.250 in = 6.35×10^{-3} m

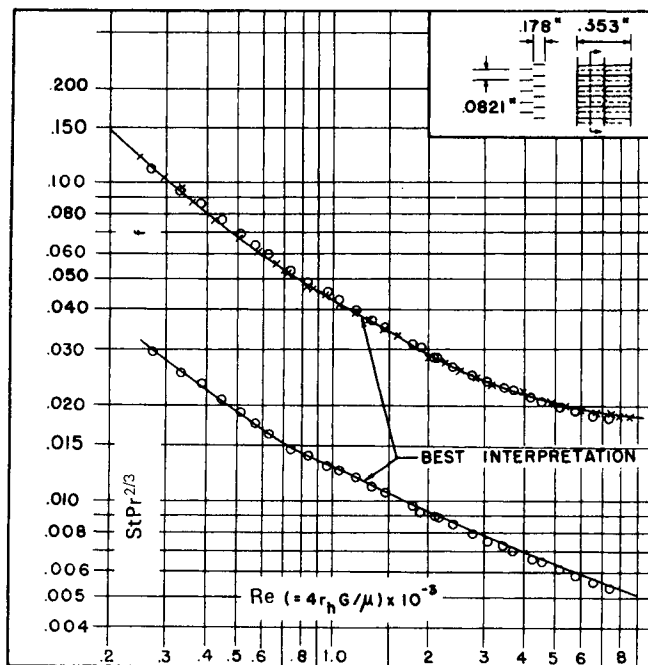
Flow passage hydraulic diameter, $4r_h = 0.00527$ ft = 1.605×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 642$ ft²/ft³ = 2106 m²/m³

Fin area (including splitter)/total area = 0.816

Fig. 10-66 Strip-fin plate-fin surface 1/6-12.18(D).



Fin pitch = 12.18 per in = 480 per m

Plate spacing, $b = 0.353$ in = 8.97×10^{-3} m

Splitter symmetrically located

Fin length flow direction = 0.178 in = 4.52×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.008648$ ft = 2.63×10^{-3} m

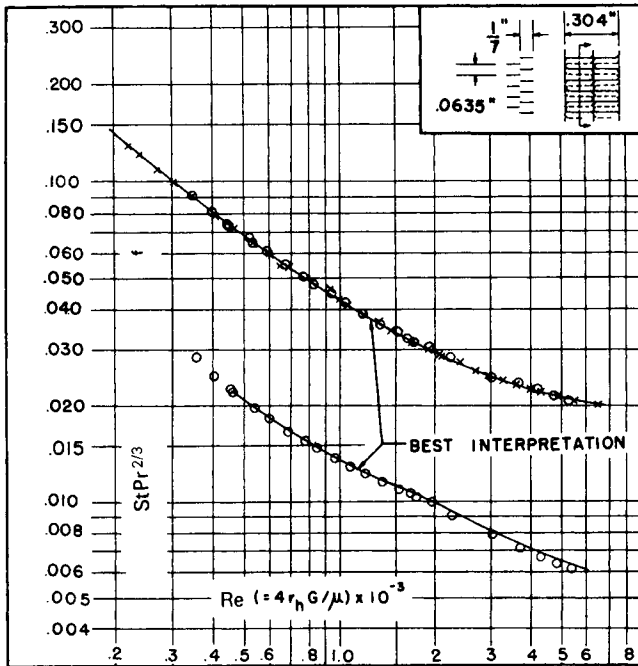
Fin metal thickness = 0.004 in, aluminum = 0.102×10^{-3} m

Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 422.4$ ft²/ft³ = 1.385 m²/m³

Fin area (including splitter)/total area = 0.847

Fig. 10-67 Strip-fin plate-fin surface 1/7 - 15.75(D).



Fin pitch = 15.75 per in = 620 per m

Plate spacing, $b = 0.304$ in = 7.72×10^{-3} m

Splitter symmetrically located

Fin length flow direction = $\frac{1}{7}$ in = 3.63×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.006790$ ft = 2.07×10^{-3} m

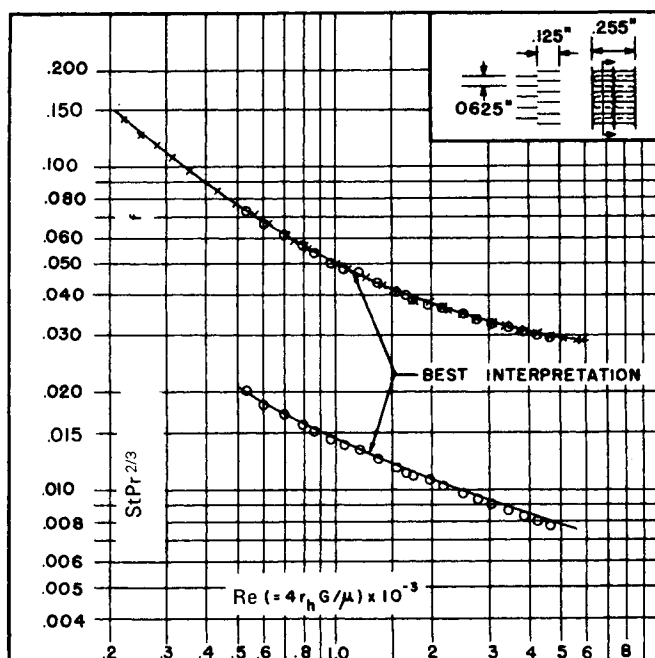
Fin metal thickness = 0.004 in, aluminum = 0.102×10^{-3} m

Splitter thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 526$ ft²/ft³ = $1,726$ m²/m³

Fin area (including splitter)/total area = 0.859

Fig. 10-68 Strip-fin plate-fin surface 1/8-16.00(D).



Fin pitch = 16.00 per in = 630 per m

Plate spacing, $b = 0.255$ in = 6.48×10^{-3} m

Splitter symmetrically located

Fin length flow direction = 0.125 in = 3.175×10^{-3} m

Flow passage hydraulic diameter, $4r_h$ 0.006112 ft = 1.862×10^{-3} m

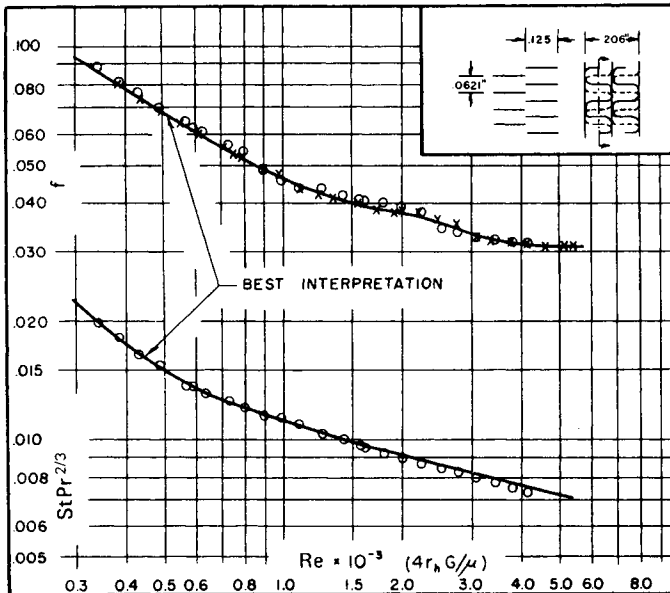
Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 549.5$ ft²/ft³ = $1,804$ m²/m³

Fin area (including splitter)/total area = 0.845

Fig. 10-69 Strip-fin plate-fin surface 1/8-16.12(D).



Fin pitch = 16.12 per in = 635 per m

Plate spacing, $b = 0.206$ in = 5.23×10^{-3} m

Splitter symmetrically located

Fin length = 0.125 in = 3.175×10^{-3} m

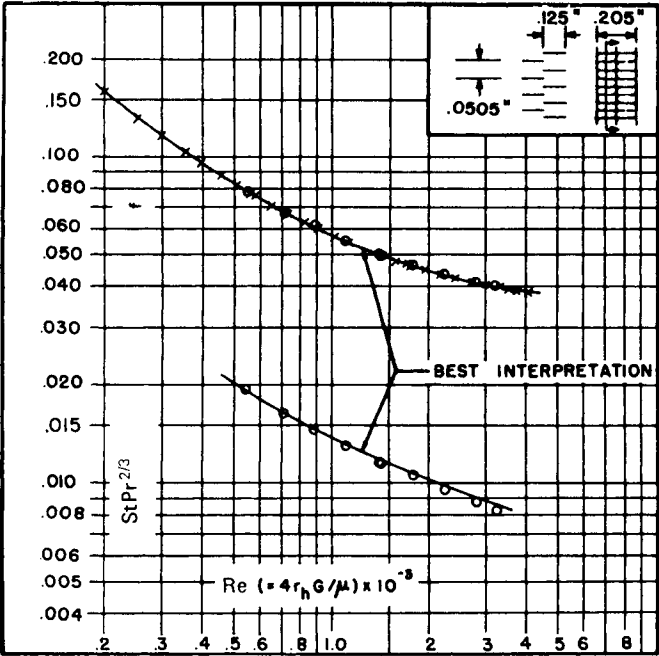
Flow passage hydraulic diameter, $4r_h = 0.00509$ ft = 1.552×10^{-3} m

Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 660$ ft²/ft³ = $2,165$ m²/m³

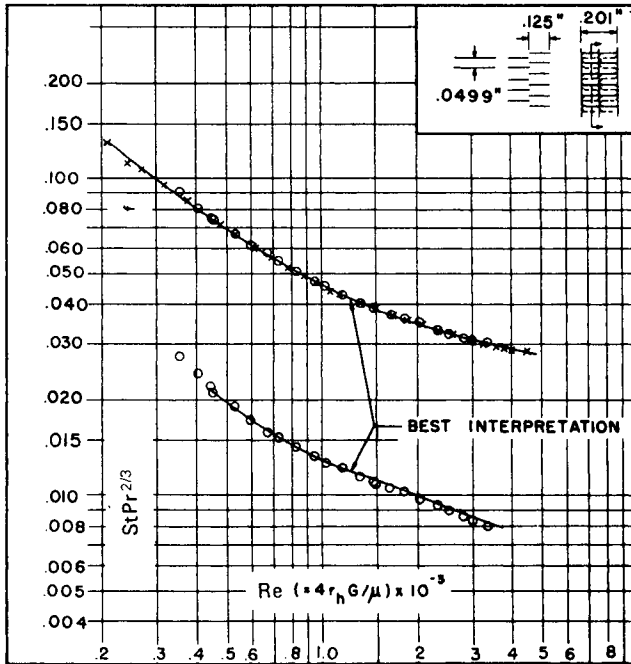
Fin area (including splitter)/total area = 0.823

Fig. 10-70 Strip-fin plate-fin surface 1/8-19.82(D).



Fin pitch = 19.82 per in = 780 per m
 Plate spacing, $b = 0.205$ in = 5.21×10^{-3} m
 Splitter symmetrically located
 Fin length flow direction = 0.125 in = 3.175×10^{-3} m
 Flow passage hydraulic diameter, $4r_h = 0.005049$ ft = 1.537×10^{-3} m
 Fin metal thickness = 0.004 in, nickel = 0.102×10^{-3} m
 Splitter metal thickness = 0.006 in = 0.152×10^{-3} m
 Total heat transfer area/volume between plates, $\beta = 680$ ft²/ft³ = $2,231$ m²/m³
 Fin area (including splitter)/total area = 0.841

Fig. 10-71 Strip-fin plate-fin surface 1/8-20.06(D).



Fin pitch = 20.06 per in = 790 per m

Plate spacing, $b = 0.201$ in = 5.11×10^{-3} m

Splitter length flow direction = 0.125 in = 3.175×10^{-3} m

Flow passage hydraulic diameter, $4r_h$, 0.004892 ft = 1.491×10^{-3} m

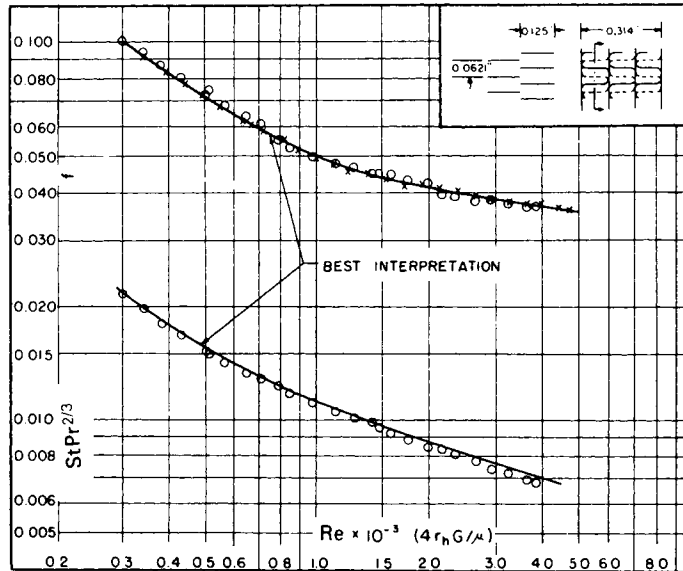
Fin metal thickness = 0.004 in, aluminum = 0.102×10^{-3} m

Splitter metal thickness = 0.006 in = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 698$ ft²/ft³ $\approx 2,290$ m²/m³

Fin area (including splitter)/total area = 0.843

Fig. 10-72 Strip-fin plate-fin surface 1/8-16.12(T).



Fin pitch = 16.12 per in = 635 per m

Plate spacing, $b = 0.314$ in = 7.98×10^{-3} m

Splitters evenly spaced

Fin length = 0.125 in = 3.175×10^{-3} m

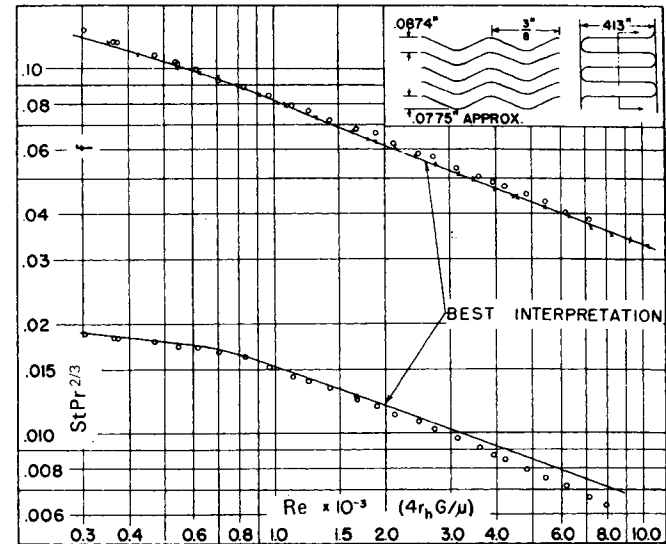
Flow passage hydraulic diameter, $4r_h = 0.00514$ ft = 1.567×10^{-3} m

Fin metal thickness 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 650$ ft²/ft³ = 2.133 m²/m³

Fin area (including splitters)/total area = 0.882

Fig. 10-73 Wavy-fin plate-fin surface 11.44-3/8W.



Fin pitch = 11.44 per in = 450 per m

Plate spacing, $b = 0.413$ in = 10.49×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.01060$ ft = 3.231×10^{-3} m

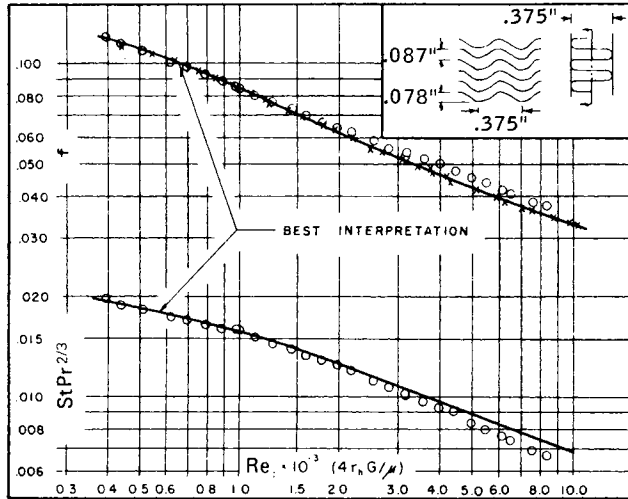
Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 351$ ft²/ft³ = $1,152$ m²/m³

Fin area/total area = 0.847

Note: Hydraulic diameter based on free-flow area normal to mean flow direction.

Fig. 10-74 Wavy-fin plate-fin surface 11.5-3/8W.



Fin pitch = 11.5 per in = 453 per m

Plate spacing, $b = 0.375$ in = 9.25×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.00993$ ft = 3.023×10^{-3} m

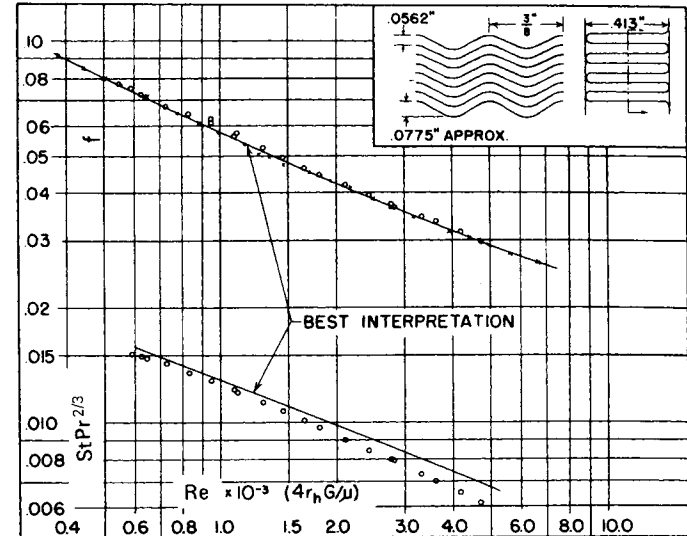
Fin metal thickness = 0.010 in, aluminum = 0.254×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 345$ ft²/ft³ = $1,138$ m²/m³

Fin area/total area = 0.822

Note: Hydraulic diameter based on free-flow area normal to mean flow direction.

Fig. 10-75 Wavy-fin plate-fin surface 17.8-3/8W.



Fin pitch = 17.8 per in = 701 per m

Plate spacing, $b = 0.413$ in = 10.49×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.00696$ ft = 2.123×10^{-3} m

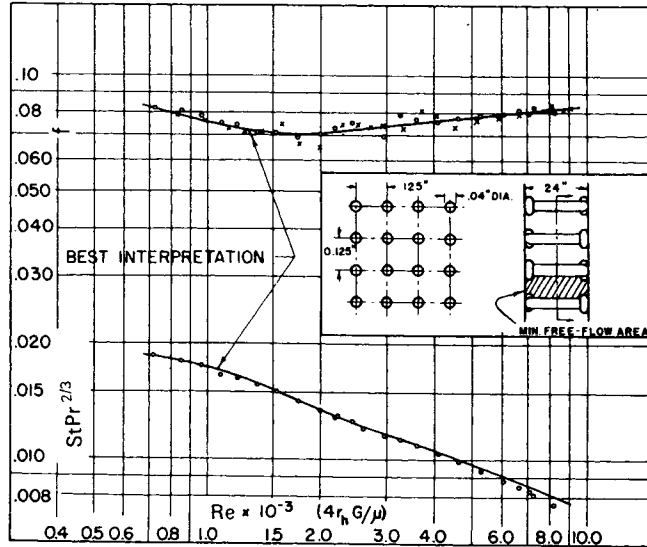
Fin metal thickness = 0.006 in, aluminum = 0.152×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 514$ ft²/ft³ = $1,686$ m²/m³

Fin area/total area = 0.892

Note: Hydraulic diameter based on free-flow area normal to mean flow direction.

Fig. 10-76 Pin-fin plate-fin surface AP-1.



Pin diameter = 0.04 in, copper = 1.02×10^{-3} m

Pin pitch parallel to flow = 0.125 in = 3.175×10^{-3} m

Pin pitch perpendicular to flow = 0.125 in = 3.175×10^{-3} m

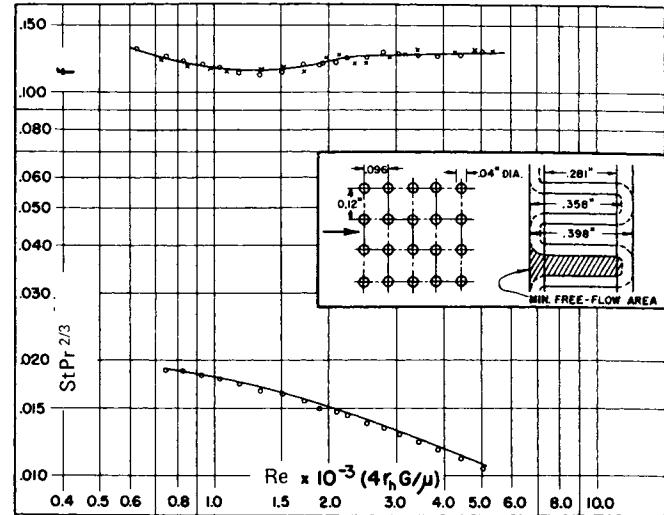
Plate spacing, $b = 0.24$ in = 6.10×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.01444$ ft = 4.404×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 188$ ft²/ft³ = 616.8 m²/m³

Fin area/total area = 0.512

Fig. 10-77 Pin-fin plate-fin surface AP-2.



Pin diameter = 0.04 in, copper = 1.02×10^{-3} m

Pin pitch parallel to flow = 0.096 in = 2.44×10^{-3} m

Pin pitch perpendicular to flow = 0.12 in = 3.05×10^{-3} m

Plate spacing, $b = 0.398$ in = 10.1×10^{-3} m

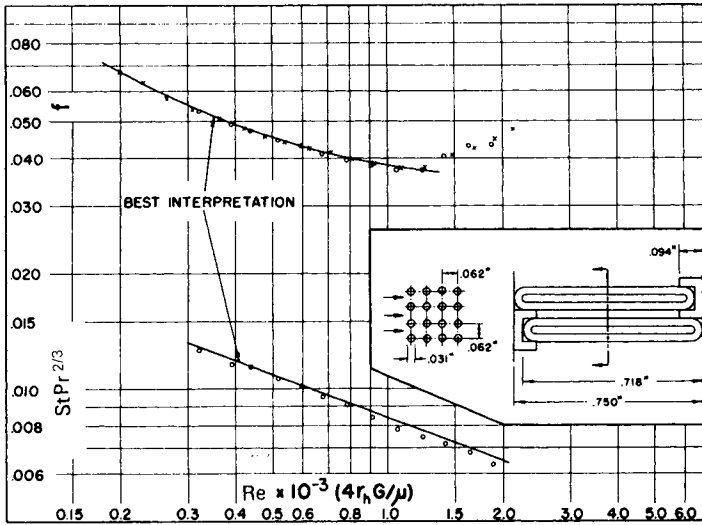
Flow passage hydraulic diameter, $4r_h = 0.01172$ ft = 3.576×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 204$ ft²/ft³ = 669 m²/m³

Fin area/total area = 0.686

Note: Plates to which the pins are attached are corrugated with the pins soldered into the corrugations.

Fig. 10-78 Pin-fin plate-fin surface PF-3.



Pin diameter = 0.031 in, aluminum = 0.79×10^{-3} m

Pin pitch parallel to flow = 0.062 in = 1.53×10^{-3} m

Pin pitch perpendicular to flow = 0.062 in = 1.53×10^{-3} m

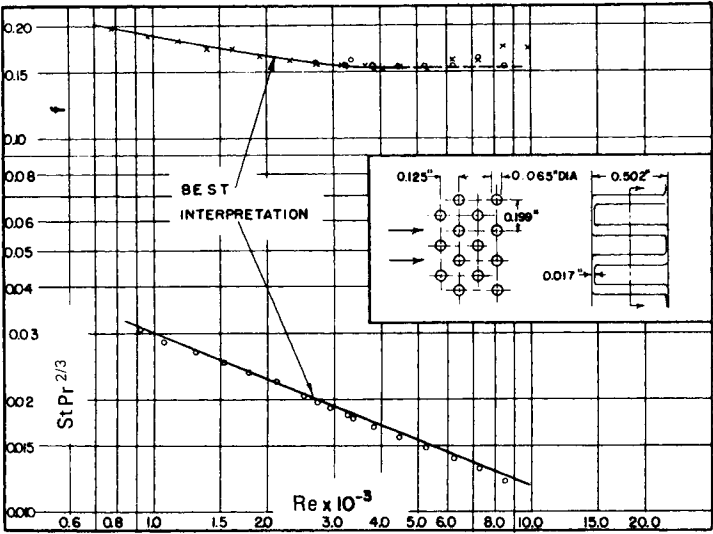
Plate spacing, $b = 0.750$ in = 19.1×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.00536$ ft = 1.636×10^{-3} m

Total heat transfer area/volume between plates, $\beta = 339$ ft²/ft³ = $1,112$ m²/m³

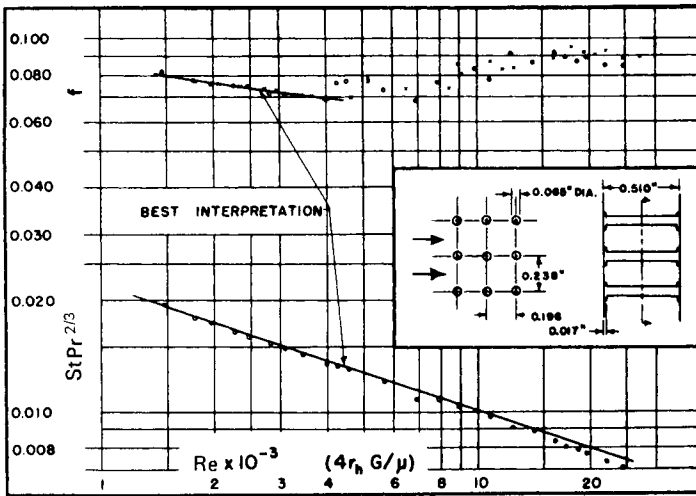
Fin area/total area = 0.843

Fig. 10-79 Pin-fin plate-fin surface PF-4(F).



Pin diameter = 0.065 in, steel = 1.65×10^{-3} m
Pin pitch parallel to flow = 0.125 in = 3.175×10^{-3} m
Pin pitch perpendicular to flow = 0.199 in = 5.055×10^{-3} m
Plate spacing, b = 0.502 in = 12.75×10^{-3} m
Flow passage hydraulic diameter, $4r_h$ = 0.0186 ft = 5.66×10^{-3} m
Total heat transfer area/volume between plates, β = 140 ft²/ft³ = 459 m²/m³
Fin area/total area = 0.704
Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-80 Pin-fin plate-fin surface PF-9(F).



Pin diameter = 0.065 in, steel = 1.65×10^{-3} m

Pin pitch parallel to flow = 0.196 in = 4.98×10^{-3} m

Pin pitch perpendicular to flow = 0.238 in = 6.05×10^{-3} m

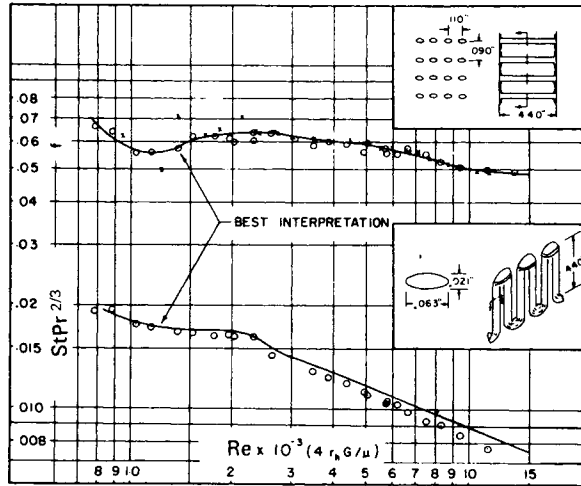
Plate spacing, b = 0.510 in = 12.95×10^{-3} m

Flow passage hydraulic diameter, $4r_h$ = 0.0297 ft = 9.042×10^{-3} m

Total heat transfer area/volume between plates, β = 96.2 ft²/ft³ = 316 m²/m³

Fin area/total area = 0.546

Fig. 10-81 Pin-fin plate-fin surface PF-10(F).



Fin pitch = 10.95 per in = 431 per m

Fin perimeter = 0.141 in, steel = 3.58×10^{-3} m

Fin cross-section area = $0.00104 \text{ in}^2 = 0.671 \times 10^{-6} \text{ m}^2$

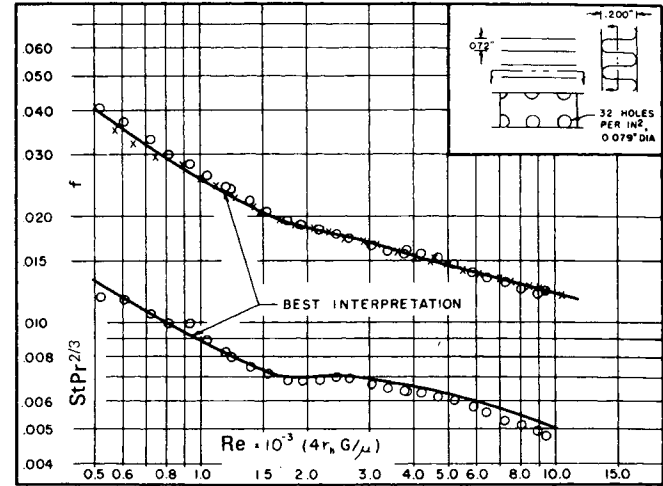
Plate spacing, $b = 0.440 \text{ in} = 11.18 \times 10^{-3} \text{ m}$

Flow passage hydraulic diameter, $4r_h = 0.01426 \text{ ft} = 4.343 \times 10^{-3} \text{ m}$

Total heat transfer area/volume between plates, $\beta = 214 \text{ ft}^2/\text{ft}^3 = 702 \text{ m}^2/\text{m}^3$

Fin area/total area = 0.693

Fig. 10-82 Perforated-fin plate-fin surface 13.95(P).



Fin pitch = 13.95 per in = 510 per m

Plate spacing, $b = 0.200 \text{ in} = 5.08 \times 10^{-3} \text{ m}$

Fin center material perforated with $0.079 \text{ in} = (2 \times 10^{-3} \text{ m})$ diameter holes spaced 32 per in^2 (4.96×10^6 per m^2) on square pattern (16% open)

Flow passage hydraulic diameter, $4r_h = 0.00822 \text{ ft} = 2.504 \times 10^{-3} \text{ m}$

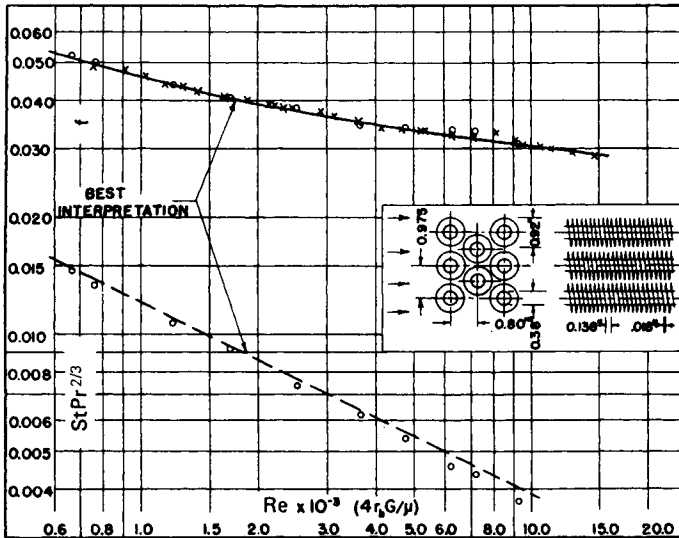
Fin metal thickness = 0.012 in , aluminum = $0.305 \times 10^{-3} \text{ m}$

Total heat transfer area/volume between plates, $\beta = 381 \text{ ft}^2/\text{ft}^3 = 1,250 \text{ m}^2/\text{m}^3$

Fin area/total area = 0.705

Area in rim of holes included in fin area

Fig. 10-83 Finned circular tubes, surface CF-7.34.



Tube outside diameter = 0.38 in = 9.65×10^{-3} m

Fin pitch = 7.34 per in = 289 per m

Flow passage hydraulic diameter, $4r_h = 0.0154$ ft = 4.75×10^{-3} m

Fin thickness (average) $t = 0.018$ in, aluminum = 0.46×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.538$

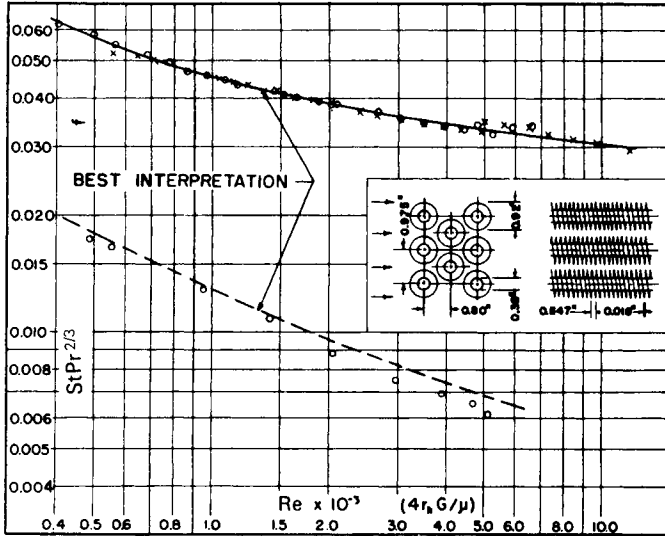
Heat transfer area/total volume, $\alpha = 140$ ft²/ft³ = 459 m²/m³

Fin area/total area = 0.892

Note: Experimental uncertainty for heat transfer results possibly somewhat greater than the nominal $\pm 5\%$ quoted for the other surfaces because of the necessity of estimating a contact resistance in the bimetal tubes.

† Fins slightly tapered.

Fig. 10-84 Finned circular tubes, surface CF-8.72.



Tube outside diameter = 0.38 in = 9.65×10^{-3} m

Fin pitch = 8.72 per in = 343 per m

Flow passage hydraulic diameter, $4r_h = 0.01288$ ft = 3.929×10^{-3} m

Fin thickness (average)† = 0.018 in, aluminum = 0.46×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.524$

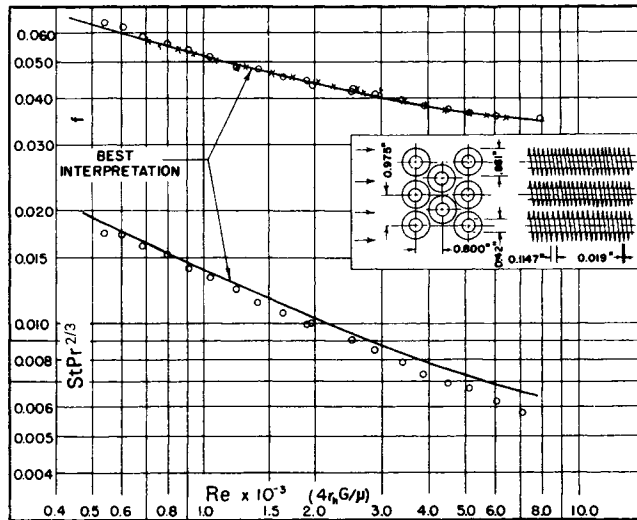
Heat transfer area/total volume, $\alpha = 163$ ft²/ft³ = 535 m²/m³

Fin area/total area = 0.910

Note: Experimental uncertainty for heat transfer results possibly somewhat greater than the nominal $\pm 5\%$ quoted for the other surfaces because of the necessity of estimating a contact resistance in the bimetal tubes.

†Fins slightly tapered.

Fig. 10-85 Finned circular tubes, surface CF-8.72(c).



Tube outside diameter = 0.42 in = 10.67×10^{-3} m

Fin pitch = 8.72 per in = 343 per m

Flow passage hydraulic diameter, $4r_h = 0.01452$ ft = 4.425×10^{-3} m

Fin thickness (average) $\dagger = 0.019$ in, copper = 0.48×10^{-3} m

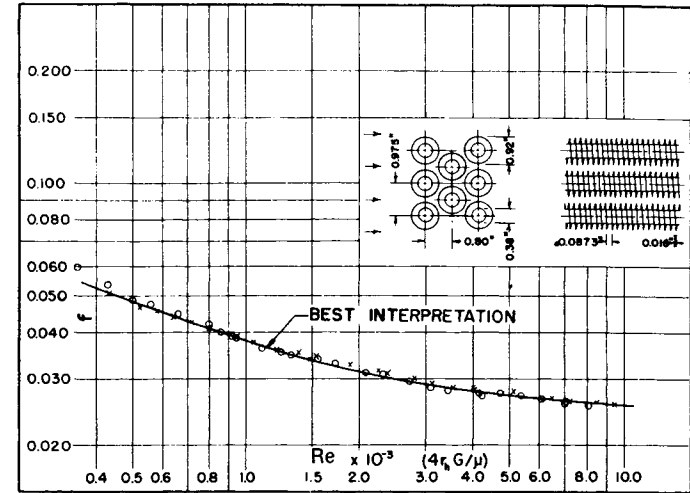
Free-flow area/frontal area, $\sigma = 0.494$

Heat transfer area/total volume, $\alpha = 136$ ft²/ft³ = 446 m²/m³

Fin area/total area = 0.876

$\dagger$ Fins slightly tapered.

Fig. 10-86 Finned circular tubes, surface CF-11.46 (friction data only).



Tube outside diameter = 0.38 in = 9.65×10^{-3} m

Fin pitch = 11.46 per in = 451 per m

Flow passage hydraulic diameter, $4r_h = 0.00976$ ft = 2.979×10^{-3} m

Fin thickness (average) $\dagger = 0.016$ in, aluminum = 0.41×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.510$

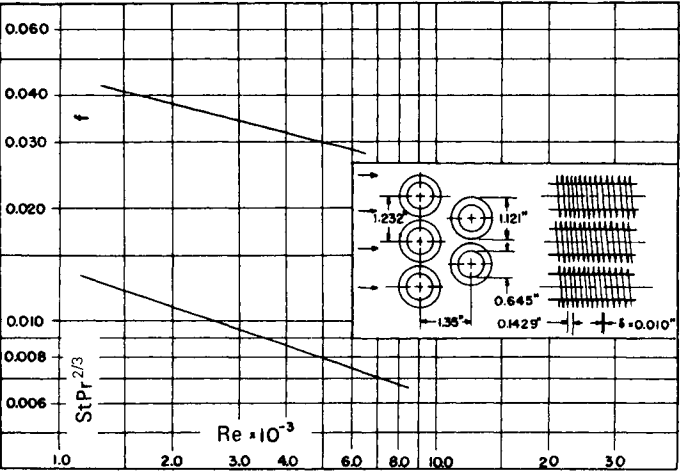
Heat transfer area/total volume, $\alpha = 209$ ft²/ft³ = 686 m²/m³

Fin area/total area = 0.931

Note: Heat transfer data not included for this surface because of the large contribution of contact resistance between the bimetal tubes and the large uncertainty in evaluating this resistance.

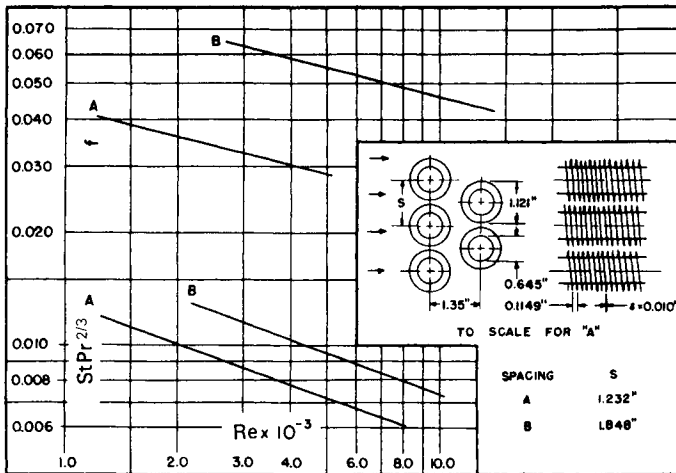
$\dagger$ Fins slightly tapered.

Fig. 10-87 Finned circular tubes, surface CF-7.0-5/8J. (Data of Jameson.)



Tube outside diameter = 0.645 in = 16.38×10^{-3} m
Fin pitch = 7.0 per in = 276 per m
Flow passage hydraulic diameter, $4r_h = 0.0219$ ft = 6.68×10^{-3} m
Fin thickness = 0.010 in = 0.25×10^{-3} m
Free-flow area/frontal area, $\sigma = 0.449$
Heat transfer area/total volume, $\alpha = 82$ ft²/ft³ = 269×10^{-3} m
Fin area/total area = 0.830
Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-88 Finned circular tubes, surface CF-8.7-5/8J. (Data of Jameson.)



Tube outside diameter = 0.645 in = 16.38×10^{-3} m

Fin pitch = 8.7 per in = 343 per m

Fin thickness = 0.010 in = 0.25×10^{-3} m

Fin area/total area = 0.862

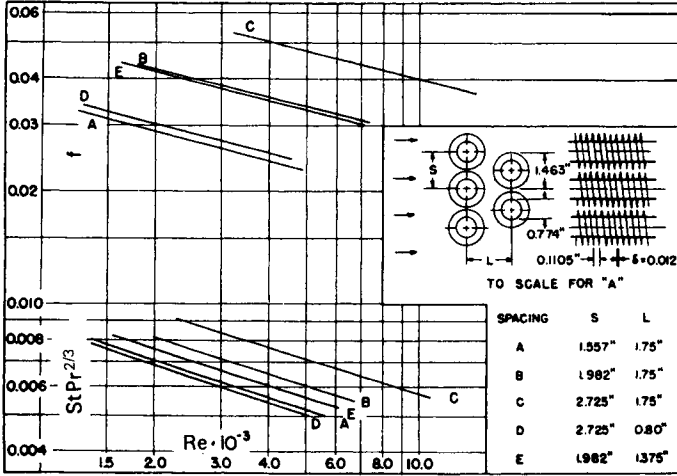
Flow passage hydraulic diameter, $4r_h = 0.01797$ $\frac{A}{B}$ 0.0383 ft. $\frac{A}{B}$ 5.48×10^{-3} 11.67×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.443$ 0.628

Heat transfer area/total volume, $\alpha = 98.7$ $\frac{A}{B}$ 65.7 ft²/ft³ 324 216 m²/m³

Note: Minimum free-flow area is in spaces transverse to flow.

Fig. 10-89 Finned circular tubes, surface CF-9.05-3/4J. (Data of Jameson.)



Tube outside diameter = 0.774 in = 19.66×10^{-3} m

Fin pitch = 9.05 per in = 356 per m

Fin thickness = 0.012 in = 0.305×10^{-3} m

Fin area/total area = 0.917

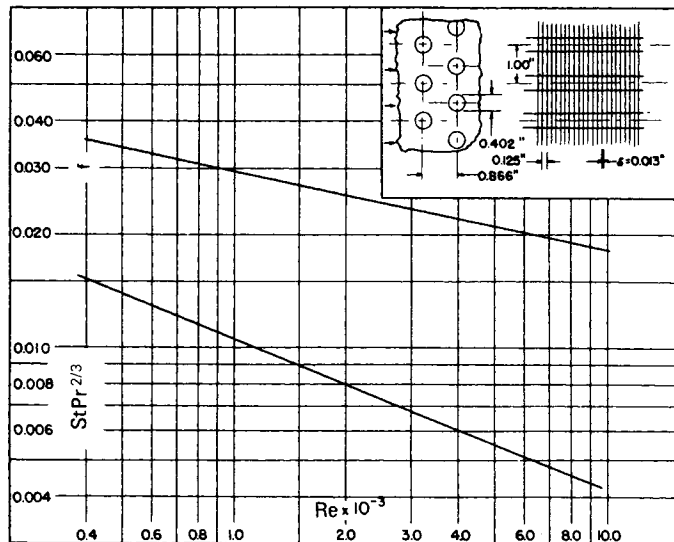
Flow passage hydraulic diameter, $4r_h$ =	A	B	C	D	E
	0.01681	0.02685	0.0445	0.01587	0.02108 ft
	5.131×10^{-3}	8.179×10^{-3}	13.59×10^{-3}	4.846×10^{-3}	6.426×10^{-3} m

Free-flow area/frontal area, σ =	0.455	0.572	0.688	0.537	0.572
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Heat transfer area/total volume, α =	108	85.1	61.9	135	108 ft ² /ft ³
	354	279	203	443	354 m ² /m ³

Note: Minimum free-flow area in all cases occurs in the spaces transverse to the flow, except for *D*, in which the minimum area is in the diagonals.

Fig. 10-91 Finned circular tubes, surface 8.0-3/8T.



Tube outside diameter = 0.402 in = 10.2×10^{-3} m

Fin pitch = 8.0 per in = 315 per m

Flow passage hydraulic diameter, $4r_h = 0.01192$ ft = 3.632×10^{-3} m

Fin thickness = 0.013 in = 0.33×10^{-3} m

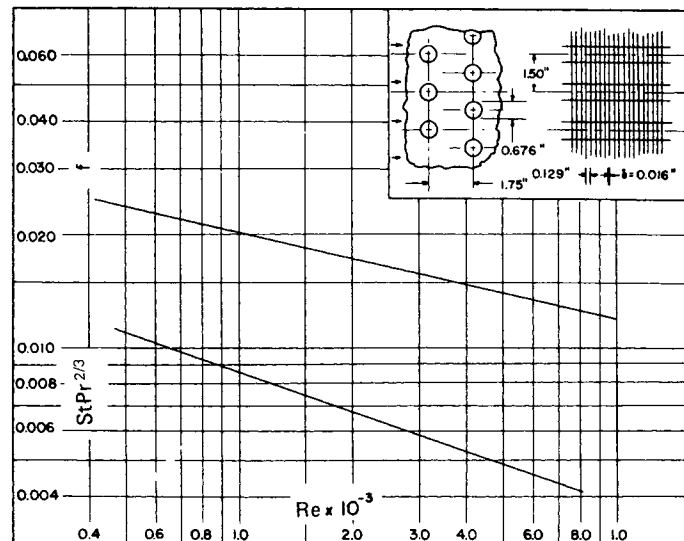
Free-flow area/frontal area, $\sigma = 0.534$

Heat transfer area/total volume, $\alpha = 179$ ft²/ft³ = 587 m²/m³

Fin area/total area = 0.913

Note: Minimum free-flow area in spaces transverse to flow.

Fig. 10-92 Finned circular tubes, surface 7.75-5/8T.



Tube outside diameter = 0.676 in = 17.17×10^{-3} m

Fin pitch = 7.75 per in = 305 per m

Flow passage hydraulic diameter, $4r_h = 0.0114$ ft = 3.48×10^{-3} m

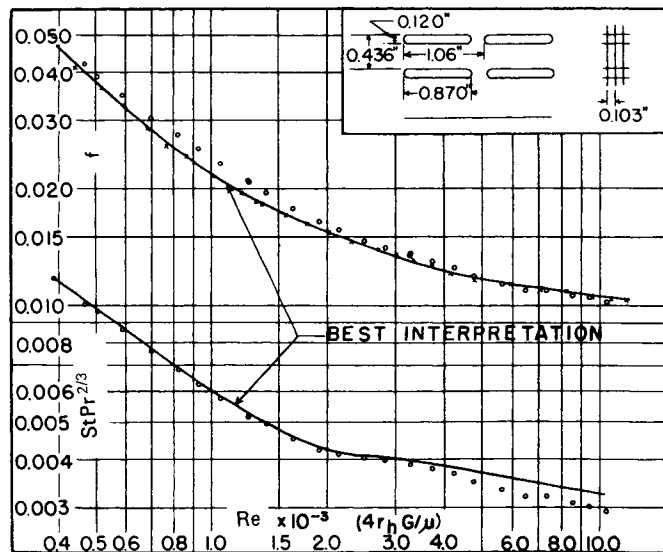
Fin-flow area/frontal area, $\sigma = 0.481$

Heat transfer area/total volume, $\alpha = 169$ ft²/ft³ = 554 m²/m³

Fin area/total area = 0.950

Note: Minimum free-flow area in spaces transverse to flow.

Fig. 10-93 Finned flat tubes, surface 9.68–0.87.



Fin pitch = 9.68 per in = 381 per m

Flow passage hydraulic diameter, $4r_h = 0.01180$ ft = 3.597×10^{-3} m

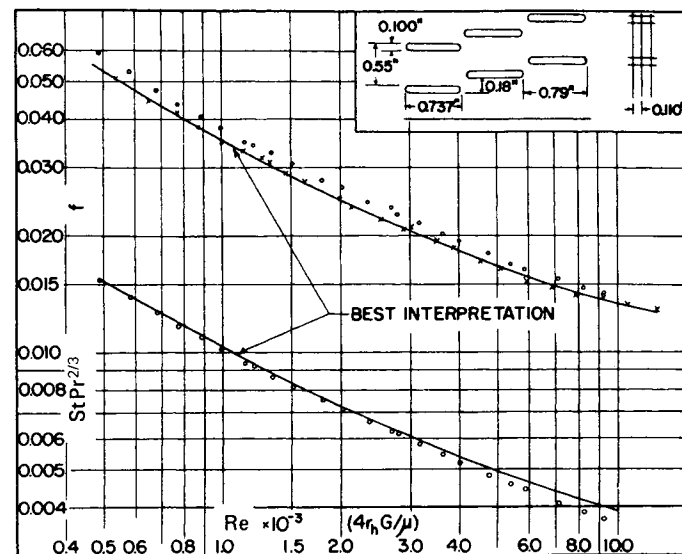
Fin metal thickness = 0.004 in, copper = 0.102×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.697$

Total heat transfer area/total volume, $\alpha = 229$ ft²/ft³ = 751 m²/m³

Fin area/total area = 0.795

Fig. 10-94 Finned flat tubes, surface 9.1–0.737-S.



Fin pitch = 9.1 per in = 358 per m

Flow passage hydraulic diameter, $4r_h = 0.01380$ ft = 4.206×10^{-3} m

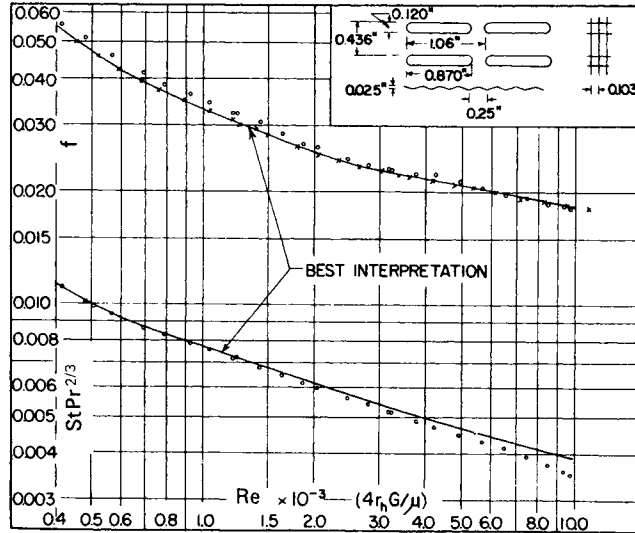
Fin metal thickness = 0.004 in, copper = 0.102×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.788$

Total heat transfer area/total volume, $\alpha = 224$ ft²/ft³ = 735 m²/m³

Fin area/total area = 0.813

Fig. 10-95 Finned flat tubes, surface 9.68–0.87-R.



Fin pitch = 9.68 per in = 381 per m

Flow passage hydraulic diameter, $4r_h = 0.01180$ ft = 3.597×10^{-3} m

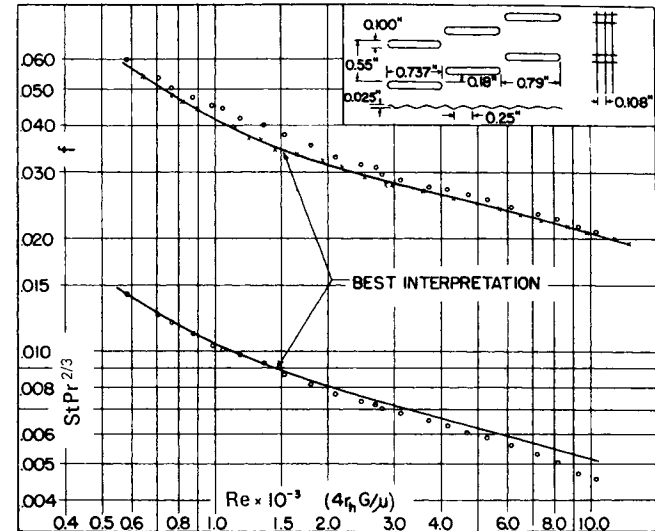
Fin metal thickness = 0.004 in, copper = 0.102×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.697$

Total heat transfer area/total volume, $\alpha = 229$ ft²/ft³ = 751 m²/m³

Fin area/total area = 0.795

Fig. 10-96 Finned flat tubes, surface 9.29–0.737-SR.



Fin pitch = 9.29 per in = 366 per m

Flow passage hydraulic diameter, $4r_h = 0.01352$ ft = 4.12×10^{-3} m

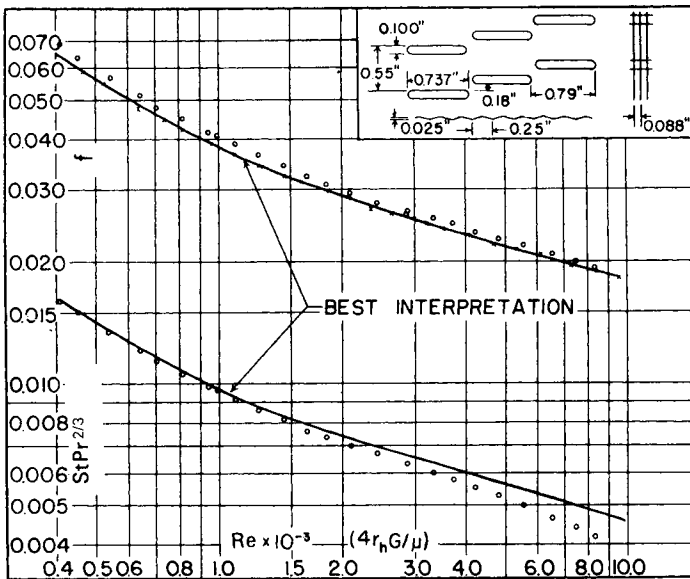
Fin metal thickness = 0.004 in, copper = 0.102×10^{-3} m

Free-flow area/frontal area, $\sigma = 0.788$

Total heat transfer area/total volume, $\alpha = 228$ ft²/ft³ = 748 m²/m³

Fin area/total area = 0.814

Fig. 10-97 Finned flat tubes, surface 11.32-0.737-SR.



Fin pitch = 11.32 per in = 446 per m

Flow passage hydraulic diameter, $4r_h = 0.01152 \text{ ft} = 3.510 \times 10^{-3} \text{ m}$

Fin metal thickness = 0.004 in, copper = $0.102 \times 10^{-3} \text{ m}$

Free-flow area/frontal area, $\sigma = 0.780$

Total heat transfer area/total volume, $\alpha = 270 \text{ ft}^2/\text{ft}^3 = 886 \text{ m}^2/\text{m}^3$

Fin area/total area = 0.845

Fig. 10-98 Crossed-rod matrices, in-line stacking; rod diameter $d = 0.375\text{ in} = 9.525 \times 10^{-3}\text{ m}$.

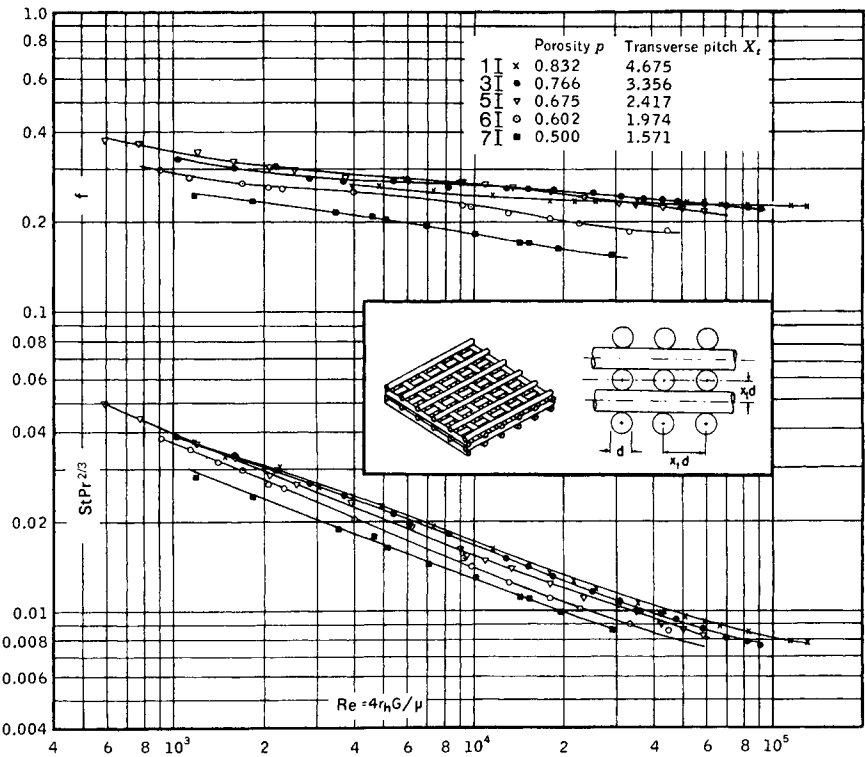


Fig. 10-99 Crossed-rod matrices, staggered stacking; rod diameter $d = 0.375 \text{ in} = 9.525 \times 10^{-3} \text{ m}$.

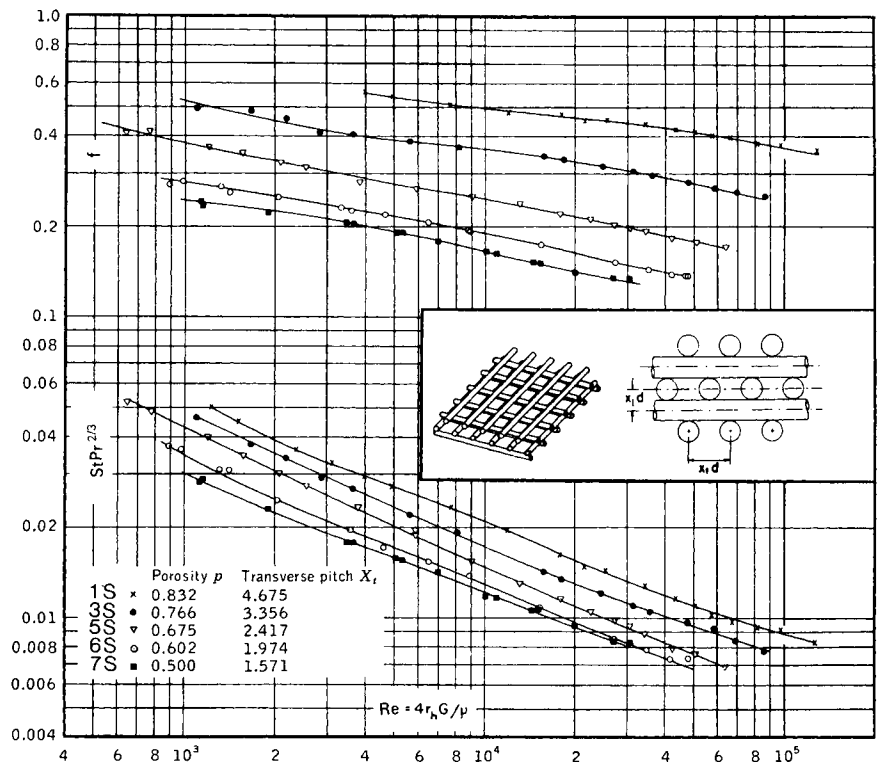


Fig. 10-100 Crossed-rod matrices, random stacking; rod diameter $d = 0.375 \text{ in} = 9.525 \times 10^{-3} \text{ m}$.

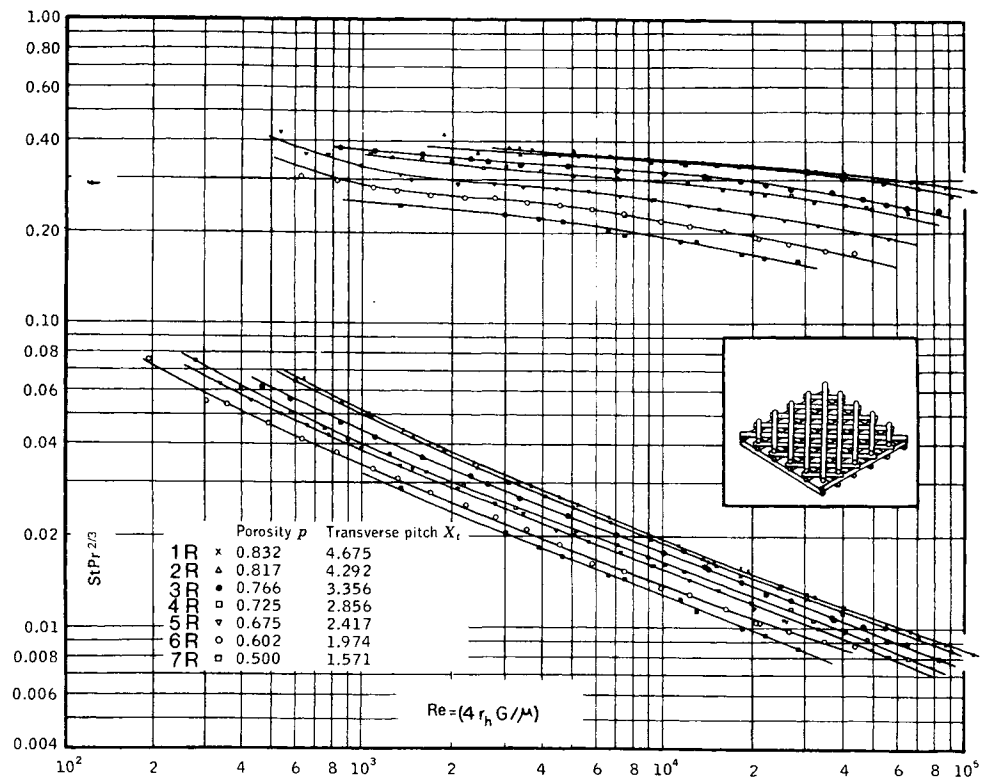
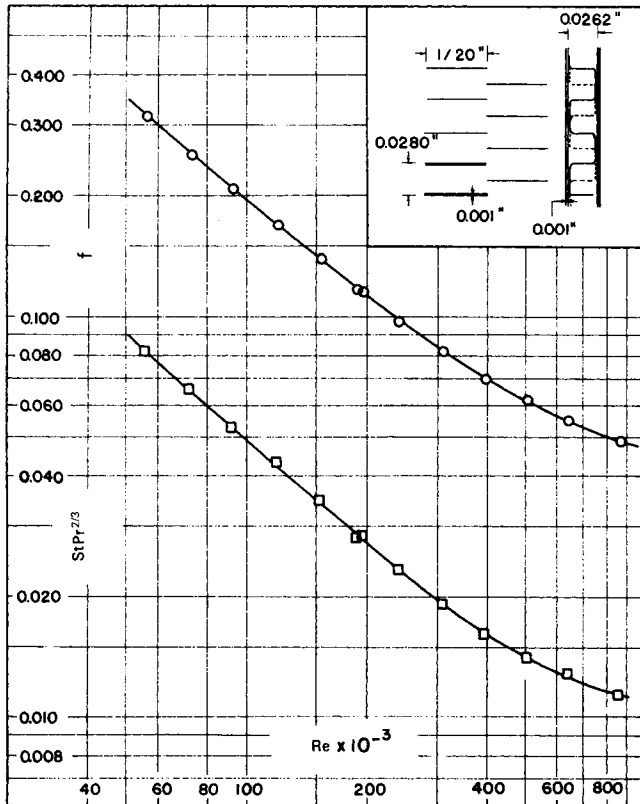


Fig. 10-101 Plate-fin matrix 501MOD.



Fin pitch = 35.74 per in = 1407 per m

Plate spacing, $b = 0.0262$ in = 0.665×10^{-3} m

Fin length = 0.05 in = 1.27×10^{-3} m

Flow passage hydraulic diameter, $4r_h = 0.001885$ ft = 0.575×10^{-3} m

Fin metal thickness = 0.001 in = 0.0254×10^{-3} m

Plate metal thickness = 0.001 in = 0.0254×10^{-3} m

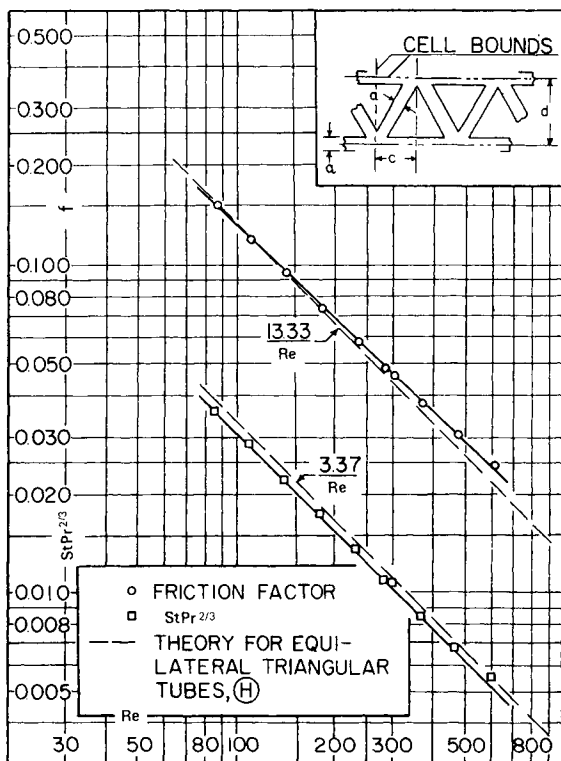
Total heat transfer area/volume between plates, $\beta = 1976$ ft²/ft³ = $6,483$ m²/m³

Total heat transfer area/volume, $\alpha = 1902$ ft²/ft³ = 6240 m²/m³

Porosity $p = 0.896$

Fins and plates unbrazed. Surface area includes curved portion of fins near the plates. Matrix is composed of a stack of fins and plates held together by pressure.

Fig. 10-102 Glass ceramic matrix 505A.



Passage count = 526 per in² = 815,300 per m²

Porosity, $p = 0.794$

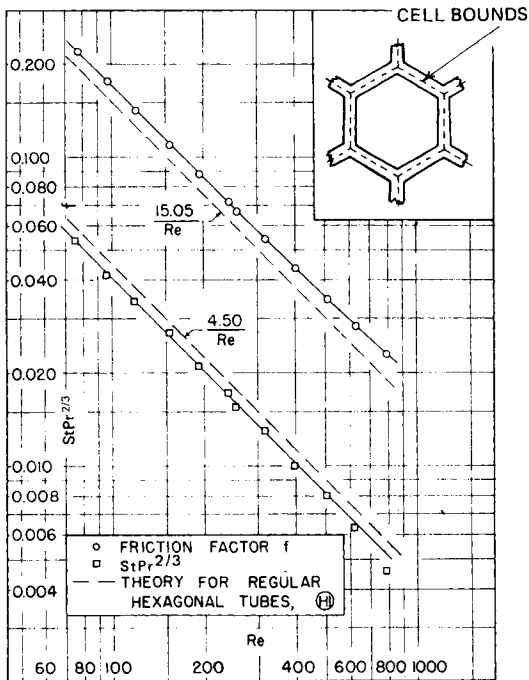
Flow passage hydraulic diameter, $4r_h = 0.00247$ ft = 0.753×10^{-3} m

Cell height/width, $d/c = 0.731$

$L/4r_h = 101$

Total heat transfer area/total volume, $\alpha = 1285$ ft²/ft³ = 4216 m²/m³

Fig. 10-103 Glass ceramic matrix 519.



Passage count = 1,683 per in² = 2,608,655 per m²

Porosity, $p = 0.8105$

Flow passage hydraulic diameter, $4r_h = 0.00197$ ft = 0.600×10^{-3} m

$L/4r_h = 127$

Total heat transfer area/total volume, $\alpha = 1,646$ ft²/ft³ = $5,400$ m²/m³

Appendix A Material Properties

To facilitate use of the design data in this book a number of charts and tables of material properties are presented in this appendix. It is not intended that this be an exhaustive compilation of properties, since such compilations are readily available, but rather the properties of a representative group of solids, liquids, and gases are presented so that feasibility studies of heat exchangers can be made without seeking information elsewhere.

The thermal conductivities of a large group of metals used in heat exchanger construction are presented in Fig. A-1. For the most part, these have been plotted from the data given by Eckert and Drake [1]. The properties are presented on the left-hand side in English units and on the right hand side in SI units.

Figures A-2 to A-17 give the properties of a number of commonly used gases and liquids in English units. The properties of the same fluids in SI units are presented in tabular form, Tables A-1 to A-14.

Figures A-2 to A-7 present the properties of a representative group of gases, all at 1 atm pressure. These are based primarily on Bureau of Standards Circular 564 [2] but have been supplemented from other sources. For most applications these gases may be treated as perfect gases for which the transport properties are independent of pressure, and the densities may be determined from the perfect gas equation of state $P/\rho = (R/M)T$, where R is the universal gas constant, $1,545 \text{ ft} \cdot \text{lb}_f/(\text{lb}_m \cdot \text{mol} \cdot ^\circ\text{R})$, and M is the molecular weight, $\text{lb}_m/\text{lb}_m \cdot \text{mol}$.

The properties of steam are given in Fig. A-8 for two different pressures. These are based on the Russian data presented by Nowak and Grosh [3].

The properties of saturated water (Fig. A-9) are based on Ref. 2. The liquid oxygen and liquid hydrogen properties (Figs. A-10 and A-11) are plotted from the data of Scott [4].

Figures A-12 to A-14 include the properties of some characteristic hydrocarbon fuels and oils, plotted from data in Ref. 5. The data on liquid metals (Figs. A-15 to A-17) have been taken from the *Liquid Metals Handbook* [6].

The tables of properties in SI units are all taken from Kays and Crawford [7].

These have in a number of cases been taken from more modern sources, and where there are minor discrepancies with the data in the figures, the tabular data should be considered more definitive.

Figures A-18 to A-20 contain correction factors that may be used to determine the density and specific heats of humid air and of the products of combustion of air and hydrocarbon fuels. In each case the factor is to be multiplied by the density or specific heat of dry air at the temperature and pressure of the mixture. Although the labels on these figures are in English units, they are in fact nondimensional and may be used with SI units.

Table A-15 contains a convenient set of conversion factors for SI and English units, as well as some of the older units whose use still persists (gallons, centipoise, miles, etc.).

REFERENCES

1. Eckert, E. R. G., and R. M. Drake: "Heat and Mass Transfer," McGraw-Hill Book Company, New York, 1959.
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3. Nowak, E. S., and R. J. Grosh: An Investigation of Certain Thermodynamic and Transport Properties of Water and Water Vapor in the Critical Region, *Argonne Natl. Lab. ANL 6064*, 1959.
4. Scott, R. B.: "Cryogenic Engineering," D. Van Nostrand Company, Inc., Princeton, N.J., 1959.
5. "Aero-space Applied Thermodynamics Manual," Society of Automotive Engineers, Inc., February 1960.
6. "Liquid Metals Handbook," R. N. Lyon, editor-in-chief, U.S. Government Printing Office, 2d ed., 1952; NAVEXOS P-733 (rev).
7. Kays, W. M., and M. E. Crawford: "Convective Heat and Mass Transfer," McGraw-Hill Book Company, New York, 1980.

Table A-1 Properties of Air at $P = 101.325 \text{ kPa}$

$M = 28.966$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$h, \text{W}/(\text{m} \cdot \text{K})$	Pr
100	3.5985	70.60 -07	19.62 -07	1.028	92.20 -04	0.787
150	2.3673	10.38 -06	43.85 -07	1.011	13.75 -03	0.763
200	1.7690	13.36 -06	75.52 -07	1.006	18.10 -03	0.743
250	1.4119	16.06 -06	11.37 -06	1.003	22.26 -03	0.724
263	1.3421	16.70 -06	12.44 -06	1.003	23.28 -03	0.720
273	1.2930	17.20 -06	13.30 -06	1.004	24.07 -03	0.717
283	1.2473	17.69 -06	14.18 -06	1.004	24.86 -03	0.714
293	1.2047	18.17 -06	15.08 -06	1.004	25.63 -03	0.712
300	1.1766	18.53 -06	15.75 -06	1.005	26.14 -03	0.711
303	1.1650	18.64 -06	16.00 -06	1.005	26.37 -03	0.710
313	1.1277	19.11 -06	16.95 -06	1.005	27.09 -03	0.709
323	1.0928	19.57 -06	17.91 -06	1.006	27.80 -03	0.708
333	1.0600	20.02 -06	18.89 -06	1.007	28.51 -03	0.707
343	1.0291	20.47 -06	19.89 -06	1.008	29.21 -03	0.706
350	1.0085	20.81 -06	20.63 -06	1.008	29.70 -03	0.706
353	1.0000	20.91 -06	20.91 -06	1.008	29.89 -03	0.705
363	0.9724	21.34 -06	21.95 -06	1.009	30.58 -03	0.704
373	0.9463	21.77 -06	23.01 -06	1.010	31.26 -03	0.703
400	0.8825	22.94 -06	26.00 -06	1.013	33.05 -03	0.703
450	0.7844	24.93 -06	31.78 -06	1.020	36.33 -03	0.700
500	0.7060	26.82 -06	37.99 -06	1.029	39.51 -03	0.699
550	0.6418	28.60 -06	44.56 -06	1.039	42.60 -03	0.698
600	0.5883	30.30 -06	51.50 -06	1.051	45.60 -03	0.699
650	0.5431	31.93 -06	58.80 -06	1.063	48.40 -03	0.701
700	0.5043	33.49 -06	66.41 -06	1.075	51.30 -03	0.702
750	0.4706	34.98 -06	74.32 -06	1.087	54.10 -03	0.703
800	0.4412	36.43 -06	82.56 -06	1.099	56.90 -03	0.703
850	0.4153	37.83 -06	91.10 -06	1.110	59.70 -03	0.703
900	0.3922	39.18 -06	99.90 -06	1.121	62.50 -03	0.702
950	0.3716	40.49 -06	10.90 -05	1.131	64.90 -03	0.705
1,000	0.3530	41.77 -06	11.83 -05	1.141	67.20 -03	0.709
1,100	0.3209	44.4 -06	13.8 -05	1.160	73.2 -03	0.705
1,200	0.2942	46.9 -06	15.9 -05	1.177	78.2 -03	0.705
1,300	0.2715	49.3 -06	18.2 -05	1.195	83.7 -03	0.705
1,400	0.2521	51.7 -06	20.5 -05	1.212	89.1 -03	0.704
1,500	0.2353	54.0 -06	22.9 -05	1.230	94.6 -03	0.704
1,600	0.2206	56.3 -06	25.5 -05	1.248	10.0 -02	0.703
1,700	0.2076	58.5 -06	28.2 -05	1.266	10.5 -02	0.702
1,800	0.1961	60.7 -06	31.0 -05	1.286	11.1 -02	0.701
1,900	0.1858	62.9 -06	33.9 -05	1.307	11.7 -02	0.700
2,000	0.1765	65.0 -06	36.8 -05	1.331	12.4 -02	0.699
2,100	0.1681	67.2 -06	40.4 -05	1.359	13.1 -02	0.696
2,200	0.1605	69.3 -06	43.2 -05	1.392	13.9 -02	0.693
2,300	0.1535	71.4 -06	46.5 -05	1.434	14.9 -02	0.688
2,400	0.1471	73.5 -06	50.0 -05	1.487	16.1 -02	0.681
2,500	0.1412	75.7 -06	53.6 -05	1.556	17.5 -02	0.673

Table A-2 Properties of O₂ at $P = 101.325$ kPa $M = 31.999$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
100	3.9906	76.80 -07	19.25 -07	0.947	90.50 -04	0.804
150	2.6186	11.27 -06	43.04 -07	0.918	13.76 -03	0.754
200	1.9557	14.65 -06	74.91 -07	0.914	18.24 -03	0.735
250	1.5598	17.77 -06	11.39 -06	0.912	22.54 -03	0.718
300	1.2998	20.67 -06	15.90 -06	0.918	26.74 -03	0.710
350	1.1141	23.37 -06	20.98 -06	0.292	30.56 -03	0.710
400	0.9749	25.89 -06	26.56 -06	0.942	34.20 -03	0.713
450	0.8665	28.28 -06	32.64 -06	0.956	37.70 -03	0.717
500	0.7799	30.54 -06	39.16 -06	0.971	41.20 -03	0.720
600	0.6499	34.70 -06	53.39 -06	1.002	48.00 -03	0.725
700	0.5571	38.50 -06	69.11 -06	1.031	54.40 -03	0.730
800	0.4874	42.10 -06	86.37 -06	1.054	60.30 -03	0.736
900	0.4333	45.40 -06	10.48 -05	1.074	66.10 -03	0.737
1,000	0.3899	48.50 -06	12.44 -05	1.090	71.70 -03	0.737
1,100	0.3545	51.40 -06	14.50 -05	1.103	77.10 -03	0.736
1,200	0.3250	54.20 -06	16.68 -05	1.115	82.10 -03	0.736
1,300	0.3000	56.90 -06	18.97 -05	1.125	87.10 -03	0.735
1,400	0.2785	59.50 -06	21.36 -05	1.134	92.10 -03	0.733

Table A-3 Properties of N₂ at $P = 101.325$ kPa $M = 28.018$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
100	3.4800	68.70 -07	19.74 -07	1.072	94.10 -04	0.783
150	2.2890	10.00 -06	43.69 -07	1.047	13.85 -03	0.756
200	1.7107	12.86 -06	75.20 -07	1.043	18.26 -03	0.734
250	1.3657	15.46 -06	11.32 -06	1.039	22.22 -03	0.724
300	1.1381	17.86 -06	15.69 -06	1.039	25.98 -03	0.714
350	0.9755	20.08 -06	20.58 -06	1.040	29.39 -03	0.711
400	0.8536	22.14 -06	25.94 -06	1.044	32.52 -03	0.711
450	0.7587	24.08 -06	31.74 -06	1.049	35.64 -03	0.709
500	0.6829	25.90 -06	37.93 -06	1.056	38.64 -03	0.708
600	0.5691	29.27 -06	51.44 -06	1.075	44.10 -03	0.713
700	0.4878	32.35 -06	66.32 -06	1.097	49.30 -03	0.720
800	0.4268	35.20 -06	82.48 -06	1.122	54.10 -03	0.730
900	0.3794	37.86 -06	99.80 -06	1.145	58.70 -03	0.739
1,000	0.3414	40.36 -06	11.82 -05	1.167	63.10 -03	0.746
1,100	0.3104	42.70 -06	13.76 -05	1.186	67.20 -03	0.754
1,200	0.2845	45.00 -06	15.82 -05	1.204	71.30 -03	0.760
1,300	0.2626	47.10 -06	17.93 -05	1.219	75.40 -03	0.761
1,400	0.2439	49.20 -06	20.17 -05	1.232	79.70 -03	0.761

Table A-4 Properties of H₂ at $P = 101.325$ kPa $M = 2.016$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
100	0.2457	42.10 -07	17.14 -06	11.16	67.60 -03	0.695
150	0.1638	55.70 -07	34.01 -06	12.61	98.60 -03	0.712
200	0.1228	67.80 -07	55.20 -06	13.52	12.80 -02	0.716
250	0.0983	79.00 -07	80.39 -06	14.03	15.60 -02	0.711
300	0.0819	89.40 -07	10.92 -05	14.27	18.15 -02	0.703
350	0.0702	99.40 -07	14.16 -05	14.38	20.33 -02	0.703
400	0.0614	10.91 -06	17.76 -05	14.49	22.12 -02	0.715
450	0.0546	11.84 -06	21.69 -05	14.49	23.89 -02	0.718
500	0.0491	12.74 -06	25.93 -05	14.50	25.64 -02	0.720
600	0.0409	14.50 -06	35.41 -05	14.53	29.10 -02	0.724
700	0.0351	16.10 -06	45.87 -05	14.60	32.50 -02	0.723
800	0.0307	17.70 -06	57.64 -05	14.71	36.00 -02	0.723
900	0.0273	19.20 -06	70.34 -05	14.83	39.40 -02	0.723
1,000	0.0246	20.70 -06	84.26 -05	14.99	42.80 -02	0.725
1,100	0.0223	22.20 -06	99.40 -05	15.17	46.20 -02	0.729
1,200	0.0205	23.60 -06	11.53 -04	15.36	49.50 -02	0.732
1,300	0.0189	25.00 -06	13.23 -04	15.57	52.80 -02	0.737
1,400	0.0175	26.30 -06	14.99 -04	15.80	56.20 -02	0.739

Table A-5 Properties of He4 at $P = 101.325$ kPa $M = 4.003$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
100	0.4878	97.80 -07	20.05 -06	5.194	73.60 -03	0.690
150	0.3252	12.50 -06	38.44 -06	5.194	96.90 -03	0.670
200	0.2439	15.10 -06	61.91 -06	5.193	11.80 -02	0.665
250	0.1951	17.60 -06	90.20 -06	5.193	13.70 -02	0.667
300	0.1626	19.90 -06	12.24 -05	5.193	15.50 -02	0.667
350	0.1394	22.20 -06	15.93 -05	5.193	17.20 -02	0.670
400	0.1220	24.30 -06	19.93 -05	5.193	18.90 -02	0.668
450	0.1084	26.40 -06	24.35 -05	5.193	20.50 -02	0.669
500	0.0976	28.40 -06	29.11 -05	5.193	22.10 -02	0.667
600	0.0813	32.20 -06	39.61 -05	5.193	25.10 -02	0.666
700	0.0697	35.90 -06	51.52 -05	5.193	28.00 -02	0.666
800	0.0610	39.40 -06	64.62 -05	5.193	30.70 -02	0.666
900	0.0542	42.80 -06	78.97 -05	5.193	33.40 -02	0.665
1,000	0.0488	46.20 -06	94.71 -05	5.193	36.00 -02	0.666
1,100	0.0443	49.40 -06	11.14 -04	5.193	38.50 -02	0.666
1,200	0.0407	52.50 -06	12.91 -04	5.193	41.00 -02	0.665
1,300	0.0375	55.60 -06	14.82 -04	5.193	43.40 -02	0.665
1,400	0.0348	58.60 -06	16.82 -04	5.193	45.70 -02	0.666

Table A-6 Properties of CO₂ at $P = 101.325$ kPa $M = 44.010$

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
220	2.4728	11.19 -06	45.25 -07	0.778	10.83 -03	0.804
250	2.1652	12.63 -06	58.33 -07	0.806	12.89 -03	0.790
300	1.7967	14.99 -06	83.43 -07	0.852	16.62 -03	0.768
350	1.5369	17.26 -06	11.23 -06	0.897	20.50 -03	0.755
400	1.3408	19.42 -06	14.48 -06	0.938	24.41 -03	0.746
450	1.1918	21.50 -06	18.04 -06	0.977	28.34 -03	0.741
500	1.0726	23.48 -06	21.89 -06	1.013	32.28 -03	0.737
600	0.8939	27.20 -06	30.43 -06	1.075	40.30 -03	0.726
700	0.7662	30.60 -06	39.94 -06	1.125	48.70 -03	0.707
800	0.6704	33.90 -06	50.57 -06	1.168	56.00 -03	0.707
900	0.5959	36.90 -06	61.92 -06	1.204	62.10 -03	0.715
1,000	0.5363	39.70 -06	74.02 -06	1.234	68.00 -03	0.720
1,100	0.4876	42.40 -06	86.96 -06	1.259	73.30 -03	0.728
1,200	0.4469	44.90 -06	10.05 -05	1.280	78.00 -03	0.737
1,300	0.4125	47.40 -06	11.49 -05	1.298	82.50 -03	0.746
1,400	0.3831	49.70 -06	12.97 -05	1.313	86.70 -03	0.753

Table A-7 Properties of Steam

$T, ^\circ\text{C}$	200	250	300	350	400	450	500	550	600
Density, kg/m^3									
$P = 100 \text{ kPa}$	0.460	0.416	0.379	0.348	0.322	0.300	0.280	0.263	0.248
$P = 7,500 \text{ kPa}$			37.4	30.8	27.1	24.4	22.4	20.7	19.3
Viscosity, $\text{Pa} \cdot \text{s}$									
$P = 100 \text{ kPa}$	16.2	18.2	20.3	22.4	24.5	26.5	28.6	30.6	32.6
$P = 7,500 \text{ kPa}$			19.7	22.1	24.4	26.6	28.8	30.9	32.9
Thermal conductivity, $\text{mW}/(\text{m} \cdot \text{K})$									
$P = 100 \text{ kPa}$	33.4	38.3	43.5	49.0	54.7	60.7	66.9	73.3	79.9
$P = 7,500 \text{ kPa}$			64.1	60.6	62.9	67.5	72.8	78.7	84.8
Specific heat, $\text{kJ}/(\text{kg} \cdot \text{K})$									
$P = 100 \text{ kPa}$	1.979	1.990	2.010	2.037	2.067	2.099	2.132	2.166	2.201
$P = 7,500 \text{ kPa}$			4.686	3.228	2.726	2.512	2.420	2.386	2.379
Prandtl number									
$P = 100 \text{ kPa}$	0.959	0.947	0.938	0.930	0.924	0.917	0.910	0.905	0.898
$P = 7,500 \text{ kPa}$			1.443	1.179	1.057	0.990	0.956	0.936	0.924

Table A-8 Properties of Saturated Liquid Water

$T, ^\circ\text{C}$	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
0.01	999.8	17.91 -04	17.91 -07	4.217	0.562	13.44
10	999.8	13.08 -04	13.08 -07	4.193	0.582	9.42
20	998.2	10.03 -04	10.05 -07	4.182	0.600	6.99
30	995.6	79.77 -05	80.01 -08	4.179	0.615	5.42
40	992.2	65.31 -05	65.80 -08	4.179	0.629	4.34
50	988.0	54.71 -05	55.37 -08	4.181	0.640	3.57
60	983.2	46.68 -05	47.48 -08	4.185	0.651	3.00
70	977.7	40.44 -05	41.36 -08	4.190	0.659	2.57
80	971.8	35.49 -05	36.52 -08	4.197	0.667	2.23
90	965.3	31.50 -05	32.63 -08	4.205	0.673	1.97
100	958.3	28.22 -05	29.45 -08	4.216	0.677	1.76
140	926.1	19.61 -05	21.17 -08	4.285	0.685	1.23
180	886.9	14.94 -05	16.85 -08	4.408	0.674	0.98
220	840.3	12.10 -05	14.40 -08	4.613	0.648	0.86
260	784.0	10.15 -05	12.95 -08	4.983	0.606	0.83

Table A-9 Properties of Saturated Liquid Oxygen

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
56	1,299	58.5 -05	45.0 -08	1.664	0.191	5.10
62	1,273	47.5 -05	37.3 -08	1.664	0.186	4.25
68	1,246	38.8 -05	31.1 -08	1.666	0.179	3.61
76	1,209	30.0 -05	24.8 -08	1.672	0.170	2.95
82	1,181	24.9 -05	21.1 -08	1.679	0.162	2.58
90.2	1,141	19.6 -05	17.2 -08	1.696	0.151	2.20
100	1,091	15.1 -05	13.8 -08	1.732	0.138	1.90
112	1,024	11.4 -05	11.1 -08	1.814	0.122	1.70
126	933	87.1 -06	93.4 -09	2.029	0.102	1.73
138	833	72.3 -06	86.8 -09	2.537	0.0838	2.19
150	675	53.7 -06	79.6 -09	5.598	0.0724	4.15

Table A-10 Properties of Saturated Liquid Parahydrogen

T, K	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
14	76.9	24.8 -06	32.2 -08	6.47	0.0746	2.15
16	75.1	19.4 -06	25.8 -08	7.36	0.0889	1.61
18	73.2	15.9 -06	21.7 -08	8.42	0.0954	1.40
20.3	70.8	13.2 -06	18.6 -08	9.66	0.0989	1.29
22	68.7	11.6 -06	16.9 -08	10.82	0.1010	1.24
24	66.0	10.1 -06	15.3 -08	12.52	0.1008	1.25
26	62.8	88.0 -07	14.0 -08	14.80	0.0984	1.32
28	59.0	76.2 -07	12.9 -08	18.48	0.0938	1.50
30	53.9	64.6 -07	12.0 -08	26.59	0.0866	1.98
32	46.0	51.3 -07	11.2 -08	65.37	0.0915	3.67

Table A-11 Properties of JP-4 Fuel

$T, ^\circ\text{C}$	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
-15	802	15.3 -04	19.1 -07	1.858	14.1 -02	20
0	789	11.2 -04	14.2 -07	1.926	14.0 -02	15
15	776	86.8 -05	11.2 -07	1.995	13.8 -02	13
30	763	70.1 -05	91.9 -08	2.063	13.7 -02	11
45	750	58.6 -05	78.2 -08	2.132	13.6 -02	9
60	737	50.4 -05	68.4 -08	2.201	13.5 -02	8
75	724	44.4 -05	61.3 -08	2.269	13.3 -02	8

Table A-12 Properties of Hydraulic Fluid (MIL-H-5606, $P = 101.325 \text{ kPa}$)

$T, ^\circ\text{C}$	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
-15	879	70.6 -03	80.3 -06	1.737	12.7 -02	962
0	868	34.2 -03	39.4 -06	1.788	12.6 -02	483
15	858	19.2 -03	22.4 -06	1.845	12.4 -02	286
30	847	12.2 -03	14.4 -06	1.905	12.1 -02	192
45	836	84.4 -04	10.1 -06	1.969	11.7 -02	143
60	826	63.0 -04	76.3 -07	2.035	11.2 -02	114
75	815	49.9 -04	61.2 -07	2.101	10.8 -02	97

Table A-13 Properties of Engine Oil (Unused)

$T, ^\circ\text{C}$	$\rho, \text{kg/m}^3$	$\mu, \text{Pa} \cdot \text{s}$	$\nu, \text{m}^2/\text{s}$	$c_p, \text{kJ}/(\text{kg} \cdot \text{K})$	$k, \text{W}/(\text{m} \cdot \text{K})$	Pr
0	898	38.3 -01	42.7 -04	1.788	14.7 -02	46 540
15	889	10.6 -01	12.0 -04	1.845	14.5 -02	13 530
30	881	36.9 -02	41.9 -05	1.905	14.3 -02	4 912
45	872	15.3 -02	17.5 -05	1.969	14.1 -02	2 130
60	863	73.2 -03	84.8 -06	2.035	14.0 -02	1 065
75	855	39.4 -03	46.1 -06	2.101	13.8 -02	598
100	840	17.2 -03	20.5 -06	2.214	13.6 -02	280
125	826	91.5 -04	11.1 -06	2.328	13.4 -02	159
150	811	56.4 -04	69.5 -07	2.440	13.2 -02	104

Table A-14 Properties of Mercury, Sodium, 22 Percent Sodium/78 Percent Potassium and Potassium

$T, ^\circ\text{C}$	0	50	100	200	300	400	500	600	700
Density, kg/m^3									
Hg	13,600	13,460	13,350	13,110	12,880				
Na	—	—	927	905	882	858	834	809	783
NaK	—	—	847	823	799	775	751	727	703
K	—	—	819	795	771	747	723	701	676
Viscosity, $\text{mPa} \cdot \text{s}$									
Hg	16.8	13.9	12.1	10.1	8.62				
Na	—	—	7.05	4.50	3.45	2.84	2.43	2.10	1.86
NaK	—	—	5.29	3.54	2.76	2.29	1.95	1.68	1.46
K	—	—	4.63	2.98	2.27	1.90	1.69	1.45	1.34
Specific heat, $\text{kJ}/(\text{kg} \cdot \text{K})$									
Hg	0.140	0.140	0.140	0.136	0.135				
Na	—	—	1.384	1.339	1.305	1.280	1.264	1.255	1.255
NaK	—	—	0.941	0.908	0.887	0.879	0.874	0.874	0.883
K	—	—	0.812	0.791	0.774	0.766	0.761	0.766	0.774
Thermal conductivity, $\text{W}/(\text{m} \cdot \text{K})$									
Hg	8.2	9.3	10.4	12.4	14.0				
Na	—	—	—	81.5	75.7	71.2	66.8	62.7	59.0
NaK	—	—	—	24.7	25.9	26.2	26.2	25.9	25.5
K	—	—	—	44.1	42.0	39.4	37.2	34.9	32.5
Prandtl number									
Hg	0.029	0.021	0.016	0.0011	0.0083				
Na	—	—	—	0.0074	0.0059	0.0051	0.0046	0.0042	0.0039
NaK	—	—	—	0.013	0.0095	0.0077	0.0065	0.0057	0.0051
K	—	—	—	0.0053	0.0042	0.0037	0.0035	0.0032	0.0036

Table A-15 Units Conversion Table

Physical quantity	Multiply by	To obtain or to convert	Multiply by	To obtain
Area				
ft ²	0.092903	m ²	10.764	ft ²
Density				
lb _m /ft ³	16.018	kg/m ³	0.062428	lb _m /ft ³
Energy				
Btu (IT)	1055.1	J	9.4782 E - 04	Btu (IT)
Btu	1054.4	J	9.4845 E - 04	Btu
cal (IT)	4.1868	J	0.23885	cal (IT)
cal	4.184	J	0.23901	cal
Energy flux				
Btu/(h · ft ²)	3.1525	W/m ²	0.31721	Btu/(h · ft ²)
cal/(s · cm ²)	4.184 E + 04	W/m ²	2.3901 E - 05	cal/(s · cm ²)
Force				
kg _f	9.80665	N	0.10197	kg _f
lb _f	4.4482	N	0.22481	lb _f
Heat-transfer coefficient				
Btu/(h · ft ² · °F)	5.6745	W/(m ² · K)	0.17623	Btu/(h · ft ² · °F)
cal/(s · cm ² · °C)	4.184 E + 04	W/(m ² · K)	2.3901 E - 05	cal/(s · cm ² · °C)
Length				
ft	0.3048	m	3.2808	ft
in	0.0254	m	39.370	in
mi (US)	1609.3	m	6.2137 E - 04	mi (US)
Mass				
lb _m	0.45359	kg	2.2046	lb _m
slug	14.594	kg	0.068522	slug
Power				
Btu/h	0.29288	W	3.4144	Btu/h
cal/s	4.184	W	0.23901	cal/s
hp (550 ft · lb _f /s)	745.70	W	1.3410 E - 03	hp
Pressure				
atm	101,325	Pa	9.8692 E - 06	atm
lb _f /in ²	6894.8	Pa	1.4504 E - 04	lb _f /in ²
Specific heat				
Btu/(lb _m · °F)	4.184	kJ/(kg · K)	0.23901	Btu/(lb _m · °F)
cal/(g · °C)	4.184	kJ/(kg · K)	0.23901	cal/(g · °C)
Temperature	(see below)			
Thermal conductivity				
Btu · in/(h · ft ² · °F)	0.14413	W/(m · K)	6.9381	Btu · in/(h · ft · °F)
Btu/(h · ft · °F)	1.7296	W/(m · K)	0.57818	Btu/(h · ft · °F)
cal/(s · cm · °C)	418.4	W/(m · K)	2.3901 E - 03	cal/(s · cm · °C)
Velocity				
ft/s	0.3048	m/s	3.2808	ft/s
Viscosity				
centipoise	0.001	Pa · s	1000	centipoise
lb _m /(h · ft)	4.1338 E - 04	Pa · s	2419.1	lb _m /(h · ft)
lb _m /(s · ft)	1.4882	Pa · s	0.67197	lb _m /(s · ft)
lb _f · s/ft ²	47.880	Pa · s	0.020885	lb _f · s/ft ²

Table A-15 Units Conversion Table (continued)

Physical quantity	Multiply by	To obtain or to convert	Multiply by	To obtain
Viscosity, kinematic				
centistoke	1.0000 E - 06	m ² /s	1.0000 E + 06	centistoke
ft ² /s	0.092903	m ² /s	10.764	ft ² /s
Volume				
ft ³	0.028317	m ³	35.315	ft ³
gal (US liquid)	3.7854 E - 03	m ³	264.17	gal (US liquid)
Volume flow rate				
ft ³ /min	4.7195 E - 04	m ³ /s	2118.9	ft ³ /min
gal (US liquid)/min	6.3090 E - 05	m ³ /s	15,850	gal (US liquid)/min
Temperature				
$T(K) = T(^{\circ}C) + 273.15$				
$T(K) = T(^{\circ}R)/1.8$				
$T(K) = [T(^{\circ}F) + 459.67]/1.8$				
$T(^{\circ}C) = [T(^{\circ}F) - 32]/1.8$				

Fig. A-1 Thermal conductivity of some metals.

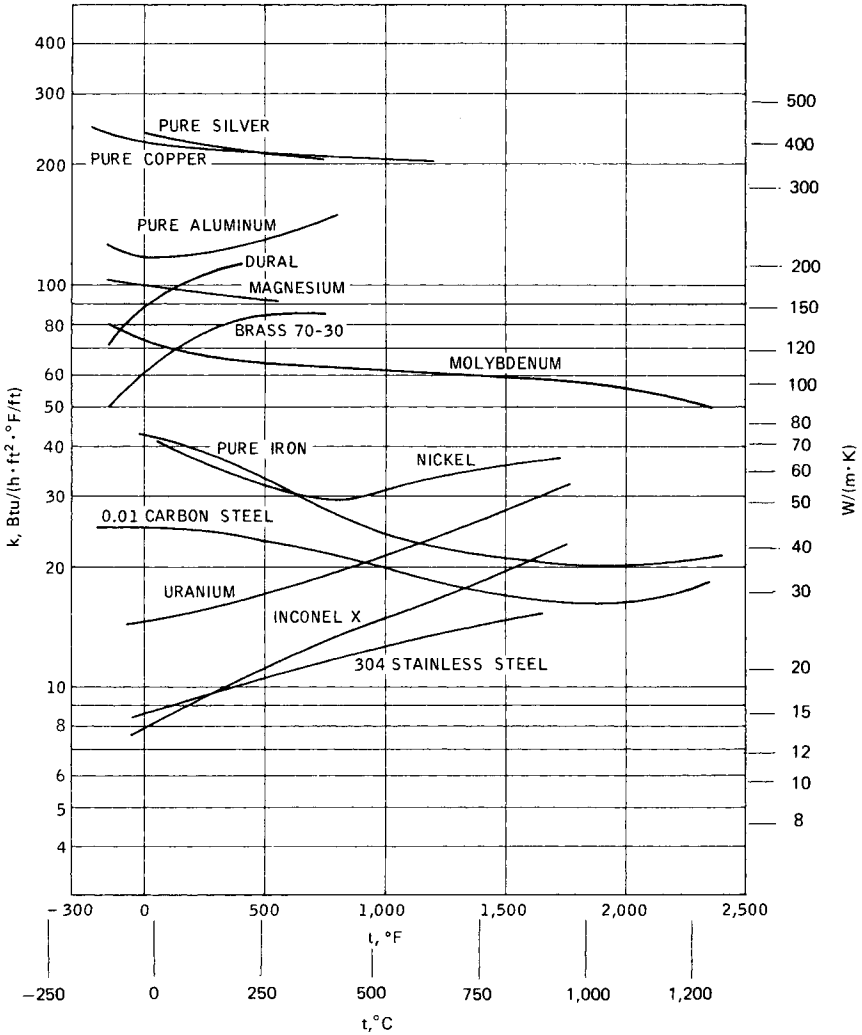


Fig. A-2 Transport properties of air at 1 atm.

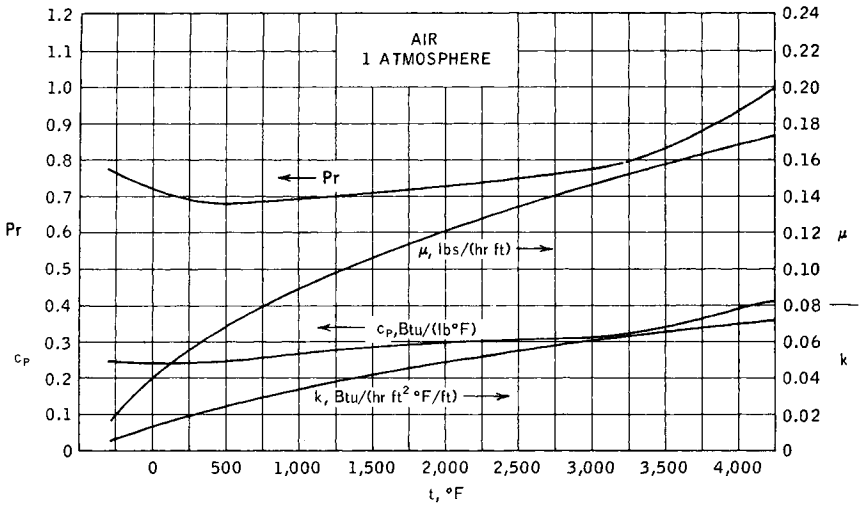


Fig. A-3 Transport properties of oxygen gas at 1 atm.

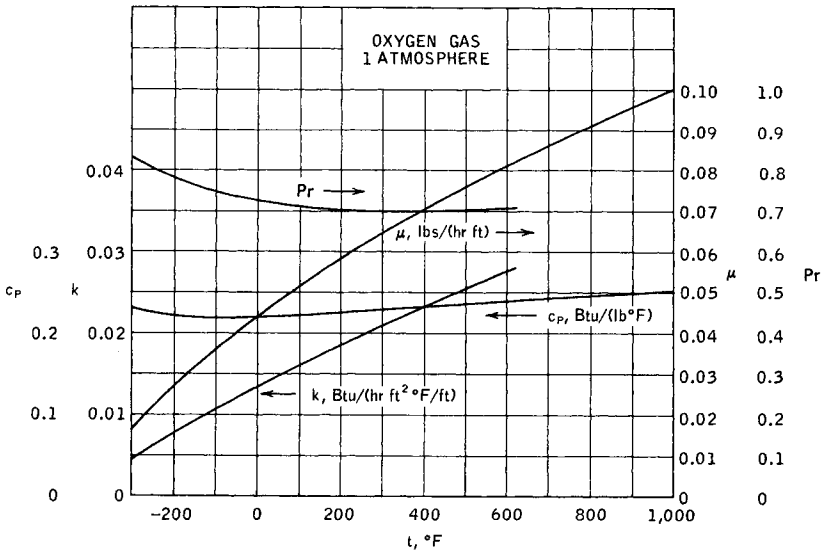


Fig. A-4 Transport properties of nitrogen gas at 1 atm.

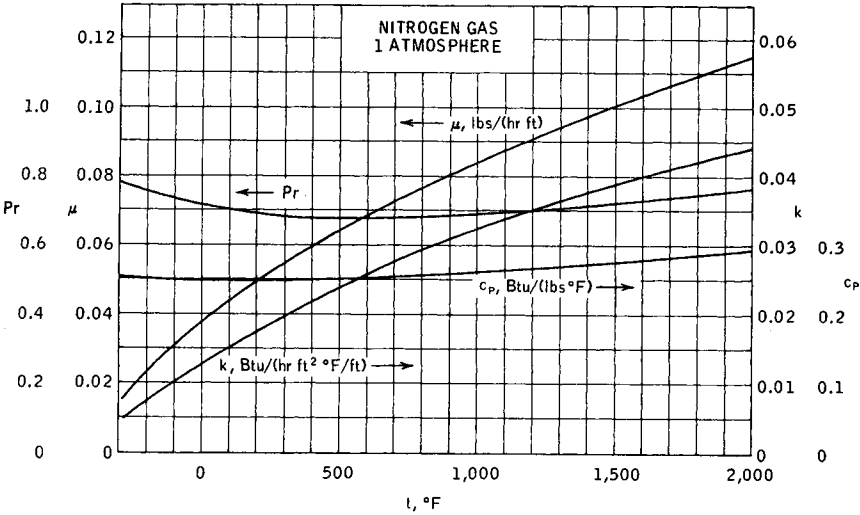


Fig. A-5 Transport properties of hydrogen gas at 1 atm.

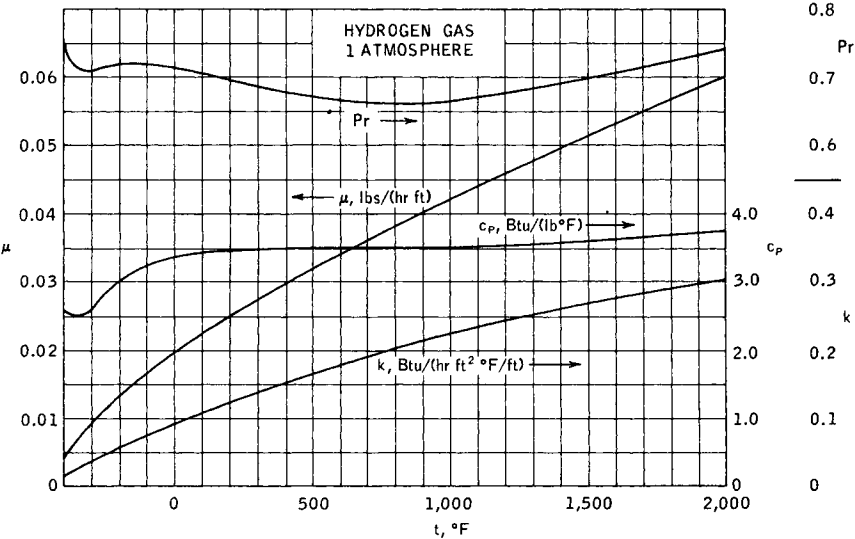


Fig. A-6 Transport properties of helium gas at 1 atm.

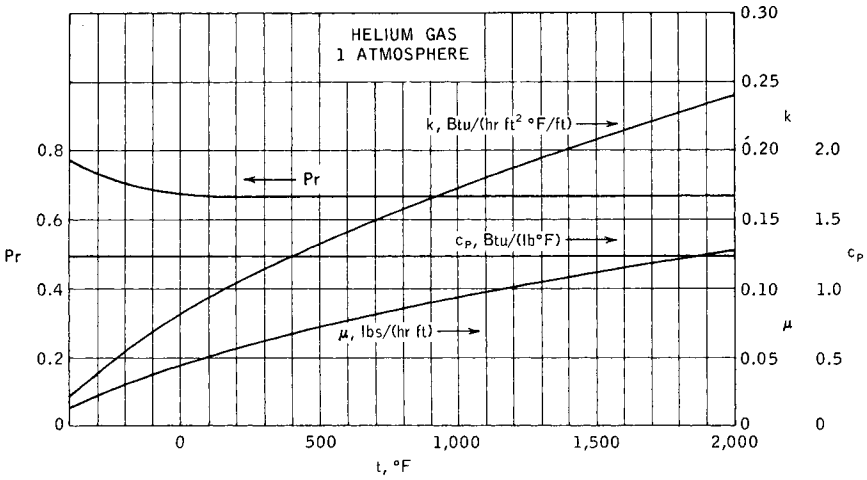


Fig. A-7 Transport properties of carbon dioxide gas at 1 atm.

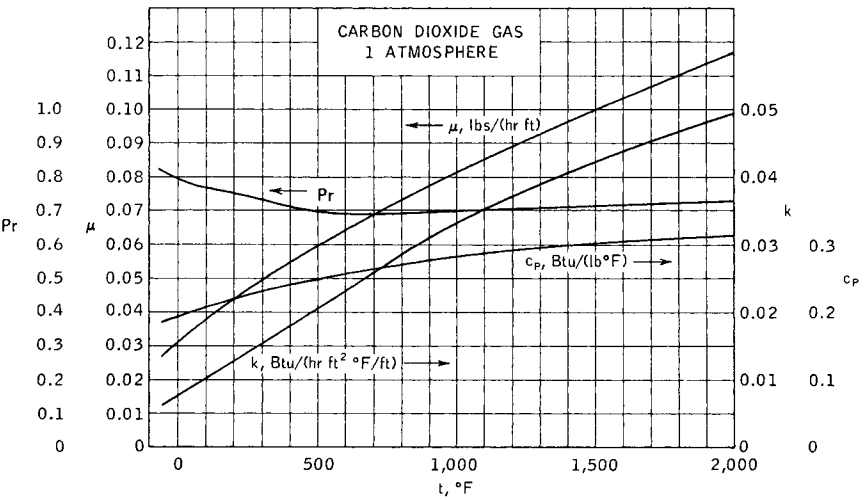


Fig. A-8 Transport properties of steam at 1 atm and at 1000 psi.

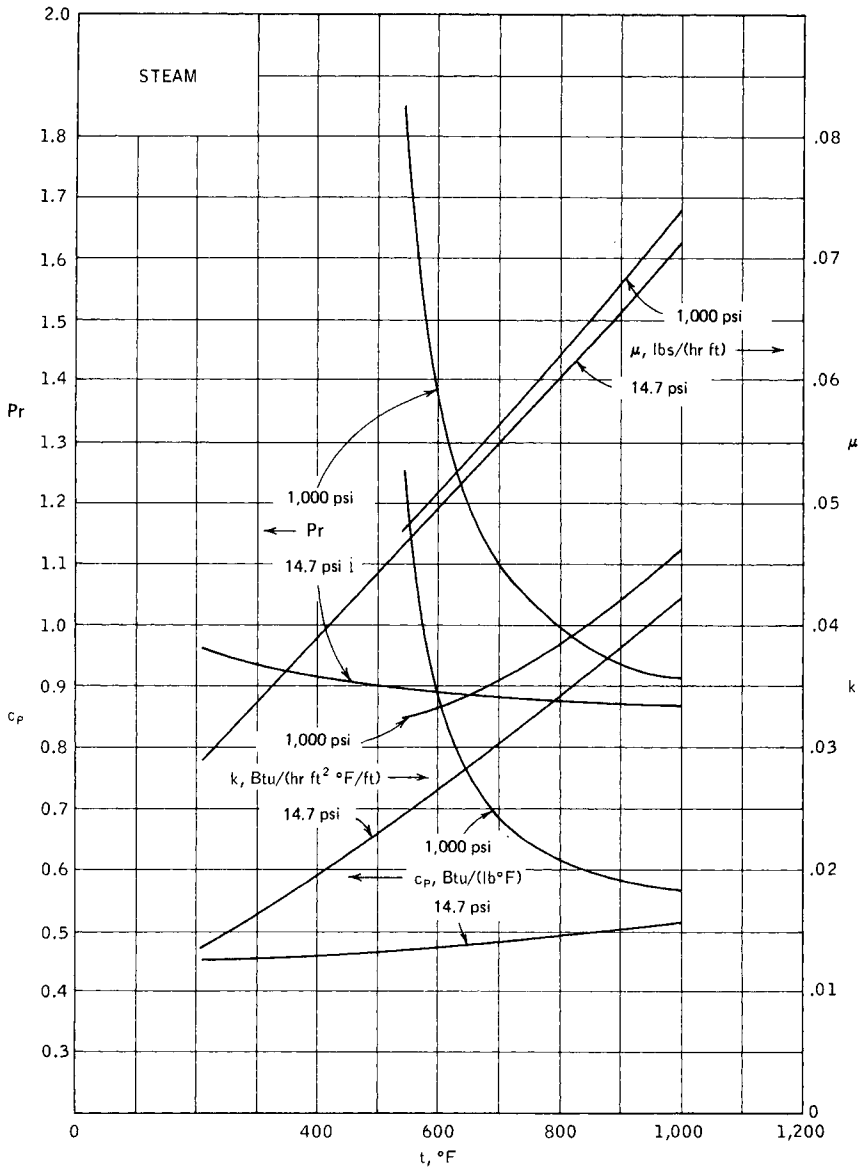


Fig. A-9 Transport properties of saturated liquid water.

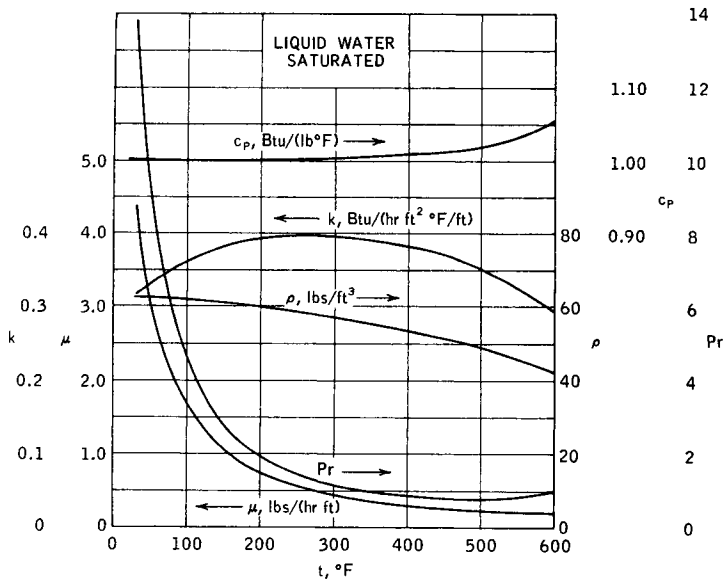


Fig. A-10 Transport properties of liquid oxygen.

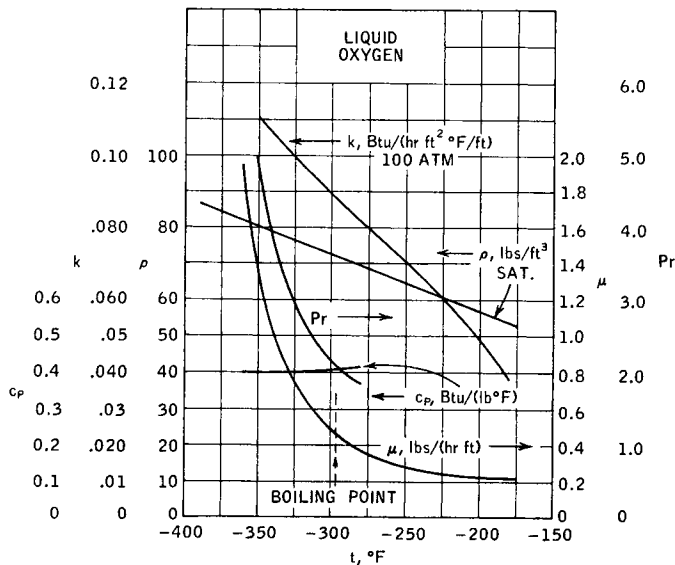


Fig. A-11 Transport properties of liquid hydrogen.

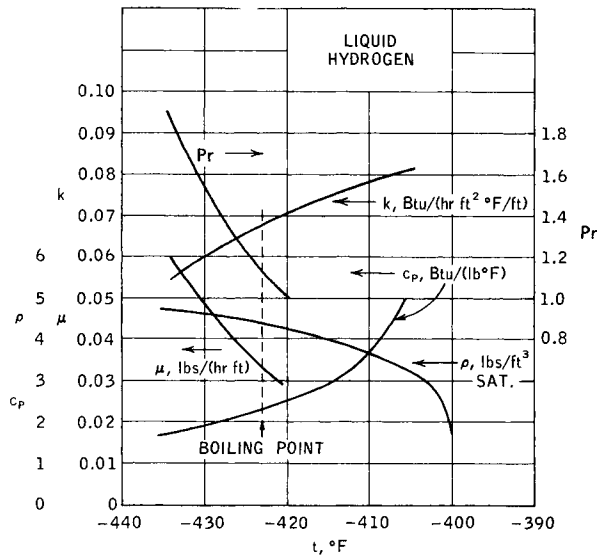
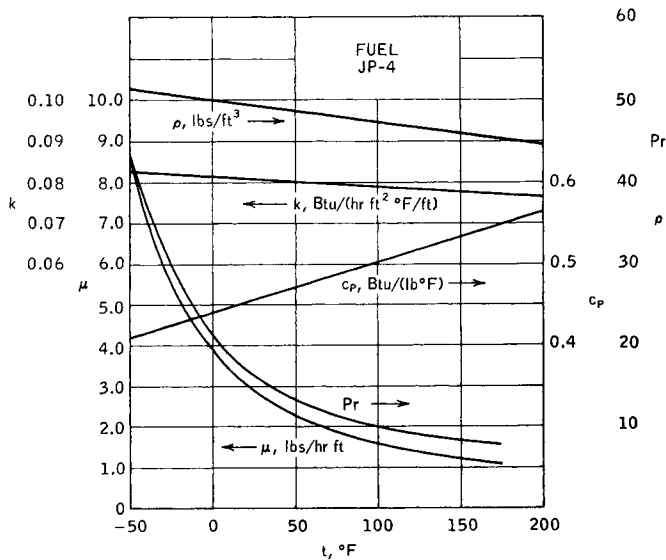


Fig. A-12 Transport properties of JP-4 fuel.



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Fig. A-13 Transport properties of hydraulic fluid MIL-H-5606.

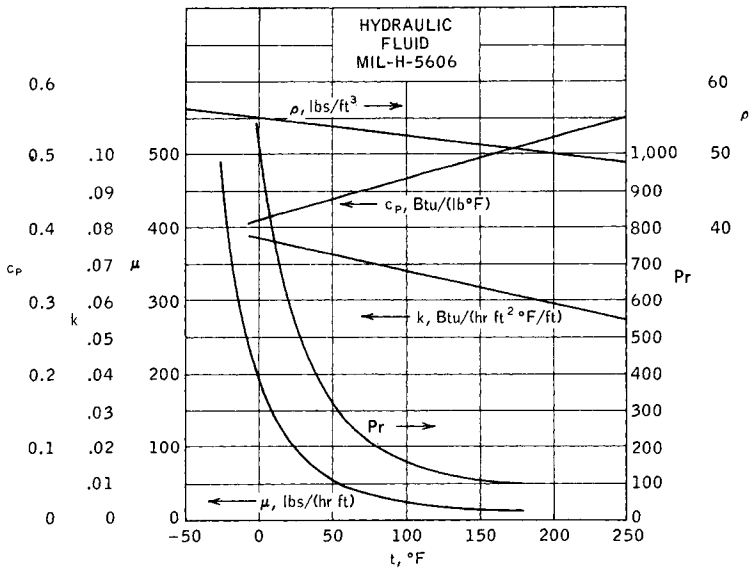


Fig. A-14 Transport properties of a typical aircraft engine oil.

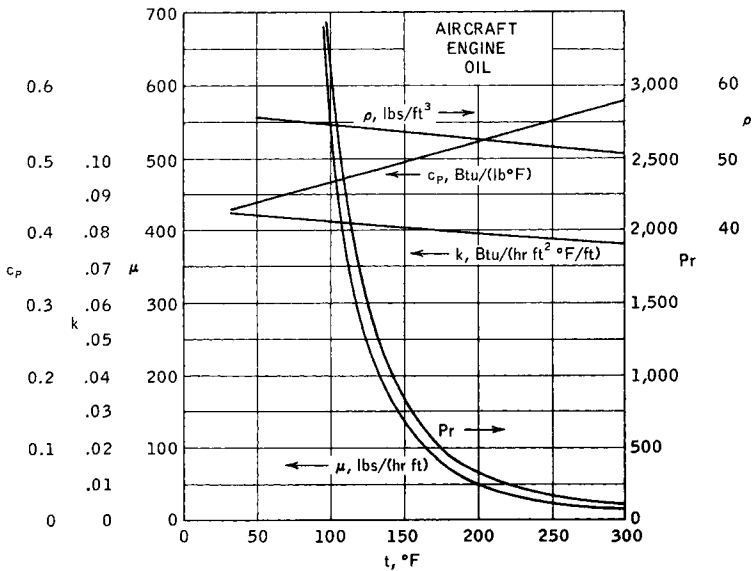


Fig. A-15 Transport properties of liquid mercury.

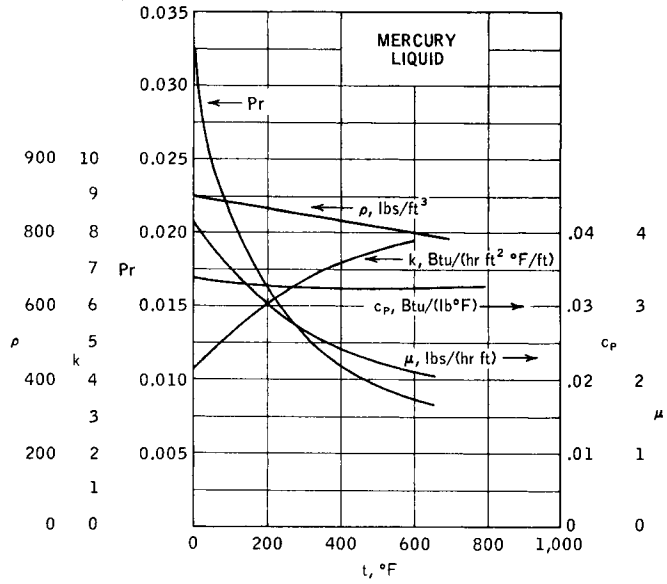
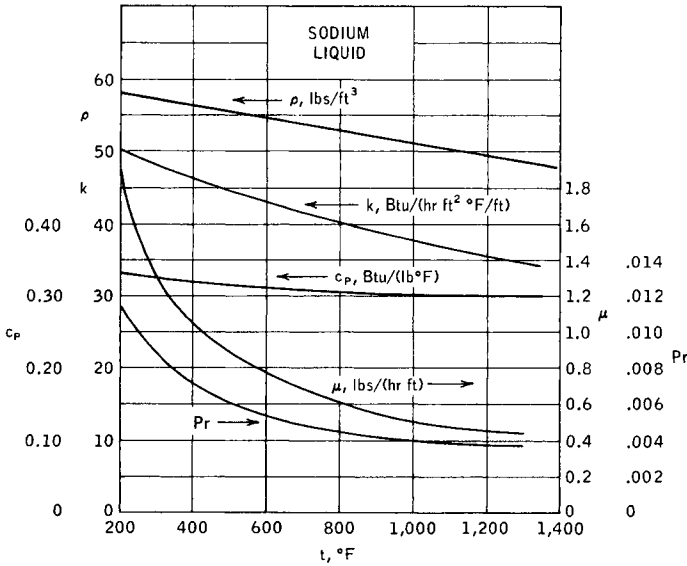


Fig. A-16 Transport properties of liquid sodium.



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Fig. A-17 Transport properties of liquid potassium.

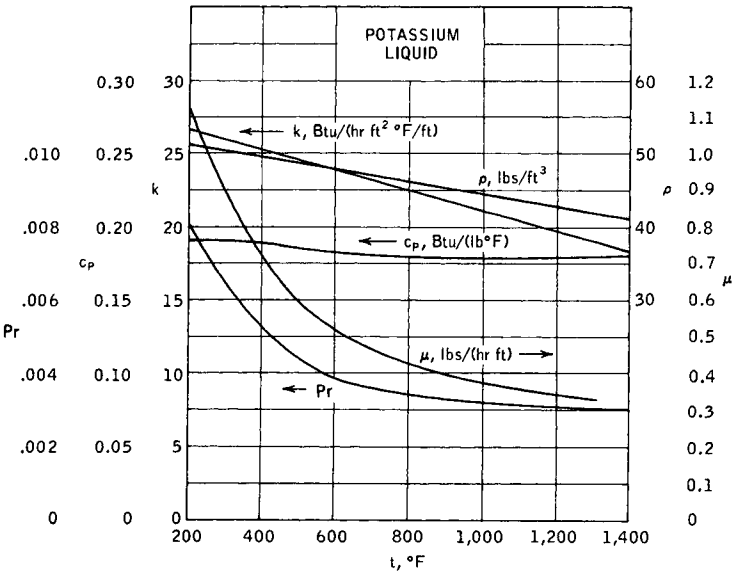


Fig. A-18 Humidity correction factors for density and specific heat. $\rho = X_{d,w} \rho_a$ $c_p = X_{c,w} c_{pa}$ (Same graph applies for SI dimensions.)

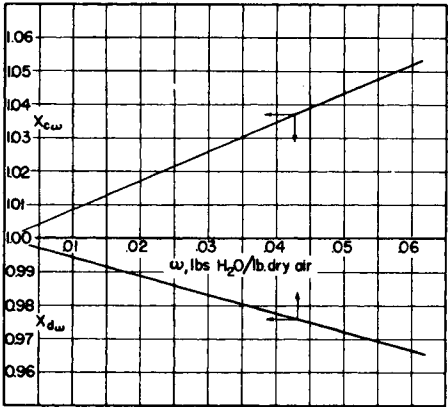


Fig. A-19 Combustion products correction factors for specific heat. $C_p = X_{c,f} c_{pa}$ (Same graph applies for SI dimensions.)

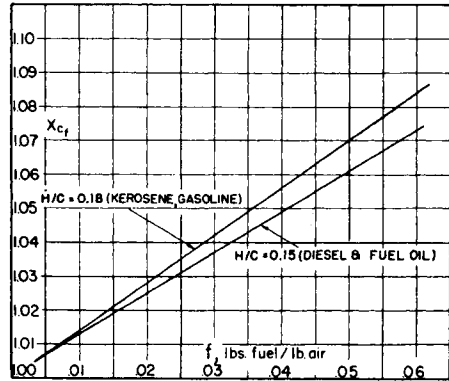
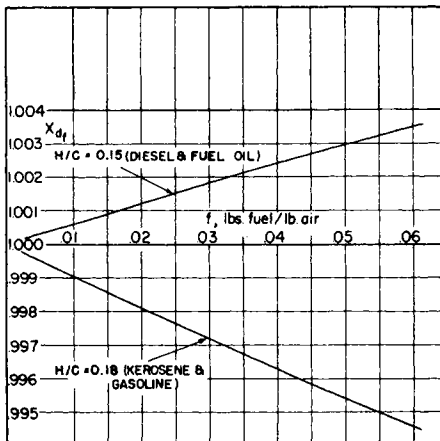


Fig. A-20 Combustion products correction factors for density. $\rho = X_{d,f} \rho_a$ (Same graph applies for SI dimensions.)



Appendix B Examples of Heat Exchanger Performance Calculations

The purpose of this appendix is to illustrate the use of the basic heat transfer and friction data presented in Chaps. 6, 7, and 10, together with the heat exchanger and flow-friction theory outlined in Chap. 2. The four exchanger applications considered are an intercooler for a gas-turbine plant and three types of gas-turbine regenerators.

Gas-turbine heat exchangers have been chosen for illustrative purposes because they provide good examples of gas-to-gas and gas-to-liquid heat exchangers where compactness is important and where much in the way of optimization can be accomplished if a detailed analysis of the exchanger performance is undertaken.

It is important for the reader to realize that the examples analyzed here represent only a partial treatment of the comprehensive design of a heat exchanger described in Chap. 2, Figure 2-12. This simplification is necessary in order to illustrate clearly to the engineer who is not completely conversant with heat exchanger design procedures the use of the contents of this publication. The experienced designer will appreciate the restricted nature of the illustrative problems and will note the omission of important considerations, such as

1. Thermodynamic basis for the selection of desired pressure drops
2. Optimum distribution of thermal resistance and flow-friction power on the two surfaces
3. Optimum design of extended surface, material thermal conductivity, and thickness
4. Uncertainty of proper thermal contact between the extended and prime surfaces
5. Allowance for fouling
6. Influence of poor flow distribution on exchanger performance
7. Plant machinery-arrangement factors, controlling in part the dimensions of the exchanger
8. Thermal stress and construction problems

Any or all of these factors may have a pronounced influence on the selection of the geometry of the particular surface to be used in the heat exchanger core element. All these factors are bypassed by the arbitrary selection of surface geometry for the illustrative problems considered here.

Sample solutions are presented for a direct-transfer-type intercooler and a direct-transfer-type regenerator in complete form; for an indirect-transfer-type regenerator in abbreviated form, showing only the application of the ε - N_{tu} approach; and for a periodic-flow-type regenerator, incorporating a screen-matrix surface, in complete form.

EXAMPLE 1 DIRECT-TRANSFER INTERCOOLER

Figure B-1 describes the core geometry of an intercooler for a 5,000-shp (3.7 MW) gas-turbine plant. Operating conditions and the heat transfer surface are specified as follows:

Operating conditions, air side:

Air-flow rate, humid	200,000 lb/h (25 kg/s)
Air humidity	0.015 lbs H ₂ O/lb _m (0.015 kg/kg)
Air entering temperature	260°F = 720°R = 127°C = 400K
Air entering pressure	39.7 lb _f /in ² abs (2.75×10^5 N/m ²)

Operating conditions, water side:

Water-flow rate	400,000 lb _m /h (800 gal/min) (50 kg/s)
Water inlet temperature	60°F (15°C, 288K)
Intercooler surface	11.32-737-SR (finned flat tube, see Fig. 10-97)

The objective of this problem is to predict, for the specified conditions and the basic heat transfer and flow-friction characteristics of the surface, (1) the intercooler effectiveness and (2) the pressure drops for both water and air flows. This prediction will be based on clean surfaces with no allowance for fouling.

The steps in the analysis require the determination, in the order given, of the following factors:

1. Surface characteristics from Table 9-4c
2. Fluid properties from Figs. A-2 and A-9
3. Reynolds numbers
4. $StPr^{2/3}$ and f from the basic characteristics of the surface
5. Heat-transfer coefficients
6. Fin effectiveness
7. Surface effectiveness
8. Overall coefficient of heat transfer
9. N_{tu} and exchanger effectiveness
10. Outlet temperatures
11. Pressure drops

These steps are carried out in detail below.

Surface Characteristics from Table 9-4c

AIR SIDE

Flow-passage hydraulic radius, $r_{h,a} = 0.00288 \text{ ft } (0.878) \times (10^{-3} \text{ m})$

Total air-side transfer area/total volume, $\alpha_a = 270 \text{ ft}^2/\text{ft}^3 \text{ } (886 \text{ m}^2/\text{m}^3)$

Fin area/total area = 0.845

Free-flow area/frontal area $\sigma_a = 0.780$

Fin-metal thickness $\delta = 0.004 \text{ in. } = 0.00033 \text{ ft } (0.10 \times 10^{-3} \text{ m})$

Fin material, aluminum with $k = 100 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{ft}) [173 \text{ W}/(\text{m} \cdot \text{K})]$

Fin length (one-half distance between tubes) $l = 0.225 \text{ in } (5.71 \cdot 10^{-3} \text{ m})$ (Fig. 10-97)

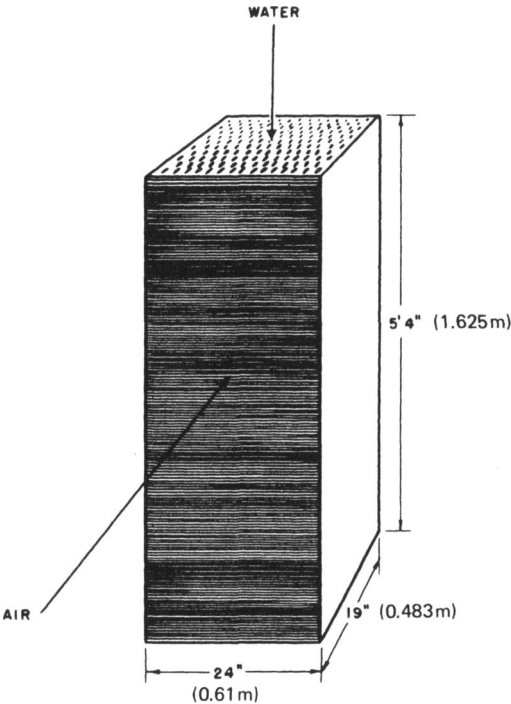
WATER SIDE

The tubes have straight sides with semicircular ends.

Outside dimensions per tube = 0.737 in by 0.10 in ($18.7 \times 10^{-3} \text{ m}$ by $2.54 \times 10^{-3} \text{ m}$)

Inside dimensions per tube = 0.717 in by 0.08 in ($18.2 \times 10^{-3} \text{ m}$ by $2.04 \times 10^{-3} \text{ m}$)

Fig. B-1 A direct-transfer intercooler for a 5,000-shp (3.7-MW) gas-turbine plant.



Frontal area associated with one tube = 0.434 in^2 ($2.8 \times 10^{-4} \text{ m}^2$) (see dimensions in Fig. 10-97)

Free-flow area of one tube = 0.0560 in^2 ($0.361 \times 10^{-4} \text{ m}^2$)

Inside periphery of one tube = 1.524 in ($38.7 \times 10^{-3} \text{ m}$)

Free-flow/frontal area $\sigma_w = 0.129$

Water-side transfer area/total volume $\alpha_w = 42.1 \text{ ft}^2/\text{ft}^3$ ($138 \text{ m}^2/\text{m}^3$)

Water-side flow-passage hydraulic radius, $r_{h,w} = 3.06 \times 10^{-3} \text{ ft}$ ($0.933 \times 10^{-3} \text{ m}$)

CORE DIMENSIONS (from Fig. B-1):

Air-side frontal area, $A_{fr} = 10.67 \text{ ft}^2$ (0.991 m^2)

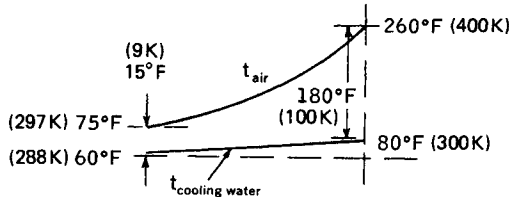
Water-side frontal area, $A_{fr,w} = 3.17 \text{ ft}^2$ (0.294 m^2)

Total heat exchanger volume $V = 16.9 \text{ ft}^3$ (0.479 m^3)

Fluid Properties

Assume, as a first approximation, an outlet air temperature of 75°F (24°C) = 535°R (297K), and an outlet water temperature of 80°F (27°C). These approximations must be verified later.

Fig. B-2 Temperature conditions in an intercooler.



In order to evaluate gas viscosity for the Re parameter, c_p for St , and Pr , a bulk average air temperature needs to be estimated (see Chap. 4). From the sketch of temperature conditions (drawn as for true counterflow, although the actual exchanger has a crossflow arrangement), Fig. B-2, it is evident that average temperatures are

$$t_a \approx 70 + \frac{180 - 15}{\log(180/15)} = 137^\circ\text{F} (58^\circ\text{C})(331\text{K})$$

$$t_w \approx 70^\circ\text{F} (21^\circ\text{C})(294\text{K})$$

From Fig. A-2 (or more accurately from tables in the literature) for dry air,

$$\mu = 0.0482 \text{ lb}_m/(\text{h} \cdot \text{ft}) (1.99 \times 10^{-5} \text{ Pa} \cdot \text{s})$$

$$Pr = 0.700$$

$$c_p = 0.240 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F}) [1.0 \text{ kJ}/(\text{kg} \cdot \text{K})]$$

To correct c_p for a humidity of $0.015 \text{ lb}_m/\text{lb}$, from Fig. A-18,

$$X_{c,w} = 1.0126$$

$$\text{and } c_p = 1.0126 \times 0.240 = 0.244 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F}) [1.02 \text{ kJ}/(\text{kg} \cdot \text{K})]$$

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Assuming a 1 percent pressure drop (to be verified later), the outlet air pressure is $39.3 \text{ lb}_f/\text{in}^2 \text{ abs}$ ($2.71 \times 10^5 \text{ N/m}^2$). The inlet and outlet air specific volumes can then be evaluated, employing the perfect gas equation of state and the correction for humidity. From Fig. A-18 the correction factor for humidity $X_{d,\omega}$ is 0.9915. It should be noted that this correction factor is applied to the density, the reciprocal of the specific volume.

$$v_1 = \frac{1}{\rho_1} = \frac{1}{X_{d,\omega} MP_1} = \frac{53.345 \times 720}{0.9915 \times 39.7 \times 144} = 6.78 \text{ ft}^3/\text{lb}_m \text{ (0.423 m}^3/\text{kg)}$$

$$v_2 = \frac{1}{\rho_2} = \frac{53.345 \times 535}{0.9915 \times 39.3 \times 144} = 5.19 \text{ ft}^3/\text{lb}_m \text{ (0.324 m}^3/\text{kg)}$$

For this application the specific volume does not vary linearly with transfer area, and v_m from Eq. (2-27a), when evaluated as an arithmetic average of v_1 and v_2 , is not a good approximation. A better approximation results from the use of a log-mean absolute temperature [Eq. (2-29)] as already evaluated in the previous calculation, t_a (log-mean average) = 137°F (59°C). Thus

$$T_{lma} = 137 + 460 = 597^\circ\text{R} \text{ (332K)}$$

Then from Eq. (2-27b),

$$\frac{v_m}{v_1} = \frac{P_1}{P_{av}} \frac{T_{lma}}{T_1} = \frac{39.7 \times 597}{39.5 \times 720} = 0.833$$

This is 6 percent lower than the arithmetic average of v_1 and v_2 . Average water properties from Fig. A-9 are

$$k = 0.347 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{ft}) [0.6 \text{ J}/(\text{m} \cdot \text{s} \cdot \text{K})]$$

$$\mu = 2.36 \text{ lb}_m/(\text{h} \cdot \text{ft}) (0.976 \times 10^{-3} \text{ N} \cdot \text{s}/\text{m}^2)$$

$$c = 1.00 \text{ Btu}/(\text{lb} \cdot ^\circ\text{F}) [4.18 \text{ kJ}/(\text{kg} \cdot \text{K})]$$

$$\text{Pr} = 6.8$$

$$v = 0.01603 \text{ ft}^3/\text{lb} \text{ } (\rho = 62.3 \text{ lb}/\text{ft}^3) (0.001 \text{ m}^3/\text{kg}) (\rho = 998 \text{ kg}/\text{m}^3)$$

Reynolds Numbers

AIR SIDE

$$G = \frac{W}{A_c} = \frac{W}{A_{fr} \sigma_a} = \frac{200,000}{10.67 \times 0.78} = 24,050 \text{ lb}_m/(\text{h} \cdot \text{ft}^2) [32.62 \text{ kg}/(\text{s} \cdot \text{m}^2)]$$

$$\text{Re}_a = \frac{4r_{ha}G}{\mu} = \frac{4 \times 0.00288 \times 24,050}{0.0482} = 5,760$$

WATER SIDE

$$G = \frac{400,000}{3.17 \times 0.129} = 978,000 \text{ lb}_m/(\text{h} \cdot \text{ft}^2)[1326 \text{ kg}/(\text{s} \cdot \text{m}^2)]$$

$$[v \approx 4.4 \text{ ft/s}(1.34 \text{ m/s})]$$

$$\text{Re}_w = \frac{4 \times 0.00306 \times 978,000}{2.36} = 5,080$$

St and f

AIR SIDE

From Fig. 10-97 for surface 11.32-.737-SR with $\text{Re}_a = 5,760$,

$$\text{StPr}^{2/3} = 0.0054$$

$$f = 0.021$$

$$\text{St} = \frac{0.0054}{0.788} = 0.00685$$

WATER SIDE

From Fig. 6-7 for turbulent flow inside of circular tubes, $\text{Re}_w = 5,080$, and $\text{Pr} = 6.8$,

$$\text{Nu} \approx 55$$

As $\mu_w/\mu_m \approx 1$, no correction need be made for viscosity variation with temperature. From Fig. 6-6, $f = 0.0094$. Note the use of circular-tube performance as an approximation for the flat tubes considered here. High accuracy for the f and h prediction is not important for the water side.

Heat Transfer Coefficients

AIR SIDE

$$h = \text{St}Gc_p = 0.00685 \times 24,050 \times 0.244$$

$$= 40.2 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})[228 \text{ W}/(\text{m}^2 \cdot \text{K})]$$

WATER SIDE

$$h = \frac{\text{Nu}k}{4r_h}$$

$$h = 55 \times \frac{0.347}{3.06 \times 10^{-3} \times 4} = 1,560 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})[8,852 \text{ W}/(\text{m}^2 \cdot \text{K})]$$

Fin Effectiveness (Air Side Only)

From Fig. 2-13, for

$$m = \sqrt{\frac{2h}{k\delta}} = \sqrt{\frac{2 \times 40.2}{100 \times 0.000333}} = 49.2 \text{ ft}^{-1} \text{ (161 m}^{-1}\text{)}$$

and

$$ml = 49.2 \times \frac{0.225}{12} = 0.923$$

$$\eta_f = 0.79$$

Overall Surface Effectiveness (Air Side Only)

From Eq. (2-3),

$$\begin{aligned}\eta_o &= 1 - \frac{A_f}{A} (1 - \eta_f) \\ &= 1 - (0.845)(1 - 0.79) = 0.823\end{aligned}$$

Overall Coefficient of Heat Transfer U (Based on Air-Side Area)

From Eq. (2-2), neglecting the very small wall resistance,

$$\begin{aligned}\frac{1}{U_a} &= \frac{1}{\eta_o h_a} + \frac{1}{(\alpha_w/\alpha_a) h_w} \\ &= \frac{1}{0.823 \times 40.2} + \frac{1}{(42.1/270) \times 1560} = 0.0343 \\ U_a &= 29.2 \text{ Btu/(h} \cdot \text{ft}^2 \cdot \text{°F)} [166 \text{ W/(m}^2 \cdot \text{K)}]\end{aligned}$$

This prediction is for a clean surface condition. A reasonable allowance for fouling could be included at this point to give a somewhat lower magnitude for U_a . However, for the sake of clarity in this illustrative problem, and because of the absence of service information on which to base a fouling factor, no such consideration will be included here.

 N_{tu} and Exchanger Effectiveness

$$\begin{aligned}\text{Air-side capacity rate: } C_a &= 200,000 \times 0.244 \\ &= 48,800 \text{ Btu/(h} \cdot \text{°F)} (25.7 \text{ kW/K)}\end{aligned}$$

$$\begin{aligned}\text{Water-side capacity rate: } C_w &= 400,000 \times 1.00 \\ &= 400,000 \text{ Btu/(h} \cdot \text{°F)} (211 \text{ kW/K)}\end{aligned}$$

From Eq. (2-7), the number of heat transfer units,

$$N_{tu, \max} = \frac{A_a U_a}{C_{\min}} = \frac{270 \times 16.9 \times 29.2}{48,800} = 2.73$$

From Fig. 2-16, for a crossflow exchanger with both fluids unmixed, and $C_{\min}/C_{\max} = 48,800/400,000 = 0.122$, the intercooler effectiveness is,

$$\varepsilon = 90 \text{ percent}$$

Outlet Temperature

From Eq. (2-6),

$$\varepsilon = \frac{C_a(t_{h, \text{in}} - t_{h, \text{out}})}{C_a(t_{h, \text{in}} - t_{c, \text{in}})}, \quad t_h = t_a$$

$$0.90 = \frac{260 - t_{h, \text{out}}}{260 - 60}, \quad t_{h, \text{out}} = 80^\circ\text{F} (27^\circ\text{C})(300\text{K})$$

Then from energy-balance considerations the outlet water temperature may be evaluated from

$$t_{w, \text{out}} - t_{w, \text{in}} = \frac{C_a}{C_w} (t_{a, \text{in}} - t_{a, \text{out}}) = 0.122 \times 180 = 21.9^\circ\text{F} (12.2^\circ\text{C}, \text{K})$$

$$t_{w, \text{out}} = 81.9^\circ\text{F} (33.8^\circ\text{C})(307\text{K})$$

The assumed outlet temperature for the evaluation of the air properties was $75^\circ\text{F} (297\text{K})$. However, it is not necessary to go through the procedure again, since the air properties will not be changed significantly from those used in the calculations.

Pressure Drops

From Eq. (2-26), neglecting entrance and exit losses, which would be very small for this type of surface and exchanger geometry,

$$\frac{\Delta P}{P_1} = \frac{G^2}{2g_c P_1} \left[(1 + \sigma^2) \left(\frac{v_2}{v_1} - 1 \right) + f \frac{A}{A_c} \left(\frac{v_m}{v_1} \right) \right]$$

AIR SIDE

$$\frac{A}{A_c} = \frac{L}{r_h} = \frac{19}{12 \times 0.00288} = 550, \quad 1 + \sigma^2 = 1.61$$

$$\frac{\Delta P}{P_1} = \left(\frac{24,050}{3,600} \right)^2 \frac{1}{2 \times 32.2} \times \frac{6.78}{39.7 \times 144} \left[1.61 \left(\frac{5.19}{6.78} - 1 \right) \right. \\ \left. + 0.021 \times 550 \times 0.833 \right] = 0.000825(-0.38 + 9.61)$$

$$= 0.0076 \text{ or } 0.76 \text{ percent}$$

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WATER SIDE

$$\frac{A}{A_c} = \frac{L}{r_h} = \frac{5.333}{3.06 \times 10^{-3}} = 1,740$$
$$v = \text{const}$$

and so,

$$\begin{aligned}\Delta P &= \frac{G^2}{2g_c} v f \frac{L}{r_h} \\ &= \left(\frac{978,000}{3,600} \right)^2 \frac{1}{2 \times 32.2} \times 0.01603 \times 0.0094 \times 1,740 \\ &= 300 \text{ lb}_f/\text{ft}^2 = 2.1 \text{ lb}_f/\text{in}^2 \quad (14.5 \times 10^3 \text{ N/m}^2)\end{aligned}$$

These calculations illustrate the use of the basic data to determine the characteristics of a given intercooler. The complete design problem, Fig. 2-12, involves considerations of choice of optimum surface, coolant flow rate, dimensions, exchanger effectiveness, and pressure drops, all of which are beyond the scope of this presentation.

EXAMPLE 2 DIRECT-TRANSFER REGENERATOR

Figure B-3 describes the core geometry of a gas-turbine regenerator for a 5,000-shp (3.7 MW) open-cycle gas-turbine plant. Operating conditions and the heat transfer surfaces are specified as follows:

Operating conditions:

Air-flow rate, humid	193,000 lb _m /h (24.3 kg/s)
Air humidity	0.015 lb _m H ₂ O/lb _m dry air (kg/kg)
Fuel-air ratio	0.015 lb _m /lb _m (kg/kg)
Fuel, fuel oil	H/C = 0.15 lb _m /lb _m (kg/kg)

Air side:

Louvered plate-fin surface	$\frac{3}{4}$ —6.06
Entering pressure	132 lb _f /in ² abs ($9.1 \times 10^5 \text{ N/m}^2$)
Entering temperature	347°F = 807°R (175°C)(448K)

Gas side:

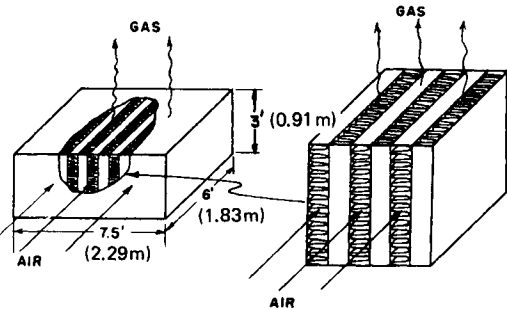
Plain plate-fin surface	11.1
Entering pressure	14.9 lb _f /in ² abs ($103 \times 10^3 \text{ N/m}^2$)
Entering temperature	805°F = 1265°R (430°C)(703K)

The objective of this problem is to determine, for the specified conditions and the basic heat transfer and flow-friction characteristics of the surface, (1) the regenerator effectiveness and (2) the pressure drops for both the air and gas sides.

The steps in the analysis require the determination of the following factors:

1. Surface characteristics, from Table 9-3
2. Heat transfer and free-flow areas
3. Fluid properties from Fig. A-2

Fig. B-3 A direct-type gas turbine regenerator for a 5,000-shp (3.7-MW) open-cycle plant.



- 4. Reynolds numbers
- 5. $StPr^{2/3}$ and f from the basic characteristics of the surface
- 6. Heat transfer coefficients
- 7. Fin effectiveness
- 8. Surface effectiveness
- 9. Overall coefficient of heat transfer
- 10. N_{tu} and exchanger effectiveness
- 11. Pressure drops

A detailed illustration of this procedure follows.

Surface Characteristics from Table 9-3a and b

	Air side	Gas side
Surface	$\frac{3}{8}$ –6.06	11.1
Plate spacing b , in	0.25 (6.35×10^{-3} m)	0.25 (6.35×10^{-3} m)
Hydraulic radius r_h , ft	0.00365 (1.11×10^{-3} m)	0.00253 (7.71×10^{-3} m)
Fin thickness δ , in	0.006 (0.15×10^{-3} m)	0.006 (0.15×10^{-3} m)
Transfer area/volume between plates β , ft ² /ft ³	256 (840 m ² /m ³)	367 (1,204 m ² /m ³)
Fin or extended area/total area	0.640	0.756

The heat exchanger core dimensions are, from Fig. B-3,

$A_{fr,a} = 22.5 \text{ ft}^2 \text{ (2.09 m}^2\text{)} \quad A_{fr,g} = 45.0 \text{ ft}^2 \text{ (4.18 m}^2\text{)}$
 $V = 7.5 \times 6 \times 3 = 135 \text{ ft}^3 \text{ (3.82 m}^3\text{)}$

Assume a plate thickness between air and gas sides

$a = 0.012 \text{ in (} 0.3 \times 10^{-3} \text{ m)}$

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and assume a fin material of high-temperature alloy steel with a thermal conductivity

$$k = 12 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}/\text{ft}) [20.8 \text{ W}/(\text{m} \cdot \text{K})]$$

from Fig. A-1.

Heat Transfer and Free-Flow Areas from Chap. 2, Eqs. (2-31)

$$\alpha_1 = \frac{A_1}{V_{\text{total}}} = \frac{b_1 \beta_1}{b_1 + b_2 + 2a}$$

AIR SIDE

$$\alpha_a = \frac{0.25 \times 256}{0.25 + 0.25 + 0.024} = 122 \text{ ft}^2/\text{ft}^3 \text{ (400 m}^2/\text{m}^3\text{)}$$

GAS SIDE

$$\alpha_g = \frac{0.25 \times 367}{0.25 \times 0.25 \times 0.024} = 175 \text{ ft}^2/\text{ft}^3 \text{ (574 m}^2/\text{m}^3\text{)}$$

The total heat transfer area on each side is given by

$$A = \alpha V$$

Consequently,

$$A_a = 122 \times 135 = 16,480 \text{ ft}^2 \text{ (1,531 m}^2\text{)}$$

$$A_g = 175 \times 135 = 23,600 \text{ ft}^2 \text{ (2,193 m}^2\text{)}$$

The ratio of free-flow area to frontal area is given by

$$\sigma = \frac{A_c}{A_{\text{fr}}} = \alpha r_h$$

$$\sigma_a = 122 \times 0.00365 = 0.445$$

$$\sigma_g = 175 \times 0.00253 = 0.443$$

The free-flow areas are then given by

$$A_c = \sigma A_{\text{fr}}$$

$$A_{c,a} = 0.445 \times 22.5 = 10.0 \text{ ft}^2 \text{ (0.929 m}^2\text{)}$$

$$A_{c,g} = 0.443 \times 45.0 = 19.9 \text{ ft}^2 \text{ (1.85 m}^2\text{)}$$

Fluid Properties

Assume as a first approximation, to be verified later, a regenerator effectiveness of 75 percent. This approximation need not be of a high order of accuracy, since

it is used only to estimate an average temperature for the evaluation of fluid properties.

From the definition of effectiveness [Eq. (2-6)] for $C_c < C_h$ and $\varepsilon \approx 75$ percent,

$$t_{a,out} \approx 691^\circ\text{F} = 1151^\circ\text{R} (366^\circ\text{C})(639\text{K})$$

$$t_{g,out} \approx 461^\circ\text{F} = 921^\circ\text{R} (239^\circ\text{C})(512\text{K})$$

Thus the bulk average temperatures are

$$t_{a,av} = \frac{691 + 347}{2} = 519^\circ\text{F} (979^\circ\text{R})(271^\circ\text{C})(544\text{K})$$

$$t_{g,av} = \frac{805 + 461}{2} = 633^\circ\text{F} (1093^\circ\text{R})(334^\circ\text{C})(607\text{K})$$

Arithmetic averages are satisfactory here because of the approximately linear variations of the fluid temperatures with flow length. For air at these bulk temperatures from Fig. A-2, or more accurately from available tables.

	Air side	Gas side
μ , $\text{lb}_m/(\text{h} \cdot \text{ft})$	0.069 ($2.85 \times 10^{-5} \text{ Pa} \cdot \text{s}$)	0.073 ($3.02 \times 10^{-5} \text{ Pa} \cdot \text{s}$)
Pr	0.67	0.67
$\text{Pr}^{2/3}$	0.766	0.766
c_p , $\text{Btu}/(\text{lb}_m \cdot ^\circ\text{F})$ for dry air	0.248 [$1.04 \text{ kJ}/(\text{kg} \cdot \text{K})$]	0.251 [$1.05 \text{ kJ}/(\text{kg} \cdot \text{K})$]

The specific heat on the air side, as given above, should be corrected for humidity, and the specific heat on the gas side, as given above, should be corrected for both the humidity and the effects of combustion of the fuel. From Figs. A-18 and A-19 for a humidity of $0.015 \text{ lb}_m \text{ H}_2\text{O}/\text{lb}_m$ (kg/kg) dry air and a fuel-air ratio of $0.015 \text{ lb}_m/\text{lb}_m$, (kg/kg)

$$X_{c,\omega} = 1.013 \quad X_{c,f} = 1.019$$

Then

$$c_{p,a} = X_{c,\omega} c_p = 0.251 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F}), (1.05 \text{ kJ}/\text{kg} \cdot \text{K})$$

$$c_{p,g} = X_{c,\omega} X_{c,f} c_p = 0.259 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F}) [1.08 \text{ kJ}/(\text{kg} \cdot \text{K})]$$

Assume as a first approximation a 2 percent pressure drop on both the air and gas sides. The air outlet pressure is then $129 \text{ lb}_f/\text{in}^2$ ($8.89 \times 10^5 \text{ N}/\text{m}^2$) and the gas outlet pressure is $14.6 \text{ lb}_f/\text{in}^2$ ($1.01 \times 10^5 \text{ N}/\text{m}^2$). The inlet and outlet air and gas specific volumes can then be evaluated from the perfect gas equation of state, employing the correction factors for humidity and the effect of the fuel. From Figs. A-18 and A-20,

$$X_{d,\omega} = 0.992 \quad X_{d,f} = 1.001$$

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It should be noted that these correction factors apply to the density ρ , the reciprocal of the specific volume v .

The specific volumes are computed as follows:

AIR SIDE

$$\begin{aligned} v_{a,1} &= \frac{1}{\rho_{a,1}} = \frac{(R/M)T_{a,1}}{X_{d,\omega}P_{a,1}} \\ &= \frac{53.345 \times 807}{0.992 \times 132 \times 144} = 2.28 \text{ ft}^3/\text{lb}_m \text{ (0.142 m}^3/\text{kg)} \\ v_{a,2} &= \frac{1}{\rho_{a,2}} = \frac{53.345 \times 1,151}{0.992 \times 129 \times 144} = 3.33 \text{ ft}^3/\text{lb}_m \text{ (0.208 m}^3/\text{kg)} \end{aligned}$$

As a fair approximation, the mean specific volume will be evaluated as the arithmetic average of the inlet and outlet specific volumes

$$v_{a,m} = \frac{v_{a,1} + v_{a,2}}{2} = \frac{2.28 + 3.33}{2} = 2.80 \text{ ft}^3/\text{lb}_m \text{ (0.175 m}^3/\text{kg)}$$

GAS SIDE

$$\begin{aligned} v_{g,1} &= \frac{1}{\rho_{g,1}} = \frac{RT_{g,1}/M}{X_{d,\omega}X_{d,f}P_{g,1}} = \frac{53.345 \times 1,265}{0.992 \times 1.001 \times 14.9 \times 144} \\ &= 31.7 \text{ ft}^3/\text{lb}_m \text{ (1.98 m}^3/\text{kg)} \\ v_{g,2} &= \frac{1}{\rho_{g,2}} = \frac{53.345 \times 921}{0.992 \times 1.001 \times 14.6 \times 144} = 23.6 \text{ ft}^3/\text{lb}_m \text{ (1.47 m}^3/\text{kg)} \\ v_{g,m} &= \frac{v_{g,1} + v_{g,2}}{2} = \frac{31.7 + 23.6}{2} = 27.65 \text{ ft}^3/\text{lb}_m \text{ (1.73 m}^3/\text{kg)} \end{aligned}$$

Reynolds Numbers

AIR SIDE

$$\begin{aligned} G_a &= \frac{W_a}{A_{c,a}} = \frac{193,000}{10.00} = 19,300 \text{ lb}_m/(\text{h} \cdot \text{ft}^2) [26.18 \text{ kg}/(\text{s} \cdot \text{m}^2)] \\ \text{Re}_a &= \frac{4r_{h,a}G_a}{\mu_a} = \frac{4 \times 0.00365 \times 19,300}{0.069} = 4,090 \end{aligned}$$

GAS SIDE

$$\begin{aligned} W_g &= (1 + f)W_a = 1.015 \times 193,000 = 196,000 \text{ lb}_m/\text{h} \text{ (24.7 kg/s)} \\ G_g &= \frac{W_g}{A_{c,g}} = \frac{196,000}{19.9} = 9,850 \text{ lb}_m/(\text{h} \cdot \text{ft}^2) [13.36 \text{ kg}/(\text{s} \cdot \text{m}^2)] \\ \text{Re}_g &= \frac{4r_{h,g}G_g}{\mu_g} = \frac{4 \times 0.00253 \times 9,850}{0.073} = 1,370 \end{aligned}$$

StPr^{2/3} and f

AIR SIDE (from Fig. 10-38)

$$\begin{aligned}\text{StPr}^{2/3} &= 0.0073 & f &= 0.0375 \\ \text{St} &= 0.00954\end{aligned}$$

GAS SIDE (from Fig. 10-26)

$$\begin{aligned}\text{StPr}^{2/3} &= 0.0045 & f &= 0.0155 \\ \text{St} &= 0.00588\end{aligned}$$

Heat Transfer Coefficients

AIR SIDE

$$\begin{aligned}h_a &= \text{St}Gc_p = 0.00954 \times 19,300 \times 0.251 \\ &= 46.1 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) [262 \text{ W}/(\text{m}^2 \cdot \text{K})]\end{aligned}$$

GAS SIDE

$$h_g = 0.00588 \times 9,850 \times 0.259 = 15.0 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) [85 \text{ W}/(\text{m}^2 \cdot \text{K})]$$

Fin Effectiveness (Fig. 2-13 or Eq. (2-4))

AIR SIDE

$$\begin{aligned}m &= \sqrt{\frac{2 \times 46.1}{12 \times 0.006/12}} \\ &= 124 \text{ ft}^{-1} (407 \text{ m}^{-1}) \\ l &\approx \frac{b}{2} = \frac{0.25/12}{2} = 0.0104 \text{ ft} (0.0032 \text{ m}) \\ ml &= 1.29 \quad \text{then} \quad \eta_{f,a} = 0.665\end{aligned}$$

GAS SIDE

$$\begin{aligned}m &= \sqrt{\frac{2 \times 15.0}{12 \times 0.006/12}} = 70.6 \text{ ft}^{-1} (231 \text{ m}^{-1}) \\ ml &= 0.733 \quad \text{then} \quad \eta_{f,g} = 0.850\end{aligned}$$

Surface Effectiveness from Eq. (2-3)

$$\eta_o = 1 - \frac{A_f}{A} (1 - \eta_f)$$

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AIR SIDE

$$\eta_{o,a} = 1 - (0.640)(1 - 0.665) = 0.786$$

GAS SIDE

$$\eta_{o,g} = 1 - (0.756)(1 - 0.850) = 0.887$$

Overall Coefficient of Heat Transfer U

From Eq. (2-2), neglecting the very small wall resistance,

$$\begin{aligned}\frac{1}{U_a} &= \frac{1}{\eta_{o,a}h_a} + \frac{1}{(A_g/A_a)\eta_{o,g}h_g} \\ &= \frac{1}{0.786 \times 46.1} + \frac{1}{(23,000/16,480) \times 0.887 \times 15.0} \\ &= 0.276 + 0.0525 = 0.0801 \\ &\quad (34.5\%) \quad (65.5\%) \\ U_a &= 12.5 \text{ Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F}) [70.9 \text{ W}/(\text{m}^2 \cdot \text{K})]\end{aligned}$$

N_{tu} and Exchanger Effectiveness

AIR-SIDE CAPACITY RATE

$$C_a = W_a c_{p,a} = (193,000)(0.251) = 48,500 \text{ Btu}/(\text{h} \cdot ^\circ\text{F}) (25.5 \text{ kW/K})$$

GAS-SIDE CAPACITY RATE

$$C_g = (196,000)(0.259) = 50,800 \text{ Btu}/(\text{h} \cdot ^\circ\text{F}) (26.7 \text{ kW/K})$$

CAPACITY-RATE RATIO

$$\frac{C_{\min}}{C_{\max}} = \frac{48,000}{50,800} = 0.955$$

From Eq. (2-7), the number of heat transfer units,

$$N_{tu,\max} = \frac{A_a U_a}{C_{\min}} = \frac{16,480 \times 12.5}{48,500} = 4.25$$

From Fig. 2-16 for a crossflow exchanger with both fluids unmixed, the effectiveness $\epsilon = 0.75$. The gas and air outlet temperatures can then be computed from the definition of effectiveness as

$$t_{a,\text{out}} = 691^\circ\text{F} (366^\circ\text{C}) (639\text{K}) \quad t_{g,\text{out}} = 477^\circ\text{F} (247^\circ\text{C}) (520\text{K})$$

Pressure Drops

The complete equation for the pressure drop [Eq. (2-26)] is given as follows:

$$\frac{\Delta P}{P_1} = \frac{G^2 v_1}{2g_c P_1} \left[(K_c + 1 - \sigma^2) + 2 \left(\frac{v_2}{v_1} - 1 \right) + \left(f \frac{A}{A_c} \frac{v_m}{v_1} \right) - (1 - \sigma^2 - K_e) \left(\frac{v_2}{v_1} \right) \right]$$

The approximate entrance and exit loss coefficients are given in Fig. 5-4 as

	K_c	K_e
Air side	0.48	0.24
Gas side	0.54	0.28

From previous calculations,

$$\left(\frac{A}{A_c} \right)_a = \left(\frac{L}{r_h} \right)_a = 1,648$$

$$\left(\frac{A}{A_c} \right)_g = \left(\frac{L}{r_h} \right)_g = 1,190$$

AIR SIDE

$$\begin{aligned} \frac{\Delta P}{P_1} &= \left(\frac{19,300}{3,600} \right)^2 \left(\frac{1}{2 \times 32.2} \right) \left(\frac{2.28}{132 \times 144} \right) \left[(0.48 + 1 - 0.445^2) \right. \\ &\quad \left. + 2 \left(\frac{3.33}{2.28} - 1 \right) + \left(0.0375 \times 1,648 \times \frac{2.805}{2.28} \right) \right. \\ &\quad \left. - (1 - 0.445^2 - 0.24) \frac{3.33}{2.28} \right] \\ &= 5.37 \times 10^{-5} (1.28 + 0.92 + 76.1 - 0.82) \\ &= 4.16 \times 10^{-3} = 0.42 \text{ percent} \end{aligned}$$

GAS SIDE

$$\begin{aligned} \frac{\Delta P}{P_1} &= \left(\frac{9,850}{3,600} \right)^2 \left(\frac{1}{2 \times 32.2} \right) \left(\frac{31.7}{14.9 \times 144} \right) \left[(0.54 + 1 - 0.443^2) \right. \\ &\quad \left. + 2 \left(\frac{23.6}{31.7} - 1 \right) + \left(0.0155 \times 1,190 \times \frac{27.65}{31.7} \right) \right. \\ &\quad \left. - (1 - 0.443^2 - 0.28) \frac{23.6}{31.7} \right] \\ &= 1.72 \times 10^{-3} (1.34 - 0.51 + 16.1 - 0.39) \\ &= 0.0285 = 2.9 \text{ percent} \end{aligned}$$

These calculations illustrate a procedure for predicting the performance of a given heat exchanger core. No allowance has been made for additional heat

transfer resistance due to fouling or to increased pressure drop due to header design and poor flow distribution. The choice of surfaces and core dimensions does not necessarily represent an optimum design. An optimum-design regenerator (see Fig. 2-12) would have more nearly equalized percent pressure drops, but design considerations such as these are beyond the scope of this presentation. It should also be noted that the low fin effectiveness on the air side (0.665) may introduce some error into the calculations, since there is question as to the accuracy of application of the fin analysis for fin effectiveness less than about 0.80 for flow in long finned passages.

EXAMPLE 3 LIQUID-COUPLED INDIRECT-TRANSFER REGENERATOR

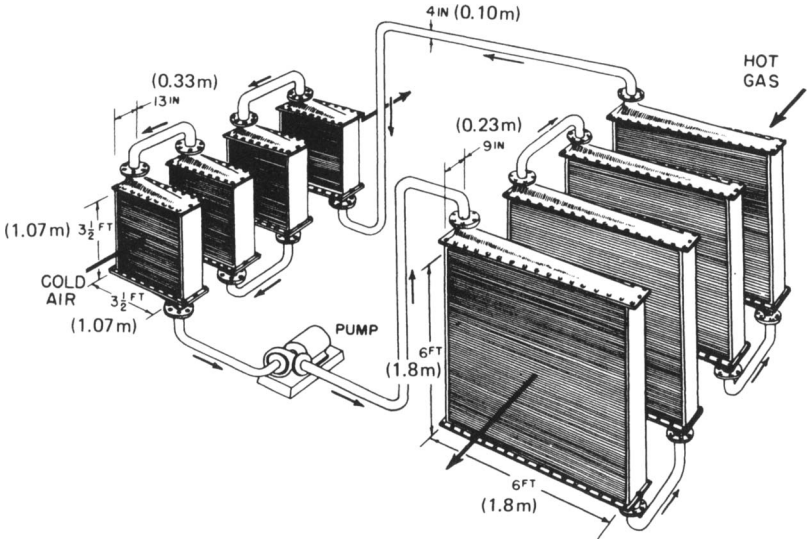
In this illustration, detailed calculations will not be presented for the air-side and gas-side exchangers, as these calculations are essentially the same as for the direct transfer-type exchangers previously considered. Emphasis in this section will be on the application of the

$$\varepsilon = \Phi\left(\varepsilon_a, \varepsilon_g, \frac{C_a}{C_g}, \frac{C_a}{C_L}\right)$$

relationships [Eq. (2-20)].

Consider the liquid-coupled indirect-transfer regenerator system pictured in Fig. B-4. Cycle conditions (same as for the previously considered direct-transfer regenerator) are given at the top of page 319.

Fig. B-4 A liquid-coupled indirect-transfer regenerator for a 5,000-shp (3.7-MW) turbine plant.



Shaft power	5,000 hp (3.7 MW)
Air-flow rate	193,000 lb/h (24.3 kg/s)
Fuel-air ratio	0.015 lb/lb (kg/kg)
Air side entering:	
Pressure	132 lb _f /in ² abs (9.1×10^5 N/m ²)
Temperature	347°F (807°R)(175°C)(448K)
Gas side entering:	
Pressure	14.9 lb _f /in ² abs (1.03×10^5 N/m ²)
Temperature	805°F (1265°R)(430°C)(703K)

The surfaces employed in each of the two exchangers and their geometrical factors are as follows, from Table 9-4c:

	Air-side unit	Gas-side unit
Surface	11.32–0.737–SR (Fig. 10-97)	9.1–0.737–S (Fig. 10-94)
Type	Finned-flat tube	Finned-flat tube
Fins per inch	11.32 (446 per m)	9.1 (358 per m)
r_h , ft†	2.68×10^{-3} (8.17×10^{-4} m)	3.25×10^{-3} (9.9×10^{-4} m)
σ , ft ² /ft ² †	0.727	0.745
α , ft ² /ft ³	270 (886 m ² /m ³)	224 (735 m ² /m ³)
$\alpha_{\text{liquid side}}$, ft ² /ft ³	42 (138 m ² /m ³)	42 (138 m ² /m ³)
Fin/total area, ft ² /ft ²	0.845	0.813
Fin thickness σ , in	0.010 (3.5×10^{-4} m)	0.010 (2.15×10^{-4} m)
Wall thickness a	0.010 (2.5×10^{-4} m)	0.010 (2.15×10^{-4} m)
Specific weight (for steel), lb _m /ft ³	63.9 (1024 kg/m ³)	54.6 (875 kg/m ³)
Material k , Btu/(h · ft ² · F/ft)	12 [21 W/(m · k)]	12 [21 W/(m · k)]

† Corrected from Table 9-4c magnitudes to allow for greater fin metal thickness.

The calculated results needed for the determination of the component effectivenesses for the air-side and gas-side units, ϵ_a and ϵ_g , respectively are summarized in the following tabulation:

	Air side	Gas side
G , lb _m /(h · ft ²)	21,430 [29 kg/(s · m ²)]	7,300 [9.9 kg/(s · m ²)]
Re	3,000	1,300
St	0.0085	0.0117
f	0.026	0.031
h , Btu/(h · ft ² · °F)	45.5 [258 W/(m ² · K)]	21.4 [121 W/(m ² · K)]
η_o	0.60	0.745
U , Btu/(h · ft ² · °F)	25.5 [145 W/(m ² · K)]	15.42 [87.5 W/(m ² · K)]
N_{tu}	7.60	7.60
$\Delta P/P_1$, %	0.40	2.4

For the four-pass arrangements pictured in Fig. B-4, the N_{tu} per pass is $7.60/4 = 1.90$ for both sides. Figure 2-16 for crossflow (unmixed), with $C_{\min}/C_{\max} \approx 1$, then yields $\epsilon_{pa} = \epsilon_{pg} = 0.60$. Equation (2-18) may now be employed to

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get the component effectiveness (for $C_L \approx C_a$, $C_L \approx C_g$)

$$\varepsilon_a, \varepsilon_g = \frac{n\varepsilon_p}{1 + (n-1)\varepsilon_p} = \frac{4 \times 0.60}{1 + 3 \times 0.60} = 0.858$$

Then, from Eq. (2-20), the overall effectiveness becomes

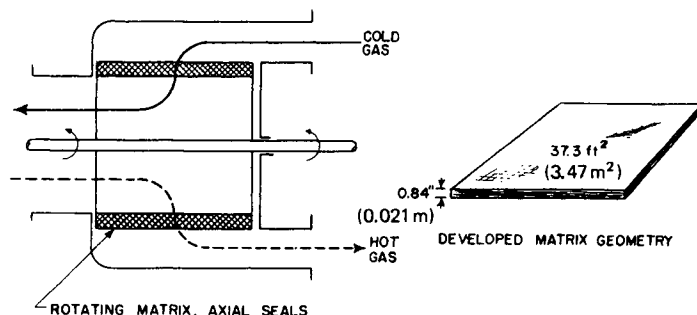
$$\varepsilon = \frac{1}{1/\varepsilon_a + 1/\varepsilon_g - C_a/C_L} = \frac{1}{1.165 + 1.165 - 1} = 0.75$$

EXAMPLE 4 THE PERIODIC-FLOW REGENERATOR

The analysis for performance of this type of regenerator is in some measure more simple than for the direct type or liquid-coupled indirect-transfer type. The same surface configuration is exposed to the air and gas flows, and no effectiveness of extended surface considerations are involved. The resulting simplifications will be evident from the following example.

Figure B-5 describes the core geometry for a stainless steel screen-matrix regenerator for a 1,000-shp (0.75-MW) open-cycle gas-turbine plant.

Fig. B-5 A rotating periodic-flow regenerator for a 1,000-shp (0.75-MW) gas-turbine plant.



FRONTAL-AREA RATIO $A_{frg}/A_{fra} = 2.5$

$$A_{frg} = 26.6 \text{ ft}^2 (2.47 \text{ m}^2), A_{fra} = 10.65 \text{ ft}^2 (0.989 \text{ m}^2)$$

SURFACE: SCREEN MATRIX $24 \times 24 - 0.014$

ROTATIVE SPEED = 26.5 r/min (0.442 r/s)

MATRIX MASS = 351 lb (159 kg)

Cycle operating conditions are as follows:

Air-flow rate (dry)	75,400 lb _m /h (9.5 kg/s)
Air humidity	Essentially zero
Fuel-air ratio	0.015 lb _m /lb (kg/kg)
Fuel oil H/C	0.15
Regenerator air side:	
Inlet pressure	47.7 lb _f /in ² abs ($3.29 \times 10^5 \text{ N/m}^2$)

Inlet temperature	333°F (793°R)(168°C)(441K)
Inlet specific volume	6.15 ft ³ /lb _m (0.384 m ³ /kg)
Regenerator gas side:	
Inlet pressure	15.2 lb _f /in ² abs (1.05 × 10 ⁵ N/m ²)
Inlet temperature	1,246°F (1,706°R)(675°C)(948K)
Inlet specific volume	41.5 ft ³ /lb _m (2.59 m ³ /kg)
Rotor speed	26.5 r/min (0.442 r/s)

The steps in the analysis require the determination of the following factors:

1. Surface characteristics and then the heat transfer and free-flow areas from the given core geometry
2. Fluid properties
3. Reynolds numbers
4. St and f
5. Heat transfer coefficients and N_{tu} 's
6. Exchanger effectiveness
7. Pressure drops

Surface Characteristics and Core Geometry

The details of the screen-matrix heat transfer surface are as follows:

Mesh per inch	24 by 24 $\frac{1}{2}$ (945 by 955 per m)
Wire diameter, in	0.0135 (3.4 × 10 ⁻⁴ m)
Thickness of one screen δ , in	0.0255 (6.9 × 10 ⁻⁴ m)
Porosity, p	0.725
Stainless steel:	
Specific heat	$c_s = 0.12 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F}) [0.5 \text{ kJ}/(\text{kg} \cdot \text{K})]$
Density	$\rho_s = 488 \text{ lb}_m/\text{ft}^3 (7,817 \text{ kg/m}^3)$

From this description, the following may be calculated:

Heat transfer area per unit volume	
α , ft ² /ft ³ , from wire diameter and mesh	980 (3,215 m ² /m ³)
Hydraulic radius r_h , ft, from Eq. (2-31h)	$0.740 \times 10^{-3} (2.26 \times 10^{-4} \text{ m})$

The flow frontal areas are

$$A_{fr,g} = 26.6 \text{ ft}^2 (2.47 \text{ m}^2), \quad A_{fr,a} = 10.65 \text{ ft}^2 (0.989 \text{ m}^2), \quad \frac{A_{fr,g}}{A_{fr,a}} = 2.50$$

and the heat transfer areas are

$$A_g + A_a = \alpha \times \text{vol} = 980 \times 37.3 \times \frac{0.84}{12} = 2,560 \text{ ft}^2 (238 \text{ m}^2)$$

$$\frac{A_g}{A_a} = \frac{A_{fr,g}}{A_{fr,a}} = 2.50$$

$$A_g = 1,830 \text{ ft}^2 (170 \text{ m}^2) \quad A_a = 730 \text{ ft}^2 (67.8 \text{ m}^2)$$

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The “effective” free-flow areas are

$$A_c = pA_{fr} \\ A_{c,g} = 19.3 \text{ ft}^2 (1.79 \text{ m}^2) \quad A_{c,a} = 7.71 \text{ ft}^2 (0.716 \text{ m}^2)$$

Then

$$\frac{A_g}{A_{c,g}} = \frac{A_a}{A_{c,a}} = \frac{L}{r_h} = 94.7$$

For round wire the ratio of solid volume to surface area is *diameter*/4. Thus the solid volume is $2,560 \times (0.0135)/(4 \times 12) = 0.720 \text{ ft}^3 (0.0204 \text{ m}^3)$. The matrix mass is then $0.720 \times 488 = 351 \text{ lb}_m (159 \text{ kg})$. This result can also be calculated using the core volume $[2.61 \text{ ft}^3 (0.0739 \text{ m}^3)]$ and the matrix solidity $(1 - p = 0.275)$, together with the material density $[\rho_s = 488 \text{ lb}_m/\text{ft}^3 (7,817 \text{ kg/m}^3)]$.

Fluid Properties

For this purpose, bulk average flow temperatures are required, and this in turn requires a preliminary estimate of exchanger effectiveness. This estimate can be quite approximate; $\varepsilon = 85$ percent will be used.

Air side	Gas side
$t_{a,in} = 333^\circ\text{F} (793^\circ\text{R})(166^\circ\text{C})(441 \text{ K})$	$t_{g,in} = 1,246^\circ\text{F} (1,706^\circ\text{R})(675^\circ\text{C})(948 \text{ K})$
$t_{a,out} = 1,108^\circ\text{F} (1,568^\circ\text{R})(598^\circ\text{C})(871 \text{ K})$	$t_{g,out} = 471^\circ\text{F} (931^\circ\text{R})(244^\circ\text{C})(517 \text{ K})$
$t_{a,av} = 720^\circ\text{F} (1,180^\circ\text{R})(383^\circ\text{C})(656 \text{ K})$	$t_{av} = 858^\circ\text{F} (1,318^\circ\text{R})(459^\circ\text{C})(732 \text{ K})$
$P_1 = 47.7 \text{ lb}_f/\text{in}^2 \text{ abs} (3.29 \times 10^5 \text{ N/m}^2)$	$P_1 = 15.2 \text{ lb}_f/\text{in}^2 \text{ abs} (1.05 \times 10^5 \text{ N/m}^2)$
$P_2 = 46.7 \text{ lb}_f/\text{in}^2 \text{ abs} (3.22 \times 10^5 \text{ N/m}^2)$	$P_2 = 14.7 \text{ lb}_f/\text{in}^2 \text{ abs} (1.01 \times 10^5 \text{ N/m}^2)$
(assumed $\Delta P = 1 \text{ lb}_f/\text{in}^2$)	(assumed $\Delta P = 0.5 \text{ lb}_f/\text{in}^2$)
$(6.9 \times 10^3 \text{ N/m}^2)$	$(3.5 \times 10^3 \text{ N/m}^2)$
$v_1 = 6.5 \text{ ft}^3/\text{lb}_m (0.41 \text{ m}^3/\text{kg})$	$v_1 = 41.5 \text{ ft}^3/\text{lb}_m (2.59 \text{ m}^3/\text{kg})$
$v_2 = 12.41 \text{ ft}^3/\text{lb}_m (0.77 \text{ m}^3/\text{kg})$	$v_2 = 23.4 \text{ ft}^3/\text{lb}_m (1.46 \text{ m}^3/\text{kg})$
$v_m = 9.27 \text{ ft}^3/\text{lb}_m (0.58 \text{ m}^3/\text{kg})$	$v_m = 32.6 \text{ ft}^3/\text{lb}_m (2.04 \text{ m}^3/\text{kg})$
$\mu = 0.077 \text{ lb}_m/(\text{h} \cdot \text{ft})^\dagger$	$\mu = 0.084 \text{ lb}_m/(\text{h} \cdot \text{ft})^\dagger$
$(3.18 \times 10^{-5} \text{ Pa} \cdot \text{s})$	$(3.47 \times 10^{-5} \text{ Pa} \cdot \text{s})$
$\text{Pr} = 0.67^\dagger$	$\text{Pr} = 0.66^\dagger$
$c_p = 0.254 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F})^\dagger$	$c_p = 0.263 \text{ Btu}/(\text{lb}_m \cdot ^\circ\text{F})^\dagger$
$(1.06 \text{ kJ}/\text{kg} \cdot \text{K})$	$[1.10 \text{ kJ}/(\text{kg} \cdot \text{K})]$

† For dry air from Fig. A-2 or, more accurately, from available tables.

‡ Includes fuel correction factor of 1.02 from Fig. A-19.

Reynolds Numbers

This calculation first requires the determination of mass velocities from $G = W/A_c$:

$$G_a = \frac{75,400}{7.71} = 9,780 \text{ lb}_m/(\text{h} \cdot \text{ft}^2) [13.26 \text{ kg}/(\text{m}^2 \cdot \text{s})]$$

$$G_g = \frac{75,400 \times 1.015}{19.3} = 3,960 \text{ lb}_m/(\text{h} \cdot \text{ft}^2) [5.371 \text{ kg}/(\text{m}^2 \cdot \text{s})]$$

Note that

$$\frac{G_a}{G_g} = \frac{A_{fr,g}/A_{fr,a}}{1 + \text{fuel-air ratio}}$$

Then from $Re = 4r_h G/\mu$,

$$Re_a = \frac{4 \times 0.740 \times 10^{-3} \times 9,780}{0.077} = 376$$

$$Re_g = \frac{4 \times 0.740 \times 10^{-3} \times 3,960}{0.084} = 140$$

St and f

From Figs. 7-8 and 7-9,

	Air side	Gas side
Re	376	140
$StPr^{2/3}$	0.057	0.088
St	0.0744	0.116
f	0.46	0.64

N_{tu} 's and Heat Transfer Coefficients

For this type of exchanger,

$$N_{tu} = StA/A_c \quad \text{and} \quad h = StGc_p$$

	Air side	Gas side
St	0.0744	0.116
A/A_c	94.7	94.7
N_{tu}	7.04	10.98
$G, \text{lb}_m/(\text{h} \cdot \text{ft}^2)$	9,780 [13.26 kg/(m ² · s)]	3,960 [5.371 kg/(m ² · s)]
$c_p, \text{Btu}/(\text{lb}_m \cdot ^\circ\text{F})$	0.254 [1.06 kJ/(kg · K)]	0.263 (1.10 kJ/(kg · K)]
$h, \text{Btu}/(\text{h} \cdot \text{ft}^2 \cdot ^\circ\text{F})^\dagger$	185 [1,050 W/(m ² · K)]	121 [687 W/(m ² · K)]

† Note these remarkably high coefficients relative to those associated with conventional surfaces.

Exchanger Effectiveness

Here the theory presented in Chap. 2 will be employed. In Figs. 2-31 to 2-38,

$$\varepsilon = \Phi \left(N_{tu,o}, \frac{C_a}{C_g}, \frac{C_r}{C_a} \right)$$

From the cycle conditions,

$$\frac{C_a}{C_g} = \frac{75,400 \times 0.254}{75,400 \times 1.015 \times 0.263} = \frac{19,150}{20,200} = 0.950$$

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The matrix capacity rate can be calculated from

$$\begin{aligned}C_r &= (r/\text{min} \times 60) \times \text{mass of matrix} \times c_s \\ &= 26.5 \times 60 \times 351 \times 0.12 = 66,900 \text{ Btu}/(\text{h} \cdot ^\circ\text{F})(35.27 \text{ kW/K})\end{aligned}$$

Then

$$\frac{C_r}{C_a} = \frac{66,900}{19,150} = 3.49$$

The overall number of transfer units from Eq. (2-23) is

$$\begin{aligned}N_{\text{tu},o} &= \frac{1}{C_a} \left[\frac{1}{\left(\frac{1}{hA}\right)_a + \left(\frac{1}{hA}\right)_g} \right] \\ &= \frac{1}{\frac{1}{N_{\text{tu},a}} + \frac{C_a/C_g}{N_{\text{tu},h}}} = \frac{1}{\frac{1}{7.04} + \frac{0.95}{10.98}} = 4.38\end{aligned}$$

The effectivenesses may be determined from Fig. 2-32:

$$\varepsilon = 0.82$$

Pressure Drops

This calculation is based on Eq. (2-26), with the porosity p in place of the flow-area contraction ratio,

$$\frac{\Delta P}{P_1} = \frac{G^2}{2g_c} \frac{v_1}{P_1} \left[(1 + p^2) \left(\frac{v_2}{v_1} - 1 \right) + f \frac{A}{A_c} \frac{v_m}{v_1} \right]$$

	Air side	Gas side
$P_1, \text{lb}_f/\text{ft}^2$	6,880 ($3.29 \times 10^5 \text{ N/m}^2$)	2,190 ($1.05 \times 10^5 \text{ N/m}^2$)
$v_1, \text{ft}^3/\text{lb}_m$	6.5 ($0.41 \text{ m}^3/\text{kg}$)	41.5 ($2.59 \text{ m}^3/\text{kg}$)
$v_2, \text{ft}^3/\text{lb}_m$	12.41 ($0.77 \text{ m}^3/\text{kg}$)	23.4 ($1.46 \text{ m}^3/\text{kg}$)
$v_m, \text{ft}^3/\text{lb}_m$	9.27 ($0.58 \text{ m}^3/\text{kg}$)	3.26 ($2.04 \text{ m}^3/\text{kg}$)
v_2/v_1	2.02	0.564
v_m/v_1	1.51	0.785
$p, \text{ft}^3/\text{ft}^3$	0.725	0.725
$A/A_c (= L/r_h)$	94.7	94.7
f	0.46	0.64
$G \text{ lb}_m/(\text{h} \cdot \text{ft}^2)$	9,780 [$13.26 \text{ kg}/(\text{m}^2 \cdot \text{s})$]	3,960 [$5.37 \text{ kg}/(\text{m}^2 \cdot \text{s})$]
$(1 + p^2) \left(\frac{v_2}{v_1} - 1 \right)$	1.56	-0.665
$f \left(\frac{A}{A_c} \right) \left(\frac{v_m}{v_1} \right)$	65.7	47.6
$\frac{G^2}{2g_c} \frac{v_1}{P_1}$	1.026×10^{-4}	3.55×10^{-4}
$\Delta P/P_1$	0.691×10^{-2}	1.67×10^{-2}
$(\Delta P/P_1)_{\text{total}}, \%$	2.36	

In summary, the exchanger effectiveness is 82 percent and the total pressure drop is 2.4 percent. It is essential to realize, however, that such important factors as duct losses, flow turning losses, carryover, and leakage have been omitted from this analysis. They will obviously depend upon the details of the final configuration.

Appendix C Derivation of ε - N_{tu} Relations

In Chap. 2, exchanger heat transfer performance theory was summarized. The purpose of this appendix is to present some of the derivations for the ε - N_{tu} relations in order to illustrate the general method of approach. These derivations will be limited to the direct-transfer counterflow, multipass overall-counterflow, and crossflow one-fluid-mixed cases, Eqs. (2-13), (2-18), (2-15), and (2-16). The derivations for the ε - N_{tu} relations for the other flow arrangements given in Chap. 2 are referenced there.

COUNTERFLOW EXCHANGER

Temperature conditions in a counterflow exchanger with $C_h < C_c$ (that is, $C_h = C_{\min}$) may be described schematically, as shown in Fig. C-1. From an energy-balance consideration,

$$dq = -C_h dt_h = -C_c dt_c \quad (C-1)$$

It is thus evident that a plot of the "operating-line," t_h versus t_c , will have a slope

Fig. C-1 Temperature conditions for counterflow with $C_h < C_c$.

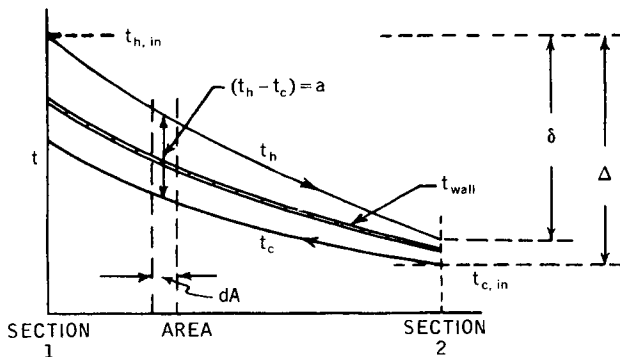
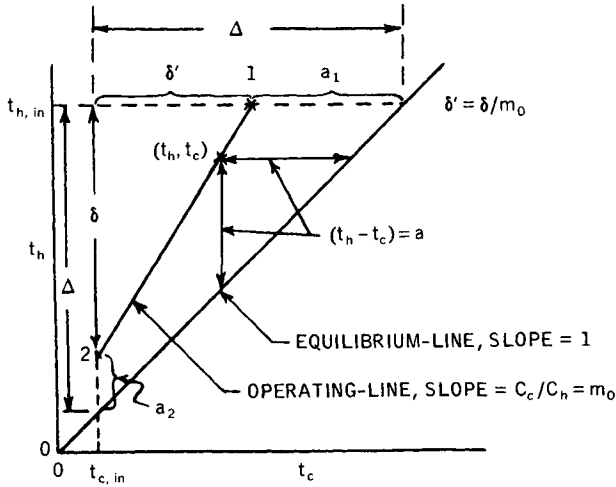


Fig. C-2 Operating-line and equilibrium-line representation for counterflow and $C_h < C_c$.



$$m_o = \frac{dt_h}{dt_c} = \frac{C_c}{C_h} = \text{const}$$

for the situation of both C_c and C_h constant. Moreover, the slope m_o is greater than unity for the assumed condition of $C_h < C_c$. This operating line is plotted in Fig. C-2. Also shown is a line specifying the conditions for thermal equilibrium between the two streams:

$$t_h = t_c \quad \text{for equilibrium}$$

From the geometry of the figure, it is evident that the vertical or horizontal intercept between these two lines, from a given operating point (t_h, t_c) , must be equal to $(t_h - t_c)$, as shown on the previous figure of temperature conditions versus area.

From the definition of ε [Eq. (2-6), Chap. 2], since $C_h = C_{min}$,

$$\varepsilon = \frac{t_{h,in} - t_{h,out}}{t_{h,in} - t_{c,in}} = \frac{\delta}{\Delta} \quad (\text{C-2})$$

where δ and Δ are defined on the operating-line–equilibrium-line plot.

In addition to the energy balance [Eq. (C-1)], a rate equation may be written for the heat transfer rate dq through the differential area dA [see Eq. (2-1)]:

$$dq = U(t_h - t_c) dA \quad (\text{C-3})$$

From Eq. (C-1),

$$d(t_h - t_c) = \left(\frac{1}{C_c} - \frac{1}{C_h} \right) dq$$

and this result, combined with Eq. (C-3), yields

$$\frac{d(t_h - t_c)}{t_h - t_c} = - \left(1 - \frac{C_h}{C_c} \right) \frac{U}{C_h} dA \quad (\text{C-4})$$

This expression may be readily integrated from the hot-fluid inlet to the outlet section to yield

$$\frac{t_{h,2} - t_{c,2}}{t_{h,1} - t_{c,1}} = \frac{a_2}{a_1} = e^{-N_{tu}(1-C_h/C_c)} \quad (\text{C-4a})$$

Inspection of the geometry of the operating-line-equilibrium-line plot will demonstrate that

$$\begin{aligned} t_{h,2} - t_{c,2} &= a_2 = \Delta - \delta \\ t_{h,1} - t_{c,1} &= a_1 = \Delta - \delta' = \Delta - \frac{\delta}{m_o} \end{aligned}$$

Thus

$$\frac{t_{h,2} - t_{c,2}}{t_{h,1} - t_{c,1}} = \frac{a_2}{a_1} = \frac{1 - \delta/\Delta}{1 - \delta/\Delta m_o}$$

where

$$m_o = C_c/C_h$$

Then, introducing Eq. (C-2),

$$\frac{a_2}{a_1} = \frac{1 - \varepsilon}{1 - (C_h/C_c)\varepsilon}$$

This result may be combined with Eq. (C-4a) and then rearranged to yield

$$\varepsilon = \frac{1 - e^{-N_{tu}(1-C_{\min}/C_{\max})}}{1 - (C_{\min}/C_{\max})e^{-N_{tu}(1-C_{\min}/C_{\max})}} \quad (\text{C-5})$$

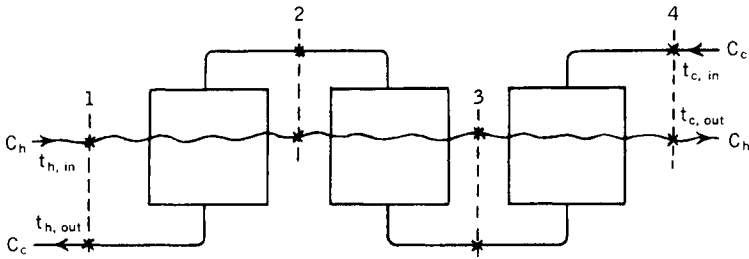
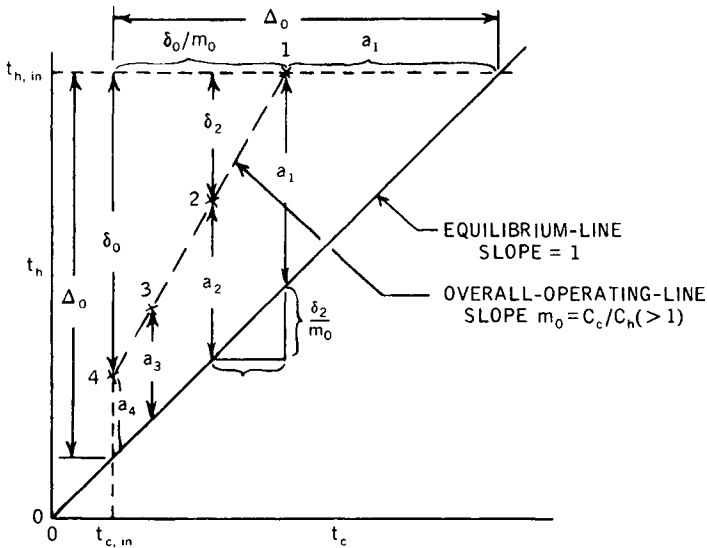
where, for this derivation, $C_{\min} = C_h$ and $C_{\max} = C_c$. However, one can demonstrate that for $C_c = C_{\min}$, the resulting equation for ε is identical to Eq. (C-5), providing only that the defining equation for ε [Eq. (2-6), Chap. 2] is properly used. Thus Eq. (2-13) of Chap. 2 has been derived.

MULTIPASS OVERALL-COUNTERFLOW EXCHANGER

In the previous demonstration the operating-line-equilibrium-line representation was useful in showing the needed algebraic relations in terms of simple geometrical relations. Nevertheless, one could have derived Eq. (C-5) with about the same ease without this representation. In the derivation of Eq. (2-18) of Chap. 2, the operating-line-equilibrium-line approach becomes of real value. Consider the three-pass ($n = 3$) overall-counterflow exchanger with $C_h = C_{\min}$ shown in Fig. C-3. Any flow configuration may be employed in any pass, providing only that each pass has the same effectiveness ε_p :

$$\varepsilon_{1-2} = \varepsilon_{2-3} = \varepsilon_{3-4} = \varepsilon_p \quad (\text{C-6})$$

Fig. C-3 Three-pass overall-counterflow exchanger.


 Fig. C-4 Operating-line and equilibrium-line representation for a multipass counterflow exchanger; $C_h = C_{\min}$, $C_c = C_{\max}$.


The operating-line–equilibrium-line representation is pictured in Fig. C-4. A broken line is used to represent the overall operating line, because the operating conditions within a pass are not necessarily those for true counterflow. Only the inlet, the outlet, and the between-pass states are assumed to correspond to counterflow conditions.

It is evident from the representation that

$$\varepsilon = \frac{\delta_o}{\Delta_o} \quad (C-7)$$

Now, as

$$\frac{a_4}{a_1} = \frac{\Delta_o - \delta_o}{\Delta_o - \delta_o/m_o} = \frac{1 - \varepsilon}{1 - \varepsilon/m_o}$$

$$\varepsilon = \frac{1 - a_4/a_1}{1 - (1/m_o)(a_4/a_1)} \quad (\text{C-8})$$

It remains to express $a_4/a_1 = \Phi(\varepsilon_p, m_o)$, and this can be achieved as follows: It is evident that ε_{1-2} bears the same relation to δ_2 and $\delta_2 + a_2$ as ε bears to δ_o and $\Delta_o = \delta_o + a_4$. Consequently, from Eq. (C-8), we may write immediately,

$$\varepsilon_{1-2} = \frac{1 - a_2/a_1}{1 - (1/m_o)(a_2/a_1)}$$

Similarly, for ε_{2-3} and ε_{3-4} ,

$$\varepsilon_{2-3} = \frac{1 - a_3/a_2}{1 - (1/m_o)(a_3/a_2)}$$

$$\varepsilon_{3-4} = \frac{1 - a_4/a_3}{1 - (1/m_o)(a_4/a_3)}$$

From Eq. (C-6) and the above, it is clear that

$$\frac{a_2}{a_1} = \frac{a_3}{a_2} = \frac{a_4}{a_3} = \frac{1 - \varepsilon_p}{1 - \varepsilon_p/m_o} \quad (\text{C-9})$$

It follows that

$$\frac{a_4}{a_1} = \frac{a_4}{a_3} \frac{a_3}{a_2} \frac{a_2}{a_1} = \left(\frac{1 - \varepsilon_p}{1 - \varepsilon_p/m_o} \right)^3 \quad (\text{C-10})$$

This result, substituted into Eq. (C-8), yields, with some rearrangement

$$\varepsilon = \frac{[(1 - \varepsilon_p/m_o)/(1 - \varepsilon_p)]^3 - 1}{[(1 - \varepsilon_p/m_o)/(1 - \varepsilon_p)]^3 - 1/m_o} \quad (\text{C-11})$$

A generalization of this three-pass exchanger result for n passes, and for the case of either $C_h = C_{\min}$ or $C_c = C_{\min}$, leads to Eq. (2-18) of Chap. 2.

Note that this equation becomes indeterminate for the case of $C_{\min}/C_{\max} = 1$. Application of L'Hospital's rule will lead to Eq. (2-18a), however, or a direct development can be made from a suitable operating-line–equilibrium-line representation.

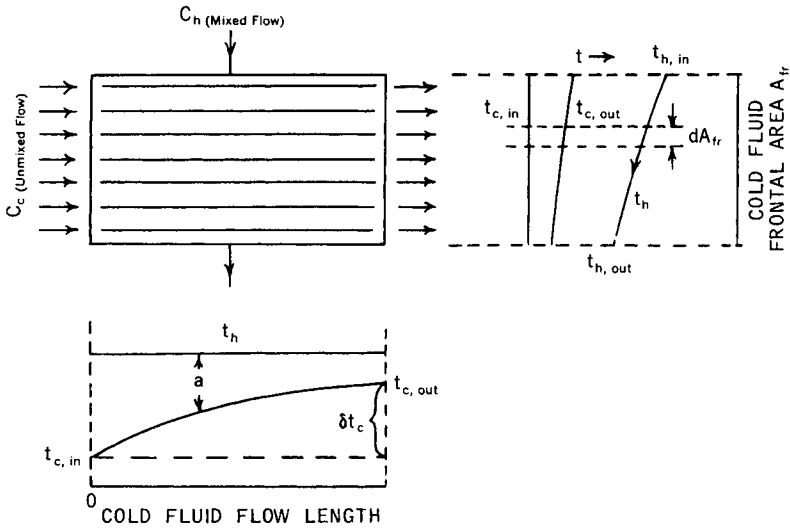
CROSSFLOW, ONE FLUID "MIXED"

This flow arrangement and the idealized temperature conditions are pictured schematically in Fig. C-5.

Uniform flow distribution for the cold stream will be specified, so that with respect to the frontal area A_{fr} ,

$$\frac{dC_c}{dA_{fr}} = \frac{C_c}{A_{fr}} = \text{const} \quad (\text{C-12})$$

Fig. C-5 Temperature conditions for crossflow, one fluid mixed, one unmixed; $C_{\text{mixed}} (= C_h) < C_{\text{unmixed}} (= C_c)$.



Moreover, a uniform distribution of the heat transfer area A will be assumed, so that

$$\frac{dA}{dC_c} = \frac{A}{C_c} \quad (\text{C-13})$$

In both cases dC_c is the differential capacity rate of the cold fluid associated with the cold-fluid frontal area dA_{fr} and the heat transfer surface dA .

Inspection of the figure for temperature conditions as a function of cold-fluid flow length suggests that a *condenser* type of effectiveness expression is applicable [see Eq. (2-13a) or (2-14a), Chap. 2].

Then, defining Γ as

$$\Gamma = \frac{\delta t_c}{t_h - t_{c, \text{in}}}$$

one may write

$$\Gamma = 1 - e^{-(UdA/dC_c)}$$

and, by Eq. (C-13),

$$\Gamma = 1 - e^{-AU/C_c} = \text{const} \quad (\text{C-14})$$

An energy balance for the hot fluid flowing over the heat transfer area associated with dA_{fr} results in

$$dq = -C_h dt_h$$

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Equation (C-14) may be employed to get another expression for dq , namely,

$$dq = \Gamma(t_h - t_{c,in})dC_c$$

Combination of the two dq expressions and separation of variables lead to

$$\frac{dt_h}{t_h - t_{c,in}} = -\Gamma \frac{dC_c}{C_h} = -\Gamma \frac{C_c}{C_h} \frac{dA_{fr}}{A_{fr}}$$

where C_c , C_h , and A_{fr} are the total magnitudes and are not variables. Integration then yields

$$\frac{t_{h,out} - t_{c,in}}{t_{h,in} - t_{c,in}} = e^{-\Gamma(C_c/C_h)} \quad (C-15)$$

Since for the case at hand $C_h = C_{min}$, it follows from the defining equation for ε [Eq. (2-6)] that

$$\varepsilon = \frac{t_{h,in} - t_{h,out}}{t_{h,in} - t_{c,in}}$$

Then from Eq. (C-15),

$$\varepsilon = 1 - e^{-\Gamma(C_c/C_h)} \quad (C-16)$$

This expression corresponds to Eq. (2-15) with

$$C_c = C_{max} = C_{unmixed} \quad \text{and} \quad C_h = C_{min} = C_{mixed}$$

and Γ from Eq. (C-14) becomes

$$\Gamma = 1 - e^{-N_{tu}(C_{min}/C_{max})}$$

With a specification of $C_h = C_{max} = C_{mixed}$, one can start with Eq. (C-15) and the appropriate definition of ε ,

$$\varepsilon = \frac{C_h(t_{h,in} - t_{h,out})}{C_c(t_{h,in} - t_{c,in})}$$

and arrive at Eq. (2-16).

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